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Improving thermal resilience to overheating against heatwaves in residential and small non-residential buildings

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Samenvatting

Klimaatverandering draagt bij aan vaker voorkomende en ernstigere hittegolven, waardoor het risico op oververhitting in gebouwen toeneemt. Oververhitting brengt ernstige gezondheidsrisico's met zich mee voor bewoners en gebruikers, waaronder hittestress, uitdroging, levensbedreigende aandoeningen zoals een hitteberoerte en zelfs sterfte. Om deze gezondheidsrisico's te beperken, is het van cruciaal belang om thermische veerkracht tegen oververhitting op te nemen als een belangrijke prestatie-eis in het ontwerp van gebouwen. In het kader van IEA EBC Annex 80 - Veerkrachtige koeling van gebouwen, verwijst thermische veerkracht naar het vermogen van een gebouw om thermische schokken te weerstaan en er tijdig van te herstellen, waarbij de ontworpen/ten minste bewoonbare thermische condities behouden blijven tijdens/na de schok.

Verschillende studies hebben indicatoren en beoordelingskaders voor thermische veerkracht ontwikkeld, maar deze benaderingen hebben beperkingen. De huidige indicatoren, zoals overschrijdingsuren richten zich voornamelijk op langdurige oververhitting en missen drempelwaarden voor kortere schokken. Ze zijn vaak afhankelijk van afwijkingen in de binnentemperatuur zonder rekening te houden met andere factoren die hittestress bij bewoners beïnvloeden. Hoewel hittestress indicatoren zoals de Standaard Effectieve Temperatuur (SET) graaduren thermische veerkracht koppelen aan de gezondheid van de bewoner en drempelwaarden bevatten voor jonge, gezonde of oudere, kwetsbare bewoners, moeten ze nog verder worden verfijnd om schokken en gebouwontwerpparameters te integreren. Kwalitatieve indicatoren (bijv. absorberend, adaptief en herstellend vermogen, en herstelsnelheid) beoordelen de veerkracht in verschillende fasen, terwijl kwantitatieve indicatoren (thermische autonomie, passief overlevingsvermogen) de veerkracht voor kortere schokken evalueren. In beide gevallen ontbreekt het echter aan duidelijke, kwantificeerbare drempelwaarden. Bestaande beoordelingskaders houden geen rekening met verschillende soorten schokken (bijv. hittegolven, installatiedefecten) of koppelen veerkracht niet aan ontwerpparameters van gebouwen. Bovendien vereisen de huidige methoden gedetailleerde simulaties, wat onpraktisch is voor het ontwerpen van gebouwen in een vroeg stadium. Er is behoefte aan een alomvattend beoordelingskader voor thermische veerkracht dat: (a) de veerkracht van verschillende soorten schokken beoordeelt, (b) de veerkracht koppelt aan kritieke ontwerpparameters, (c) de gezondheid en het comfort van de bewoners integreert door middel van drempelwaarden voor hittebelasting, en (d) toegankelijk en gebruiksvriendelijk is voor architecten en ontwerpers. Dit raamwerk zou de evaluatie van thermische weerstand in verschillende gebouwontwerpen mogelijk moeten maken en effectieve ontwerpwijzigingen om de weerstand te verbeteren moeten benadrukken.

Als eerste stap naar de ontwikkeling van een allesomvattend beoordelingskader voor thermische veerkracht en het koppelen van de beoordeling van thermische veerkracht aan schokken, werd een parametrische studie uitgevoerd met behulp van toekomstige weergegevens voor hittegolven, gebaseerd op het RCP8.5 klimaatscenario. De studie richtte zich op extreme hittegolven in klimaatzone 4A (Gent), met hittegolven van

verschillende duur en intensiteit. Door de mate van schok te kwantificeren, vergeleek de studie de impact van verschillende soorten schokken, waaronder externe schokken (hittegolven) en systeemschokken (bv. falen van koelsystemen, falen van zonnewering, falen van natuurlijke nachtventilatie). De bevindingen toonden aan dat hittegolven het meest kritieke type schokken zijn voor gebouwen, omdat ze het effect van passieve koelstrategieën ondermijnen en leiden tot langdurig ongemak voor bewoners.

De tweede stap in de ontwikkeling van een allesomvattend beoordelingskader van de thermische veerkracht is de identificatie van de meest invloedrijke ontwerpparameters (van zowel gebouwen als systemen) die de thermische veerkracht tijdens extreme hittegolven beïnvloeden. Een onzekerheidsanalyse onthulde de variatie in thermische veerkracht als gevolg van variaties in het ontwerp. Een gevoeligheidsanalyse identificeerde dat ontwerpparameters, zoals de ratio venster-geveloppervlakte, de koelcapaciteit en de werking van passieve strategieën (bijv. natuurlijke nachtventilatie, zonwering), een aanzienlijke invloed hebben op de thermische veerkracht van een gebouw.

Uiteindelijk werd, als laatste stap in de ontwikkeling van een uitgebreid beoordelingskader voor thermische veerkracht, een indicator voor thermische veerkracht ontwikkeld. Deze indicator werd ontwikkeld door dynamische simulaties, gevoeligheidsanalyses en regressiemodellen te combineren. Er werd een beslissingsboom ontwikkeld om systematisch de thermische veerkracht te beoordelen en de SET graaduren te voorspellen voor a) diverse gradaties van schokken (van hoog over gemiddeld tot laag), b) verschillende gebouwtypes (woon- en kantoorgebouwen), c) gradaties in thermische massa (zwaar, gemiddeld en licht) en d) verschillende koelstrategieën (zonwering, natuurlijke nachtventilatie, koeling). Hierbij werd ook rekening gehouden met invloedrijke ontwerpparameters zoals de ratio venster-geveloppervlakte, de eigenschappen van de gebouwschil en beglazing, ventilatiedebiet, enz. Voor elke tak in de beslissingsboom werd een regressiemodel opgesteld. Dit biedt een gebruiksvriendelijk kader voor het beoordelen van de thermische veerkracht van een gebouw tijdens de vroege ontwerpfase.

De voorspellende capaciteit van de ontwikkelde indicator voor thermische veerkracht werd geëvalueerd aan de hand van simulaties van woon- en kantoorgebouwen uit de praktijk. Hieruit bleek dat de ontwikkelde indicator nauwkeurig is om te voorspellen of een gebouw al dan niet veerkrachtig is tegen oververhitting bij een hittegolf. Dit uitgebreide beoordelingskader voor thermische veerkracht biedt een vereenvoudigd maar robuust kader voor architecten en ontwerpers om de thermische veerkracht in de vroege ontwerpfasen te beoordelen.

Summary

Climate change is contributing to more frequent and intense heatwaves, which increase the risk of indoor overheating in buildings. Overheating risk poses serious health risks to occupants, including heat stress, dehydration, life-threatening conditions like heatstroke and even mortality. To mitigate these health risks, it is crucial to incorporate thermal resilience to overheating as a key performance indicator in building design. Within the framework of IEA EBCs Annex 80- Resilient Cooling of Buildings, thermal resilience refers to a building's ability to withstand and recover in timely manner from thermal shocks, maintaining its designed/at least habitable thermal conditions during/post shock period.

Several studies have developed thermal resilience indicators and assessment frameworks, but these approaches have notable limitations. Current indicators, such as exceedance hours, unmet-degree hours, and indoor overheating degree, primarily focus on long-term overheating and lack defined thresholds for shorter shocks. They tend to rely on indoor temperature deviations without accounting for other factors that influence occupant heat stress. While heat-stress indicators like Standard Effective Temperature degree.hours (SET-Dh) link thermal resilience to occupant health and include thresholds for young, healthy, or older, vulnerable occupants, they still need further refinement to integrate shock impacts and building design parameters. Qualitative indicators (e.g., absorptive, adaptive, and restorative capacities, recovery speed) assess resilience across phases, while quantitative indicators (thermal autonomy, passive survivability) evaluates resilience for shorter shocks. However, both lack clear, quantifiable thresholds. Existing frameworks fail to account for different shock types (e.g., heatwaves, system failures) or lacks link resilience to building design parameters. Furthermore, current methods require detailed simulations or data analysis, which are impractical for early-stage building design. A comprehensive thermal resilience framework is needed—one that: (a) assesses resilience across various shock types, (b) links resilience to critical design parameters, (c) incorporates occupant health and comfort through heat-stress thresholds, and (d) is accessible and user-friendly for architects and designers. This framework should enable the evaluation of thermal resilience across different building designs and highlight effective design changes to improve resilience.

As first step towards developing a comprehensive thermal resilience assessment framework and linking thermal resilience assessment to shocks, a parametric study was conducted using future weather data for heatwaves, based on the RCP8.5 climate scenario. The study focused on extreme heatwaves in climate zone 4A (Ghent), with heatwaves of varying durations and intensities. By quantifying the degree of shock (doS), the study compared the impact of different types of shocks, including external (heatwaves) and system shocks (e.g., cooling system failure, solar shading system failure, natural night ventilation failure). The findings revealed that heatwaves are the most critical shock type for buildings, as they overwhelm passive cooling strategies and lead to prolonged occupant discomfort.

Once the most critical shock type has been identified, as second step towards development of a comprehensive thermal resilience assessment framework, it is crucial to identify the most influential design parameters (both buildings' and systems') that impact thermal resilience during the extreme heatwaves. An uncertainty analysis revealed the variation in thermal resilience performance due to design variations and sensitivity analysis identified that design parameters, such as window-to-wall ratio, cooling capacity, and the operation of passive strategies (e.g., natural night ventilation, solar shading), significantly impact a building's thermal resilience.

Ultimately, as the final step towards development of a comprehensive thermal resilience assessment framework, a thermal resilience indicator (TRI $_{SET\text{-}Dh}$), was developed to assess and improve residential and office buildings' thermal resilience to overheating during heatwaves. TRI $_{SET\text{-}Dh}$ was developed by combining dynamic simulations, sensitivity analysis, and regression models. A decision tree was developed to systematically assess thermal resilience and predict SET-Dh across –(a) multiple shock severity (high, medium and low),(b) building type (residential and office), (c) thermal mass (heavy, medium and light), and strategies (solar shading, natural night ventilation, cooling), and incorporating influential design parameters like window-to wall ratio, envelope and glazing properties, ventilation rates etc. TRI $_{SET\text{-}Dh}$ is based on a decision tree and regression models based on each branch of the decision tree providing a user-friendly framework for assessing a building's thermal resilience during the early design phase. The predictive capacity of the developed TRI $_{SET\text{-}Dh}$ was evaluated against simulations of real-life cases of residential and office buildings to test the accuracy of the developed TRI $_{SET\text{-}Dh}$. Key findings showed that buildings optimized with balanced window-to-wall ratios, thermal mass, solar shading, and ventilative cooling performed better during heatwaves, with a combination of passive and active cooling strategies providing the highest thermal resilience. This comprehensive thermal resilience assessment framework offers a simplified yet robust framework for architects and designers to assess thermal resilience in the early stages of design.

Nomenclature

Abbreviations

ACF	Air conditioning failure
AHU	Air handling unit
BMS	Building monitoring system
BPS	Building performance simulation
CCOR	Climate change overheating resistivity
C3S	Copernicus climate change service
CORDEX	Coordinated regional climate downscaling experiment
DCV	Demand controlled ventilation
DWD	Deutscher Wetterdienst
EBC	Energy in Buildings and Communities Programme
ECMRWF	European Centre for Medium-Range Weather Forecasts
EPBD	Energy Performance of Buildings Directive
ES	External shock
EU	European union
HHIH	Heat index hazard hours
HPC	High performance computing
HVAC	Heating ventilation air conditioning
HW	Heat wave
IDEAS	Integrated district energy assessment by simulation
IEA	International Energy Agency
IEC	Indirect evaporative cooling
IPCC	Intergovernmental panel on climate change
KPI	Key performance indicator
LHS	Latin hypercube sampling
MAE	Mean absolute error
MCA	Monte Carlo Analysis
MET	Metabolic rate
NOAA	National oceanic and atmospheric administration
NNV	Natural night ventilation
NS	No shock
nZEB	Net zero energy building
OEF	Overheating escalation factor
OTR	Office thermal resilience
PI	Proportional integral
PHS	Predicted heat strain
PMV	Predicted mean vote
PO	Power outage
PPD	Predicted percentage dissatisfied
PS	Passive survivability
PTS	Predicted thermal sensation

pyDOE	Design of experiments for Python
RCM	Regional climate model
RCP	Representative concentration pathways
RMSE	Root mean square error
RO	Research objectives
RSS	Residual sum of squares
SA	Sensitivity analysis
SF	Shading failure
SRC	Standardized regression coefficient
TA	Thermal autonomy
TLR	Test lecture rooms
TMY	Typical meteorological year
TRI	Thermal resilience index
TZ	Thermal zones
UA	Uncertainty analysis
VEKA	Flemish energy & climate agency
VF	Ventilation failure
WCRP	World climate research program
WMO	World meteorological organization
WHO	World health organization
XMY	Extreme meteorological year

Parameters

AWD	Ambient warmness degree [°C]
A_{window}	Effective opening area of window for NNV operation as [%] of floor area
CDD	cooling degree.days [°C.days]
CLO	clothing
CO_2	Carbon Dioxide
Dh	degree.hours [°C.h]
doS	Normalized degree of shock [-]
doES	Normalized degree of external shock [-]
doSS	Normalized degree of system shock [-]
EH	Exceedance hours [h]
HI	Heat index [°C]
$H_{comfortable}$	number of hours indoor temperature is within the defined comfort range [h]
H_{total}	Total number of occupied hours in a given time period [h]
IOD	Indoor overheating degree [°C]
Q_c	Cooling capacity [W/m^2]
RH	Relative humidity [%]
RH_{TZ}	Relative humidity of the thermal zone [%]
SHGC	Solar heat gain coefficient [-]
SET	Standard effective temperature [°C]
SET-Dh	Degree.hours above $SET_{Threshold}$ [°C.h]

$SET_{threshold}$	Threshold value of SET[°C]
SET (t)	SET at time t[°C]
S_t	Shading Threshold [W/m²]
$\overline{T_{oa,ES}}$	Average outdoor dry bulb temperature during external shock [°C]
$T_{oa,TMY}$	Average outdoor dry bulb temperature during typical meteorological year [°C]
$\overline{T_{ra,NS}}$	Average zone air temperature during no shock [°C]
$\overline{T_{ra,VF}}$	Average zone air temperature during ventilation failure [°C]
T_{ref}	Average indoor/ outdoor temperature during no shock period [°C]
T_{shock}	Average indoor/ outdoor temperature during shock [°C]
$\overline{T_{sa,CSF}}$	Average supply air temperature during cooling system failure [°C]
$T_{sa,NS}$	Average supply air temperature during no shock [°C]
$T_{set\ cool}$	Cooling set point [°C]
T_{TZ}	Temperature of the thermal zones [°C]
$\overline{T_{wn,NS}}$	Average window internal surface temperature during no shock [°C]
$T_{wn,SF}$	Average window internal surface temperature during shading failure [°C]
t_{abs}	Absorptivity time [h]
t_{rec}	Recovery time [h]
t_{op}	Total operational hours of the system in one day [h]
t_{shock}	Shock duration [h]
$t_{shock\ max}$	Duration of the longest shock [h]
UDH	Unmet degree.hours [°C.h]
UTCI	Universal thermal climate index [°C]
$U_{glazing}$	U-value of glazing [W/m²K]
U_{wall}	U-value of wall [W/m²K]
WBGT	Wet-bulb globe temperature [°C]
WWR	Window-to-wall ratio [%]

Symbols

$\propto$	constant weighting factor
$\propto_{IOD}$	Overheating escalation factor
i	Occupied hour counter
ρ	Density [kg/m³]
η	Efficiency [%]
n50	Airtightness [1/h]
N_{occ}	Occupied period
θ	Orientation[°]
T	Temperature [°C]
t	Time [h]
Δ	Difference
z	Zones
TL_{comf}	Thermal comfort temperature limit
$\dot{V}$	Ventilation rate [m³/h]
Z	Total number of zones in the building

Subscripts

a	air
avg	average
comf	comfort
ed	mean outdoor temperature
fr	free running
max	maximum
min	minimum
oa	outdoor air
op	operational (without any shock/interruption)
ra	room air
rmo	running mean outdoor temperature
sa	supply air
TZ	thermal zone
wn	window
z	zone counter

Chapter 1: Introduction

Chapter 1 lays the foundation of this dissertation by summarizing the state-of-the-art and providing the necessary background and scope of the research. Through this analysis, key research gaps are identified, which directly lead to the formulation of the research objectives. The chapter concludes with a concise overview of the thesis structure, setting the stage for the subsequent chapters.

1.1 Climate change, heatwaves and related mortality in Europe

Climate change and extreme heat are critical issues faced globally. Globally, 2024 was 0.72°C warmer than the 1991–2020 average, and 1.60°C warmer than the pre-industrial level, making it the first calendar year to exceed 1.5°C above that level [1]. Since the 1980s, Europe has been warming at twice the global average rate, making it the fastest-warming continent on Earth [2]. Europe has witnessed in the past two decades, 18 of the warmest years and an increase in the frequency and severity of heatwaves[3]. The Intergovernmental Panel on Climate Change (IPCCs) Special Report on global warming of 1.5°C foresees an increase in severity and longevity of heatwaves[4]. These prolonged and intense heatwaves are emerging as a significant natural hazard, posing substantial risks and challenges to human health, buildings, and building-related systems. Since 1970, extreme heat has been the leading cause of weather- and climate-related deaths in Europe [2]. Of the 30 deadliest heatwaves, 23 have occurred since 2000, including five in the past three years. Heatwaves in 2003 [5], 2019 [6] and 2022[7] each claimed an estimated 55,000 to 72,000 lives. Heat-related deaths in the World Health Organizations'(WHOs') European Region have surged by 30% in the last 20 years[2]. By 2050, such heatwaves could become more frequent, potentially raising heat-related mortality by 257% [8].

1.2 Overheating risk in buildings and health impacts on occupants

During these extreme heatwaves, buildings need to shelter occupants from the outdoor environment, making it crucial for buildings to sustain habitable thermal conditions. Currently to reduce carbon emissions and combat climate change, buildings often implement energy-efficient measures. Energy efficient year-round solutions include having highly insulated and airtight envelopes [9,10], improved glazing [11], and high insulation [10]. For the cooling season, solar shading [12] and cooling systems (such as such as ground source cooling [13], natural night ventilation (NNV) [14] and evaporative cooling [15]) are installed. The main goal of energy efficient buildings are to meet thermal comfort needs at reduced operational energy costs [16]. However, studies showed that overheating risk has become frequent in these energy efficient buildings during heatwaves[17]. Ren et al. [18] stated that an "over emphasis on energy efficiency" can undermine a building's capacity to maintain comfortable indoor temperatures during heatwaves, as features such as high insulation and airtight construction may increase the risk of overheating. Overheating risks in buildings, leading to higher cooling demands, are mainly influenced by building design parameters (like glazing ratio, high insulation rate, increased airtightness, internal heat gains and energy performance) [19][20]. A study on Dutch dwellings shows poorly ventilated dwellings are vulnerable to overheating and are the most sensitive to climate change, particularly if their windows are not well protected against direct solar radiation [21]. Overheating risk in buildings significantly impacts human health by elevating core-body temperatures and disrupting essential bodily functions, leading to heat-related illnesses such as sleep deprivation, heat stroke, heat-stress and even mortality [22][23]. Highly insulated and airtight buildings are prone to overheating; however, these buildings can mitigate overheating through strategies like solar

shading, ventilation, and night cooling, demonstrating that energy-efficient buildings can achieve both low energy use and mitigate overheating risk with proper design.

1.3 Thermal resilience to overheating

Buildings are typically designed based on expected weather patterns and operational conditions, such as occupancy loads and solar heat gains. However, throughout their lifespan, they often face sudden shocks and unforeseen events, like heatwaves, that disrupt indoor thermal conditions and lead to overheating [24,25], posing significant risks to occupants' health [26,27]. Despite the growing threat of climate change, designs fail to address the escalating risks of overheating, leaving buildings ill-equipped to ensure safe and comfortable conditions during extreme heat events. The rising frequency and intensity of heatwaves underscore the urgent need to enhance the thermal resilience of buildings to overheating to protect occupants and reduce health-related impacts. Resilience is a concept widely applied in disaster risk management. It refers to "the ability of a system and its components parts to anticipate, absorb, accommodate, or recover from the effects of a hazardous event in a timely and efficient manner, including through ensuring the preservation, restoration, or improvement of its essential basic structures and functions", as defined by IPCC [28]. Thermal resilience of buildings is defined as an ability of the building and its systems to withstand disruptions caused by extreme weather events (such as heatwaves and cold snaps), power failure, change in use (increase or decrease of occupancy and internal heat gains) and atypical conditions (failure of solar shading, cooling systems, ventilation system etc.); to maintain the capacity to function and recover to its designed/at least habitable thermal conditions in a timely manner without causing thermal-stress on occupants[29][30]. Thus, apart from energy performance, thermal resilience to overheating is gaining importance to assess building performance [31][32] and should be considered as a primary function of the building [33].

1.4 Thermal resilience assessment methods

In recent years, multiple studies focused on presenting the concept of thermal resilience to overheating and evaluating buildings' thermal resilience. Miller et al. [34] offered a model focused on resilient cooling for people and buildings, while Hong et al. [35] identified 10 key questions for advancing thermal resilience research across technological, social, and policy dimensions. Tavakoli et al. [36] reviewed cooling strategies in low-energy buildings, emphasizing the need for more research on resilience of passive cooling.

Numerous studies assessing and improving thermal resilience of buildings to overheating focused on-(a) developing qualitative aspects of thermal resilience, (b) developing assessment frameworks to evaluate thermal resilience of buildings and (c) evaluating thermal resilience using existing heat-stress, thermal resilience and overheating indicators.

Studies focusing on developing qualitative aspects of thermal resilience, for instance, Zhang et al. [37] outlined four resilience criteria—absorptive, adaptive, restorative capacity, and recovery speed—employing them to qualitatively assess various cooling

strategy's resilience performance. However, this approach missed quantification and determination of standardized thresholds of qualitative thermal resilience aspects. Development of assessment frameworks as done by Homaei et al.[38]_introducing a benchmarking framework and multi-phase metric for quantifying thermal resilience during cold snaps and power outage demonstrating the significant impact of design upgrades and resilience enhancement strategies. Ji et al. [39] extended this methodology for heatwaves and developed a Thermal Resilience Index (TRI) framework using the Standard Effective Temperature (SET) to link the thermal resilience performance with occupants' heat stress. These frameworks by Homaei et al.[38] and Ji et al.[39] did not focus on linking thermal resilience assessment to building and system design parameters and needs detailed simulation for assessing thermal resilience thus making them non applicable in early design stage. A simulation study by Ismail et al. [40] developed a simplified metric model for office thermal resilience (OTR) that can quantify the thermal resilience of non-residential buildings during power disruptions and reflects the influence of building design parameter on thermal resilience. While this assessment framework used a simple regression modelling technique to assess thermal resilience, it did not link thermal resilience to heat-stress and cannot be used to assess thermal resilience for different intensity of shocks. Furthermore, several studies [41][42][43][44][45] evaluated buildings' thermal resilience during various shocks (such as heatwaves and power outages) using existing heat-stress, overheating and thermal resilience indicators. Sengupta et al. [46] attempted to quantify the recovery of a Belgian dwelling with passive strategies such as natural night ventilation (NNV) and solar shading during a heatwave. Another study on an educational building with indirect evaporative cooling (IEC) highlighted the uncertainty in recovery speed[47]. The existing overheating indicators used in these studies highlight a gap in standardized threshold for short shocks. These indicators focus on long-term overheating and do not include heat-stress of occupants.

In summary, the key challenges and research gaps in existing thermal resilience assessment methods-(a) currently there is no consensus on thermal resilience assessment methods. Studies evaluating thermal resilience often use existing long-term overheating indicators that do not capture the buildings response to sudden short-term shocks and have no thresholds defined to assess thermal resilience during these short-term shocks (b) often thermal resilience assessment do not include heat-stress indicators that is crucial to evaluate the impact on occupants' health (c) thermal resilience assessment frameworks do not include impact of building and system design parameters and (d) evaluating thermal resilience is not considered during early design phase (crucial to integrate thermal resilience to climate-proof buildings and systems) due to the complexity of calculation of the existing assessment methods which require detailed dynamic building performance simulation and analysis.

These gaps in current thermal resilience assessment methods highlight a need for a comprehensive thermal resilience assessment framework that captures the building's response to sudden thermal shocks, incorporate heat-stress indicators to evaluate occupants' health, integrate design (buildings' and systems') parameters, and provide tools for evaluating resilience early in the design phase to ensure buildings are climate-

proof and adaptable to future extreme conditions. This research focuses on residential and small non-residential buildings, which are more prone to overheating due to higher occupant density and limited cooling infrastructure. These building types are crucial for societal well-being, directly affecting the comfort, health, and productivity of a substantial portion of the population.

1.5 Research Objectives (ROs)

The main objective of this research is to gain insight, assess, and improve thermal resilience to overheating of residential and small non-residential buildings. Four key research objectives (ROs) have been identified, with each aligning with the chapters (2 to 5) of this dissertation:

RO 1: *Gain insight into thermal resilience of buildings to overheating*
The first objective is to understand the concept of thermal resilience to overheating in buildings and critically evaluate existing assessment methods.

RO 2: *Identify and quantify shocks that impact buildings' thermal resilience performance.*
The second objective is to identify and quantify the shocks that impact buildings' thermal resilience to overheating. Shocks include heatwaves, power outages, and system failures (e.g., shading, ventilation, cooling). .

RO 3: *Identify the most influential design parameters of buildings and systems that impact thermal resilience of buildings to overheating*
The third objective is to identify the key design parameters of buildings' and systems', that significantly influence thermal resilience to overheating during heatwaves.

RO 4: *Develop a comprehensive thermal resilience assessment framework to assess and improve buildings' thermal resilience to overheating during heatwaves*
The fourth objective is to develop a comprehensive assessment framework to address the limitations of existing methods and improve the evaluation of residential and non-residential buildings' thermal resilience to overheating during heatwaves. It provides a standardized, step-by-step methodology for assessing resilience.

1.6 Scope of the research

The thermal resilience assessment framework developed in this research is designed to be applicable globally, provided that key inputs such as (a)weather files (especially heatwave data), (b) building type (construction method, occupancy pattern, occupant density, internal loads etc.), and, (c) temperature management strategies (such as solar shading, cooling systems, and their control mechanism) are known. However, the framework has only been evaluated in a Belgian climate context for residential and office buildings. The methodology is adaptable to different climate zones and building typologies, making it a versatile methodology for assessing and improving thermal resilience of buildings to overheating against heatwaves in various geographical contexts.

1.7 Structure of the thesis

Figure 1-1 illustrates the structure of this dissertation.

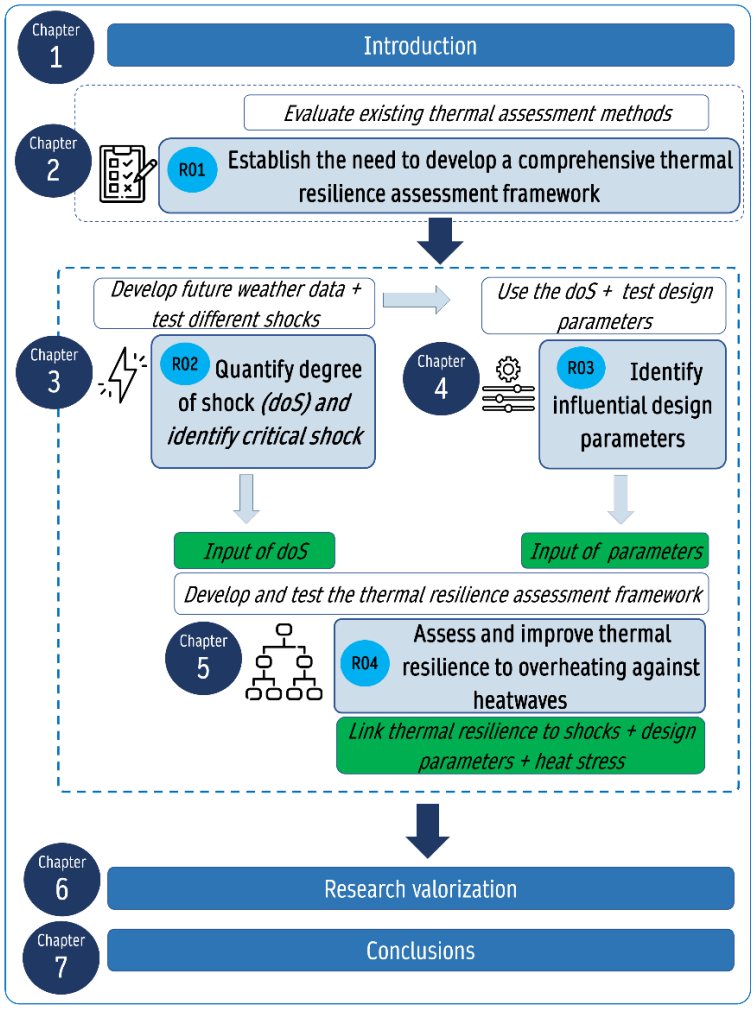


Figure 1-1. Structure of the thesis

Chapter 1 identifies the research gaps and defines the objectives of this dissertation. Chapter 2 provides a comprehensive review of the state-of-the-art. Chapters 3 to 5 focus on the development of a thermal resilience assessment framework, with each chapter addressing a specific research objective (RO 2 to 4). Chapter 6 outlines the valorization plan, and the dissertation concludes with key findings and future directions in Chapter 7.

Chapter 1: Introduction- Sets up the scene for the research by introducing briefly thermal resilience to overheating in buildings. In this chapter, the research gaps are identified, and research objectives are formulated. Finally, the structure of the thesis is laid out.

Chapter 2: Thermal resilience of buildings to overheating: concept and assessment - It examines the concept of thermal resilience to overheating in buildings and its importance as a performance indicator for building assessment. Existing assessment methods are evaluated by (a)simulation studies and (b) by evaluating how these existing assessment methods perform. Finally, their limitations are identified. This chapter aligns with RO1 and emphasizes the need for a comprehensive thermal resilience assessment framework that integrates impact of shocks, occupant heat-stress, and design parameters.

Chapter 3: Identifying and quantifying shocks that impact buildings' thermal resilience performance. This chapter aligns with RO2. This chapter establishes the first step in developing the assessment framework by linking thermal resilience to shocks. It introduces methods to identify and quantify various shocks and determines the most critical shock type, which is integrated into the framework for robust assessment.

Chapter 4: Determining the most influential design parameters that impact buildings' thermal resilience to overheating. This chapter aligns with RO3. Integrating impact of design parameters on thermal resilience assessment framework is addressed by identifying key building and system design parameters that influence thermal resilience during critical shocks, such as heatwaves and incorporates a heat-stress indicator in the framework. Chapter 4 describes step by step method to quantify the impact of the selected design parameters and combines findings from Chapter 3 (shocks) to refine the assessment framework.

Chapter 5: Development and testing of a comprehensive thermal resilience assessment framework- Chapter 5 details the development and validation of the comprehensive thermal resilience assessment framework. Testing of the framework is conducted on case study buildings for different heatwave shocks demonstrating the application of the developed framework in assessing and improving buildings' thermal resilience. This chapter aligns with RO4.

Chapter 6: Research valorization- This chapter explores the impact of the research highlighting its potential applications in building design, policy-making, and innovative solutions to enhance buildings' thermal resilience against overheating.

Chapter 7: Future outlook and conclusions- The final chapter synthesizes link of the developed framework addressing each research objective, presents the main conclusions, addresses limitations, proposes future research directions, and underscores the study's contributions to improving thermal resilience of buildings to overheating.

Chapter 2: Thermal resilience of buildings to overheating: concept and assessment

Chapter 2 aims to give insight into the concept and assessment methods of thermal resilience of buildings to overheating. The chapter begins by discussing the impacts of climate change and heatwaves on human health and underscores the importance of mitigating indoor overheating risks. The concept of resilience in built environment is introduced followed by defining the thermal resilience of buildings to overheating. A review of existing indicators for indoor overheating, heat stress and thermal resilience (including quantitative and qualitative indicators, as well as thermal resilience assessment frameworks), is conducted. Simulation study on case study buildings evaluates thermal resilience performance of the buildings and eventually assesses the existing indicators. This chapter focuses on identifying shortcomings in the existing assessment methods, leading to the proposal for a comprehensive thermal resilience framework addressing the limitations of current approaches.

This chapter is based on the following publications:

*[1] W. Miller, A. Machard, E. Bozonnet, N. Yoon, D. Qi, C. Zhang, A. Liu, **A. Sengupta**, J. Akander, A. Hayati, M. Cehlin, O.B. Kazanci, R. Levinson, Conceptualising a resilient cooling system: A socio-technical approach, City Environ. Interact. 11 (2021) 100065. https://doi.org/10.1016/j.cacint.2021.100065.*

*[2] C. Zhang, O.B. Kazanci, R. Levinson, P. Heiselberg, B.W. Olesen, G. Chiesa, B. Sodagar, Z. Ai, S. Selkowitz, M. Zinzi, A. Mahdavi, H. Teufl, M. Kolokotroni, A. Salvati, E. Bozonnet, F. Chtioui, P. Salagnac, R. Rahif, S. Attia, V. Lemort, E. Elnagar, H. Breesch, **A. Sengupta**, L.L. Wang, D. Qi, P. Stern, N. Yoon, D.I. Bogatu, R.F. Rupp, T. Arghand, S. Javed, J. Akander, A. Hayati, M. Cehlin, S. Sayadi, S. Forghani, H. Zhang, E. Arens, G. Zhang, Resilient cooling strategies – A critical review and qualitative assessment, Energy Build. 251 (2021) 111312. https://doi.org/10.1016/J.ENBUILD.2021.111312.*

*[3] **A.Sengupta**, D. Al Assaad, M. Steeman, H. Breesch, Thermal resilience to overheating assessment in a Belgian educational building with passive cooling strategies during heatwaves and power outages, E3S Web Conf. 396 (2023). https://doi.org/10.1051/e3sconf/202339601018. +*

*[4] H. Vanwynsberghe, **A. Sengupta**, H. Breesch, Evaluating thermal resilience to overheating in a Belgian apartment in shock scenarios, (2021). 42nd AIVC-10th TightVent-8th Venticool Conference-Rotterdam, Netherlands. 5-6 October,2022. +*

*[5] **A.Sengupta**, J. Kerckaert, M. Steeman, H. Breesch, Which design parameters impact the resilience to overheating in a typical apartment building? 43nd AIVC-11th TightVent-9th Venticool Conference-Copenhagen, Denmark. 4-5 October,2023 . +*

*[6] **A.Sengupta**, P. Proot, T. Trioen, H. Breesch, M. Steeman, Impact of future climate on the performances of ground-source cooling system, CLIMA 2022 : 14th REHVA HVAC World Congress, Rotterdam, Netherlands, 22-25 May 2022. https://proceedings.open.tudelft.nl/clima2022/article/view/326. +*

*[7] **A.Sengupta**, H. Breesch, D. Al Assaad, M. Steeman, Evaluation of thermal resilience to overheating for an educational building in future heatwave scenarios, Int. J. Vent. 22 (2023) 366–376. https://doi.org/10.1080/14733315.2023.2218424+*

*[8] **A.Sengupta**, J. Deleu, B. Lucidarme, M. Steeman, H. Breesch, Assessing Thermal Resilience To Overheating In An Office Building, 41st AIVC-ASHRAE IAQ 2020 Conference, Athens, Greece, 4-6 May 2022+*

*[9] **A.Sengupta**, M. Steeman, H. Breesch, Analysis of Resilience of Ventilative Cooling Technologies in a Case Study Building, ICRBE Procedia. 1 (2020) 1–10. https://doi.org/10.32438/icrbe.202041. +*

+These publications evaluated the thermal resilience of case study buildings against overheating using existing thermal resilience indicators

2.1 Introduction

While chapter 1 introduced briefly the concept of thermal resilience, chapter 2 presents the state-of-the-art of thermal resilience of buildings to overheating and current assessment methods. Chapter 1 outlines the four main research objectives (ROs). Chapter 2 focuses on RO1- to gain insight into the concept of thermal resilience of buildings to overheating and its existing assessment methods.

Section 2.2 explores the relationship between climate change and heatwaves, highlighting their adverse effects on human health, which underscores the importance of enhancing thermal resilience in buildings. While section 2.3 describes the heat related mortality and heat-stress on occupants during heatwaves, section 2.4 emphasizes the critical need to assess and mitigate indoor overheating to prevent heat-related health issues for building occupants. Sections 2.5 and 2.6 discuss the concept of resilience within the built environment, focusing on the thermal resilience of buildings and existing assessment indicators. Section 2.7 presents a simulation-based study examining various building types and systems under different shocks using existing thermal resilience assessment methods. This section also identifies the shortcomings in current thermal resilience assessment methods and outlines how a comprehensive thermal resilience framework can address these deficiencies.

2.2 Climate change and heatwaves

Climate change and associated heatwaves are critical issues faced globally. IPCC defines climate extreme as the occurrence of a value of a weather or climate variable above (or below) a threshold value near the upper (or lower) ends of the range of observed values of the variable[48]. The IPCC special report on global warming of 1.5°C [4] warns that limiting global warming to 1.5°C requires rapid, unprecedented changes in all sectors to avoid severe consequences like extreme weather (i.e. heatwaves and cold snaps), rising sea levels, and biodiversity loss, while the 2023 IPCC Synthesis Report [49] emphasizes that global temperatures have already risen by 1.1°C due to human activities through emissions of greenhouse gases. The report urges accelerated efforts to cut emissions and mitigate escalating risks to ecosystems, human health, and infrastructure. Future warming depends on greenhouse gas emissions, with human activities adding 11 billion metric tons of carbon yearly—outpacing natural absorption and raising CO_2 levels [49]. The IPCC's Representative Concentration Pathways (RCPs) [4] are four greenhouse gas concentration trajectories that models possible climate futures, based on varying levels of emissions and mitigation efforts: RCP2.6 (low emissions), RCP4.5 and RCP6.0 (medium emissions), and RCP8.5 (high emissions). RCP8.5 represents the "business-as-usual" scenario with high emissions and minimal mitigation, leading to the most severe warming and associated impacts. Figure 2-1 illustrates the different RCP scenarios and the projected global temperature increase relative to the 1901-1960 average. Recent warming is occurring at a rate significantly exceeding the long-term average, with some regions experiencing temperature increases of over 0.5°C per decade. The 2023 Global Climate Report from NOAA's National Centers for Environmental Information indicates that each month of 2023

ranked among the seven warmest for its respective month, with the second half of the year (June–December) experiencing record-breaking heat. Notably, July, August, and September saw global temperatures exceed 1.0°C above the long-term average, marking the first instance in NOAA's historical record where any month surpassed this threshold [50].

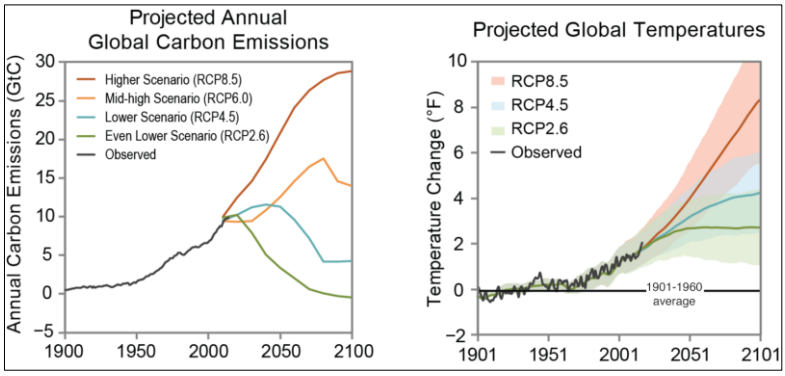


Figure 2-1. Hypothetical pathways of carbon emissions or RCPs throughout the twenty-first century based on different possible energy policies and economic growth patterns (left) and projected temperature increase relative to the 1901-1960 average depending on the different RCPs (right)(page 7 [51])

The World Meteorological Organization (WMO) warns that Europe is warming twice as fast as the global average since 1980s [52]. As seen in Figure 2-2, in 2023, Europe recorded its second-warmest year on record, with temperatures ranging from 2.48 to 2.58°C above pre-industrial levels.

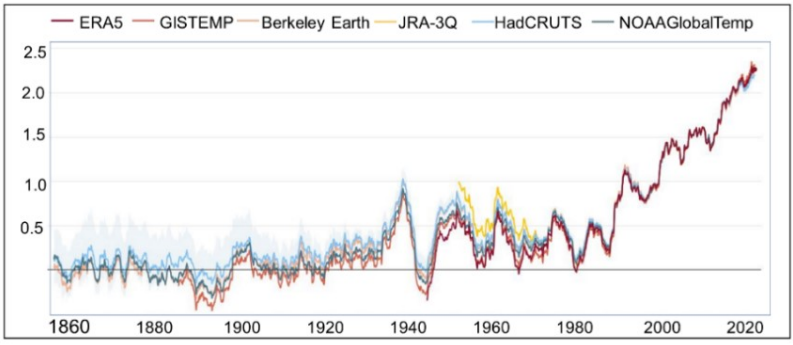


Figure 2-2. Average temperature over European land relative to the average for the 1991–2020 reference period and as an increase above the 1850–1900 average, according to six different datasets Data source: Six temperature datasets covering all or part of 1850-2023 , Credit: C3S/ECMWF (Page 23 [2])

The three warmest years for Europe have all occurred since 2020, and the ten warmest years have been recorded since 2007. Although 2023 was 0.13 to 0.17°C cooler than

the record-setting year of 2020, it still ranked among the top ten warmest years for most of the continent. Notably, many regions in southeastern, western, and central Europe experienced their warmest year on record. Overall, temperatures in Europe exceeded average levels for 11 months of the year, with September marking the highest temperatures recorded for that month [2]. As seen in Figure 2-3, if the current trend of global greenhouse gas emissions continues i.e., RCP8.5, by 2100 an increase of the outdoor temperature of approximately 5°C is expected in Belgium [53].

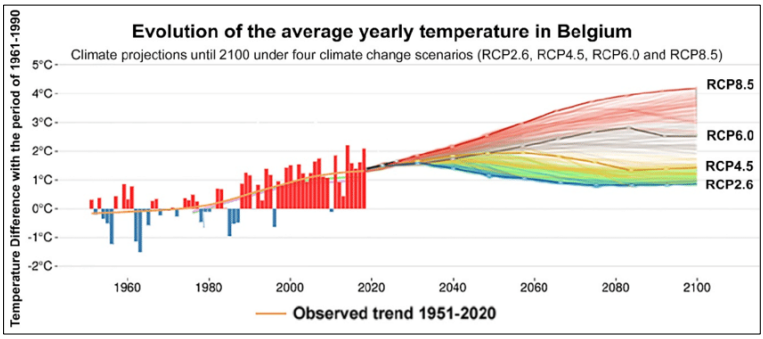


Figure 2-3. Evolution of the average annual temperature in Belgium for the different RCP scenarios (Page 2 [54]*)*

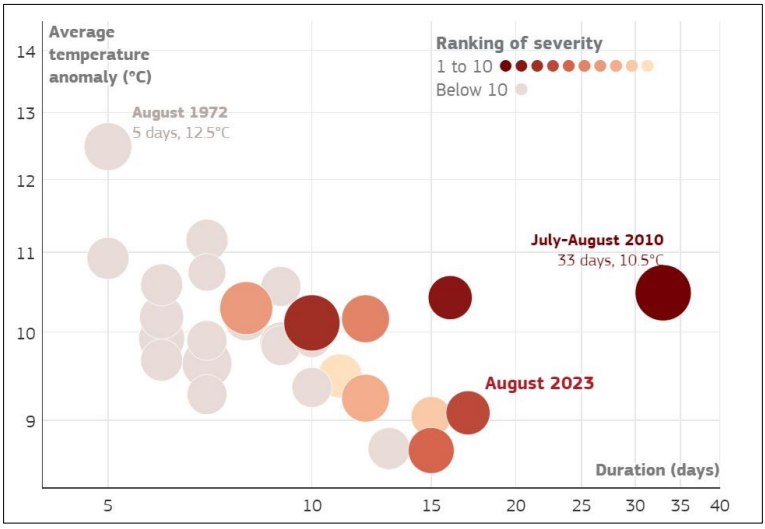


Figure 2-4. The 30 most severe heatwaves in Europe, 1950–2023. The size of a circle is proportional to the area affected by the corresponding heatwave. Heatwaves ranked as more severe are indicated by darker colors, and grey indicates those with a ranking of severity below 10, Data source: DWD , Credit: DWD/C3S/ECMWF(Page 20 [2]) [1]

[1] *Note : Data presented here are derived from the European State of the Climate 2023 report, based on findings available as of 2023. The 2024 report has not yet been released and is therefore not included in this thesis.*

Climate change has increased the frequency and severity of heatwaves, even though definitions of heatwaves vary by region, reflecting local climate and societal factors. For example, the European State of the Climate 2023 defines a heatwave as 'at least three consecutive days where daily minimum and maximum temperatures exceed the 95th percentile for that day (1991–2020 baseline)' [55], while the National Crisis Centre of Belgium defines it as 'five consecutive summer days with temperatures of at least 25°C, including three days above 30°C' [56]. Figure 2-4 shows the top 30 severe heatwaves in Europe (1950–2023), highlighting their increasing severity in recent times.

2.3 The impact of extreme weather on the human health

2.3.1 Heat-related mortality during heatwaves

The rising global temperatures and increasing frequency and intensity of heatwaves causes heat-stress and heat-related mortality. Since 1970, heatwaves have been the leading cause for climate-related deaths in Europe, with a sharp rise in fatalities observed since 2000 [2].

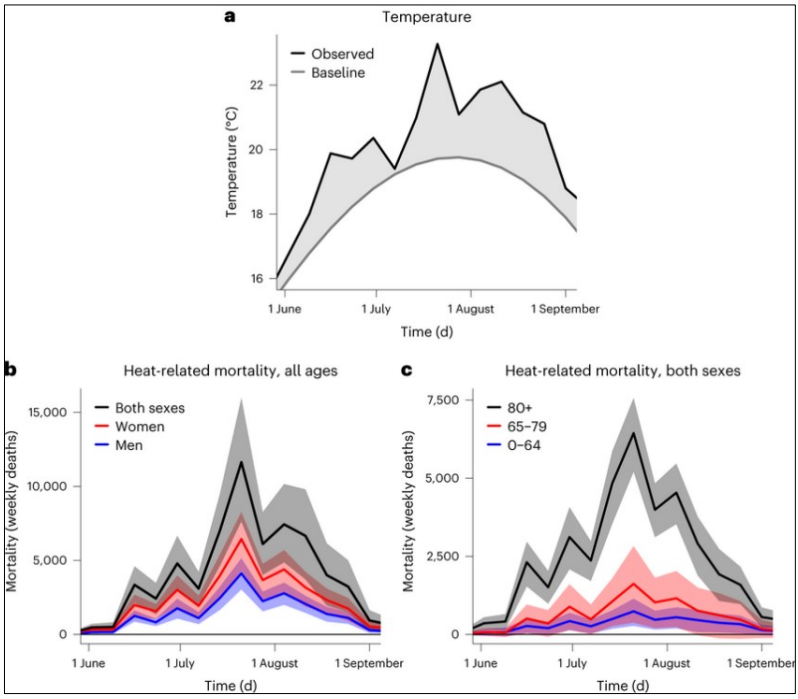


Figure 2-5. Weekly temperature and heat-related mortality numbers in Europe during the summer of 2022 (Page 1860 [7]) [2]

[2] *Note : (a) Weekly average temperatures across Europe: the grey line shows the 1991–2020 baseline, the black line shows observed values, and grey shading represents temperature anomalies (b), Weekly heat-related deaths across Europe: black shows totals, red for women, blue for men (c)blue for ages 0–64, red for 65–79, black for 80+ with 95% confidence intervals (shading).*

Heatwaves in the summers of 2003 [5], 2010 [57], 2019 [6], and 2022 [7] alone are estimated to have caused between 55000 and 72000 fatalities [58]. In July 2023, the World Health Organization's (WHOs) Europe Regional Office declared the climate crisis and heatwaves a public health emergency for the first time. Over the past 20 years, heat-related mortality has risen by approximately 30%, with 94% of nearly 1000 monitored European regions reporting an increase in heat-related deaths [59]. EuroHEAT estimated that the increase in mortality ranges from 7.6% to 33.6% during heatwaves by analyzing the impact of long heatwaves in nine European cities (Athens, Barcelona, Budapest, London, Milan, Munich, Paris, Rome and Valencia) [60]. Several other studies showed an increased mortality rate during heatwaves [56,59,61–64] and concluded that by 2050, heat-related mortality could increase by 257% [8]. For example, Figure 2-5 illustrates the heat related mortality in Europe during summer of 2022 [7].

2.3.2 Heat-stress during heatwaves

Apart from increased mortality, heatwaves impact human health by inducing heat-stress. Heat-stress of occupants' during heatwaves is becoming a critical public health issue as it exacerbates vulnerabilities in human health. Heat-stress refers to a condition in which the human body struggles to regulate its temperature due to prolonged exposure to elevated heat and environmental factors [65]. Extremely high temperatures can cause heatstroke, heat exhaustion and other heat-related diseases [66]. Heat-stress typically results from a combination of factors such as high ambient temperatures, high humidity, physical exertion, and inadequate hydration. The body's heat balance, or human heat budget, depends on the relationship between the heat produced internally and the heat exchanged with the surrounding environment. This balance is regulated by the thermoregulatory system, which includes the hypothalamus, sweat glands, circulatory system, and skin, as well as physiological factors and behaviors like hydration, activity level, rest, and clothing choices. Symptoms of heat-stress can include an elevated core body temperature, increased heart rate, rapid breathing, excessive sweating, nausea, and dizziness. In more severe cases, it can lead to heat-related illnesses such as heat exhaustion or heat stroke—a medical emergency that can be life-threatening if not treated promptly [67].

A number of studies focused on developing indicators to assess heat-stress. Apart from assessing skin and core body temperature, heart rate and hydration levels for assessing heat-stress in outdoor conditions, indicators such as the wet-bulb globe temperature (WBGT), measures the combined effects of temperature, humidity, wind speed, and solar radiation. Heat Index (HI) recommended by NOAA, combines air temperature and relative humidity in shaded areas to reflect what the temperature feels like to the human body [43]. Another key indicator is the Universal Thermal Climate Index (UTCI), which evaluates how the body perceives heat by considering air temperature, radiant temperature, wind speed, and humidity [68,69]. The UTCI provides a "feels-like" temperature and categorizes thermal stress into levels based on physiological responses, aiding in the management of heat stress in contexts like occupational settings, outdoor activities, and urban planning. Figure 2-6 illustrates the percentage of days per year with extreme heat-stress (feels like outdoor temperature exceeding

46°C) averaged over European land. Table 2-1 shows the different outdoor temperature ranges and associated heat-stress levels [67].

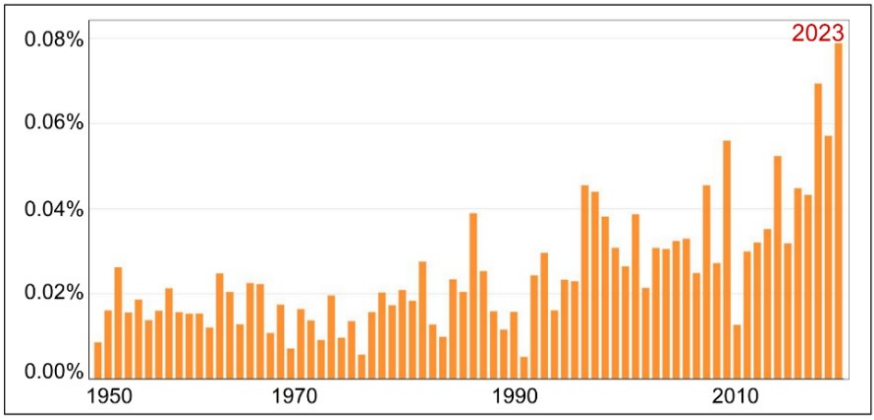


Figure 2-6. Percentage of days with 'extreme heat stress' (UTCI exceeding 46°C), for European land, for each year between 1950 and 2023 highlighting the growing prevalence of heat stress and its implications for health during heat events, Data: ERA5-HEAT daily maximum Universal Thermal Climate Index (UTCI), Credit: C3S/ECMWF [70]

Table 2-1. Outdoor temperature and the heat-stress levels (Credit: C3S/ECMWF [67])

Stress Category	Outdoor Temperature range [°C]	Physiological response
No stress	*9 -26*	*Averaged sweat rate* of >100 g h^{-1}.Plateau in rectal temperature time gradient.*
Moderate heat stress	*26-32*	*Change in sweat rate, rectal and skin temperature. Occurrence of sweating at 30 minutes. Mild stress may be experienced, but the body's ability to regulate temperature is generally effective. Individuals may feel some discomfort but can adapt without major physiological strain.*
Strong heat stress	*32-38*	*Averaged sweat rate of >200 g h^{-1}. Rectal temperature increases after 30 minutes' exposure.*
Very strong heat stress	*38-46*	*Low core-skin temperature gradient. Increase in rectal temperature after 30 minutes' exposure. Increased strain on the body's thermoregulatory mechanisms. Individuals may experience discomfort, and prolonged exposure could lead to mild thermal strain.*

Extreme heat stress	*Above 46*	*Increase in rectal temperature time gradient. Steep decrease in total net heat loss. Average sweat rate >650 g h^{-1}.Severe strain on the body's thermoregulation. This category is associated with an increased risk of heat-related health issues, and precautions are necessary to avoid heat-related illnesses and heat stroke.*

2.4 Indoor overheating risk in buildings

During heatwaves, rising temperatures within buildings elevate the indoor overheating risk, leading to heat stress for occupants [21]. This indoor overheating risk is further intensified as heatwaves often coincide with power outages due to grid strain from peak electricity demand [71][72]. Longer summers and extreme heatwaves can still put significant pressure on relatively stable and well-managed electrical grid, increasing the risk of power outages. High electricity demand for cooling, combined with potential reductions in power generation efficiency (e.g., thermal power plants struggling with cooling water shortages), can strain the grid. In European context, a study conducted by Golub et al. [73] demonstrated the vulnerability of the German power sector to climate change and provided a qualitative and quantitative analysis of emerging risks from extreme weather events (droughts and extreme temperatures). Additional study conducted by Bras et al. [74] quantified the impact of extreme weather events on power system technologies and analyzed how this impact has evolved over time (past three decades) across European regions. These studies demonstrate the potential risk of pressure on power generation systems in European context during extreme weather events such as heatwaves. Apart from investigating heat related mortality and heat-stress in outdoor conditions, there is an increasing focus on mitigating overheating risk in buildings. Heat stress experienced by occupants is directly related to the overheating risk in buildings, as excessive indoor temperatures can impair comfort, health, and productivity. When a building's thermal environment fails to control heat effectively, occupants are more likely to experience heat stress, highlighting the need for improving building envelope and HVAC systems to mitigate overheating risk in buildings. Buildings are designed to shield occupants and maintain comfortable thermal conditions during typical weather conditions. However, buildings are increasingly vulnerable to indoor overheating and rising cooling demands—even in moderate climates like Belgium [19][20]. Indoor overheating involves a complex interaction between factors such as the building and system's characteristics (envelope, glazing, insulation, ventilation, solar exposure, shading, cooling strategies and their capacity), occupant behavior, occupants' activity levels, and their susceptibility to heat stress [19][20], and create a multifaceted challenge in managing indoor temperatures and safeguarding occupant health [75].

Current building design prioritizes energy efficiency to reduce operational CO_2 emissions, but these design principles often conflict with indoor overheating risks. Highly insulated and airtight buildings are more prone to indoor overheating. A study on Dutch dwellings shows that poorly ventilated dwellings are vulnerable to indoor

overheating and are the most sensitive to climate change, particularly if their windows are not well protected against direct solar radiation [21][46]. While buildings with airtight and highly insulated envelopes delay the onset of indoor overheating, they can accumulate excess heat that cannot completely be removed by the passive cooling strategies [76]. Tillson et al. [77] predicted that with a 1.4°C rise in summer temperatures, 99% of UK homes would face medium to high indoor overheating risks. There has been an increase in research that focused on indoor overheating risk assessment through building simulation modeling [21][78][79], assessing building performance in future weather scenarios [80][81], building design scenario analysis [20][82], regulation and adaptation of building design [20][83][26], advanced technologies (passive and active) [76][84], occupant behavior [85], and post-occupancy measurements and evaluations [86]. Studies showed that indoor overheating has become a recurring problem in these highly insulated and airtight buildings with poor ventilation[71][87]. Further, studies were conducted to assess the impact of passive strategies to mitigate indoor overheating during heatwaves and the results indicate that low-energy cooling strategies that are effective today, might fail under long-term climate change or extreme thermal shocks like heatwaves and power outages [21][88][89][90]. To climate -proof the buildings against heatwaves and mitigate heat related health impacts on occupants, apart from thermal comfort and energy efficiency, it is essential to assess and improve buildings' thermal resilience to overheating.

2.5 Resilience of the built environment

Resilience is a key concept in disaster risk management, particularly regarding natural hazards like earthquakes, floods, and landslides [91][92]. According to the IPCC [64], resilience refers to "the ability of a building or system to anticipate, absorb, accommodate, or recover from shocks in a timely and efficient manner while preserving, restoring, or improving essential structures and functions." Disaster risk management for hazards such as earthquakes, fires, floods, and energy supply disruptions is integrated into the built environment through building codes and standards. For instance, as seen in Figure 2-7, a population's resilience to extreme thermal shock involves four stages: (1) exposure to the shock, (2) responding by developing coping and resistance capacities, (3) recovering from the shock, and (4) enhancing resilience by anticipating risks and adopting adaptation techniques.

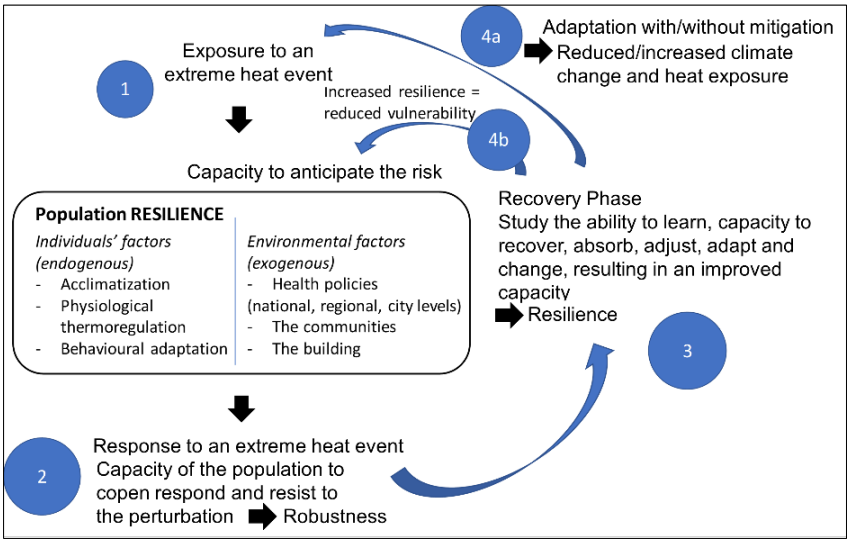


Figure 2-7. Framework for resilience to temperature hazard based on [93]

Table *2-2* lists out the definitions and the key concepts of resilience of built environment. Resilience is often associated with a response to a shock. The definitions of resilience in the context of buildings and built environment highlight several key concepts:

- Withstanding and absorbing shocks: Resilience involves the capacity of buildings and systems to endure severe weather, natural disasters, or other external stresses without immediate failure. This includes both robustness and the ability to "absorb" shocks/disturbances to minimize damage.
- Timely and efficient recovery: Resilient buildings are expected to recover functionality promptly after the shock, restoring essential functions or adapting to new conditions to prevent prolonged impacts on occupants.
- Functional continuity: A key aspect of resilience is the capacity of buildings and systems to maintain or quickly resume their core functions during and after shocks, supporting safety and usability without significant delays.
- Adaptation and flexibility: Resilient systems are not static; they must adapt to evolving threats, adjusting their operations in response to changing environmental conditions.
- Holistic integration: Resilience in buildings is not only about structural integrity but also includes the alignment of support systems (physical and institutional) and the active role of people within the environment, enabling both social and physical adaptability.

In summary, resilience involves withstanding and absorbing the shock and recovering swiftly to restore functionality under adverse conditions [62].

Table 2-2. Definitions for the concepts of resilience of built environment.

Definition	Characteristics	Threats	Scale	Year
"Building resilience is defined as a building's ability to withstand severe weather and natural disasters along with its ability to recover in a timely and efficient manner if it does incur damages." [91]	*Withstand, recover, rapidity*	*Climate and natural disasters*	*Building*	*2013*
"A resilient building is a building that not only is robust but also can fulfill its functional requirements (withstand) during a major disruption. Its performance might even be disrupted but has to recover to an acceptable level in a timely manner in order to avoid disaster impacts." [92]	*Withstand, absorb, recover, rapidity*	*Climate extremes*	*Building*	*2019*
"A resilient built environment as one designed, located, built, operated, and maintained in a way that maximizes the ability of built assets, associated support system (physical and institutional) and the people that reside or work within the built asset, to withstand, recover from, and mitigate the impacts of threats." [94]	*Withstand, recover, mitigate*	*Natural hazards (geo-hazards and hydro-meteorological hazards)*	*Built environment*	*2008*
"The intrinsic ability of the built environment to react positively before, during and after the presence of the adversely exogenous input (e.g., landslides), i.e., the ability to absorb external disturbances, in order to maintain the system's original states or reach a new set of steady states for serving its normal functionalities." [95]	*Vulnerability, adaptive capacity, recoverability*	*Geo-environmental hazards*	*Built environment*	*2017*
"Resilient urban energy system needs to be capable of "planning and preparing for", "absorbing", "recovering from", and "adapting" to any adverse events that may happen in the future. The complex, dynamic, and adaptive systems (e.g. cities) would not necessarily return to an equilibrium state." [96]	*Preparation, absorption, recovery, adaptation*	*Disruptions in energy supply*	*Urban energy systems*	*2016*
"Resilience can be understood as the ability of the system to reduce the chances of a shock, to absorb a shock if it occurs (abrupt reduction of	*Robustness, redundancy, resourcefulness, rapidity*	*Earthquake*	*Community (technical,*	*2003*

performance) and to recover quickly after a shock (re-establish normal performance).” [97]			*organization, social, economic)*	
“The ability of a system and its component parts to anticipate, absorb, accommodate, or recover from the effects of a hazardous event in a timely and efficient manner, including through ensuring the preservation, restoration, or improvement of its essential basic structures and functions.” [28]	*Anticipate, absorb, accommodate, recover*	*Climate extremes and disasters*	*System*	*2012*
“Resilience is a continuous process starting from a reliable initial condition, followed by a vulnerability-survivability state after a disruptive event and eventually a recoverability phase aimed at achieving a new stable equilibrium condition.” [98]	*Vulnerability-Survivability, recoverability*	*Flood*	*Infrastructure system*	*2012*

2.6 Thermal resilience to overheating in buildings: definition and assessment methods

Recent studies highlight a growing interest in thermal resilience as KPI for building performance, with research on "resilience" in buildings and systems significantly increasing, especially since 2019 [99][31][32].The following section presents the definition and the existing thermal resilience assessment methods.

2.6.1 Thermal resilience to overheating: definition

Thermal resilience of buildings is defined as an ability of the building and its systems to withstand disruptions caused by extreme weather events (such as heatwaves and cold snaps), power failure, change in use (increase or decrease of occupancy and internal heat gains) and atypical conditions (failure of solar shading, cooling systems, ventilation system etc.); to maintain the capacity to function and recover to its designed/at least habitable thermal conditions in a timely manner without causing thermal-stress on occupants.

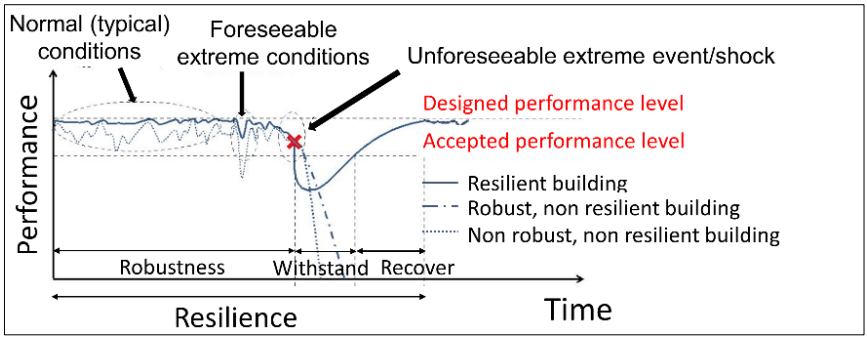


Figure 2-8. Different phases of resilience (Source-Attia et al. [100])

Moazami et al. [100] proposed a framework for defining the protection against climate uncertainty defining the robust and resilient buildings. Figure 2-8 illustrates the performance curve and the key phases for resilience and robustness defined by Moazami et al. [100] and later updated by Attia et al. [25] for thermal resilience. The performance curve that fluctuates slightly under normal conditions and even during foreseeable events, experiences a significant decline during an unforeseen extreme event. This curve also highlights the different phases of resilience- Vulnerability reflects a building's susceptibility to adverse conditions while resistance and robustness measure its ability to withstand and function under stress, either short-term or sustained—and the subsequent recovery process, which together define overall resilience. Notably, a robust building is not inherently resilient. Recoverability, on the other hand, emphasizes the speed and effectiveness of returning to comfortable or designed conditions after extreme events, which is critical for occupant well-being In the context of thermal resilience to overheating, this performance curve represents how a thermal shock, such as a heatwave, leads to an increase in indoor temperature. A

thermally resilient building can not only withstand this shock but also recover back to its designed or at least habitable thermal conditions in a timely manner.
Based on the definition and key aspects of resilience, a resilient building and system is the one which can anticipate the shock, withstand the shock with a minimum deviation from the design condition during the shock period and can recover back to its designed state in a timely manner without causing any heat-related impact on its occupants. To achieve future-proof buildings, it is crucial to design buildings and systems that can withstand shocks and reduce the impact of shocks on occupants' health in a building. This is known as 'thermal resilience to overheating' in buildings.

2.6.2 Thermal resilience to overheating assessment methods

IEA EBC Annex 80 [101] conducted several studies evaluating the thermal resilience of existing buildings [37–40] utilizing existing indoor overheating indicators, heat-stress indicators, qualitative and quantitative thermal resilience indicators and assessment frameworks. In a recent study Siu et al.[104] broadly grouped existing indicators used to assess thermal resilience in simulation studies and showed two broad classifications- (a) indicators identifying conditions that may cause thermal stress (such as temperature, WBGT, HI, SET) and (b) indicators of thermal resilience (degree.hours, time of exceedance, recovery time etc.). Numerous building performance studies have focused on developing indicators and frameworks to assess thermal resilience performance of buildings [36,45,105–109]. Based on existing literature and the KPIs listed by Annex 80 in the dynamic simulation guideline for the performance testing of resilient cooling strategies [110], the assessment methods can be broadly categorized into- (i) indoor overheating risk assessment indicators (ii) heat-stress assessment indicators and (iii) thermal resilience assessment methods (indicators and frameworks).

2.6.2.1 Indoor overheating risk assessment indicators

Several studies focused on developing [21] and reviewing [111][112] overheating risk indicators. Commonly used metrices for assessing overheating risk in buildings such as unmet degree hours (UDH) [43][113][108], exceedance hours (EH) [20][88][114], indoor overheating degree (IOD), ambient warmness degree (AWD) and overheating escalation factor ($\propto_{IOD}$)[99][115][116][117], were utilized to assess thermal resilience to overheating in buildings. In recent studies, indoor overheating assessment indicators were used to assess thermal resilience of buildings to overheating [99][45][117][39]. Additionally, within IEA Annex 80s' thermal resilience simulation framework [110], which aimed in assessing thermal resilience of buildings during future climate and heatwave periods, the following indoor overheating risk indicators were included:

Exceedance hour (EH)

EH refers to the percentage of occupied hours when the indoor operative temperature has crossed a threshold temperature [118]. Method A as described in Annex F of the EN 16798-1 also describes calculation of exceedance hours [119]. The annual percentage of occupied hours when the indoor operative temperature is above certain thresholds

(e.g. 25°C, 26°C or 28°C) are calculated. 5% of the occupied hours above the threshold temperature limit in a year is considered acceptable while lower than 3% is considered good.

Unmet degree hour (UDH)

UDH analogous to that of temperature weighted exceedance hours, a metric defined in ASHRAE Standard 55–2017 [120] quantifies the magnitude and duration of indoor operative temperatures that fall outside the acceptable comfort range. Calculation of UDH is based on the indoor cooling setpoint or $T_{threshold}$. The UDH metric weights each occupied hour that the temperature of a conditioned zone exceeds a certain threshold by the number of degrees Celsius by which it surpasses that threshold. Compared with the average temperature or unmet hours (which would be a simple count of how many hours a temperature threshold is violated), UDH provides a more complete picture of the overall history of temperature exceedance. UDH are calculated as follows:

$$UDH = \int_{t_i}^{t_2} [T(t) - T_{threshold}]\, dt \qquad \text{(Equation 2-1)}$$

where T is the indoor air temperature [°C]; $T_{threshold}$ is the temperature threshold [°C]; t is time [h]. $T_{threshold}$ is the indoor cooling set-point.

Overheating Escalation Factor ($\propto_{IOD}$)

Hamdy et al. [21] proposed metrices such as IOD, AWD and $\propto_{IOD}$ to assess overheating risk in buildings due to climate change. $\propto_{IOD}$ metric is used to estimate the sensitivity of buildings to overheating [21]. It represents the variation in the indoor temperatures when they exceed a chosen thermal comfort temperature limit in a given time period as a consequence of the severity of outdoor warmness. The severity of outdoor warmness is quantified using $AWD_{18℃}$. The Overheating escalation factor is defined as:

$$\propto_{IOD} = \frac{IOD}{AWD_{18℃}} \qquad \textit{(Equation 2-2)}$$

IOD quantifies the overheating risk, taking into account both the intensity and the frequency of indoor overheating [21]. The intensity is quantified by the temperature difference (ΔT) between the free-running indoor operative temperature (T_{fr}) and a chosen thermal comfort temperature limit (TL_{comf}), whereas the frequency is calculated by integrating the intensity of overheating during the occupied period (N_{occ}) into the different building zones (z) to present the overall overheating in a building. The free-running indoor operative temperature represents the indoor temperature of the zone when no heating or cooling systems are operating. IOD is represented as :

$$IOD = \frac{\sum_{Z=1}^{Z} \sum_{i=1}^{N_{occ}(Z)} [(T_{fr,i,z} - TL_{comf,i,z}) * t_{i,z}]}{\sum_{Z=1}^{Z} \sum_{i=1}^{N_{occ}(Z)} t_{i,z}} \qquad \text{(Equation 2-3)}$$

where z is the building zone counter, i is the occupied hour counter, t is the time step (typically it is 1 h), Z is the total number of zones in a building, N_{occ} (z) is the total

occupied hours in a given calculation period, T_{fr} is the free-running indoor operative temperature at the time step *i* in the zone *z*, and TL_{comf} is the comfort temperature limits at the time step *i* in the zone *z*. Two types of TL_{comf} are used to assess overheating risks in free-running buildings-fixed or adaptive temperature limits.

Ambient Warmness Degree (AWD 18°C) evaluates the severity of outdoor warmness by averaging the Cooling degree hours calculated for a base temperature of 18°C during the summer hours when the outdoor air temperature is not lower than 18°C [21]. AWD is represented by:

$$AWD_{18^{\circ}C} = \frac{\sum_{i=1}^{N}[(T_{a,i}-T_b)*t_i]}{\sum_{i=1}^{N} t_i} \quad \text{(Equation 2-4)}$$

where T_a is the outdoor dry-bulb air temperature, T_b is base temperature set at 18°C, *N* is the number of occupied hours such that $T_{a,i} > T_b$ in the summer season, and *t* is the time step (1 h). Only positive differences are taken into the summation ($T_{a,i}$ - T_b). The base temperature of 18°C was chosen because this value is lower than every minimum summer comfort temperature limit. $AWD_{18^{\circ}C}$ is higher than zero for every climate scenario in which the outdoor air temperature is higher than the minimum summer comfort temperature limit for at least 1 h. $AWD_{18^{\circ}C}$ provides an averaged quantification of the warmness of a given climate scenario that considers both the accumulation of amplitude and the duration of each warmness occurrence. For this reason, it measures the severity of a yearly weather dataset and is used independently of the time evolution or the assumptions used to build the weather file.

The Climate Change Overheating Resistivity (CCOR)

Rahif et al.[121] developed the climate change overheating resistivity (CCOR) metric to couple the outdoor and indoor environments quantifying the climate change overheating resistivity of cooling strategies in buildings. The CCOR shows the rate of change in the IOD with an increasing AWD due to the impact of climate change. It can be calculated using the linear regression methods assuming linearity between the IOD and AWD.

$$\frac{1}{CCOR} = \frac{\sum_{Sc=1}^{SC=M}(IOD_{Sc}-\overline{IOD})\times(AWD_{Sc}-\overline{AWD})}{\sum_{Sc=1}^{Sc=M}(AWD_{Sc}-\overline{AWD})2} \quad \text{(Equation 2-5)}$$

Where S_c is the weather scenario counter, *M* is the total number of weather scenarios, and $\overline{IOD}$ and $\overline{AWD}$ are the averages of total IODs and AWDs. CCOR > 1 means that the building is able to suppress the increasing outdoor thermal stress due to climate change, and CCOR < 1 means the building is unable to suppress increasing outdoor thermal stress due to climate change.

2.6.2.2 Heat-stress assessment indicators

The evaluation of thermal comfort and overheating risks in buildings requires the determination of an appropriate thermal comfort model. The thermal comfort in buildings can be divided into two major models: (a) static and (b) adaptive. The static

comfort models in comfort standards are primarily based on Fanger's widely recognized PMV/PPD model (Predicted Mean Vote and Predicted Percentage Dissatisfied), developed in the 1970s[122] [123]. The PMV model is most reliable for air-conditioned spaces and does not apply well to sleeping individuals or the elderly, especially in discomfort ranges. On the other hand, adaptive thermal comfort is relevant for naturally ventilated or free-running buildings, particularly in climatic conditions where the outdoor monthly mean temperature ranges from 10°C to 33.5°C [123] and threshold is calculated based on mean running temperature of outdoor conditions (see equation 2.6 and 2.7). In free-running buildings, occupants engage in sedentary activities (with metabolic rates from 1 to 1.3 met) and have the flexibility to adapt their environment, such as adjusting clothing (0.5 to 1 clo), opening windows, using fans, or moving around. These adaptive behaviors allow occupants to maintain comfort even when indoor temperatures are a few degrees higher than in air-conditioned spaces. Adaptive comfort model assumes that occupants can effectively adjust to varying environmental conditions, which leads to a higher acceptable range of operative temperatures in naturally ventilated buildings.

$$T_{rmo} = (1-\alpha)\{T_{ed-1} + \alpha T_{ed-2} + \alpha^2 T_{ed-3} + \cdots\}$$

(Equation 2-6)

$$T_{rmo} = \left(\frac{1}{3.8}\right)\{T_{ed-1} + 0.8T_{ed-2} + 0.6T_{ed-3} + 0.5T_{ed-4} + 0.4T_{ed-5} + 0.3T_{ed-6} + 0.2T_{ed-7}\}$$

(Equation 2-7)

Where, T_{rmo} is the running mean outdoor temperature (°C), α is a constant weighting factor, typically between 0 and 1, T_{ed-n} is the mean outdoor temperature (T_{ed}) on previous days (n is the number of days ago). However, these models focus on maintaining acceptable thermal comfort conditions under normal operating conditions. During extreme thermal shocks, such as heatwaves or sudden temperature spikes, these comfort-based models fall short as outdoor temperature during heatwaves often fall outside the defined upper range of the model acceptability. In such scenarios, it is crucial to shift from comfort models to heat stress models, as the priority during these shocks is not just thermal comfort but survivability. Heat stress models, which consider factors such as air temperature, mean radiant temperature, relative humidity, air velocity, clothing (clo) and metabolic rate (met) of occupants are more relevant for evaluating impact of extreme thermal shocks on occupants. Researchers have used terms like "discomfort index," "stress index," and "heat index" to describe the link between environmental conditions and human thermal perception, studied through biometeorological, physiological, psychological, and other perspectives. Over time, more than 160 human thermal bioclimatic metrics have been introduced in the scientific literature and in standards [124]. Most research aiming to develop indices to assess human thermal perception, evaluate outdoor climate, and predict thermal health risks or discomfort-related events such as morbidity, absenteeism, and even crime. Thus, linking heat-stress during extreme shocks to thermal resilience performance assessment is crucial. Recent studies assessing thermal resilience of buildings to overheating have employed heat-stress indicators such heat index hazard hours (HIHH)

[45] deploying heat index [43] and standard effective temperature degree.hours (SET-DH) [43] deploying standard effective temperature (SET) [102] and linking human health risk due to high temperature exposure.

Heat Index hazard hours (HIHH)

HIHH is widely used to assess heat stress. The heat index (HI), also known as the apparent temperature, combines air temperature and relative humidity in shaded areas to reflect what the temperature feels like to the human body [43]. When the relative humidity is high, the evaporation rate of water is reduced. This means that heat is removed from the body at a lower rate, causing it to retain more heat than it would in dry air. Heat index hazard hours (HIHH) are calculated as follows:

$$HHIH = \int_{t_1}^{t_2} f(HI(t) - HI_{threshold})dt \qquad \text{(Equation 2-8)}$$

where f(x)=1 if x>0, or 0 otherwise, HI is the heat index of the zone [℃] and t is the time [h]. Figure 2-9 illustrates the correlation between core body temperature and health condition by Pisacane et al. [125] and between SET and physiological state by Parsons et al. [126].

Standard Effective Temperature degree hour (SET-Dh)

SET-Dh weights each hour that the SET of a thermal zone exceeds a certain $SET_{threshold}$ by the number of degrees Celsius by which it surpasses that threshold. The SET is defined as the temperature of still air of an imaginary indoor environment at 50% relative humidity and where the mean radiant temperature equals the air temperature, in which an imaginary occupant performing the same actual activity and wearing clothing standardized for the activity level, has the same skin heat loss, heat stress (skin temperature) and heat strain (skin wettedness) as in the actual environment. Parsons [126] in a study showed the correlation between SET and physiological state (see Figure 2-9). SET is based on the two-node physiological model developed by Gagge et al. [127] and recommended by ASHRAE Standard 55 [123], and is a widely accepted metric. To assess thermal resilience, SET-Dh is utilized by several studies [128]. Hourly SET is calculated with parameters such as air temperature, mean radiant temperature, air velocity, relative humidity, clothing insulation (clo), and metabolic rate (met)of occupants [129]. Set-Dh [128] is defined as :

$$SET - Dh = \sum_{t=1}^{N} (SET(t) - SET_{threshold}) * \Delta t \qquad \text{(Equation 2-9)}$$

Where, SET-Dh represents the total degree-hours above $SET_{threshold}$, $SET(t)$is the SET at time t, and Δt is the duration of each time interval (typically one hour).

For calculation of hourly SET values, dynamic clo and met can be considered for different building types depending on their function [88]. Table 2-3 shows the dynamic met and clo for different building types to calculate the hourly SET. In residential settings, 1 met corresponds to light activities such as sitting or reading, while 0.7 met is associated with sleep or reclining. For office and school environments, 1.1 met is typical for light desk work or sedentary activities. For residential settings, 0.5 clo is used

for light, casual clothing like a t-shirt and shorts, and 1.38 clo corresponds to heavier, insulated sleepwear. In office and school environments, 0.57 clo reflects standard attire like a shirt and trousers [120].

Table 2-3. Dynamic met and clo for hourly (Source- Laouadi et al. [88])

Building type	Metabolic rate (met) and clothing (clo)
Residential	*1 met & 0.5 clo (wake);0.7 met & 1.38 clo (sleep)*
Office	*1.1 met & 0.57 clo*
School	*1.1 met & 0.57 clo*

To effectively assess and improve thermal resilience of buildings to overheating, it is essential to link thermal resilience with heat stress on occupants. This approach is vital for protecting occupant health during extreme thermal shock, such as heatwaves. Heat stress indicators like SET-Dh are particularly suited for this purpose, as they integrate multiple key parameters—air temperature, relative humidity, mean radiant temperature, clo, and met. These parameters directly influence how heat is perceived and experienced by the human body. What sets SET-Dh apart as an optimal heat stress indicator is its ability to account for dynamic changes in occupant conditions, such as clothing and metabolic rate, which vary depending on activity levels (e.g., day versus night) and building type. Furthermore, SET-Dh is aligned with specific thresholds already defined in the literature, which enhances its practical applicability and accuracy in assessing heat stress risks across different environments and conditions [128].

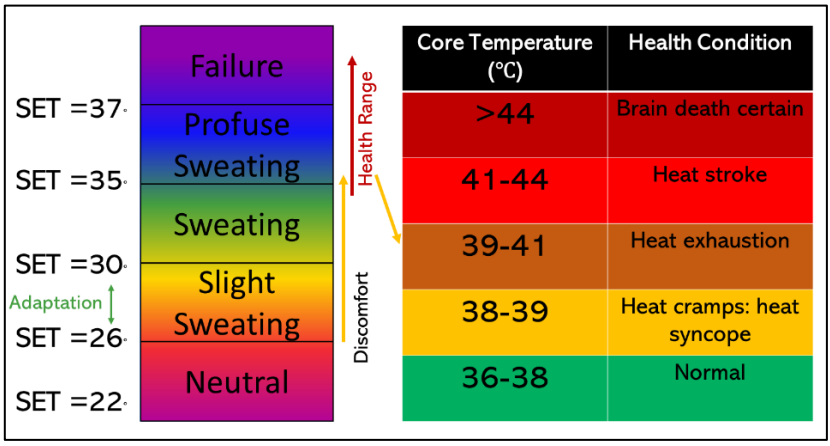


Figure 2-9.Correlation between core body temperature and health conditions [125] and between SET and physiological state [126]

2.6.2.3 Thermal resilience assessment methods

Thermal resilience indicators are designed to assess how buildings respond to thermal shocks, particularly in extreme heat shocks. While Siu et al. [104] categorize these indicators into three broad aspects—(i) the ability to prevent buildings from reaching

critical temperatures (e.g., passive survivability, temperature at various timestamps), (ii) the ability to reduce exposure to thermal stress (e.g., thermal autonomy), and (iii) the ability to recover to a habitable state (e.g., recovery time)—the current state-of-the-art in thermal resilience assessments can be broadly classified into three types: (a) quantitative indicators, (b) qualitative indicators, and (c) thermal resilience assessment frameworks.

(a) Quantitative indicators

Quantitative indicators are typically derived from building performance simulations or data collected through measurement campaigns. Existing quantitative thermal resilience indicators include (i) Thermal autonomy and (ii) Passive survivability

Thermal Autonomy (TA)

Thermal Autonomy (TA) is the percentage of occupied hours when indoor operative temperature is inside limits of thermal comfort without intervention of active systems [130]. While there is no universal threshold, a common benchmark is 80% thermal autonomy is accepted, meaning that a building is expected to maintain designed conditions passively for at least 80% of the time [131]. However, the specific threshold may vary depending on the building's design goals, climate, and the thermal comfort criteria being applied. TA is expressed as:

$$TA = \left(\frac{H_{comfortable}}{H_{total}}\right) \times 100 \qquad \text{(Equation 2-10)}$$

Where $H_{comfortable}$ is the number of hours during which the indoor temperature is within the defined comfort range and H_{total} is the total number of occupied hours in a given period .

Passive Survivability (PS)

Passive survivability (summer) refers to the time (in hours) from when cooling is shut off to when the indoor operative temperature reaches 30 °C from the original cooling set-point [132]. There are no specific thresholds for how long a building and system should take to reach 30°C to have a high or low passive survivability.

(b) Qualitative indicators

Zhang et al. [37] assessed resilient cooling strategies during two main thermal shocks-(a) heatwaves and (b) power outages and proposed four key qualitative criteria-(i) absorptive capacity, (ii) adaptive capacity (iii) restorative capacity and (iv)recovery speed. Adaptive capacity is the ability of a building or system to adjust to adverse conditions by learning, modifying configurations, and improving flexibility, particularly when absorptive capacity is exceeded. Restorative capacity refers to the ability to recover to normal or improved operation after a disruption, such as using night cooling to eliminate accumulated heat. While these capacities are vital, thermal resilience assessment should integrate key criteria such as absorptive capacity and recovery speed, focusing on how buildings and systems absorb and recover from shocks effectively. Resilience capacities are categorized as high, moderate, or low. High

indicates the ability to maintain or increase cooling capacity during thermal shocks like heatwaves or power outages, moderate maintains capacity most of the time, and low reflects a decrease in capacity under such conditions. Recovery speed is also classified qualitatively as high (within an hour), moderate (several hours), or low (a day or more).

Absorptive capacity

The absorptive capacity is the degree to which a system is able to absorb the impacts of shock and minimize their consequences with little effort. For example, heavy thermal mass in a building (capacitance) can absorb unwanted solar gains and can minimize and/or delay the air temperature increase in the building without the use of cooling energy.

Recovery speed

The recovery speed is the speed of the recovery process. Recovery may be accelerated if absorption activities are well implemented and the system can quickly mobilize and effectively use all the resources at its disposal. For example, the speed with which night cooling can remove heat from the building's thermal mass and restore the building to its desired condition depends on the ventilation flow rate and the outdoor air temperature.

(c) Thermal resilience assessment frameworks

Apart from developing quantitative and qualitative thermal resilience indicators, a number of studies focused on developing resilience assessment frameworks.

Thermal Resilience Index (TRI)

Homaei et al. [24] introduced a benchmarking framework and multi-phase metric for quantifying thermal resilience during cold snaps and power outage demonstrating the significant impact of design upgrades and resilience enhancement strategies. Ji et al. [39] (as seen in Figure 2-10) extended the methodology developed by Homaei et al. [24] for heatwaves and developed a Thermal Resilience Index (TRI) using the SET to link the thermal resilience performance with occupants' predicted heat strain (PHS)-a response to dynamic indoor thermal conditions.

TRI involves 3 main phases-(a) preparation phase - collection and analysis of climate data (heatwave files), development of building simulation model/indoor condition measurement (b) development of the resilience trapezoid that evaluates thermal resilience to heatwaves using the indoor temperature profiles (in this case hourly SET linking thermal resilience to heat-stress) and (c) the calculation of the zone or building level's thermal resilience and classify from Class F (lowest resilience) to Class A+ (highest resilience), reflecting relative improvements in indoor thermal conditions. This approach enables the overall resilience of a building, as well as the resilience of individual thermal zones, to be assessed.

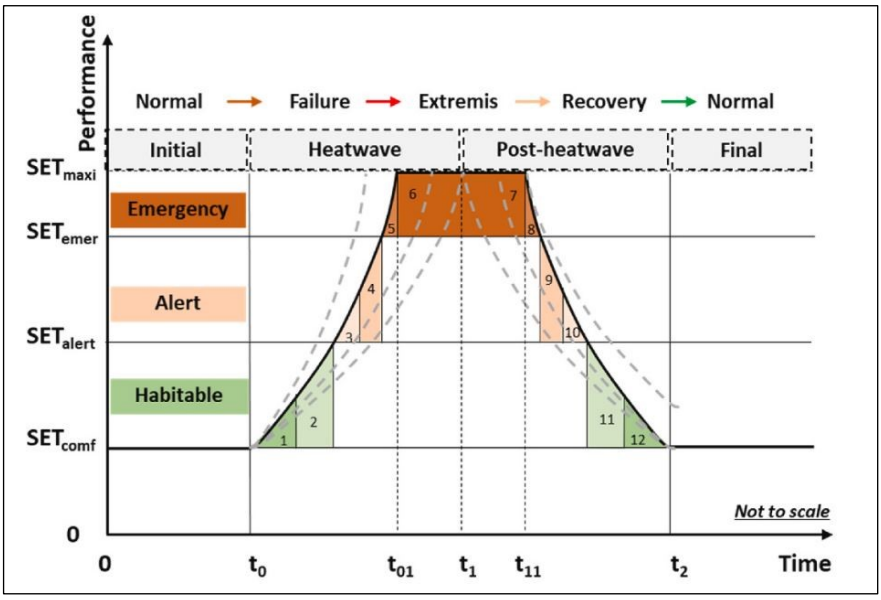


Figure 2-10. Building thermal resilience curve based on resilience trapezoid developed by Ji et al. [39]

Office thermal resilience (OTR)

In this study, Ismail et al. [40] developed a simplified metric for office thermal resilience (OTR) to quantify thermal resilience in office buildings during power outages. The model evaluates thermal habitability and passive survivability, focusing on the number of safe hours before loss of productivity during power outage. OTR involves developing a simple regression model with building parameters (envelope and glazing properties, geometry and occupancy profile) and predicting passive survivability hours during a power outage. While this study developed a simplified regression method to include design parameters, they did not link heat-stress to thermal resilience assessment. Furthermore this framework cannot be used to assess thermal resilience of various building types against several shocks with growing intensities.

2.6.3 Shortcomings in existing thermal resilience assessment methods

Existing thermal resilience indicators have several shortcomings, lacking a comprehensive framework that integrates the following (a) quantification and comparison of different types of thermal shocks, such as varying heatwave durations and severities, to assess their distinct impacts on building resilience. For example, a 3-day severe heatwave impacts buildings differently than a milder 10-day heatwave, yet existing resilience assessment frameworks overlook this variation (b) consideration of building and system design parameters such as envelope and glazing properties, HVAC systems and their capacity, and thermal mass which influences thermal resilience. Identifying most influential design parameters and then intervention on these parameters to improve thermal resilience of buildings are often overlooked in existing

thermal resilience framework (c) linking resilience assessments to heat-stress indicators to mitigate health risks for occupants. Heat-stress indicators like SET-Dh consider air velocity, relative humidity, and clothing are underutilized in resilience studies, neglecting the health implications of extreme conditions (d) benchmarking resilience performance across different building types for standardized evaluation. There is no standardized way to compare resilience for diverse buildings, such as office spaces versus residential units, under similar thermal shocks. This lack of benchmarking complicates the identification of best practices for different building categories and finally,(e) simplified calculation methods suitable for early design phases, eliminating the need for complex simulations or measurement campaigns, enabling practical use by architects and engineers. Current frameworks often rely on detailed energy simulations or complex experiments, which are impractical during the early design stages. A straightforward metric that architects and engineers can easily apply, such as regression-based tools, would bridge this gap and encourage widespread adoption. The shortcomings of the existing thermal resilience assessment methods (indicators and assessment methods from the state-of the-art) are summarized in Table 2-4. They are as follows:

Table 2-4. Overview of the existing thermal resilience assessment methods

Category	Indicators	Evaluation
Indoor overheating risk assessment indicators	*EH, UDH,* $\propto_{IOD}$*, CCOR*	– *Do not capture the impacts of short-term temperature shocks, rather focuses on long-term overheating* – *Lacks defined thresholds- to assess thermal resilience for short-term shocks* – *No link with heat-stress of occupants* – *Rely on temperature profiles alone and ignores parameters such as relative humidity, air velocity, clothing and metabolic rate of occupants which impacts heat-stress*
Heat-stress assessment indicators	*HHIH, SET-Dh*	– *Links heat-stress to thermal resilience assessment (SET-Dh, HI)* – *Offers critical insights into the relationship between thermal resilience and occupants' health* – *Estimates how long individuals can safely remain in a building without experiencing heat-related health issues* – *Still a gap in linking heat-stress with shock types* – *Still a gap in linking heat-stress with design parameters*
Quantitative thermal resilience indicators	*TA, PS*	– *Lack explicit thresholds to quantify thermal resilience performance* – *Gap in linking quantitative indicators to shock types* – *Gap in linking building and system design parameters*
Qualitative thermal resilience indicators	*Absorptive capacity, Adaptive capacity, Restorative capacity, Recovery speed*	– *Lack of quantification of the qualitative indicators* – *Lack of defined thresholds to assess thermal resilience* – *Gap in linking qualitative indicators to design parameters and shock types and shock intensities*
Thermal resilience assessment frameworks	*TRI, OTR*	– *Gap in linking thermal resilience assessment to growing shock intensities shock* – *No comprehensive framework exists that can quantify a building's thermal resilience across multiple shock types*

	– *Gap in linking thermal resilience performance to building and system designs* – *Absence of quantification of impact of design parameters' on thermal resilience leads to limiting the ability to create targeted strategies for enhancing resilience* – *Lack of an easy to calculate and comprehend thermal resilience framework* – *Often necessitates comprehensive building performance simulations or extensive analysis of measurement data* – *Impractical for architects and designers to use during the early stages of a project* – *OTR is an easy to use regression model, yet it is still not linked to multiple shocks and heat-stress indicators*

2.7 Thermal resilience assessment of case study buildings with existing indicators

A simulation study was set up to assess thermal resilience performance of case study buildings. Thermal resilience performance was assessed for different building typologies (schools, single family dwellings, apartment and offices) equipped with various temperature management strategies such as NNV, solar shading, floor cooling, indirect evaporative cooling (IEC) etc. utilizing the existing thermal resilience indicators during different types of disruptions (heatwaves, power outage, solar shading failure, NNV failure. Dynamic building performance simulations were performed adopting the simulation framework of IEA Annex 80 [110]. Table 2-5 summarizes the different building typology, resilient cooling strategies tested, type of disruption/ shocks tested and finally the existing thermal resilience indicators used for thermal resilience assessment.

Table 2-5. Building, systems and shock typologies tested and the existing thermal resilience indicators utilized for each cases

Building type	cooling strategies	Shock type	Existing Thermal resilience indicator utilized	Conclusions
Dwelling	*Solar shading, NNV*	*Heat wave, solar shading failure and combine*	*Recovery speed*	*First attempt of a resilience indicator (recovery speed) to assess NNV and solar shading failure, heat wave or combination of both. It takes 3.4 days for the indoor temperature to return below 25°C, in case of shading failure during a heatwave compared to 9 hours where the building is equipped with solar shading and NNV during a heatwave. [41]*
Dwelling	*Radiant floor cooling*	*Future climate scenarios*	*EH*	*Indoor overheating is strongly influenced by occupancy, with acceptable comfort achieved when the upper limit is raised to 26°C, except in the long-term future weather scenario where current cooling capacity may fail to keep the building within this threshold temperature limit. Solar shading significantly improves comfort, especially in highly glazed spaces (high WWR), ensuring better performance across all climate scenarios (present and future). [133]*
Apartment	*Thermal mass, WWR, solar shading, NNV*	*Heatwave*	*Adaptive thermal comfort, UDH, EH*	*Low thermal resilience is most likely in those buildings with high WWR, no solar shading, and lighter thermal mass. WWR has the greatest impact on thermal resilience, followed by thermal mass. Solar shading and passive cooling strategies like NNV improve resilience, though NNV is less effective during heatwaves. Buildings with heavy thermal mass and lower WWR perform better in shorter heatwaves. Future work should assess other design factors like building orientation, insulation, glazing, and active cooling strategies to enhance thermal resilience. [134]*
Apartment	*NNV, Air conditioning*	*Heatwave, power outage and combine*	*Recovery speed, UDH*	*NNV improves the apartment's resilience, but additional passive or active cooling is necessary to withstand future heatwaves. Recovery is significantly accelerated with NNV, while solar shading further enhances thermal resilience. Removing shading worsens thermal comfort and increases unmet degree hours (UDH),*

				especially during heatwaves. Increasing WWR increases heat gains, worsening thermal resilience, while decreasing WWR reduces ventilation, slowing recovery. Air conditioning offers rapid recovery but consumes high electricity, risking outages, so it should be used cautiously alongside passive strategies. [134]
Office	*Solar shading, Geothermal heat pump used for cooling*	*Power outage*	*TA, PS*	*Thermal autonomy results showed the building has very low resilience, with indoor temperatures consistently exceeding acceptable limits in summer, making cooling systems essential. Passive survivability analysis revealed that during a warm week with power failure, temperatures would reach 30°C in 8 hours and take two nights to return to normal, though sun blinds and NNV can mitigate overheating and speed up recovery time. [135]*
Educational building	*Solar shading, NNV, IEC*	*Heatwave, power outage, solar shading failure, NNV failure, IEC failure*	*PS, Recovery speed, UDH,EH, IOD,AWD,* $\propto_{IOD}$	*Thermal resilience is influenced by high airtightness and WWR, with passive cooling strategies maintaining temperatures below 24°C under typical conditions. However, during power outages or heatwaves, thermal zones with heavy thermal mass retain heat longer, leading to slower recovery, indicating the building's low thermal resilience. [42] [117]*

These studies shown in Table 2-5 revealed significant variations in thermal resilience across building types, influenced by design features, cooling strategies, and shock types. Dwellings are highly vulnerable to overheating during heatwaves, especially if solar shading systems fail. NNV significantly improves cooling, but its effectiveness in improving thermal resilience is higher when combined with shading. For instance, under shading failure and a heatwave, it takes 3.4 days to reduce indoor temperatures below 25°C, compared to 9 hours with functional shading and NNV (89% faster). In dwellings, radiant floor cooling enhances comfort and thermal resilience in most scenarios but struggles to maintain comfort in when tested for long-term future climate. Solar shading, particularly in glazed spaces, remains critical for improving thermal resilience. Thermal resilience in apartments depends heavily on design. High WWR and light thermal mass increase overheating risks, while heavy thermal mass and reduced WWR improve thermal resilience (during shorter shocks). Air conditioning offers rapid recovery but is energy intensive and thus should complement passive measures. Offices have low thermal resilience, with indoor temperatures exceeding 30°C within 8 hours during power outages and taking two nights to recover. Passive measures like solar shading and NNV help but are insufficient alone, highlighting the need for integrated cooling strategies. Educational buildings stay below 24°C under normal conditions but struggle during heatwaves. Heavy thermal mass slows recovery (during longer shocks), and failures in shading or cooling systems worsen overheating. Assessment of thermal resilience performance of the case study buildings shows that air-tight, highly insulated buildings and buildings with high WWR have low thermal resilience to overheating. Buildings equipped with cooling strategies such as NNV, floor cooling and IEC improve thermal resilience performance during heatwaves. These studies also point out the importance of building and system design parameters in manipulating buildings' thermal resilience performance.

2.8 Conclusions

Chapter 2 aims to introduce thermal resilience to overheating and evaluate existing thermal resilience assessment methods and indicators. The key findings are :

- Climate change is driving more frequent and intense heatwaves, increasing the risk of indoor overheating in buildings. This overheating can lead to serious health impacts on occupants, including heat-stress, dehydration, and potentially life-threatening conditions such as heatstroke. Incorporating thermal resilience to overheating as a building performance KPI is essential to ensure safer indoor conditions, protect occupant health, and adapt to evolving climate risks. 'Thermal resilience to overheating' is the extent to which these buildings and their systems can withstand the thermal shock, and recover back to the original designed performance'.
- Researchers have classified these indicators into types focusing on indoor overheating, heat-stress, and resilience assessment, each designed to measure different aspects of thermal comfort and resilience. Several studies were focused on developing thermal resilience indicators (quantitative and qualitative) and assessment frameworks. A simulation study was conducted on case study buildings utilizing existing

indicators to assess thermal resilience of different types of buildings with cooling strategies during shocks such as heatwaves, power outage, cooling strategy failures and combinations of these shocks. While, evaluation of thermal resilience performance of buildings showed that buildings would need combination of passive strategies to be thermally resilient during severe shocks and importance of building and system design parameters in improving thermal resilience performance, the study also highlighted the shortcomings of the current indicators.

- Evaluation of existing thermal resilience indicators and assessment frameworks (from literature) revealed –(a) current thermal resilience indicators give an indication of the deviation of indoor conditions and the time for which this deviation occurred but do not often link with heat-stress to quantify thresholds based on impact of this deviation on health (b) existing qualitative indicators gives insight into different phases of resilience- adaptive capacity, absorptive capacity, restorative capacity and recovery speed, but these qualitative indicators are not quantified and no thresholds were formulated (c) some of the existing quantitative indicators have threshold limits while others lack threshold to quantify thermal resilience performance (such as passive survivability, IOD, AWD, $\propto_{IOD}$) (d) currently resilience performance are often calculated based on detailed building performance simulation and/or on analysis of measurement data (from building measurement campaigns). Thermal resilience assessment requires expertise and thus is not considered during early design phase of buildings and systems (e) no existing resilience assessment framework links resilience performance with shocks and thus buildings' thermal resilience performance cannot be assessed for different shock types to see which level of shocks the buildings' design can endure and finally, (f) these existing frameworks do not have a link with design parameters and thus thermal resilience performance cannot be linked to design parameter interventions (which parameter intervention would lead to improvement in thermal resilience during shocks ?) Any assessment of thermal resilience would require new set of simulations and analysis of data to check whether a change in any building and system design improved or worsened thermal resilience performance of buildings.
- Thus, there is a need for a holistic thermal resilience framework that integrates (a) assessment of thermal resilience across multiple shock types (b) establishes clear relationships between critical building design parameters and thermal resilience, (c)integrate—impacts on occupant health and thermal comfort and include quantifiable thresholds for safe occupancy durations and (d)is user-friendly and easily calculable for architects and designers, enabling informed decision-making and comparative analysis of different building designs to enhance overall resilience. Chapter 3 lays the first step of the development of this resilience assessment framework by quantifying shocks that impact thermal resilience in buildings to overheating and aims to identify the most critical shock.

Chapter 3 : Identifying and quantifying shocks that impact buildings' thermal resilience performance

Chapter 3 aims to identify and quantify various shocks that challenge buildings' thermal resilience performance. This is the first step towards development of the comprehensive thermal resilience assessment framework (as described in Chapter 1). This chapter includes identifying and quantifying the shocks by degree of shock (*doS*). *doS* is developed for different shock types-heatwaves, system failures and power outages. The chapter also describes the development of typical and extreme weather datasets to quantify *doS* for heatwaves. *doS* was employed to compare the impact of different types of shocks on thermal resilience of buildings to overheating. The most critical type of shock identified in Chapter 3 is utilized in Chapter 4 to identify the most influential building and system design parameters impacting thermal resilience.

This chapter is based on the following publications:

*[1] **A. Sengupta**, D. Al Assaad, J.B. Bastero, M. Steeman, H. Breesch, Impact of heatwaves and system shocks on a nearly zero energy educational building: Is it resilient to overheating?, Build. Environ. 234 (2023). https://doi.org/10.1016/j.buildenv.2023.110152.*

*[2] Anaïs Machard, Agnese Salvati, Mamak P. Tootkaboni, Abhishek Gaur, Jiwei Zou, Liangzhu Leon Wang, Fuad Baba, Hua Ge, Facundo Bre, Emmanuel Bozonnet, Vincenzo Corrado , Xuan Luo, Ronnen Levinson, Sang Hoon Lee, Tianzhen Hong, Marcello Salles Olinger, Rayner Maurício e Silva Machado, Emeli Lalesca Aparecida da Guarda, Rodolfo Kirch Veiga, Roberto Lamberts, Afshin Afshari, Delphine Ramon, Hoang Ngoc Dung Ngo, **Abantika Sengupta**, Hilde Breesch, Nicolas Heijmans, Jade Deltour, Xavier Kuborn, Sana Sayadi, Bin Qian, Chen Zhang, Ramin Rahif, Shady Attia , Philipp Stern & Peter Holzer, Typical and extreme weather datasets for studying the resilience of buildings to climate change and heatwaves, (2024) 1–21. https://doi.org/10.1038/s41597-024-03319-8. +*

+Within the framework Annex 80, following the methodology developed within the Weather Data Task Force, future TMY and Heatwave files were developed for ASHARE climate 4A-Ghent

3.1 Introduction

Chapter 2 explored the concept of thermal resilience of buildings to overheating and their current assessment methods. Reviewing existing assessment methods highlighted critical short comings and demonstrated the need for developing a comprehensive thermal resilience assessment framework addressing the current gaps. The first step of the development of this comprehensive assessment framework includes integration of shocks. Chapter 3 aims to address this link between thermal resilience assessment and shocks. It involves identifying and quantifying various thermal shocks to assess their relative impact on buildings' thermal resilience performance. By comparing different types of thermal shocks, this chapter aims to pinpoint which thermal conditions pose the greatest threat to thermal resilience of buildings to overheating, providing crucial insights into the most critical shocks for future-proofing buildings. Together, these analyses lay the groundwork for a more targeted and effective approach to enhancing the thermal resilience of buildings, ensuring that they are better equipped to withstand and recover from severe shocks. This chapter aims to answer the question- 'Resilience to what?" This chapter focuses on the RO2- to identify and quantify shocks that impact buildings' thermal resilience performance.

Chapter 3 is organized as follows: Section 3.2 explores the state-of-the-art, examining various shocks affecting thermal resilience of buildings to overheating, emphasizing gaps in comparing different shock types and highlighting the importance of an integrated approach to compare these shocks, identify the most critical ones, and understand their specific impacts on buildings' thermal resilience. Section 3.3 introduces the development of the normalized degree of shock (*doS*) to quantify different shock types, while Section 3.4 focuses on creating future weather data files (both typical and heatwave scenarios) to enable the determination of *doS* for HWs. Section 3.5 describes the methodology, including the case study building, its model, and the experimental setup for validating the simulation model. Section 3.6 outlines the scenarios for the parametric study, with Section 3.7 presenting the corresponding results. Finally, Section 3.8 discusses the findings, and Section 3.9 concludes the chapter.

3.2 State-of-the-art

Energy efficient buildings and their systems succeed in providing comfortable environments under expected weather and operational conditions (i.e., occupancy loads, solar heat gains, etc.), which is not always the case. Throughout their lifetime, buildings can face sudden shocks and unforeseen events, that cause the indoor thermal environment to deviate from the designed comfort conditions by causing over or underheating [24,25], impacting occupants' health [26,27]. Buildings face indoor overheating risk during warmer and longer summers [136], and during heatwaves lasting from a few days to several weeks[8]. Moreover, during longer summers and especially during heatwaves electric grids are under extreme pressure, leading to frequent power outages. This further exacerbates overheating risk in buildings [72,73]. Apart from heatwaves and power outages, buildings also face other shocks related to

failure in system operation. This includes failure of air conditioning [137] and envelope strategies [138], which lead to indoor overheating during the cooling season. Systems' failure is a common occurrence in buildings especially with increasing system complexity. It can be due to faulty sensors, failure or absence of efficient fault diagnostics techniques and poor or irregular maintenance [139–141].

Within the framework of IEA EBC's Annex 80 [101] several studies evaluating thermal resilience of building, tested impacts of shocks such as heatwaves, system failures and power outages or a combination of these shocks [37–40]. Donovan et al.[114] evaluated the thermal resilience of passive cooling strategies under extreme weather conditions, demonstrating that strategies combining multiple measures maintained comfort for 57–95% of occupied hours without mechanical cooling but faced limitations under future extreme conditions. Ji et al. [102] evaluated the thermal resilience against a heatwave of a naturally ventilated care facility with interior blinds and found that simultaneously increasing window opening area size, replacing blinds with external shading, applying green roofs and NNV, can improve the buildings' thermal resilience. Lomas et al.[142] evaluated resilience of a naturally ventilated building and showed advanced natural ventilation strategies demonstrate higher thermal resilience to future heatwaves compared to single sided natural ventilation using less energy than mechanical ventilation. While these studies focused on evaluating thermal resilience during heatwaves, a number of studies evaluated thermal resilience assessment during heatwave accompanied by power outages. Sengupta et al.[103] evaluated thermal resilience of an educational building during heatwaves and power outages and demonstrated that passive cooling strategies exhibiting moderate to high recovery capacity during typical climate scenario perform poorly during heatwaves and power outages. Sun et al.[43] evaluated passive cooling measures for improving residential heat resilience during heatwaves and power outages in disadvantaged communities, and found that solar-control window films and roof insulation are the most effective, reducing unmet degree-hours by up to 51% with power and 20% without, while natural ventilation also significantly mitigates overheating risks. Additionally, some studies focused on failure of systems during heatwaves. Sengupta et al. [41] evaluated the thermal resilience of a Belgian dwelling during the heatwave of 2020 with and without natural night ventilation and solar shading. Their results showed that with natural night ventilation, the building recovers 89% faster during the heatwave and decreases maximum temperatures indoors by 4.3°C compared to the scenario of solar shading and natural night ventilation failure.

While assessing thermal resilience, these studies focused on one or a combination of type of shocks. However, they did not compare thermal resilience of buildings during the same shock of increasing intensity. For example, they did not compare impact of heatwaves of varying severity or duration and assess thermal resilience of buildings during a short severe shock compared to a milder longer heatwave. Moreover, for power outage scenario, power outage was considered for a 24h period starting on the hottest day of the heatwave or no electricity from grid during the entire duration of heatwave [143]. The studies evaluating system failure did not assess thermal resilience during

system shocks of varying duration and varying time of occurrence. There is a lack of quantifying shocks in a common framework to compare their impact on buildings' thermal resilience and identify the most critical shock that impacts thermal resilience of buildings to overheating.

3.3 Shocks' classification and quantification

A shock is defined as a sudden disturbance that causes the building to shift partially or completely from its designed conditions for a certain amount of time. Thus, it challenges the resilience performance of buildings [100]. Figure 3-1 illustrates the temporal evolution of outdoor and indoor temperature during a shock and the different aspects of thermal resilience [25,100]. As seen in Figure 3-1, a shock such as a heatwave starts at t_1 and ends at t_3. This is denoted as the duration of the shock itself (t_3-t_1). However, due to the thermal mass and insulation of building envelope, the buildings' thermal response has a delay compared to starting and ending time of the shock itself, which either delays the onset of overheating when the shock starts and even keeps the indoor thermal condition overheated even after the shock has ended. The impact of shock on the indoor space may be longer than the shock duration itself. As seen in Figure 3-1, when the shock occurs, the indoor temperature rises to max indoor temperature (T_{max}), causing certain levels of discomfort [144,145]. The period during which the temperature rises to T_{max} from a defined threshold temperature is denoted as shock absorptivity time (t_2-t_1). When the shock ends (at t_3) and/or aa recovery action becomes effective, the building recovers from this shock back to eventually thermally comfortable levels (below threshold level)[26]. This period is denoted as recovery time of the shock (t_4-t_2). The absorptivity, recovery time and T_{max} depend on several factors:

(1) Type of shock, severity and duration
(2) Building characteristics (e.g., thermal mass, insulation, airtightness, glazing ratio, space volume etc.).
(3) Heating ventilation and air conditioning systems, their control strategies

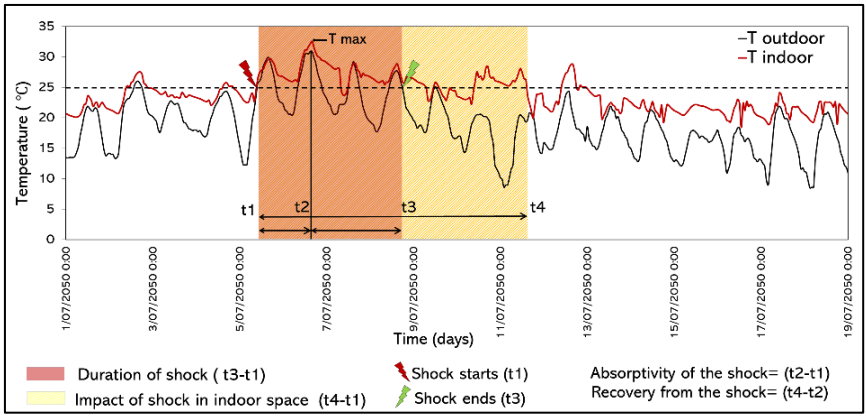


Figure 3-1. Temporal evolution of indoor temperature during a shock

To assess and compare the impact of shocks and shock combinations on the thermal resilience of a building to overheating, it is crucial to define, classify and quantify different shocks. Shocks were classified depending on how they impact the thermal environment as will be seen below. To quantify them, this study introduces the normalized degree of shock (doS) index in:

$$doS = \underbrace{\frac{T_{shock} - T_{ref}}{T_{ref}}}_{Severity} \times \underbrace{\frac{t_{shock}}{t_{ref}}}_{Duration} \qquad \textit{(Equation 3-1)}$$

The *doS* combines the severity and duration of the shock. The aspect of shock severity expressed in equation 3.1 is a quantification of the shock origin and overheating causality rather than a quantification of shock consequence on the room air temperature, which can be influenced by multiple parameters simultaneously and not just the shock. Thus, severity is the increase of a shock characteristic temperature T_{shock} compared to the reference characteristic temperature T_{ref} under normal expected conditions. The characteristic temperatures depend on the type of shock. The duration is the ratio of t_{shock} (i.e., the duration of the shock (h) during an observation period (Figure 3-1) and t_{ref} (i.e., a reference period depending on the type of shock). Two types of shocks were identified as follows:

(1) **External shocks (ES):** ES are events occurring outside the building and cause a significant temporary increase in the outdoor temperatures averaged over t_{shock} ($\overline{T_{oa,ES}}$) compared to typical meteorological year (TMY) temperatures averaged over the same period ($\overline{T_{oa,TMY}}$). ES cause overheating by increasing the heat gains through the building envelope (i.e., walls and glazing) and reducing the efficiency of mechanical and ventilative cooling strategies (e.g., evaporative cooling that relies on outdoor wet and dew point temperatures, natural ventilation that relies on outdoor dry bulb temperatures and diurnal temperature differences). ES include climate change events such as heatwaves (HW) [6], and disaster events (e.g., nearby fire increasing temperatures near the building) [146]. ES was quantified by the normalized degree of ES ($doES$), defined by equation 3.2:

$$doES = \underbrace{\frac{\overline{T_{oa,ES}} - \overline{T_{oa,TMY}}}{\overline{T_{oa,TMY}}}}_{Severity} \times \underbrace{\frac{t_{shock}}{t_{shock,max}}}_{Duration} \qquad \textit{(Equation 3-2)}$$

In equation 3.2, $t_{shock,max}$ is the duration of the longest ES that is predicted to possibly occur. ES can last from a few hours to multiple days.

Note that the $doES$ can be applied to both overheating events due to heatwaves but also underheating events due to cold snaps. ES can be applied to any type of climate zone with knowledge of the meteorological data for the reference case (TMY) and shock case from a weather station or a weather data prediction model. The current approach to normalization was chosen to ensure a consistent comparison of shocks across different shock scenarios. For ES, time is normalized to the highest ES that can occur in next 100 years to represent an extreme boundary condition that defines the maximum

possible disruption duration. This allows for a clear benchmarking of how long a building/system is exposed to a given shock relative to the most severe expected event. On the other hand, the current approach does not normalize severity to the highest possible shock type because the goal is to capture the relative deviation in outdoor temperature between typical conditions (TMY) and extreme heatwave conditions, rather than an absolute upper bound. *doS* formulation is designed to reflect how much a given event deviates from typical conditions, making it a relative measure rather than an absolute one. Normalizing severity to the highest possible temperature (e.g., from the worst climate scenario) could be misleading, as it would anchor all severity values to a single extreme case, potentially dampening variations in shocks of different magnitudes. This would reduce the ability to differentiate between moderate and extreme heatwaves, as all shocks would be compared to a distant upper bound rather than their respective baselines. By keeping severity as the deviation between typical and extreme conditions, the method ensures that it remains contextually relevant for different climates and scenarios, preserving its ability to indicate how much a particular event deviates from what is normally expected rather than an absolute extreme.

(2) **System shocks (SS):** SS are partial or complete failure in the operation of building systems in managing the space cooling loads through the envelope or by relying on a working fluid (i.e., air, water). They occur due to faulty feedback sensors in case of automated systems [140], deterioration of components (e.g., motors, valves, dampers, belts) [147], irregular preventive maintenance [148] and inadequate fault detection diagnostics [141]. Depending on the nature of the impaired system, SS can cause overheating differently. In this study, three sub-types of SS were identified:

I- Envelope strategy failure (ESF) is a failure of any active envelope strategies. This includes any type of external or internal shading failure (SF), failure of window cooling in a solar chimney) causing an increase in the solar heat gains through the envelope. Thus, the average internal window surface temperature over t_{shock} during ESF ($\overline{T_{wn,ESF}}$) increases with respect to the average internal window surface temperature over the same period ($\overline{T_{wn,NS}}$) in the case of no shock (NS). ESF shocks can be quantified by $doSS_{ESF}$ in equation 3.3:

$$doSS_{ESF} = \underbrace{\frac{\overline{T_{wn,ESF}} - \overline{T_{wn,NS}}}{\overline{T_{wn,NS}}}}_{Severity} \times \underbrace{\frac{t_{shock}}{t_{op}}}_{Duration} \qquad \textit{(Equation 3-3)}$$

In equation 3.3, t_{op} refers to the designed operation time of the system. Note ESF can cause underheating (e.g., failure of window heating using electric strips). The $doSS_{ESF}$ can still be applied in that case. The $doSS_{ESF}$ cannot be applied to fixed envelope elements (i.e., insulation, glazing) as the latter cannot be associated with a failure.

II- Cooling system failure (CSF) (passive or active) installed in a building's air handling unit (AHU). Such failures (i.e., partial, or complete failure of fans, sensor failures, disruption in pumps and water supply) may lead to incomplete or lack of cooling of the

characteristic fluid (*cf*) that drives the cooling load removal from a zone. This leads to an increase in its average temperature over t_{shock} ($\overline{T_{cf,CSF}}$) compared to its average supply temperature for NS ($\overline{T_{cf,NS}}$). CSF can be quantified by $doSS_{CSF}$ in equation 3.4:

$$doSS_{CSF} = \underbrace{\frac{\overline{T_{cf,CSF}} - \overline{T_{cf,NS}}}{\overline{T_{cf,NS}}}}_{Severity} \times \underbrace{\frac{t_{shock}}{t_{op}}}_{Duration} \qquad \textit{(Equation 3-4)}$$

The $doSS_{CSF}$ can be applied to any cooling system including all-air cooling systems (i.e., adiabatic coolers, split air conditioning units, fan coil units and chillers) and hydronic systems (i.e., chilled ceilings, beams). In the case of all-air systems, the cooling fluid is the supplied air (sa) to a zone ($\overline{T_{cf}}$ = $\overline{T_{sa,}}$) while for hydronic systems, the cooling fluid is the supplied water (sw) ($\overline{T_{cf}}$ = $\overline{T_{sw}}$). System failure can also be associated with failure of a heating system causing underheating events (e.g., all-air heating systems such as AHU with heating coils, hydronic heating systems (radiators, heated floors). Equation 3.4 is still applicable.

III- Ventilation system failure (VF): This includes failure of both natural day ventilation and/or natural night ventilation (NNV) through automated windows and failure of a mechanical ventilation system (i.e., partial or complete failure of fans). These strategies provide cooling through supply of outside air. The resulting average room temperature over t_{shock} ($\overline{T_{ra,VF}}$) during the system operation times would be higher than in the case of NS ($\overline{T_{ra,NS}}$) . VF can be quantified by $doSS_{VF}$ in equation 3.5:

$$doSS_{VF} = \underbrace{\frac{\overline{T_{ra,VF}} - \overline{T_{ra,NS}}}{\overline{T_{ra,NS}}}}_{Severity} \times \underbrace{\frac{t_{shock}}{t_{op}}}_{Duration} \qquad \textit{(Equation 3-5)}$$

The $doSS_{VF}$ can be applied to ceiling fans as the latter promote mixing and influence room air temperature.

Both $doES$ and $doSS$ can be used to quantify ES and SS in any type of building where preserving certain temperatures is a requirement for thermal comfort and occupant wellbeing (residential, non-residential, commercial buildings). For this study, degree.hours [K.h] was used as the resilience key performance indicator [43]. It accounts for the occupied hours during which the zone temperature exceeds a pre-defined threshold by 1K [149]. For this study 24°C was chosen as the overheating threshold for European buildings according to the CIBSE TM52 standard [150].degree.hours is calculated as follows:

$$degree.hours = \int_1^2 [T(t) - T_{threshold}]\, dt \qquad \textit{(Equation 3-6)}$$

where T is the indoor operative temperature (°C); $T_{threshold}$ is the 24°C temperature threshold [°C]; t is time (h). The acceptable threshold according to the same standard is equal to 6 [K.h.]

To compare different shocks having different durations (especially ES), Method A as described in Annex F of the EN 16798 [119] was selected. Following this method, the percentage of occupied hours when the zone operative temperature is above 24°C was

evaluated. A percentage of occupied hours below 5% is considered as acceptable and below 3% is considered good.

3.4 Development of typical and extreme weather datasets for studying the resilience of buildings to climate change and heatwaves

Step-by-step method to develop future typical meteorological year (TMY) and heatwave (HW) files for Ghent, Belgium are presented. These HW files were utilized to quantify *doES* and assess thermal resilience of buildings during varying heatwaves. In recent studies assessing thermal resilience, HW files and extreme meteorological year (XMY) files are utilized. Recent studies [151][81][152] have used regional climate model (RCM) data to generate TMY, HW and XMY files for cities across Europe and North America, often incorporating adjustments for temperature, humidity, solar irradiance, and specific thresholds for heatwaves. However, future weather files based on bias-corrected RCM predictions are not easily available to the building scientific community; therefore, a large-scale international collaborative effort by IEA Annex 80- Resilient Cooling for Buildings [101] was made to curate and produce extreme weather data covering major global cities subject to extreme heat hazards by adopting a standardized procedure. For this research, within the framework of IEA Annex 80, weather files (TMY and HW) were developed for ASHRAE climate 4A-Ghent. Building simulation weather files ready to be used by building researchers and practitioners to carry out building energy simulations were developed.

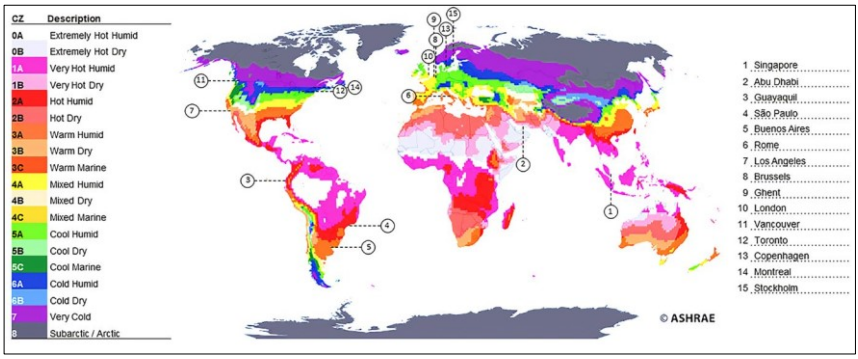


Figure 3-2. 15 cities selected according to ASHRAE classification [156]

As seen in Figure 3-2, the dataset covered 15 cities representative of the ten climate zones of ASHRAE classification [153], focusing on areas where climate change is expected to significantly increase cooling demand [154], prioritizing those with large and growing populations. The majority are in Europe, North America, and Asia, reflecting both data availability and the high frequency of heatwave events recorded in these regions over the last decade[155]. These files are novel in the following respects:

(a) weather datasets were prepared to employ a consistent methodology over 15 cities distributed across the globe in different continents and climate types for 10

climate zones worldwide, as defined by the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) 169–201314 [153]

(b) the future weather files were prepared directly from regional climate model (RCM) simulation results and hence are able to account for complex future changes such as heatwaves in the climate variables projected for each city;

(c) the use of a multivariate bias-correction method was employed to correct the bias associated with the regional climate model simulations;

(d) the reference typical years (TMY) and extreme heatwave (HW) files were provided for building energy, overheating and thermal resilience applications; and

(e) bias-corrected multi-year projections were also made available for additional research and other applications.

Multi-year projections of the TMY and HW files were developed for three periods: (i)historical 2010s (2001–2020), (ii) future mid-term 2050s (2041–2060), and (iii)future long-term 2090s (2081–2100). The files were generated under the RCP 8.5 scenario (selected to evaluate the worst case possible in a resilience and adaptation context) for future periods and were bias-corrected using multi-year observational data for each city. This dataset was among the first to assemble globally uniform bias-corrected RCM projections, incorporating complex future changes, particularly in the frequency, duration, and magnitude of extremes. The flow chart in Figure 3-3 illustrates the steps adopted to generate the weather files. In step 1, raw climate data were extracted for the different weather variables that dominantly affect the thermal performance of buildings for historical (2010s) and two future periods (2050s and 2090s) (20 years for each period). In step 2, these raw climate data were bias-corrected using observations of the different weather variables for the specific locations. In step 3, the weather files were assembled from the multiyear bias-adjusted datasets to generate (a) TMYs based on the EN ISO15927-4 standard [157]and (b) heatwave years (HWYs). The HWYs were based on the method proposed by [158] to detect the heatwaves on a Coordinated Regional Climate Downscaling Experiment (CORDEX) dataset and already tested for building performance simulations in [159].

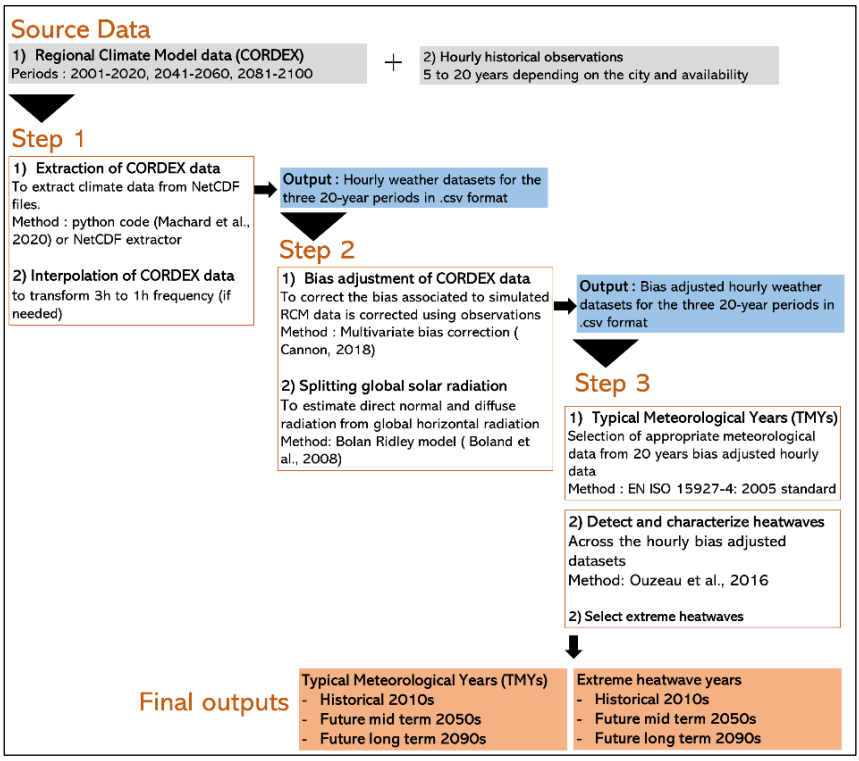


Figure 3-3. Methodology used for the weather datasets generation[160]

Step 1- Extraction and interpolation of CORDEX data : The World Climate Research Program (WCRP)[161], initiated the CORDEX[162] project to evaluate, harmonize and improve climate data projections. The future weather datasets were created using multi-years climate model data from the CORDEX project for historical and future periods (20 years for each period), for RCP 8.5 scenario [57][58], (see [165] for the methodology).

Step 2- Bias adjustment of CORDEX data: The raw climate model data, issued from the CORDEX database, were bias-corrected[166] using observational data of the different weather variables for the specific locations (For this study observational data between 2015-2020 for Ghent was obtained from KMI)[167][168]. Multi-variate bias-correction method adjusts RCM simulations by aligning their statistical properties (e.g., mean, variance, correlations) with observed climate data, ensuring more accurate representation of temperature, humidity, and other variables (solar irradiance). This process corrects systematic biases while preserving inter-variable relationships crucial for realistic climate projections. After bias correction of the historical period (2010s) with observed data, the same bias correction was applied for the two future scenarios (2050s and 2090s).

Step 3- Extract TMY and HWY data: The weather files were assembled from the multi-year bias-adjusted datasets to generate: TMYs for 2010s, 2050s and 2090s based on

the EN ISO 15927-4 standard [169]. To select HWs from the 20-year dataset, the method proposed by Ouzeau et al.[158] was used. The method was validated for France, comparing the new criteria with previous heatwaves detection criteria. This work builds on the methodology initiated by Machard et al.[170] to assemble future weather files including heatwave for building energy and thermal performance simulations from CORDEX data. The method is based only on the temperature to detect HWs. While humidity is known to be an important variable affecting heat stress in hot humid regions, and solar radiation is known to greatly impact indoor overheating, for the purpose of the study a common method was chosen for all the cities. The proposed method has the advantage of being applicable to different cities worldwide since it is based on relative thresholds and not absolute thresholds. As seen in Figure 3-4 three temperature thresholds are used to detect HWs: The S_{pic} threshold (99.5 percentile of the historical multi-year period) is used to detect a temperature peak and a potential HW. The S_{deb} threshold (97.5 percentile) is used to calculate the HW duration and severity (°C.days over the threshold). If the temperature goes under S_{deb} threshold during three consecutive days, the HW stops. The S_{int} threshold (95 percentile) is used to end the heatwave if the temperature drops below the S_{int}. In Belgium, if the temperatures exceed 25°C for a minimum of 5 days and if among these 5 days, temperatures exceed 30°C for a minimum of 3 days, then the event can be considered as a HW [56].

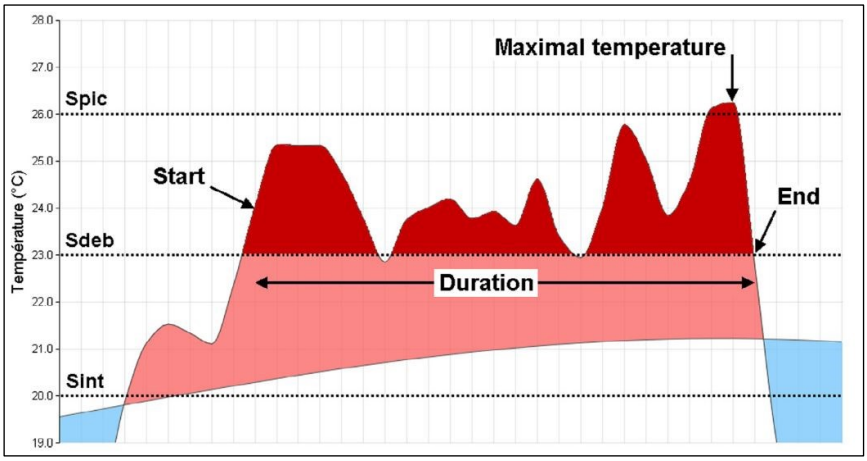


Figure 3-4. Characterization of a heat wave from a daily mean temperature .[158]

Nevertheless, from the detected set of HWs for all three time periods it is possible to add additional criteria to select the extreme heatwaves (XTRM HWs). All the HWs detected from the three 20 year periods can be characterized by three criteria:(a) its intensity (maximum daily mean temperature °C reached during the HW), (b) duration (days) and,(c) severity (aggregated temperature above the S_{deb} threshold in [°C.days]). Using these three criteria, XTRM HWs were detected for each period. Since the purpose of the dataset is to carry out resilience assessments, the three most XTRM HWs from each period were selected to study the buildings thermal resilience performance. Figure

3-5 illustrates the HWs detected for all three periods (2010s, 2050s and 2090s) for Ghent, Belgium. The center of each circle corresponds to the maximum daily temperature during the HW period (y axis) and the duration (in days)(x axis). The area of the circle represents severity i.e., the degree.days for each HW. Adopting the above-mentioned methods in Figure 3-3, a TMY (2010s) and 6 heatwave (historical intense, historical severe-longest, mid-term intense, mid-term severe-longest, long-term intense, long-term severe-longest) files for Ghent was developed and utilized for this study. For Ghent, the most severe and longest HWs coincide within each time period. This means that during the contemporary period, the most severe HW is also the longest, and the same applies to the mid-term and long-term future i.e., the most severe HW during the mid-term is also the longest HW during mid-term and the most severe HW during long-term is also the longest HW during long-term period. As a result, instead of identifying nine distinct HWs across the three periods, only six unique HWs were observed.

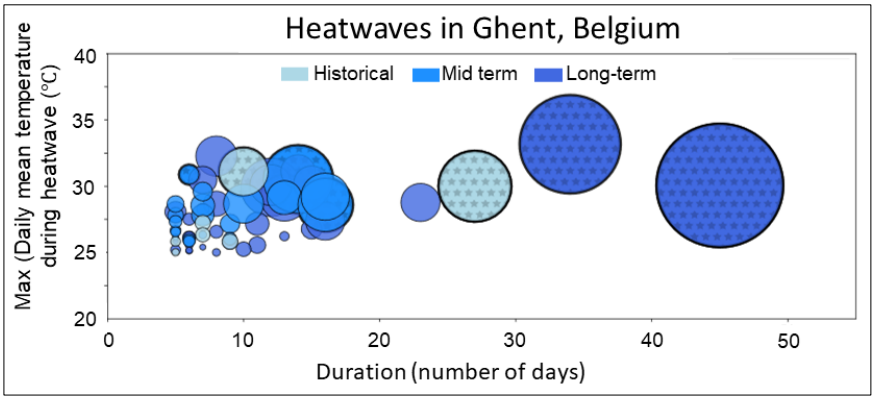


Figure 3-5. Detected heatwaves for Ghent, Belgium for all 3 periods

3.5 Methods

Figure 3-6 illustrates the methodology flowchart which illustrates the steps followed throughout this chapter. The first step was to model in Modelica, the case study building equipped with different passive cooling strategies (NNV, solar shading and IEC). The case study building, its technical installations and their control were described in this step. Step 2 consisted of the validating the model experimentally in one of the case study rooms for a base case of no shock (NS) and a shock case. This step is important to make sure the model's temperature predictions were reliable to conduct the next step of this study which is the parametric analysis (Step 3). The analysis was conducted for different *doES* and *doSS* and combinations of the latter. Finally, the resilience performance of the different cooling strategies were discussed in Step 4, as well as comparisons of different shock types, and assessment of the impact of thermal mass to explain some of the discrepancies between the two zones.

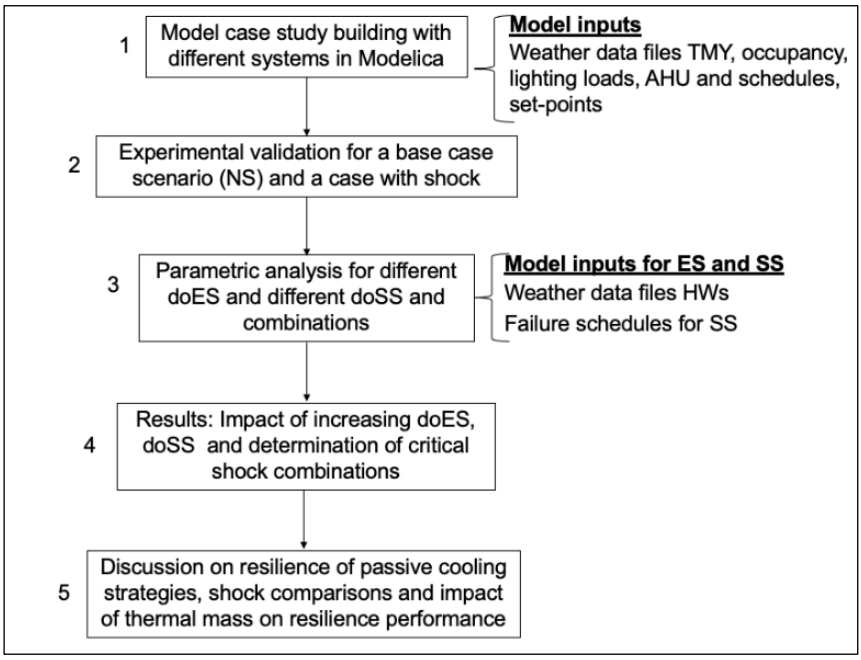


Figure 3-6. Methodology flowchart

3.5.1 Building and Systems' description

The case study building is a real-use educational building, situated at the Ghent Campus, KU Leuven, Belgium. The building consists of two test lecture rooms (TLRs), E120 (1st floor) and E220 (2nd floor) (13.28 m length × 10.69 m width × 2.7 m height) that can each seat a maximum of 80 students. The Air Handing Unit (AHU) [171] serving the lecture rooms resides in a technical room on the top floor. Figure 3-7 illustrates a schematic of the building and its AHU. The building was designed according to Passive House Standard [172]. The two TLRs are identical zones with different air tightness. The airtightness n_{50} of E120 and E220 is 0.41 h^{-1} and 0.29 h^{-1} respectively. E120 has a brick external wall with exterior insulation and E220 has light-weight timber frame external wall. Both exterior walls have the same U-values of 0.15 $W/m^2.K$. Both lecture rooms have concrete slab floor with cellulose insulation and have U-values of 0.15 $W/m^2.K$. The roof is wooden frame with cellulose insulation and have U-value of 0.14 $W/m^2.K$ [173]. The composition of the envelope was presented in Table 3-1 .

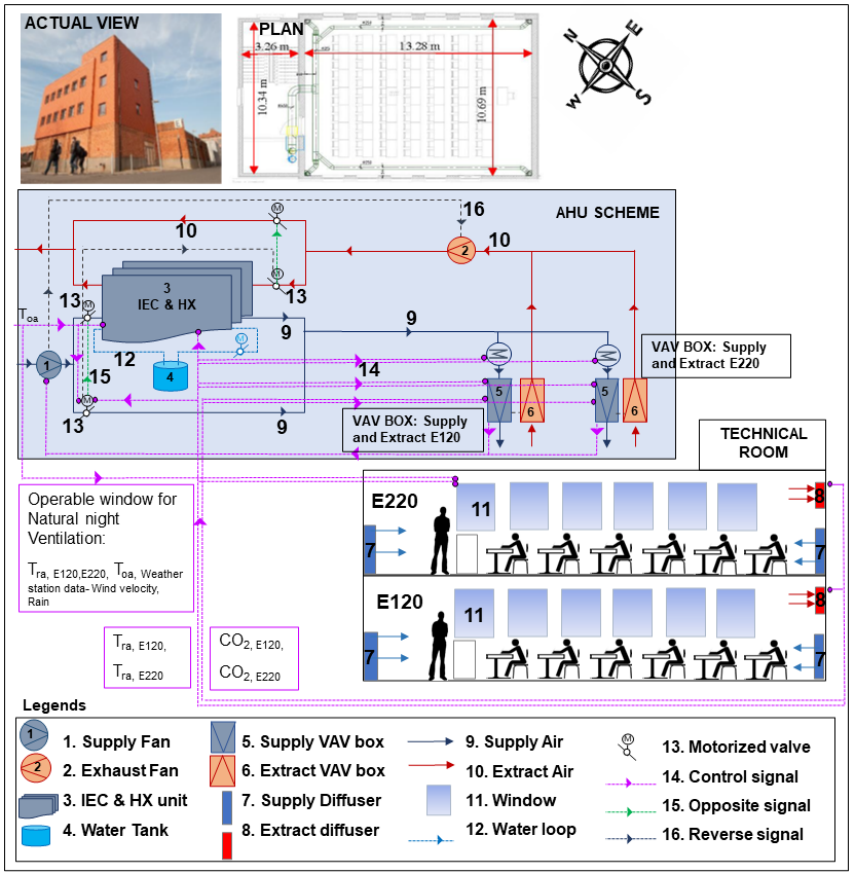


Figure 3-7. Schematic of the test lecture building consisting of E120, E220 and the AHU

Each TLR has ten triple glazed windows (U-value = 0.65 W/m^2.K, window-to-wall ratio = 26.5%, window-to-floor ratio = 13%): six on the southwest facade and four on the northeast façade (Figure 3-7). All windows have manually operable external screens. The screens on the southwest façade were additionally automated (shading ON for 15 minutes when the global solar radiation on the windows is above 250 W/m^2).

The AHU operates from Monday to Friday from 7:30 a.m. to 6:00 p.m. and is shut off during the weekends as there are no classes. The building is equipped with an all-air system with balanced mechanical ventilation. The air is supplied via a displacement ventilation system in both lecture rooms (Figure 3-7). The supply and extract fans supply a minimum airflow of 400 m^3/h and a maximum of 4400 m^3/h. Four variable air volume (VAV) boxes regulated by Proportional Integral (PI) signals control the airflow of the demand-controlled ventilation (DCV) system based on competing CO_2 and temperature demands [174]. The PI control compares the level of deviation of each room temperature from its 22°C setpoint and the level of deviation of each room's CO_2 concentration from an assigned threshold of 1000 ppm [175].

During the cooling season in the daytime, the TLRs are passively cooled by an IEC from 7:30 a.m. until 6:00 p.m., having a maximum capacity of 13.1 kW. It is composed of wet channels equipped with cooling pads wetted with water originating from a feeder tank. A secondary flow of air – in this case the exhaust air from the TLRs, flows through the cooling pads. The clean outdoor air also enters the AHU through a supply fan. If the IEC operation is triggered, all the supply flow is completely diverted through the IEC and the flow is maximum. The IEC operation is regulated via a control block consisting of two types of controls: an ON/OFF controller and a hysteresis controller. The control block takes as input measurements of the TLRs' room air temperatures ($T_{ra,E120}, T_{ra,E220}$) and the outdoor dry bulb temperature T_{oa}. The IEC is triggered if one of the following conditions is true:

a) ON/OFF controller detects that T_{oa} ≥ 22°C or

b) Hysteresis controller detects that max($T_{ra,E120}, T_{ra,E220}$) ≥ 24.5°C.

Once triggered, the hysteresis controller will keep the IEC in operation until max ($T_{ra,E120}, T_{ra,E220}$) ≤22.5°C.

The supply VAV boxes control the flow rate for each lecture depending on the demand and cooled air is supplied to E120 and E220.

The building is also equipped with NNV for pre-cooling of the TLRs during night-time [176]. NNV is activated from 10:00 p.m. to 6:00 a.m. from 1st April to 31st October, by opening of 10 motorized windows on both southwest and northeast façades [176]. The total effective operable area of these windows is 4% of the floor area. Once open, the window will remain open for at least 15 minutes if all of the following conditions are met simultaneously:

a) Max ($T_{ra,E120}$ or $T_{ra,E220}$) ≥ 22°C and ≥ than the outdoor air temperature T_{oa} by 2°C

b) Maximum zone temperature ($T_{ra,E120}, T_{ra,E220}$) of the previous day ≥ 23°C

c) T_{oa} >12°C

d) RH < 70%

e) There is no rainfall and the wind velocity < 10 m/s measured at a height of 10 m from ground level.

Table 3-1. Wall, ceiling, and floor composition (outside to inside) of E120 and E220 and n50 values.

TLR	Component	Material	U value	Component	Material	U value
Lecture room E120 (n_{50} = 0.41 h^{-1})	*External wall composition*	*Cement panel 12 mm* *Wooden framework 40 mm* *Wood fiber insulation 18 mm* *Cellulose insulation 300 mm* *Brickwork 140 mm* *Plasterboard 10 mm*	*0.15 W/m²K*	*Floor composition*	*Gypsum plate 10 mm* *Cellulose insulation 360 mm* *Concrete slab 320 mm* *Screed 60 mm* *Linoleum 3 mm*	*0.15 W/m²K*
Lecture room E220 (n_{50} = 0.29 h^{-1})	*External wall composition*	*Cement panel 12 mm* *Wooden framework 40 mm* *Wood fiber insulation 18 mm* *Cellulose insulation 300 mm* *Gypsum board 12.5 mm*	*0.15 W/m²K*	*Floor composition*	*Linoleum 3 mm* *Screed 60 mm* *Concrete slab 320 mm* *Cellulose insulation 300 mm* *Wooden frame 50 mm* *Rockwool insulation 40 mm*	*0.15 W/m²K*
	Roof composition	*EPDM rubber 22 mm* *Cellulose insulation 320 mm* *Wooden frame 100 mm* *Rockwool insulation 40 mm*	*0.14 W/m²K*			

Table 3-2. Class schedules in a) E120 and b) E220.

a) E120			
Classes	**Start**	**End**	**Duration (minutes)**
Class 1	*8:15 a.m.*	*9:45 a.m.*	*90'*
Class 2	*9:45 a.m.*	*11:15 a.m.*	*90'*
Class 3	*11:15 a.m.*	*12:45 p.m.*	*90'*
Break	*12:45 p.m.*	*13:30 p.m.*	*45'*
Class 4	*13:30 p.m.*	*15:00 p.m.*	*90'*
Class 5	*15:00 p.m.*	*16:30 p.m.*	*90'*
b) E220			
Classes	*Start*	*End*	*Duration (minutes)*
Class 1	*8:15 a.m.*	*11:15 a.m.*	*180'*
Class 2	*11:15 a.m.*	*12:45 p.m.*	*90'*
Break	*12:45 p.m.*	*13:30 p.m.*	*45'*
Class 3	*13:30 p.m.*	*15:00 p.m.*	*90'*
Class 4	*15:00 p.m.*	*16:30 p.m.*	*90'*

3.5.2 Modelica model and inputs

A simulation model of the building was developed using Modelica [177] in the Dymola environment (2020x) [178] and using the "Integrated District Energy Assessment by Simulation" (IDEAS) [179] library (Version v2.2.1). Modelica is an open-source, equation-based modelling language with embedded libraries like IDEAS containing validated sub-models that can accurately represent the building dynamics, HVAC systems, the control strategies as well as varying events with different time scales. Figure 3-8 illustrates the Modelica model of the building as seen in Dymola. Equation 3.7 represents the energy balance in the zone model:

$$V_{ra} C_{p,a} \frac{dT_{ra,z}}{dt} = \dot{Q}_{a,z} C_{p,a} \left(T_{supp} - T_{ra,z}\right) + \dot{Q}_{inf,z} C_{p,a} \left(T_{oa} - T_{ra,z}\right) + Q_i + Q_{envelope}(t) \qquad \textit{(Equation 3-7)}$$

The left side of equation 3.7 represents the transient heat stored in the lecture room. The 1st term on the right-hand side represents the net convective heat flow due to the building ventilation and energy systems, where $C_{p,a}$ is the specific heat of air, T_{supp} is the supply air temperature and $T_{ra,z}$ is the zone temperature. The second term is the convective heat flow due to infiltration through the building envelope (i.e., cracks, window opening), where T_{oa} is the dry bulb temperature. The 3rd term on the right-hand side Q_i is the sensible heat generation by occupants and lighting. $Q_{envelope}(t)$ is the convective heat exchange between the lecture room and its envelope. As seen in Figure 3-8 each lecture room was modelled as a separate thermal zone by the zone component (2) and the envelope components (outer walls (19), windows (20, 22), partitions (21)). The building envelope use equations described in [180] to compute the varying radiative heat exchange and convective heat transfer coefficients between the component and the zone air temperature. The floor of E120 (lower lecture room) and partition wall separating both lecture rooms to the staircase was modelled as adiabatic.

The zone model was considered to achieve well-mixed and homogeneous conditions of temperature.

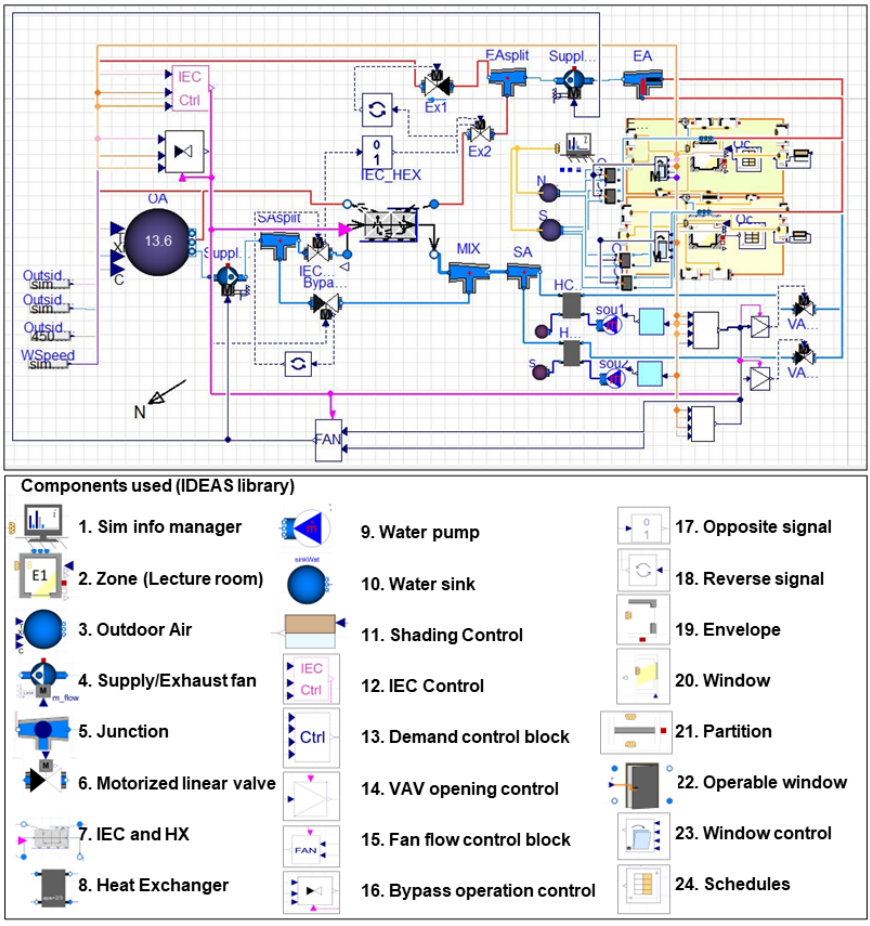


Figure 3-8. Modelica Model of the test lecture room building consisting of the building envelope, and systems + controls

As seen in Figure 3-8 using the IDEAS library, the hydronic heating system for each classroom was modeled by the interconnected components 8, 9 and 10 which represent the air-to-water heat exchanges (HX), water pump, and water sink respectively. The HX is a counterflow plate HX with the heating coils modeled using the effectiveness-NTU method. The water pump is an ideal flow source of water with fixed temperature of 60°C. The flow rate is regulated with the PI control block 13 to maintain the classroom setpoint of 22±0.5°C. The cooling system of the building was modeled by the interconnected component 7, (IEC) and its control block component 12, which controls its operation according to the description in section 3.5.1. The IEC model is a validated model of a Menerga AHU, composed of dry and wet channels. The supply air passes in the dry channels where it is sensibly cooled. The exhaust air stream passes in the wet

channels, where it absorbs the heat released from the supply air in the dry channels and the evaporated water before being discharged to the ambient as hot humid air. The model is based on the effectiveness-NTU method and solves the heat and mass balance equations in the dry and water channels of the IEC. It takes as input the UA value, the heat exchange efficiency η and outputs the final supply air temperature T_{sa} after exiting the dry channels (T_{supp}) and the final exhaust air temperature and its humidity content after exiting the wet channels. When the IEC is not needed (during winter and sometimes during early or late summer), component 7 operates as an HX for heat recovery between the supply and return air. To regulate the fraction β of air going through the heat recovery and the fraction $1-\beta$ going through the bypass, component 6 (a two-way linear valve) and component 5 (junction to split the flow) were used. The percentage of opening of the valves was controlled using component 16: Bypass Control. The flow rate is directly proportional to the valve lift at a constant differential pressure. The NNV was modelled via the operable window components (22), which are connected to the outdoor air component on one side and the zone model on another side. The operable window model is based on power law relationships that calculate the airflow depending on the pressure difference in both sides of the component [181]. The outdoor air component (3) computes the wind pressure coefficients around the building. Depending on the outside and zone conditions, the window control (23) opens or closes the windows. The windows (20) have an internal shading PI control (11) that takes as input the solar radiation on the envelope surface from the weather data file to be 0= OFF or 1=ON.

The model inputs were the weather data in the Sim Info Manager component (1) and the occupancy schedules (24). The sensible and latent heat generation from occupants were 75 W and 45 W respectively during sedentary activities [182]. The occupancy schedule of the two lecture rooms was taken as input from the building monitoring system (BMS). Occupancy in the lecture rooms was tracked by an Acurity 3D vertical optical counting system installed in the lecture rooms 1 m from the door entrance [183]. The two TLRs, were typically used from Monday to Friday, between 8:15 a.m. and 6:00 p.m. following the class schedule in Table 3-2. Figure 3-9 represents the weekly occupancy schedule from 8th to 15th of June 2021. This week was chosen for the simulations since the full classes schedules were conducted each day of the week for each TLR (see Table 3-2). Since the effect of the simulated shocks will be monitored for several weeks during and after the shock, this occupancy schedule was repeated periodically. Sensible heat generation due to lights was defined in Modelica as 700 W in each zone model. The lights only turned on when occupants were present.

Simulations were run with 1 minute time step to compute the operations AHU, NNV and shading apart from the temperature variations. The simulations were run from seven days prior to stabilize the solution on the period of shock, and fifteen days after to check the extend of shock impact on thermal conditions of the TLRs. Each simulation period varies according to the duration of each shock type. For SS it is 24 hours (simulation time = 1 week prior + shock duration =1 day + 2 weeks post shock). For ES it depends on the duration of HWs (simulation time= 1 week prior + HW duration + 15 days post

shock). To solve for the system of non-linear equations, the RADAU II, an implicit Runge-Kutta method, with a tolerance of 10^{-8} was used. This method is a single-step method, where the solution at the current time step is only affected by information from the previous time step [184]. Thus, RADAU II solver was able to handle complex equations and compute the solutions with higher accuracy in comparison to methods such as Euler or Dassl.

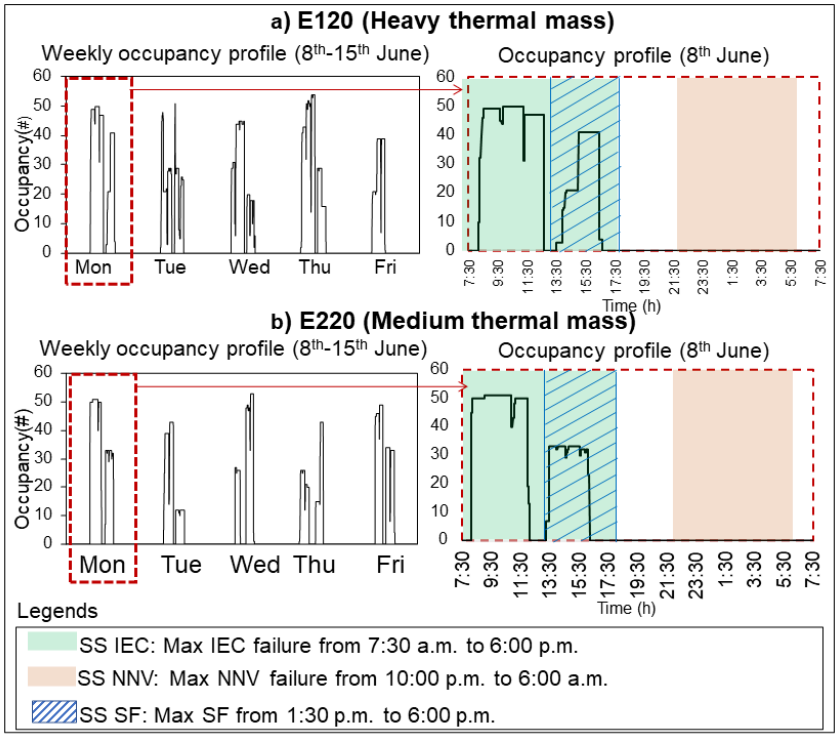


Figure 3-9. Weekly occupancy schedules in a) E120 and b) E220 (8th-15th June) and a close up of the occupancy schedule on the 8th of June when SS was inflicted

3.5.3 Experimental model validation

Figure 3-10 illustrates the experimental setup in lecture room E120. Experiments were conducted in E120, using thermal manikins to validate the numerical predictions of the Modelica model (thermal fields, operation of IEC and NNV) under normal operating conditions without shocks (NS =base case) and a shock case (30-minute power outage). Thermal manikins were used instead of occupants in the experiment to provide controlled input for internal gains in the experimental set up as well as the simulation model.

The experimental method of Al Assaad et al. [63,64], employing heated cylindrical thermal manikins, was used to validate the numerical predictions of the Modelica model ($T_{ra,E120}$, operation of IEC and NNV). 21 fully enclosed cylindrical thermal manikins made of 3 mm thick aluminum sheets were used to simulate the occupancy. These

manikins were equipped with a 75 W heat source to simulate the sensible heat released by occupants during sedentary activities [182]. The manikins had a surface area of 1.8 m^2, and had a height of 1.1 m [185]. Thermal manikins were used instead of occupants in the experiment to provide controlled input for internal gains in the experimental set up as well as the simulation model. The actual occupancy schedules were later on used for the parametric study. The manikins were uniformly distributed across E120 to reduce spatial gradients in temperature distribution [182]. Temperatures were monitored at the room exhaust diffuser at 1-minute intervals using VAISALA GMW90 sensors [187] that measure temperatures between -5°C and 55°C with an accuracy of ±0.5°C. Apart from these measurements, continuous logging of the parameters like supply and return air temperatures, operation signal of the AHU, IEC, NNV, VAVs etc., was done using the BMS [188]. The Modelica model was validated under normal operating conditions without shocks (no shock NS=base case) and a shock case (30-minute power outage) [186]. Two experimental scenarios were conducted during the 2nd week of August 2021 and lasted one day each. No classes were scheduled during that period. The base case was conducted on the 8th of August. The second case is a shock scenario of 30 min power outage (PO) shock and was conducted on the 14th August 2021. The PO was manually inflicted by turning off the AHU for 30 minutes starting 3:00 p.m. In both cases, the manikins' heat sources were turned on 12:00 p.m. and turned off at 5:00 p.m. This type of shock is a combination of two SS (failure of IEC, and ventilation failure- VF). It was chosen due to its severe impact, ease of implementation and since triggering an ES is not possible in a real case study.

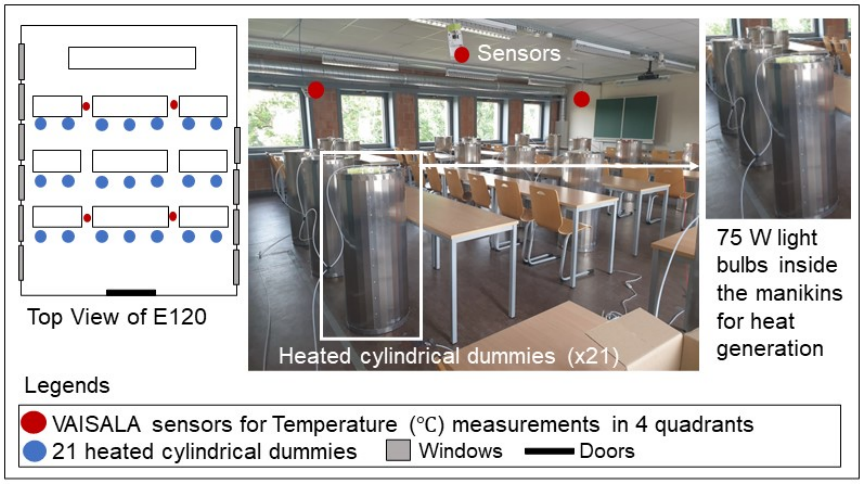


Figure 3-10. Experimental setup in E120 with 21 heated cylinders and measuring equipment.

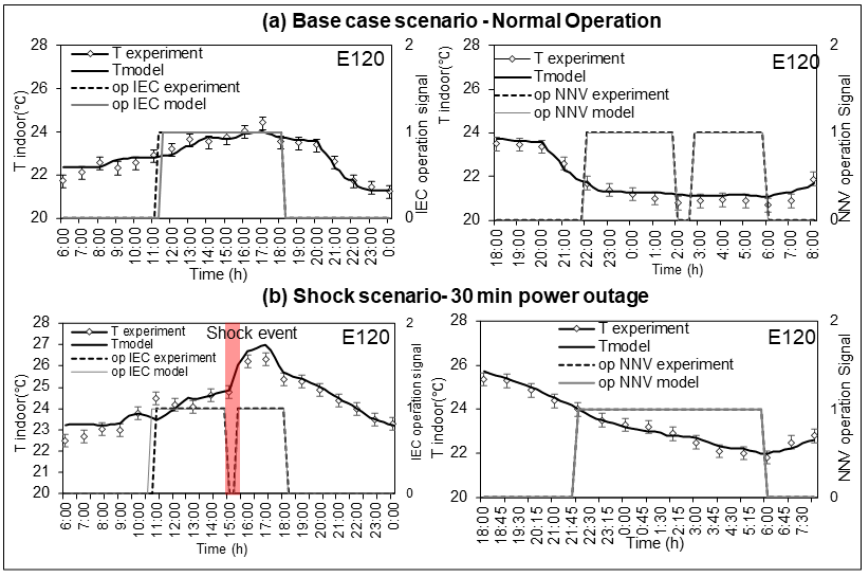


Figure 3-11. Comparison of the predicted and measured temperatures of E120, operation of IEC and NNV for: (a) normal operation and (b) 30 minute power outage

Figure 3-11 shows a comparison of predicted vs. measured $T_{ra\ E120}$, operation of IEC and NNV during - (a) base case and (b) 30 min power outage. For normal operation (Figure 3-11 (a)), $T_{ra\ E120}$ increased from 22.3±0.5°C at 8:15 a.m. reached a maximum of 24.4±0.5°C at 3:30 p.m. due to heat build-up (internal gains due to occupants, equipment and solar gains). Experimentally, the IEC was triggered at 11:07 a.m., due to outdoor conditions. IEC is deactivated at 6:00 p.m. After the manikins were turned off at 5:00 p.m., $T_{ra\ E120}$ decreased gradually to reach 22.5±0.5°C at 9:00 p.m.. NNV operation started at 10:00 p.m. $T_{ra\ E120}$ decreased by 1.3°C. NNV was inactive for 30' from 2:00 a.m. to 2:30 a.m. as $T_{ra\ E120}$ reached below 21.5°C. The same temperature trend and operation of IEC and NNV can also be observed in the simulation model. There are two conditions that the results of the simulation model must meet to be considered validated-(a) The MAE (Mean absolute value of error) < 1°C and (b) RMSE (Root mean squared error) <1.5°C [189]. The MEA and RMSE between the simulated model and the monitored $T_{ra\ E120}$ is 0.75°C and 0.95°C. Good agreement was obtained between predicted and measured values of $T_{ra\ E120}$ (maximum relative error of 4.6%). During 30' power outage (Figure 3-11.(b)), 21 manikins were activated at 1:00 p.m. IEC was activated in experimental setup at 10:50 a.m. as T_{oa} reached 22°C. Due to IEC activation, $T_{ra\ E120}$ dropped by ±0.5°C at 11:00 a.m.. After the manikins were activated at 1:00 p.m., $T_{ra\ E120}$ increased from 22.7±0.5°C to reach 24.4±0.5°C at 3:00 p.m. A power outage (AHU shut-down) was inflicted from 3:00 p.m. to 3:30 p.m. IEC operation was restored at 3:30 p.m., but $T_{ra\ E120}$ increased to 26.0±0.5°C at 5:00 p.m. due to insufficient IEC operation and high wet bulb temperature. The manikins were turned off at 5:00 p.m. NNV was activated at 10:00 p.m. and $T_{ra\ E120}$ dropped by 0.5°C. The

same temperature trend and operation of IEC and NNV can also be observed in the simulation model. MEA and RMSE between the simulated model and the monitored $T_{ra\ E120}$ is 0.95°C and 1.10°C respectively. Good agreement was obtained between predicted and measured values of $T_{ra\ E120}$ and the measured vs simulated IEC and NNV operation times (maximum relative error of 1.2%).

3.6 Parametric study

A parametric study was conducted using the validated building model to assess and compare the impact of ES, SS and combinations of them. For benchmarking, base case scenarios with no shock (NS) were defined for each shock type.

3.6.1 Selection of ES

TMY and HW files are needed as input to the Modelica model to assess ES. Development of weather files were described in section 3.4. For each period, the most extreme heatwaves (intense, severe, and longest) were selected, as shown in Table 3-3 and Figure 3-12.

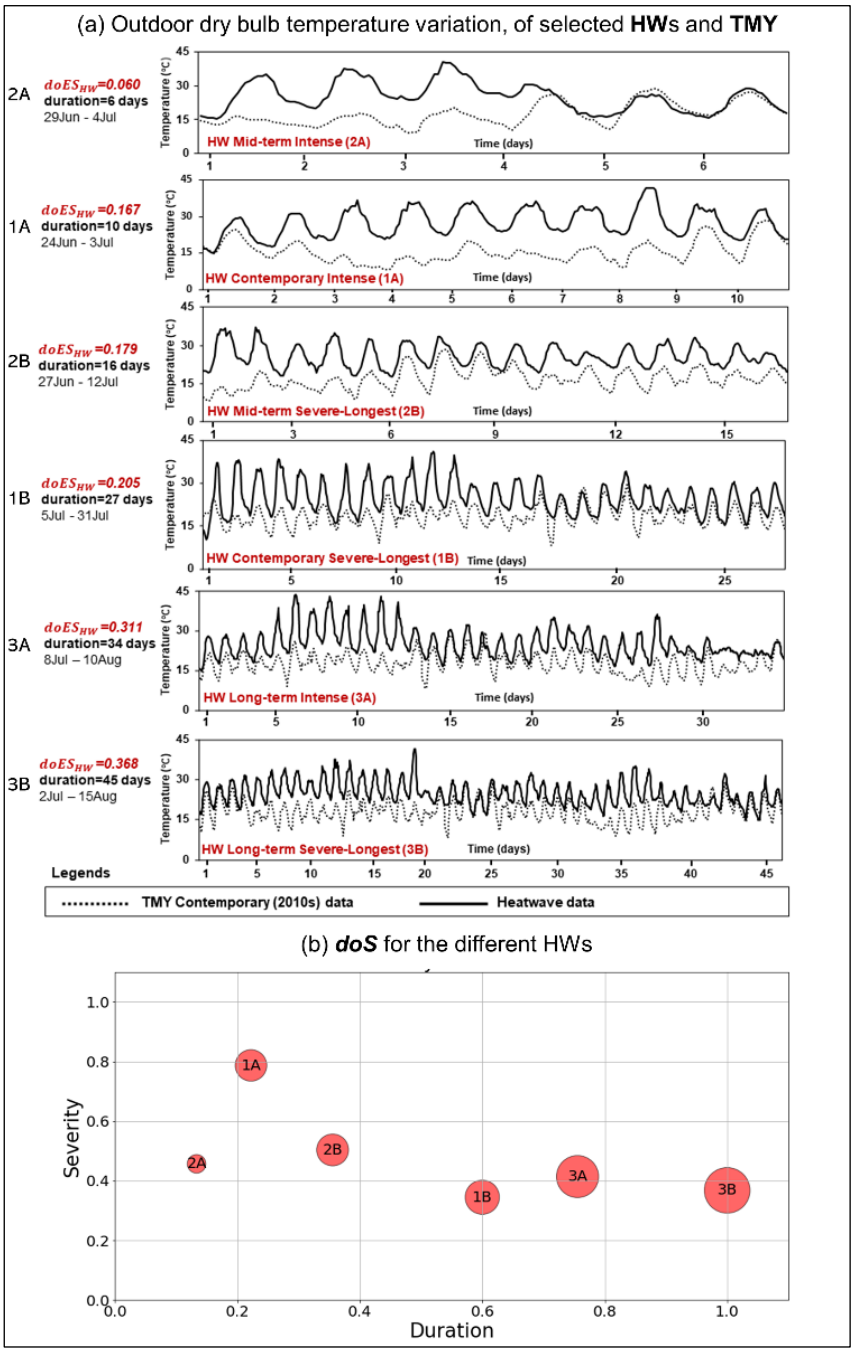


Figure 3-12. Outdoor dry bulb temperature variation, of selected six extreme heatwaves and corresponding TMY period and, (b) doS (diameter of each circle) with duration in days (x -axis) and the mean temperature during HWs (y-axis left)

Table 3-3. Selected HWs for Ghent

Heatwave period and ID	Characteristics of HWs[1]						
	$\overline{T_{oa,HW}}$ (average temperature during heatwave) [°C]	T_{ref} (average outdoor temperature during corresponding TMY period) [°C]	Peak temperature during heatwave [°C]	Peak temperature during TMY period [°C]	t_{shock} (Duration of heatwave [days]	$t_{shock\ max}$ (duration of the longest heatwave [days]	$doES_{HW}$[2]
Contemporary 1A	26.8	15	41.8	28.7	10 (June 24 - July 3, 2020)	45	0.167
Contemporary 1B	25	18.6	40.9	29.3	27 (July 5 - July 31,2002)	45	0.205
Mid-term **2A (doS_{low})***	24.9	17.1	40.4	28.7	6 (June 29 - July 4, 2043)	45	**0.060***
Mid-term **2B (doS_{medium})***	26.3	17.5	37	28.6	16 (June 27 - July 12,2056)	45	**0.179***
Long-term 3A	25.3	17.9	43.8	29.8	34 (July 8 - Aug 10,2097)	45	0.311
Long-term **3B (doS_{high})***	25.7	18.8	41.7	29.4	45 (July 2 - Aug 15,2090)	45	**0.368***

[1] Corresponding TMY weather data files (reference case) have the same duration and occurrence period as HWs

[2] $\overline{T_{oa,ES}}$ in equation 3.2 corresponds to $\overline{T_{oa,HW}}$. t_{shock} (days) is the duration of the HW being evaluated, whereas $t_{shock,max}$ (days) is the longest predicted heatwave in the span of the next 100 years (45 days)

*Selected for development and testing the thermal resilience assessment framework in Chapter 4 and 5

3.6.2 Selection of system shocks (SS)

To evaluate the impact of increasing degree of SS, Shading failure (SF) ($doSS_{SF}$), IEC failure ($doSS_{IEC}$) and NNV failure ($doSS_{NNV}$) were inflicted upon the two TLRs in the beginning of the week (Monday -June 8th). For each of the SS, the benchmark case is a base case with NS having the same duration as each shock case. Table 3-4 presents the different SS simulation scenarios. For SS the max duration of each SS is equivalent to its full operational period. For each SS, the shock was increased from 30' to its maximum operation duration with an increment of 30' (maximum operation time is 630' for IEC, 270' for SF and 480' for NNV). For IEC, the shock started at 7:30 a.m. as it is the starting time of the AHU. When the conditions for IEC operation are met (section 3.5.1), IEC is operational between 7:30 a.m. and 6:00 p.m. Thus maximum duration of IEC failure is the entire duration of its operation i.e. 630'. For SF, the shock started at 1:30 p.m. since solar radiation increases (>250 W/m^2) on the southwest façade starting from 1:30 p.m. The shading operating schedule from the building monitoring system also demonstrated activation of shading during summer from 1:30 p.m. onwards. Thus, maximum operation period for solar shading is between 1:30 p.m. and 6:00 p.m. Similarly, for NNV failure, if the conditions for window opening are met (see section 3.5.1), the shock starts at 10:00 p.m. (i.e., start time of NNV operation) and maximum shock duration is 480' (10:00 p.m. to 6:00 a.m.). $\overline{T_{sa,IEC}}$, $\overline{T_{sa,NS}}$, $\overline{T_{wn,SF}}$, $\overline{T_{wn,NS}}$, $\overline{T_{ra,NNV}}$ and $\overline{T_{ra,NS}}$ are outputs from the validated Modelica model.

To assess the most critical time of SS occurrence, each SS was inflicted at different times during system operation. IEC failure can occur during any time period between (7:30 a.m. to 6:00 p.m.), the time of shock occurrence can have different effects depending on the occupancy and thermal storage. SF can occur during any time period between (1:30 p.m. to 6:00 p.m.) and the time of shock occurrence can have different effects depending on the solar radiation on the glazing. NNV failure can occur during any time between (10:00 p.m. to 6:00 a.m.) and the time of shock occurrence can have different effects depending on the difference due to outdoor and indoor conditions (temperature, RH (%), wind speed and direction) and also on the thermal storage of the two lecture rooms. Table 3-4 illustrates the different times of occurrence of each type of SS.

3.6.3 Shock combinations

Shock combinations were assessed to quantify how critical a shock is if it is accompanied by another shock (e.g., power outages being common during heatwaves). However, in this research to quantify the doS of different shock combinations, simple addition or multiplication of the individual doS values were not selected because combination of shocks are not interdependent, and their effects do not scale linearly. Adding doS values assumes independent contributions, which would overestimate or underestimate the total impact, while multiplication would unrealistically suggest an exponential effect. Moreover, the response to multiple failures is inherently non-linear—cooling and ventilation failures, for instance, amplify overheating risks differently depending on factors such as climate, insulation, and thermal mass. The

combined impact of these shocks is often greater than the sum of individual effects due to cascading interactions. To accurately capture these dependencies, further research would be necessary. Such as, a more sophisticated, weighted non-linear model would correctly capture the doS of a combined shock taking into account weights for individual shocks. However, defining appropriate interaction terms necessitates further research. For this study, the doS of individual shocks were noted in Table 3-5 without defining the combined doS. For the shock combination of ES+SS scenario, the mid-term (2050s) HW (ID = 2A; $doES_{HW}$ = 0.06, Table 3-3) was selected. This HW was selected since it is in line with the past and recent HWs in Belgium in terms of daily average outdoor air temperature and in duration (i.e., most relevant for current times and buildings). The HW (ID = 3B, $doES_{HW} = 0.368$), with highest impact on the two TLRs was not selected since it was predicted to occur in the long-term future scenario (2081-2100) when the current building stocks are assumed to have undergone improvement in construction techniques, energy efficient cooling strategies and other overheating mitigation strategies. SS was inflicted on the hottest day of the HW (in this case-1st July). For a SS + SS combination, the maximum shock duration for each SS was inflicted. Table 3-5 gives an overview of the different chock combinations simulation scenarios.

Table 3-4. Parametric study table for SS

Type of shock	Max shock duration (minutes)	Time of occurrence	Calculated doS	Justification [1 $\Delta(\mathrm{doSS_{SF}}) = 0.006$ equivalent to 30' increment of SF, 2 $\Delta(\mathrm{doSS_{IEC}}) = 0.015$ equivalent to 30' of IEC failure, 3 $\Delta(\mathrm{doSS_{NNV}}) = 0.001$ equivalent to 30' of NNV failure]
NS	-	-	-	*No shock occurs (base case)*
			SS: SF: Failure of automated shading	
$[doSS_{SF}]^1$	*270*	*1:30 p.m.*	*[0.006 - 0.323] for both TLRs*	*0.006 corresponds to 30' of SF starting from 1:30 p.m. and 0.323 corresponds to max* $\mathrm{doSS_{SF}}$ *of 270 minutes: 1:30 p.m. - 6:00 p.m.*
	180	*3:00 p.m.*	*[0.002 - 0.323] for both TLRs*	
			SS - VF: Failure of IEC	
$[doSS_{IEC}]^2$	*630*	*7:30 a.m.*	*[0.015 - 0.127] for both TLRs*	*0.015 corresponds to 30' of IEC failure starting from 7:30 a.m. and 0.127 corresponds to max* $\mathrm{doSS_{IEC}}$ *of 630': 7:30 a.m. - 6:00 p.m.* $\mathrm{doSS_{IEC}}$ *range varies in the two TLRs due to different occupancy level and the demand for the supply air.*
	495	*9:45 a.m.*	*[0.004 - 0.127] for E120* *[0.005 – 0.127] for E220*	
	180	*3:00 p.m.*	*[0.005 - 0.127] for E120* *[0.011 – 0.127] for E220*	
			SS - VF: Failure of NNV	
$[doSS_{NNV}]^3$	*480*	*10:00 p.m.*	*[0.001 - 0.047] for E120* *[0.001 -0.021] for E220*	*0.001 corresponds to 30'of NNV failure starting from 10:00 p.m.. 0.047 and 0.021 are max* $\mathrm{doSS_{NNV}}$ *for 480': 10:00 p.m.- 6:00 a.m. for E120 and E220 respectively.* $\mathrm{doSS_{NNV}}$ *range varies in the two TLRs due to different heat gains during the day (occupancy) and difference in air-tightness for both the TLRs*
	360	*12:00 a.m.*	*[0.0009 - 0.047] for E120* *[0.0008 - 0.021] for E220*	
	240	*2:00 a.m.*	*[0.00075- 0.047] for E120* *[0.0006 - 0.021] for E220*	
	120	*4:00 a.m.*	*[0.0005 - 0.047] for E120* *[0.0003 - 0.021] for E220*	

Table 3-5. Parametric study table for Shock combinations

Type of shock 1	Type of shock 2	Maximum shock duration	Calculated doS for both TLRs	Time of occurrence
NS	*NS*	*-*	*0.00*	*-*
		ES and SS combinations[1]		
HW	*IEC*	*HW= 6 days; IEC= 630'*	$doES_{HW}$ *= 0.060;* $doSS_{IEC}$ *= 0.127*	*IEC : 7:30 a.m.*
HW	*SF*	*HW= 6 days; SF= 270'*	$doES_{HW}$ *= 0.060;* $doSS_{SF}$ *= 0.323*	*SF : 1:30 p.m.*
HW	*NNV**	*HW= 6 days; NNV= 480'*	$doES_{HW}$ *= 0.060;* $doSS_{NNV}$ *= 0.050*	*NNV : 10:00 p.m.*
HW	*IEC and VF (AHU failure)*	*HW= 6 days; IEC = 630'; VF= 630'*	$doES_{HW}$ *= 0.060;* $doSS_{VF}$ *= 0.168;* $doSS_{IEC}$ *= 0.127*	*Ventilation system and IEC failure : 7:30 a.m.*
HW	*IEC and SF*	*HW= 6 days; IEC = 630' ; SF= 270'*	$doES_{HW}$ *= 0.060;* $doSS_{IEC}$ *= 0.127;* $doSS_{SF}$ *= 0.323*	*IEC : 7:30 a.m.* *SF : 1:30 p.m.*
HW	*IEC and NNV**	*HW= 6 days; IEC = 630' ;NNV= 480'*	$doES_{HW}$ *= 0.060;* $doSS_{IEC}$ *= 0.127;* $doSS_{NNV}$ *= 0.05*	*IEC : 7:30 a.m.;* *NNV : 10:00 p.m.*
HW	*SF and NNV**	*HW= 6 days; SF=270' NNV= 480'*	$doES_{HW}$ *= 0.060;* $doSS_{SF}$ *= 0.323;* $doSS_{NNV}$ *= 0.050*	*SF : 1:30 p.m.;* *NNV : 10:00 p.m.*
HW	*IEC, SF and NNV**	*HW= 6 days; IEC = 630'; SF= 270'; NNV= 480'*	$doES_{HW}$ *= 0.060;* $doSS_{SF}$ *= 0.323;* $doSS_{IEC}$ *= 0.127;* $doSS_{NNV}$ *= 0.050*	*IEC : 7:30 a.m. ;* *SF : 1:30 p.m. ;* *NNV : 10:00 p.m.*
HW	*VF, IEC and NNV* (AHU and NNV* failure)*	*HW= 6 days; VF= 630'; IEC = 630' ;NNV= 480'*	$doES_{HW}$ *= 0.060;* $doSS_{VF}$ *= 0.168;* $doSS_{IEC}$ *= 0.127;* $doSS_{SF}$ *= 0.323*	*Ventilation system and IEC failure : 7:30 a.m. ; SF : 1:30 p.m.* *NNV: 10:00 p.m.*

HW	*VF, IEC, SF and NNV* (Power outage)*	*HW= 6 days; VF= 630'; IEC = 630' ;SF= 270' ;NNV= 480'*	*$doES_{HW}$ = 0.060; $doSS_{VF}$ = 0.168; $doSS_{IEC}$ = 0.127; $doSS_{SF}$ = 0.323; $doSS_{NNV}$ = 0.05*	*Ventilation system and IEC failure:7:30 a.m.; SF : 1:30 p.m. ; NNV : 10:00 p.m.*
		SS and SS combinations[2]		
IEC	*VF*	*VF= 630'; IEC = 630'*	*$doSS_{VF}$ = 0.168*	*Ventilation system and IEC : 7:30 a.m.*
IEC	*SF*	*IEC = 630'; SF= 270'*	*$doSS_{SF}$ = 0.323*	*IEC : 7:30 a.m. ; SF : 1:30 p.m.*
IEC	*NNV**	*IEC = 630' ;NNV= 480'*	*$doSS_{NNV}$ = 0.050*	*IEC : 7:30 a.m. ; NNV : 10:00 p.m.*
SF	*NNV**	*SF= 270' ;NNV= 480'*	*$doSS_{NNV}$ = 0.050*	*SF : 1:30 p.m. ; NNV : 10:00 p.m.*
IEC	*SF and NNV**	*IEC = 630' ;SF= 270' ;NNV= 480'*	*$doSS_{SF}$ = 0.323; $doSS_{NNV}$ = 0.050*	*IEC : 7:30 a.m. ; SF : 1:30 p.m. ; NNV : 10:00 p.m.*
VF	*IEC and SF*	*VF= 630'; IEC = 630'; SF= 270'*	*$doSS_{VF}$ = 0.168 ; $doSS_{SF}$ = 0.323*	*Ventilation system and IEC failure : 7:30 a.m.; SF : 1:30 p.m.*
VF	*IEC, SF and NNV* (Power outage)*	*VF= 630' ; IEC = 630' ; SF= 270' ; NNV= 480'*	*$doSS_{VF}$ = 0.168; $doSS_{SF}$ = 0.323; $doSS_{NNV}$ = 0.050*	*Ventilation system and IEC failure : 7:30 a.m. ; SF failure :1:30 p.m. ; NNV:10:00 p.m.*

*[1] SS (one or more types) occur on the hottest day of the 6 day long HW period. ; [2] One or more system shocks occur on the beginning of a week in summer period (Monday 8th June); * Common value of $doSS_{NNV}$*

3.7 Results

This section first presents the base case scenario for SS and for ES, then presents the impact of increasing the degree of shock for ES and SS and finally reports the impact of different shock combinations.

3.7.1 Base case scenario (no shocks)

For the base case scenario with no shock (NS), the contemporary TMY (2010s) file was selected which can properly demonstrate the most typical pattern of performance during recent summers. TMY (2010s) weather scenario has an average dry bulb temperature of 10.9°C with a maximum temperature of 39.1°C in July. It is characterized by 178 cooling degree days (CDD 18°C). The average global horizontal radiation yearly and for summer period are 125 Wh/m^2 and 192 Wh/m^2 respectively. Figure 3-13 illustrates the dry bulb temperature and global horizontal radiation variations between 1st-15th June in the TMY (2010s) weather scenario. Outdoor temperature variations on 8th June (Monday) represents a typical summer day of Belgian climate [80]. Average dry bulb temperature is 17°C with maximum of 21.6°C.

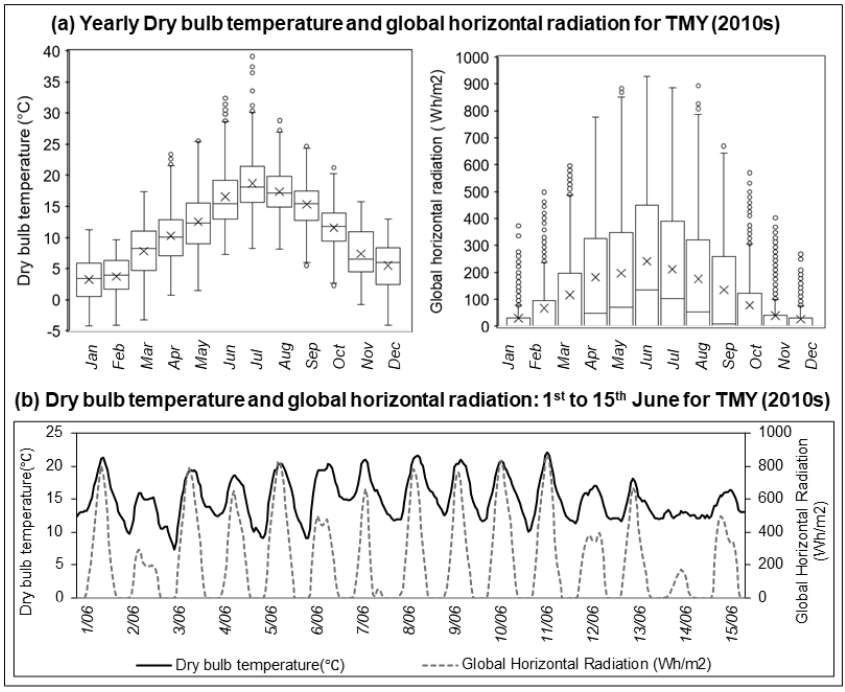


Figure 3-13. Dry bulb temperature and global horizontal radiation variation monthly and between 1st and 15th June in TMY (2010s) weather data

Figure 3-14 illustrates the temporal evolution of T_{ra} for both TLRs, occupancy, IEC and NNV operation of the base case scenario during a normal operation with NS (doS = 0.00) on 8th June. $\overline{T_{oa\ TMY}}$ on 8th June was 19.2°C with a maximum of 21.6 °C at 3:00 p.m. The temporal trend of T_{ra} in both TLRs was proportional to their respective

occupancy schedules, T_{oa} and the solar gains. During occupancy period (8:15 a.m. to 4:30 p.m.), $\overline{T_{ra\ E120}}$ and $\overline{T_{ra\ E220}}$ were 23.8°C and 24.3°C respectively and $T_{ra,\max\ E120}$ of 24.5°C was reached at 4:25 p.m. while $T_{ra,\max\ E220}$ of 24.8°C at 4:10 p.m. At 8:15 a.m., T_{ra} started to increase by 1.5°C in both TLRs due to gradual increase in occupancy (46 students for E120 and 48 students for E220). The IEC was triggered at 10:50 a.m. as $T_{ra,220}$ reached 24.5°C. However, T_{ra} kept increasing due to internal loads despite IEC operation and reached 24.7°C at 12:30 p.m. During the break at 12:30 p.m., there was a drop of 0.3°C in both TLRs due to the sudden decrease of occupants. IEC was unable to lower the temperature to the setpoint due to high occupancy, solar gains especially on the south west façade and high RH. After 6:00 p.m., T_{ra} in both TLRs kept increasing until 10:00 p.m. due to thermal storage effect of the envelope. Once NNV was active, T_{ra} for both E120 and E220 decreased below 24.0°C within 30'. $\overline{T_{ra\ E120}}$ and $\overline{T_{ra\ E220}}$ during night (10:00 p.m. to 6:00 a.m.) were 22.4°C and 24.0°C respectively. Daytime and night-time $T_{ra,\max}$ of E220 was higher than E120 by 0.7°C and 2.0°C respectively. $degree.hours$ were 0.64 and 3.84 for E120 and E220 respectively (< 6, acceptable, no overheating).

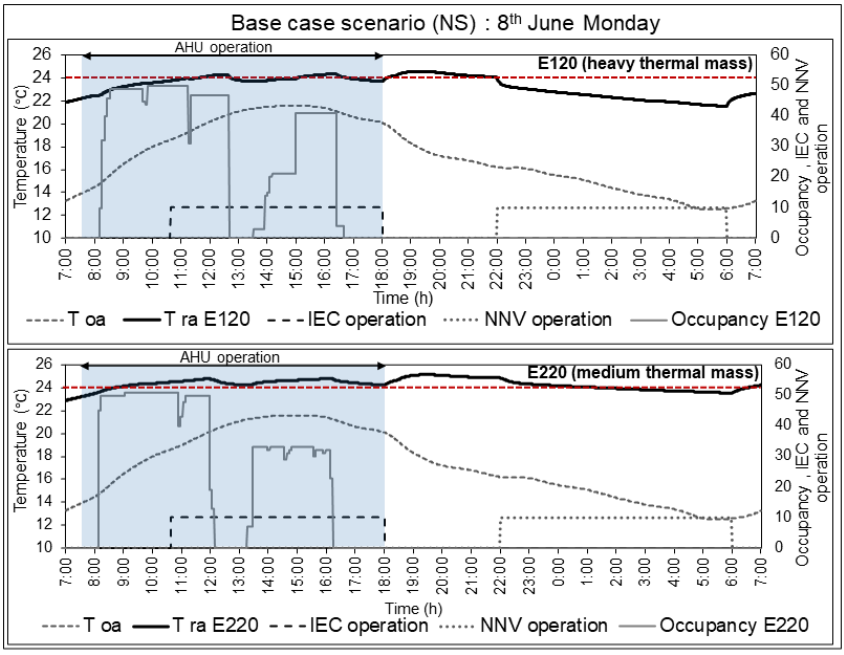


Figure 3-14. Temporal evolution of T_{oa}, T_{ra} for E120 and E220, occupancy of both TLRs and operation of IEC and NNV on June 8th for case of NS

Figure 3-15 illustrates the (0.4% and 2.8%) of occupied hours above a 24°C threshold for E120 and E220 respectively base case NS scenarios, used to compare the impact of the 6 extreme HWs (Table 3-3). Each HW will be assessed for the same duration as TMY (2010s). As $T_{oa\ TMY}$ gets warmer in July and August compared to June,

degree.hours for each TLR increased. In all 6 base case scenarios, (%) of occupied hours above 24°C were below 3%. The results demonstrated that in typical period, there was no overheating in both TLRs. degree.hours in E220 was on average 335% higher than E120. With operation of NNV at night, E120 had lower peak T_{ra} compared to E220 by 2°C- 5°C.

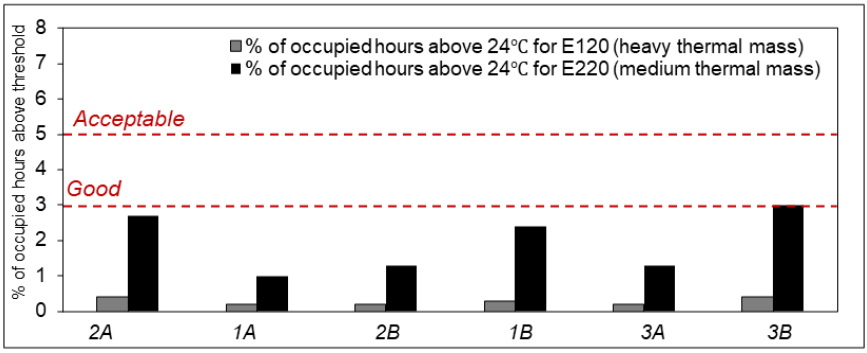


Figure 3-15. Percentage of occupied hours above a 24°C threshold for E120 and E220 for the NS case. The corresponding duration in TMY (2010s) weather data is the same as the 6-day HW

3.7.2 Impact of increasing *doS*

It is crucial to assess (a) how each type of shock impacts the two TLRs while doS increases and (b) the most critical time of shock occurrence. To assess impact of each shock type (ES and SS), doS was increased from 0 (NS) to doS_{max} and critical time of occurrence of SS was evaluated.

3.7.2.1 External Shock (ES)

Figure 3-16 illustrates the evolution of degree.hours for both TLRs for 6 extreme HWs (Table 3-3) in increasing order from $doES_{HW\ min}$ to $doES_{HW\ max.}$ ($\Delta(doES_{HW}) = 0.308$).

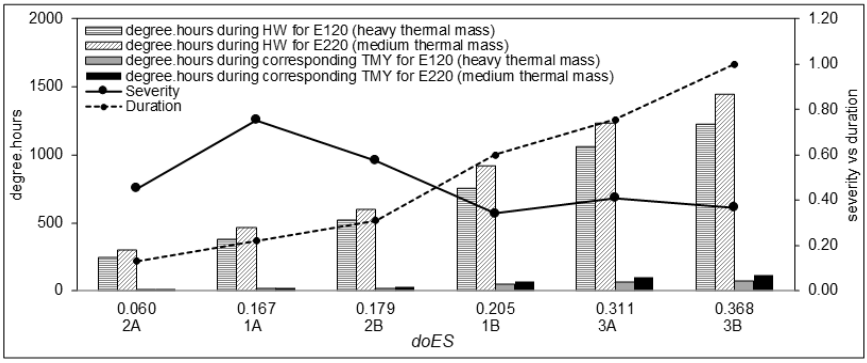


Figure 3-16. Comparison of impact of $doES_{HW}$ of 6 extreme HWs for Ghent in E120 and E220

The impact of the intensity; duration, and severity of HWs (Table 3-3) had a significant influence on the $doES_{HW}$. Thus, the mid-term intense HW (ID = 2A) had the lowest impact on the two TLRs and long-term severe HW (ID = 3B) had the highest impact. The rate of increase of each $doES_{HW}$ is proportional to the rate of increase of degree.hours in both TLRs. There were significant increase in degree.hours during the HWs compared to their respective NS scenarios. For ES, with $doES_{HW\ min}$. (HW ID = 2A) degree.hours increased by 132× (1.4 degree hours to 187.4 degree hours) and 6× (9.6 degree.hours to 56.6 degree.hours) compared to degree.hours in NS scenario for E120 and E220 respectively. During $doES_{HW\ max}$. (HW ID = 3B) degree.hours for E120 and E220 increased by 36× (34.3 degree hours to 1234.8 degree hours) and 7× (195.8 degree.hours to 1370.5 degree.hours) respectively compared to degree.hours in NS scenario. 100% of the occupied hours were beyond the 24°C threshold when the building was evaluated against a $doES_{HW\ max}$. When the duration of HW increased from 6 (ID = 2A) to 10 days (ID = 1A), $doES_{HW}$ increases 2.8× whereas degree.hours increase 1.5× for both TLRs. As the duration and severity of HWs increases from 10 days and 18°C.days (ID = 1A) to 45 days and 46.1°C.days (ID = 3B), $doES_{HW}$ increases by 1.2× for each HW, whereas degree.hours increased 1.4× for each HW. For both TLRs, $doES_{HW\ max}$ had 5× more impact than $doES_{HW\ min}$. During all the HWs, 80% of the occupied hours were above the 24°C threshold, significantly above the 5% acceptable limits. During HWs, E120 had 1.2× less degree.hours compared to E220.

3.7.2.2 System shocks (SS)

The impact of increasing $doSS$ and varying time of occurrence for IEC, SF and NNV has been illustrated in Figure 3-17. Table 3-4 gives an overview of the different SS simulation scenarios. Each SS has been increased by a duration of 30 minutes from NS to $doSS_{\ max}$.

(a) Increasing doSS

If the IEC failure occurred from 7:30 a.m. to 6:00 p.m., degree.hours in E120 and E220 increased by 5.7× and 4× respectively compared to no IEC failure. With the increase in shock duration, the severity of the shock also increased due to heat build-up in the lecture rooms without cool supply air. $\Delta(doSS_{IEC})$ increased by 2× when shock duration increases from 0' (NS) to 30'. Additionally, $\Delta(doSS_{IEC})$ increased by 1.5× for each additional 30' shock from 30' to 180', 1.2× from 180' to 390' and 1.1× increase from 390'to 630'. For each 30' increase in shock duration, degree.hours in both TLRs increased by 1.2×. For E120, a 630' IEC failure still resulted in less than 6 degree.hours. However, for E220, 3 hours of IEC failure already violated this threshold.

If the SF occurred from 1:30 p.m. to 6:00 p.m., degree.hours in both TLRs increased by 4× compared to no SF. As duration of SF increased, severity of the shock increased due to high solar gains through the envelope. $\Delta(doSS_{SF})$ increased by 1.5× when the shock duration increased from 0' (NS) to 30'. Additionally, $\Delta(doSS_{SF})$ increased by 1.1× for each additional 30' shock from 30' to 270'. For each 30' increase in shock duration, degree.hours in both TLRs increased by 1.1×. For E120, 270' SF still resulted in less than 6 degree.hours. However, for E220, 45' of IEC failure already violated this threshold. If NNV failure occurred from 10:00 p.m. to 6:00 a.m., degree.hours in E120

and E220 increased by 4.5× and 3× respectively compared to no NNV failure. With increase in shock duration, severity of the shock also increased due to lack of ventilative cooling. Both lecture rooms are unable to flush out the heat build-up during the day due to internal and solar gains. $doSS_{NNV}$ for E120 was higher than E220. For both TLRs, until 120' shock, there was no increment in $doSS_{NNV}$. From 120' to 480', for each additional 30' of NNV failure, $\Delta(doSS_{NNV})$ increased by 1.2×. This resulted in 1.3× increase in degree.hours for each 30'. For E120, 480' NNV failure still resulted in less than 6 degree.hours. However, for E220, 160' of NNV failure already violated this threshold.

(b) Most critical time of shock occurrence:

Each SS can occur at any time within its operation period. For both TLRs, in the case of SF, IEC and NNV failures, the highest degree.hours were obtained when the shock occurred at the start of the operational time of the system (i.e., IEC failure at 7:30 a.m., SF at 1:30 p.m. and NNV failure at 10:00 p.m.).

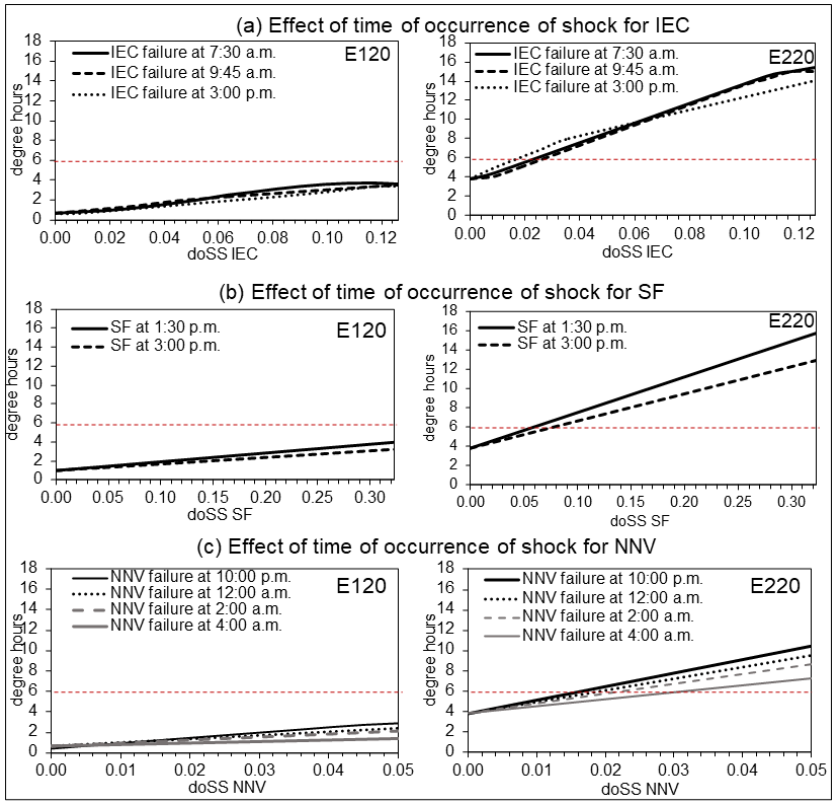


Figure 3-17. Impact of SS (IEC, SF and NNV) and corresponding degree.hours for E120 and E220

For IEC failure, in both TLRs, degree.hours were respectively 1.2× and 1.3× lower when the shock occurred at 9:45 a.m. and 3:00 p.m. instead of 7:30 a.m. It might be noted that until a 270' shock, the most critical time of occurrence was at 9:45 a.m. due to its high occupancy. But due to higher heat build-up in indoor spaces, shock starting at 7:30 a.m. proved to be most critical. For SF, in both TLRs, there was an average 1.4× increase in degree.hours when the shock occurred at 1:30 p.m. instead of 3:00 p.m. NNV failure occurring at 10:00 p.m. was the most critical time of shock for both TLRs. For example, in E120, NNV failure at 10:00 p.m., led to 1.2×, 1.1× and 1.5× increase in degree.hours compared to a failure occurring at 12:00 a.m., 2:00 a.m. and 4:00 a.m. respectively.

3.7.3 Impact of shock combinations

Figure 3-18 shows the impact of different combinations of ES (HW, ID = 2A,Table 3-3) + SS (SF, IEC and NNV) failure during the whole operation period (maximum duration of shock) and combinations of them at the most critical times of occurrence) and SS+SS (

Table 3-5). The degree.hours for the different ES+SS and SS+SS were plotted in increasing order from least to most severe.

According to Figure 3-18(b), in both TLRs, IEC+SF resulted in higher degree.hours than IEC+NNV failure, followed by SF+NNV failure. For example, in E120, degree.hours for IEC+SF failure were 1.1× higher than IEC+NNV and SF+NNV failure respectively. Thus combinations having the least impact are the ones without IEC failure (i.e., lack of cooling). The worst is AHU failure (i.e., lack of cooling and ventilation: no load removal). Combining three SS resulted in 3× higher degree.hours than two SS except for IEC+VF. IEC+VF+SF was the most critical followed by IEC+VF IEC +SF+NN. For example, in E120, degree.hours for IEC+VF+SF failure were 1.7× and 2.7× higher than IEC+VF and IEC+SF+NNV failures respectively. For three combined SS, the worst is also the one with AHU failure. Combining all SS resulted in the highest degree.hours. For example, in E120, degree.hours for IEC+VF+SF+NNV failure were 3.6× higher than those for IEC+SF+NNV.

According to Figure 3-18 (c), in both TLRs, HW+IEC failure resulted in higher degree.hours than HW+SF, followed by HW+NNV failure. For example, in E120, degree.hours for HW+IEC failure were 1.1× higher than HW+SF and HW+NNV failure respectively. Combining two SS with the HW resulted in 1.3× higher degree.hours than single SS+HW with HW+(IEC+SF) being the most critical followed by HW+(NNV+SF), HW+(IEC+NNV) and HW+(IEC+VF). For example, in E120, degree.hours for HW+(IEC+SF) failure were 1.07×, 1.04× and 1.03× higher than HW+(NNV+SF), HW+(NNV+IEC) failures and HW+(IEC+VF) respectively. Adding an additional SS to the ES+SS combinations resulted in 1.02× higher degree.hours with HW+(IEC+SF+VF) being more critical than HW+(IEC+SF+NNV). For example, in E120, degree.hours for HW+(IEC+SF+NNV) failure were 1.01× higher than HW+(IEC+VF+SF).

According to Figure 3-18, combinations of the same number of ES+SS were always more critical than those of SS+SS. For example, in E120, degree.hours for HW+IEC failure was 15× higher than IEC+VF. The degree.hours for HW+IEC+SF was 9.2× higher than IEC+VF+SF failure and degree.hours for HW+IEC+SF+NNV were 7.1×higher than

IEC+SF+NNV+VF. Note that the degree.hours in E220 were always higher than in E120. For example, degree.hours during HW+IEC+VF+SF+NNV is 1.1× higher in E220 than E120.

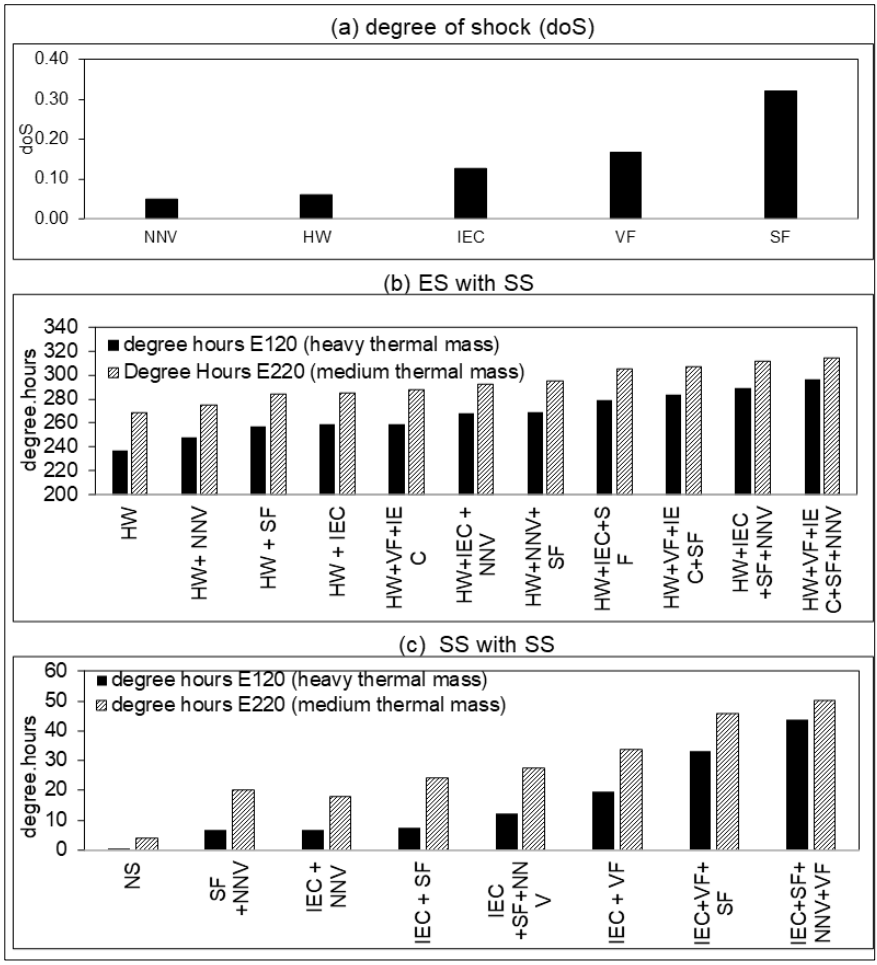


Figure 3-18. Effect of combined SS+SS and ES+SS shocks on thermal comfort and thermal resilience to overheating in E120 and E220

3.8 Discussion

3.8.1 Impact of shocks on nZEB non-residential buildings

Assessing of the impact of ES on the two TLRs demonstrated that during HWs, combinations of passive cooling strategies working in their full capacity cannot remove excess heat build-up and there was evident overheating in both TLRs. The effectiveness of evaporative coolers relies heavily on the outdoor psychrometric conditions. Since HWs result in increased wet bulb and dew point temperatures, the cooling capacity of evaporative coolers (i.e., direct, indirect and dew point evaporative coolers) is

significantly reduced. The same applies for strategies that rely on diurnal temperature differences that are extremely reduced during a HW. Example of such strategies are NNV used in this case study and radiative cooling panels. The same applies for envelope shading strategies, which effectiveness reduces considerably with increased solar irradiation on windows during HWs. Note that for NNV, its effectiveness not only relies on diurnal temperature gradients but also on the design, favorable wind directions and velocities, window opening area. To be maximally efficient, NNV relies on stack effect establishment. This is not the case in the current case study building. Moreover, the windows' opening area was limited to 4% of the floor area.

Failure of different systems proved that they deliver their highest cooling efficiency at the beginning of their operational period as they don't allow for significant heat to build up inside the space. For example, if the IEC failure was to occur later at 9:45 a.m. or 3:00 p.m., TLRs would have been partially cooled by the IEC since the start of its operation in the morning until a failure occurred. The decrease in the occupancy during break would also contribute to lower violations. SF occurring later throughout the day would protect the space from the highest solar irradiation at noon. If the NNV failure would have occurred at a later period, the partial operation of NNV from 10:00 p.m., until the time of occurrence of the shock would have already ventilated the two TLRs, lowering the $\overline{T_{ra\,E120}}$ and $\overline{T_{ra\,E220}}$.

3.8.2 Comparison of different shock types

To compare the different shock types, a common range of *doS* for ES and SS was chosen. As outlined in Table 3-3, the highest *doS* value of 0.368 corresponds to the most severe and longest heatwave predicted for Ghent (ASHRAE climate type 4A) during the period 2001-2100. For different system shocks (SS), the highest *doS* values observed are as follows: 0.127 for IEC failure (630'), 0.323 for solar shading failure (270'), and 0.047 for NNV failure (480'). While each SS type has its own *doS* value, they are not directly comparable across types, as they reflect different failure modes and durations.

To allow for comparison, a common *doS* range of [0-0.368] was selected, and the impact of each SS type was extrapolated to this maximum value of 0.368. This means that, for the highest *doS* of 0.368:

- IEC failure corresponds to 3 days of failure in lecture room E120 and 5 days of failure in E220.
- Solar shading failure corresponds to 1.5 days of failure in E120 and 3 days of failure in E220.
- NNV failure corresponds to 12 consecutive nights of failure in E120 and 18 consecutive nights of failure in E220.

Figure 3-19 compares the impact of the same *doS* on the two TLRs due to different shock types. According to Figure 3-19, across the entire *doS* range and for both TLRs, ES was more critical than all SS types by several orders of magnitude. For SS, NNV failure was the most critical followed by IEC failure and SF. For example, in E120, for $doS = 0.368$, ES resulted in 110× higher degree.hours compared to IEC failure, 290× higher than SF and 31× higher than NNV failure. Similarly, for E220, for *doS*

$= 0.368$, ES resulted in 43× higher degree.hours than IEC failure, 75× higher than SF and 22× higher than NNV failure. This also explains why during shock combinations, any ES+SS was more critical than SS+SS (Figure 3-19).

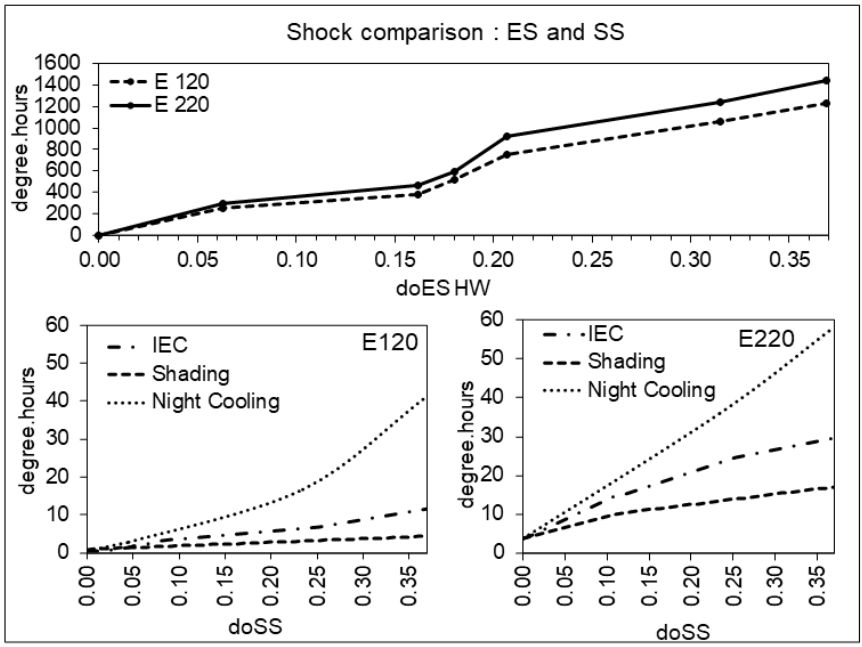


Figure 3-19. Comparison of the effect of ES, and SS in E120 and E220

3.8.3 Impact of thermal mass

The results showed that during the case of NS and during the cases of ES and SS as well as combinations of them, the temperature violations in E220 were higher than in E120. Comparing both classrooms, this can be since E220 has a lighter thermal mass than E120. However, it can also be since E220 has higher heat gains due to the presence of the roof. However, despite having the same number of average occupants over the week (32 occupants in E120 and 34 occupants in E220), the number of occupied hours in E120 was higher than E220 (35h vs 19 h for E120 and E220 respectively). Thus, conclusions on the impact of thermal mass cannot be immediately drawn and the two classrooms cannot be compared. Additional simulations were conducted where the thermal mass was varied in each classroom between light brick (ρ = 850 kg/m^3), medium brick (ρ = 1400 kg/m^3), and heavy brick (ρ = 1850 kg/m^3). The specific capacity of the different bricks remained the same (C_ρ = 840 J/kg.K). The brick was located at the interior layer of the building envelope wall in E120. The rest of the parameters were not varied.

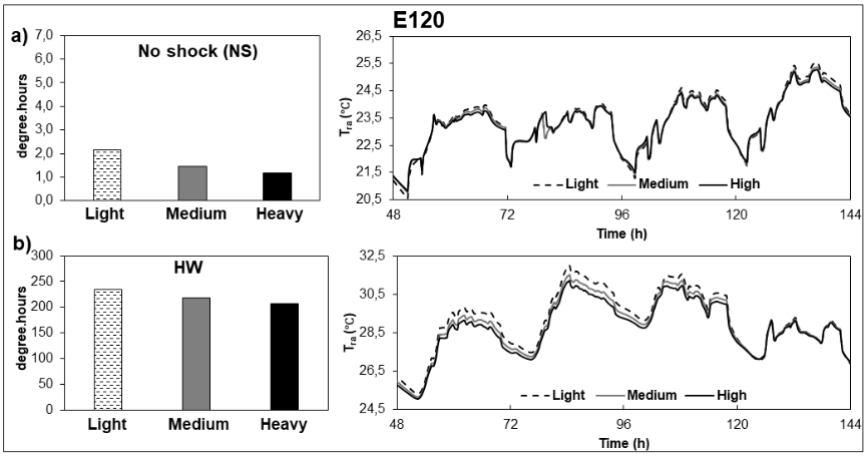


Figure 3-20. Illustration of the degree.hours (left) and temporal room air temperature variation (right) for different thermal mass for the case of a) no shock (NS) and b) HW (ID=2A) for E120

The simulations were conducted for the case of No shock (NS), and the case of an ES as they are the worst types of shock when compared to SS. The HW (ID = 2A,Table 3-3) was selected. Figure 3-20 illustrates the degree.hours and temporal temperature profile in E120 for a) NS and b) HW for light, medium and heavy thermal mass. Results showed that both for the cases of NS and HW, increasing the thermal mass density from light to heavy reduced the room air temperatures during the day and thus reduced the unmet cooling degree.hours. This is since the heavier thermal mass was able to store more thermal energy, providing inertia against temperature fluctuations. However, the damping efficiency of the thermal mass was reduced in the HW scenario as compared to NS. For example in E120, for the case of NS, the degree.hours decreased by 32.4% when switching from light to medium thermal mass and further by 20% when switching to a heavy thermal mass. However, during a HW, the degree.hours decreased by 7% when switching from light to medium thermal mass and further by 4.7% when switching to a heavy thermal mass. This can be due to the extremely high outdoor temperatures during both night-time and daytime encountered during a HW and the reduction in diurnal temperature differences.

The damping efficiency of thermal mass was also less pronounced due to its higher heat gains from the roof. For the case of NS, the degree.hours decreased by 21.7% when switching from light to medium thermal mass and further by 14.2% when switching to a heavy thermal mass. During a HW, the degree.hours decreased by 6.3% when switching from light to medium thermal mass and further by 4% when switching to a heavy thermal mass. Thus, the thermal mass has an impact on the thermal resilience performance of nZEB during overheating events, however less pronounced than during reference scenarios with NS. The discrepancies found between the classrooms can be attributed in part to the thermal mass but also to the increased solar gains in E220 given the presence of the roof.

3.9 Conclusions

In Chapter 3, *doS* was introduced to classify, quantify and compare the impact of different shock types on the thermal resilience to overheating in two lecture rooms. The key conclusions were summarized below:

- Development of future weather files: For evaluating external shocks, future weather files (typical and heatwave) were developed for RCP8.5 scenario for ASHRAE climate zones within the framework of IEA EBC Annex 80-Resilient cooling of buildings. Adopting this framework for developing weather files, within this research, TMY and heatwave files were developed for climate zone 4A (Ghent) and was utilized to assess thermal resilience of buildings to overheating. For each time period- 2010s, 2050s and 2090s, three extreme heatwaves- most severe, most intense and longest were detected and tested to assess thermal resilience of buildings under extreme conditions.
- Quantifying *doS*: Quantifying *doS* for different shock types (external shocks an system shocks) aids in shock comparison and identification of most crucial shock type. This methodology developed for classifying and quantifying external shocks (heatwaves) and system shocks (failure of solar shading, natural night ventilation, and indirect evaporative cooling system) is applicable to any type of building equipped with any type of cooling strategy, with knowledge of (a) weather data and (b) occupancy profiles. The aspect of shock severity is a quantification of the shock origin and overheating causality rather than a quantification of shock consequence on the room air temperature, which can be influenced by multiple parameters simultaneously and not just the shock. Thus, severity is the increase of a shock characteristic temperature T_{shock} compared to the reference characteristic temperature T_{ref} under normal expected conditions.
- Characteristic temperature selection for *doS* quantification: The characteristic temperatures depends on the type of shock. For shading failure (SF), the internal window surface temperature is chosen because shading primarily influences surface heat absorption, which directly affects heat transfer into the zone. For cooling system failure (CSF) the cooling fluid temperature is selected as it determines cooling capacity. In all-air systems, this corresponds to the supplied air temperature (T_{sa}), while in hydronic systems, it is the supplied water temperature (T_{sw}). For ventilation failure (VF) the average room air temperature over the shock period ($T_{ra,VF}$) is used, as ventilation failures impact the ability to remove excess heat, affecting indoor conditions. While these shocks influence different thermal components of the building, they remain comparable within the *doS* framework because each represents the relative deviation from its reference state, rather than an absolute temperature value. This approach ensures that *doS* remains meaningful across different failure modes. Temperature is chosen as a primary indicator as it directly reflects the thermal consequences of system failures, affecting overheating risk and indoor comfort. However, for ventilation failure, room temperature is also influenced by other factors such as internal gains, thermal mass, and building characteristics. While temperature deviation still provides insight into the impact of ventilation failure, it does not isolate the system's direct effect. A

more direct characteristic temperature for ventilation failure could be based on supply and exhaust temperature differences, or metrics related to air exchange effectiveness and heat removal capacity.

- *doS* quantification for shock combinations: Shock combinations such as power failure (combination of different system shocks such as failure of ventilation system, cooling system, solar shading) or power failure during heatwaves (system shock combination with external shock) impact thermal resilience of buildings. In this research, to quantify *doS* of shock combinations, simple addition or multiplication of the individual *doS* values were not selected because combination of shocks are not interdependent, and their effects do not scale linearly. Adding *doS* values assumes independent contributions, which would overestimate or underestimate the total impact, while multiplication would unrealistically suggest an exponential effect. Moreover, the response to multiple failures is inherently non-linear—cooling and ventilation failures, for instance, amplify overheating risks differently depending on factors such as climate, buildings' insulation, and thermal mass. The combined impact of these shocks is often greater than the sum of individual effects due to cascading interactions. To accurately capture these dependencies, further research would be necessary. Example, a more sophisticated, weighted non-linear model would correctly capture the *doS* of a combined shock taking into account weights for individual shocks. Thus, quantifying the *doS* of combined shocks requires further research to understand the individual impact of each shock and the nature of their interaction, whether they amplify, dampen, or modify each other's effects.
- Impact of shock types: Impact of various shock types (external and system) on thermal resilience was evaluated. For external shocks, extreme heatwaves predicted to occur in next 100 years (end of the century) was evaluated. For external shocks, heatwaves with the longest duration are the most critical, as prolonged high temperatures cause cumulative heat build-up in buildings, reducing their ability to dissipate heat. More than shorter but intense heatwaves, extended heat exposure overwhelms passive cooling, increases overheating risk, and prolongs occupant discomfort/heat-stress, making long-duration heatwaves particularly challenging for thermal resilience of buildings. For system shocks- increasing *doS* (increasing duration of shock), time of occurrence of shock was tested. System shocks occurring at the start of the operation period and lasting throughout are more critical than those during peak occupancy because they prevent the building from stabilizing its indoor conditions from the outset. This leads to prolonged exposure to suboptimal conditions, increased thermal stress, and potential system overload. In contrast, shocks during peak occupancy, while impactful, occur when the building has already reached operational conditions and may have some thermal inertia to mitigate short-term disruptions.
- Identifying the most critical shock type: This study primarily focused on heatwaves and their impact on thermal resilience to overheating, with the application of the *doS* of system failures (SS) serving as an extension to determine which shocks are more critical for buildings. The impact of external shocks (heatwaves) is significantly higher than any system shocks. Heatwaves in future climate scenarios can be

considered as the most extreme shock for buildings. Heatwaves impact the outdoor climate conditions by simultaneously increasing dry bulb, wet bulb, and dew point temperatures as well as solar gains. They also reduce the diurnal differences in ambient air temperatures between day and night. Heatwaves will not only significantly increase envelope gains but renders inefficient passive cooling strategies which performance relies on ambient outdoor conditions.

- *doS* for heatwaves: To assess buildings' resilience to overheating against most critical shock type, different heatwaves should be assessed (e.g., short and intense which will impact the absorptivity of the shock or long and severe which will impact the recovery of the shock). In long-term future, as heatwaves get longer in duration and higher in intensity, the buildings' thermal resilience to overheating is compromised and the passive cooling strategies fail to remove the heat build-up in the indoor space during severe heatwaves. Thus including *doS* of heatwaves in the thermal resilience assessment framework will aid in assessing thermal resilience of buildings for different heatwave types (short and severe vs longer mild). However, in the *doS* quantification for heatwaves, further research is required to include the time of occurrence of peak temperature and minimum temperature during a heatwave as it dictates the absorptivity and recovery aspects of thermal resilience assessment. The minimum temperature during a heatwave is an important parameter when evaluating the overall impact of the shock, particularly in terms of human comfort and health. While much of the focus tends to be on maximum daytime temperatures, nighttime temperatures can significantly influence the thermal resilience of both buildings and occupants. In fact, the ability of a building to cool down during the night is a key factor in preventing heat stress, especially in buildings with passive cooling strategies such as natural night ventilation. If the minimum temperature remains high during the night, it can prevent the building from recovering thermally, leading to an accumulation of heat that could exacerbate discomfort and health risks for occupants. For vulnerable populations, such as the elderly or those with pre-existing health conditions, prolonged exposure to high nighttime temperatures can increase the likelihood of heat-related illnesses. Thus, while maximum daytime temperatures often dominate heatwave impact assessments, incorporating minimum nighttime temperatures in future research for doS quantification for heatwaves could provide a more comprehensive understanding of how heatwaves affect building systems and human health. This would allow for better predictions of thermal resilience and the development of more effective mitigation strategies, especially in climates where nighttime cooling is critical to maintaining comfort. Therefore, both parameters are crucial in thermal resilience assessment, as they determine the balance between heat absorption and recovery capacity.
- Applicability of *doS* for external shocks in other climate zones: Within the framework of IEA Annex 80, future TMY and heatwave data files were generated for all ASHRAE climate zones. A recent study quantitatively assessed the thermal resilience of passive cooling strategies during heatwaves across different climate zones, calculating the *doS* for extreme heatwaves in ASHRAE climate zones 2A, 3A, 3B, 4A, and 6A [190]. The study showed, *doS* values ranged from [0.025-0.405] for the studied varying climate

zones. It is important to note that *doS* values for external shocks are not directly tied to a specific climate type; rather, they are calculated based on the relative characteristics of each climate zone. Therefore, *doS* itself is not inherently linked to a particular climate type and can be used to quantify *doS* for external shocks for different climate types with knowledge of TMY and HW data.

- Impact of thermal mass: The time lag effect of thermal mass in buildings diminishes during heatwaves. Thus, buildings with heavy thermal mass doesn't improve the thermal resilience performance of buildings during longer heatwaves, as is the case during typical conditions. The discrepancies found between the classrooms can be attributed in part to the thermal mass but mostly to the increased solar gains in classroom in upper floor given the presence of the roof.

Chapter 3 paved way for quantifying different shocks and comparing their impact on thermal resilience of buildings. Since heatwaves were identified as most critical shocks, development of thermal resilience framework will include heatwaves of varying *doS* and in chapter 4 will identify which building and system parameters dominates the thermal resilience performance during increasing *doS* of heatwaves. Including *doS* in the thermal resilience framework will bridge the missing gap of assessing thermal resilience of a building during *doS* in a single framework that is currently evaluated to be missing (as seen in Chapter 2).

Chapter 4 : Determining the most influential design parameters that impact buildings' thermal resilience to overheating during heatwaves

Chapter 4 aims to identify the most influential design parameters (buildings and systems) that impacts buildings thermal resilience to overheating. In Chapter 3, thermal shocks were identified and quantified by *doS*. Impact of various types of shock were compared, and heatwaves were concluded to be the most severe shock type that impacts thermal resilience of buildings. In the second step to develop a comprehensive thermal resilience assessment framework, leveraging various *doS* (identified in Chapter 3), it is crucial to identify the most influential building and system design parameters impacting thermal resilience. Additionally, for increasing *doS*, the impact of these design parameters on thermal resilience will be quantified.

This chapter is based on the following publication:

*[1] **A. Sengupta**, D. Al, O. Berk, J. Shinoda, H. Breesch, M. Steeman, Building and system design's impact on thermal resilience to overheating during heatwaves : An uncertainty and sensitivity analysis, Build. Environ. 265 (2024) 112031. https://doi.org/10.1016/j.buildenv.2024.112031.*

4.1 Introduction

As demonstrated in Chapter 2, existing thermal resilience frameworks fail to establish a clear connection between resilience assessment and specific building and system design parameters. They lack guidance on targeted interventions during design to enhance thermal resilience to overheating. Developing a comprehensive thermal resilience assessment framework requires identifying the most influential design parameters that impact building's thermal resilience to overheating. Building on the findings of Chapter 3, this chapter utilizes increasing *doS* for heatwaves to pinpoint critical design parameters and quantify their impacts on thermal resilience under varying heatwave intensities. RO3 aligns with this chapter-to identify the most influential design parameters of buildings and systems that impact thermal resilience of buildings to overheating.

Section 4.2 reviews existing studies that uses sensitivity methods to identify impactful parameters in building energy and thermal comfort assessments. Section 4.3 then describes the step-by-step process used in this chapter to identify the most impactful parameters. This process begins with defining the thermal shocks to be assessed, determining the thermal resilience KPIs, developing and selecting a reference Belgian apartment, and finally identifying the design parameters and their assessed ranges. Section 4.4 presents the results of the uncertainty and sensitivity analysis. Finally, sections 4.5 summarizes its key conclusions.

4.2 State of the art

Chapter 3 introduced the degree of shock (*doS*) to compare impact of various shocks on buildings' thermal resilience, demonstrating that heatwaves pose the greatest challenge. Underlining the growing frequency and severity of heatwaves in future climate scenarios and their significant impact on thermal resilience of buildings to overheating, this chapter aims to- (a) quantify and determine the most influential design parameters that impact thermal resilience performance for varying *doS* of heatwaves and (b) quantify thermal resilience aspects-shock impact, absorptive and recovery aspects and evaluate the uncertainty of these three aspects during varying *doS* due to change in design variations. To achieve these aims, it is crucial to comprehend the intricate relationship between design parameters and thermal resilience performance and thus uncertainty and sensitivity analysis was utilized. Hopfe [191] demonstrated that integrating uncertainty and sensitivity analysis and decision-making protocols into Building Performance Simulation (BPS) enhances its utility in design phases by guiding decisions. One sensitivity study on building stock energy models proposed enhancing through efficient input sampling [192], while another identified weather and occupancy [193] as key drivers of energy consumption. Zeferina et al. [194] used global sensitivity analysis to find that ventilation rate highly impacts HVAC demand in large office buildings. Another study pointed out uncertainties in standardized conditions for energy assessments in Belgian schools, emphasizing factors like set point temperature and occupancy profiles [195]. Beyond energy performance, uncertainty and sensitivity analysis methods are utilized to evaluate

thermal comfort and overheating risks. One such study found that energy-efficient buildings, while better with proper ventilation, are more prone to overheating without it, especially in future climates [23]. Another study on Canadian schools identified optimal designs that balance overheating risk, energy use, and lighting needs under extreme climates [196]. Breesch et al. [197] highlighted factors like internal heat gains and glazing properties as key to thermal comfort in naturally ventilated buildings. Finally, Machard et al. [159] noted that while current ventilation strategies mitigate overheating, future climate warming will reduce their effectiveness, requiring additional measures like reducing glazing percentage and enhancing roof albedo during the design phase. Thus, even though several studies employed uncertainty and sensitivity analysis method to determine the influential design parameters affecting buildings' energy performance, thermal comfort and overheating risks, no such study has been previously conducted to determine most influential design parameters (buildings and systems) on three aspects of thermal resilience-shock impact, absorptive and recovery aspects and compared them for varying *doS.* In conjunction with the *doS* metric (developed in chapter 3) and identification of the most influential design parameters affecting buildings' thermal resilience (Chapter 4), the establishment of a comprehensive assessment framework will enable the testing and enhancement of thermal resilience to overheating in contemporary buildings and systems (Chapter 5).

4.3 Methods

Figure 4-1 outlines the methodology for identifying the most influential design parameters (buildings + systems). Step 1 defines the shock for thermal resilience assessment, followed by Step 2, which establishes the KPIs. Step 3 details the Modelica model development for a reference Belgian apartment. Step 4 covers techniques for uncertainty and sensitivity analysis. In Step 5, simulations are run, and thermal resilience is assessed with KPIs determined in step 2. Step 6 presents the uncertainty analysis results, showing the impact of design parameter variations. Finally, Step 7 conducts sensitivity analysis to identify the most influential design parameters and their effects on thermal resilience during varying *doS*.

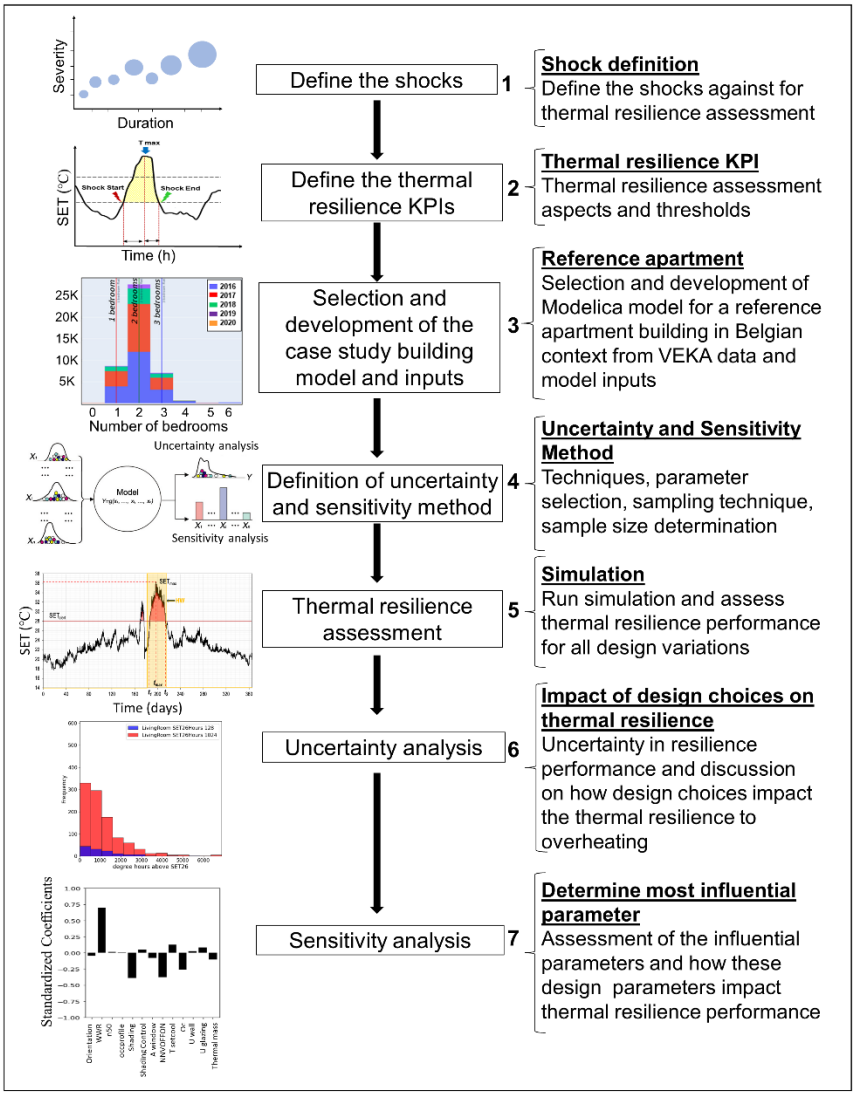


Figure 4-1. Methodology to identify the most influential design parameters (buildings + systems) that impact thermal resilience performance to overheating

4.3.1 Shock definition

Chapter 3 described the methodology for development of the heatwave files. From the six extreme HWs predicted to occur in Ghent, Belgium, over the next 100 years (2001-2100) (see Table 3-3 and Figure 3-12), 3 representative *doS* is selected to evaluate thermal resilience and identify the most influential design parameters. The three varying *doS* applied in this study represent *–(a) doS $_{low}$= 0.060 (b) doS $_{medium}$=0.179 and (c) doS $_{high}$= 0.368.* These three *doS* represent an increasing severity and varying type (short-severe, mild-longer and longest-severe) of heatwaves and are adequate to (a)

identify the most influential design parameters during varying representative shock types and (b) determine whether these critical parameters of thermal resilience will shift in importance during varying *doS*.

4.3.2 Thermal resilience assessment KPIs

During any shock, a building's ability to adapt, absorb, and recover from extreme heat highlights the challenge of thermal resilience. This concept includes the capacity of both the building and its occupants to withstand, absorb, and recover from shocks.

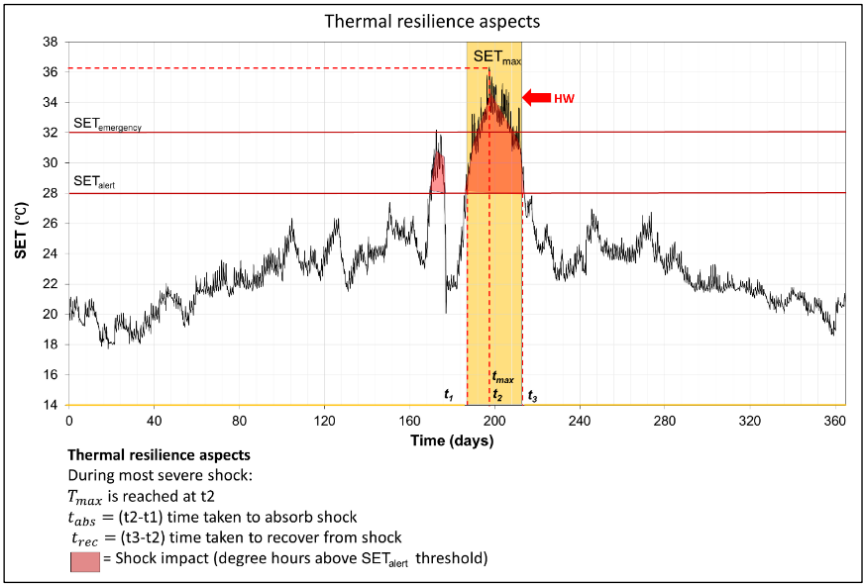


Figure 4-2. Visualization of three aspects of thermal resilience

In this study, three aspects of thermal resilience were quantified and evaluated: (a) Shock impact employing heat-stress indicator SET-Dh, measuring the duration and severity of discomfort; (b) Absorptivity time (t_{abs}) the period during which temperature rises from a pre-defined heat-stress threshold to its peak; and (c) Recovery time (t_{rec}) the duration needed to return to the pre-defined heat-stress threshold level after the shock ends or recovery actions are implemented. Figure 4-2 illustrates the three thermal resilience KPIs assessed in this study. Note that in a year, multiple overheating events/shocks may occur in the building, but for this study, thermal resilience performance during the most severe shock will be evaluated. Previously qualitatively assessed absorptivity and recovery aspect of thermal resilience is quantified and evaluated in this chapter to determine how design parameters influence these aspects of resilience.

4.3.2.1 Resilience aspect 1: Shock impact (SET-Dh)

Section 2.6.2.2 describes heat-stress indicator SET-Dh to assess thermal resilience of buildings to overheating. In this research, the modeling approach for calculation of SET

involved determining input parameters from dynamic building energy simulations performed in Modelica. Yearly simulations were conducted with 15-minute time step to capture variations in parameters such as air temperature, relative humidity, mean radiant temperature and air velocity (dependent on the ventilation rate). Simplifying SET calculations by assuming constant values for certain parameters (met and clo) can be acceptable in stable environments but may reduce accuracy in dynamic or complex scenarios, depending on the required precision and context. Assumptions of parameters such as metabolic rate of the occupants and clothing were based on type of building-(residential and office) and (b) level of activity (wake or sleep). These values were adopted from [88] (see Table 2-3). These input parameters that are necessary to calculate SET was used from the dynamic simulations and were calculated using the pythermalcomfort Python package[129].

The first step for the SET-Dh calculation is the determination of SET threshold (Equation 2-9) The ASHRAE 7-point predicted thermal sensation (PTS) scale can be used to quantify occupants' perceived thermal sensation within a space. It ranges from -3 (cold) to +3 (hot). Studies were conducted to develop the relationship between the SET and the PTS for an average population [198]. Three SET thresholds (SET_{comf}, SET_{alert}, SET_{emer}) can be determined in relation to occupants' heat stress [199][200]. For an average population, when PTS = 0 (neutral), SET_{comf} = 24.1 °C; when PTS = 1 (slightly warm), SET_{alert} = 28.1°C; and when PTS = 2 (warm), SET_{emer} = 32.1 °C. To calculate SET-Dh, in this study SET_{alert} threshold was adopted as $SET_{threshold}$ as it is calculated for average population. Ji et. al [201] developed several regression models for PTS for different climates and reflect the adaptability of different populations using ASHRAE Global Thermal Comfort Database. Assaad et. al [190] in a recent study to quantitatively evaluate thermal resilience of passive cooling strategies during heatwaves, calculated SET_{alert} for different climate zones using the method proposed by [201], but to ensure a consistent basis for evaluating resilience performance across these diverse climates, adopted an averaged SET_{alert} threshold of 28.1°C. In second step, a threshold should be adopted for the SET-Dh to determine buildings' thermal resilience performance. Laouadi et. al [88] determine the threshold of 230 [°C.h] based on per occupant and on SET threshold of 30°C. The SET-DH threshold is climate-related and is not an absolute number across all climates. Instead, it is specific to the applied climate and context. The 230°h threshold for SET-DH is based on studies that assess thermal discomfort and overheating risks. However, its applicability depends on the climate, building type, and occupant adaptation-(i)in warmer climates, the threshold may need to be adjusted higher because occupants are more acclimatized to heat (ii) in cooler climates, lower thresholds might be appropriate since people are less adapted to high temperatures (iii)Urban vs. rural settings can also influence applicability due to heat island effects and ventilation differences and (iv) The threshold is also dependent on the health of the occupants (young, healthy vs. old, sick).

While to assess heat-stress per occupant may be more accurate, assessing zones rather than individuals is suitable for building performance evaluation rather than personal exposure. However, the model used in Modelica does not account for occupant-specific

thermal conditions, it's important to recognize the limitations inherent in the software itself. Modelica operates with lumped parameters, providing a single value for air temperature and relative humidity, which assumes a uniform environment across the entire zone. The absence of stratification or individual thermal comfort outputs is simply a constraint of the modeling approach, not a flaw in the methodology. In this context, it was necessary to make assumptions that each occupant in the zone experiences the same thermal environment, as this is the only viable option within the given constraints. To accurately predict SET for each individual occupant, more complex software such as CFD would indeed be required. CFD simulations can account for localized temperature variations, airflow, and humidity, delivering a more precise prediction of SET. Since the SET-Dh threshold proposed by Laouadi et al. [88] was based on body-dehydration rate and developed for both young-healthy and old-sick occupants, this threshold of 230 [°h] was adopted in this study.

4.3.2.2 Resilience aspect 2: Shock absorption-Absorptivity time (t_{abs})

Zhang et al. [37] qualitatively defined absorptive capacity as the degree to which a building and system is able to absorb the impacts of the shock and minimize their consequences with little effort. In this study, absorptivity time (t_{abs}) is quantitatively defined as:

$$t_{abs} = t_{peak} - t_{start} \qquad \textit{(Equation 4-1)}$$

Where: t_{abs} is the absorptivity time, t_{peak} is the time when the peak SET is reached and t_{start} is the time when the SET-alert threshold of 28.1°C is first exceeded, provided that this threshold has not been exceeded for at least 24 hours prior [88]. Higher t_{abs} indicates that the building and the occupant remains in a heat-stress period for longer duration, thus demonstrating lower thermal resilience. As seen in Figure 4-2 during the most severe shock period, the building starts to absorb heat from t_1 as indicated by the violation of the SET-$_{alert}$ threshold (28.1°C). The temperature continues to rise, reaching T_{max} at t_2 and recovering from shock at t_3. However, there can be a less severe/ shorter shock period before the evaluated heatwave but during this shorter shock there is at least 24-hour period between the smaller shock and the evaluated heatwave when the indoor temperature does not exceed the SET-$_{alert}$ threshold. This period when SET alert threshold is not violated for at least 24h is considered a rest period and the counter of the t_{abs} re-starts. Thus, the actual t_{abs} for this shock is calculated starting from t_1 and reaches peak at t_2 and recovers at t_3.

4.3.2.3 Resilience aspect 3: Shock recovery-Recovery time

Similarly, in [37] recovery speed was defined as the speed of the recovery process. In this study recovery time (t_{rec})is qualitatively defined as:

$$t_{rec} = t_{below} - t_{peak} \qquad \textit{(Equation 4-2)}$$

Where: t_{rec} is the recovery time, t_{peak} is the time when the peak SET is reached and t_{below} is the time when the SET first drops back below the SET-$_{alert}$ threshold (28.1°C) and remains below this threshold for at least 24 hours [88]. Higher t_{rec} indicates that

the building and the occupant remains in a heat-stress period for longer duration as the building and the temperature management strategies takes longer to bring the indoor conditions back to designed level, thus demonstrating lower thermal resilience. The calculation of t_{below} stops once the hourly SET does not exceed the SET-$_{alert}$ threshold for a continuous period of at least 24 hours, allowing occupants sufficient recovery time from the shock [88].

t_{abs} and t_{rec} is not dependent on the building's construction but rather on their impact on the occupants. t_{abs} is calculated when the hourly SET crosses the SET $_{alert}$ threshold to reach the SET $_{max}$, signaling that occupants are already experiencing thermal stress and discomfort. The longer the exposure to thermal stress (t_{abs}), the higher the likelihood of discomfort and heat-related health risks for the occupants. Similarly, t_{rec} is the time taken for the temperature to drop from the SET $_{max}$ back down to below the SET $_{alert}$ threshold, indicating recovery from heat stress. A shorter t_{rec} indicates a quicker recovery, which is desirable for reducing prolonged discomfort. In short, both shorter t_{abs} and t_{rec} are preferred for occupant well-being.

4.3.3 Reference apartment building, simulation model and model inputs

4.3.3.1 Development and selection of the reference apartment in Belgian context

Residential buildings (single-family, terraced, multi-family houses and apartments [202]) constitute 60-89% of the building stock in Europe [203]. Several studies analyzing the indoor overheating risk in different residential buildings have concluded that apartments are at higher risk of overheating due to their high density, unavailability of integrating cross-ventilation in the design, heavy thermal mass and high WWR [113,204–207]. Thus, within the framework of the Flanders Innovation and Entrepreneurship in the Flux 50 Project 'ReCOver++: Improving resilience of buildings to overheating [208], KU Leuven, Ghent University and Renson [209], developed unique floor plans for reference apartments in Belgium, while maintaining some degree of freedom (floor area, number of bedrooms, open/closed kitchen, number of surfaces exposed to external conditions, level of apartment-ground, middle or upper floor), in order to perform parametric study. The floor plans are based on commercial information of Flemish Energy & Climate agency (VEKA) [210] data from 2016 to 2020 (new buildings) available on the internet and is evaluated against guidelines, when available (as illustrated in Figure 4-3).

The habitable rooms and kitchen are arranged to have an "L shape layout". There are floor plans having multiple bedrooms, ranging from 1 to 3 (See Figure 4-4). The surface of the living room, kitchen, utility room and bathroom increase in function of the number of bedrooms (see Table 4-1). The surface of the bedrooms and toilet remains (nearly) identical. As a consequence, the façade area also increases in function of the number of bedrooms. The rooms are kept as much as possible at the same position, allowing to maintain the same interaction between adjacent rooms (flow due to internal leakages). The number of façades is also variable, ranging from 1 to 3, resulting in a different flow pattern and thermal flux through the building envelope. Façades can have an orientation to the east, the south and/or the west. Only the single bedroom flat has

façades oriented to the east and west. Orientation can be varied to evaluate the impact of building orientation.

Multi-family dwellings (in Belgium) typically have an open plan living area. However, a set of floor plans has been made for the living room and kitchen separated by a wall. The utility room is situated next to the kitchen. On the floor plan, the door of the utility room is placed towards the hall. However, this door could also be placed towards the kitchen. This can be an extra variable when performing a parameter study. To summarize, variation in floor plan layout can be achieved by means of – (a) variation in number of bedrooms (b) variation in number of facades (c) variation in kitchen type and (d) variation in door position of the utility room. Based on the VEKA data, the different configuration of the apartments with varying floor area was developed (see Table 4-1).

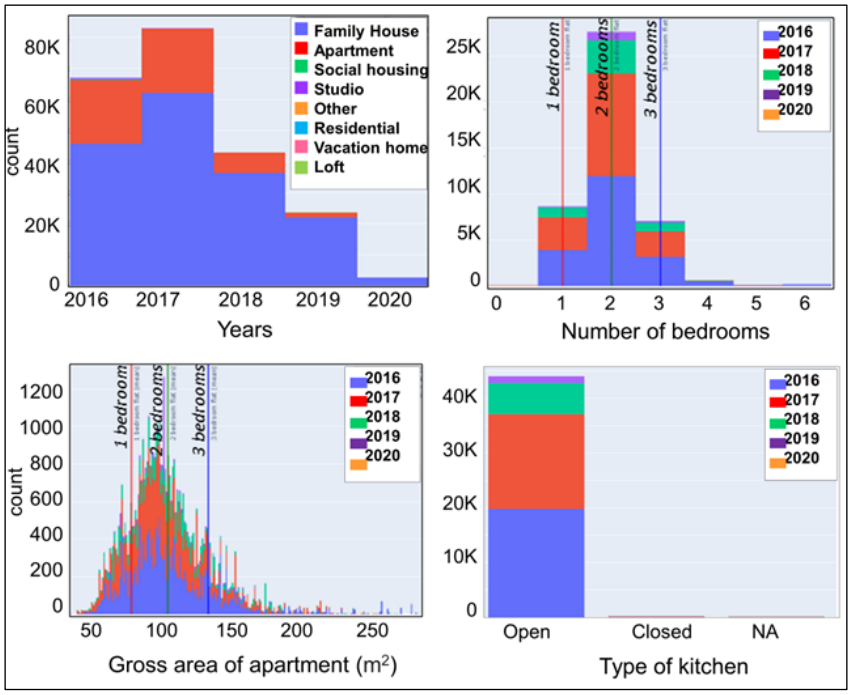


Figure 4-3.VEKA data from 2016 to 2020 (new buildings) showing the typology of buildings, number of bedrooms and gross area of each apartment [210]

To evaluate thermal resilience performance, a two-bedroom reference apartment with open kitchen and living room in an intermediate floor level (situated between a lower and upper floor to test the most representative typology) was selected. The selection of the apartment typology and design was based on the new buildings data from the (VEKA) [210] in accordance with the Energy Performance and Buildings directive (EPBD) guidelines [211]. The selected reference apartment has a brick wall with external insulation resulting in a u-value of 0.24 W/m^{2}K and has heavy thermal mass according to the EN ISO 13790 [72]. Table 4-2 shows building envelope properties (also describes the different construction possibilities for varying thermal mass) for the reference

apartment in line with the EPBD guidelines [212]. Figure 4-5illustrates the plan of the reference apartment showing the four thermal zones (TZs). Table 4-3 shows the characteristics of the selected two-bedroom apartment.



Figure 4-4. Typology for the 1-, 2- and 3-bedroom apartments (open kitchen + 2 side external wall)

4.3.3.2 HVAC Systems

During a typical heating/cooling season the TZs maintain constant comfortable temperatures by utilizing the heating and cooling systems. The apartment is equipped with balanced mechanical ventilation with heat recovery with a total airflow of 200 m^3/h calculated according to the NBN D50-001 [213]. For temperature management, heating and cooling loads were calculated for typical weather data developed using the method described in [160] for ASHRAE climate zone 4A. Heating load was calculated according to NBN EN 12831-1 ANB:2020 [214] to maintain the setpoint of 20±0.5°C during heating season in all TZs and the cooling load of the building was calculated according to VDI 2078 [215] and set points of the living room and bedrooms were 26±0.5°C and 24±0.5°C respectively for the cooling season. The reference apartment is equipped with automated solar shading (external blinds) that are deployed when the solar radiation on the surface wall is higher than 250 W/m^2. Furthermore, the reference apartment is equipped with automated control for NNV, which activates motorized windows from 10:00 p.m. to 6:00 a.m. between April 1st and October 31st. The total effective operable area of these windows is 4% of the floor area [216].

Table 4-1. Net floor area (m^2) of each space in the different variations of the 1, 2 and 3 bedroom apartments

Net floor surface area [m^2]						
	1 bedroom flat		2 bedroom flat*		3 bedroom flat	
	open	*closed*	***Open****	*closed*	*open*	*closed*
living room (closed plan)	*-*	*28.1*	*-*	*30.7*	*-*	*35.2*
kitchen (closed plan)	*-*	*6.3*	*-*	*7.9*	*-*	*10.4*
living room + kitchen (open plan)	*34.7*	*-*	***38.9****	*-*	*46.0*	*-*

bedroom 1	*11.6*	*11.6*	***11.9****	*11.9*	*14.0*	*14.0*
bedroom 2	*-*	*-*	***11.6****	*11.6*	*11.0*	*11.0*
bedroom 3	*-*	*-*	*-*	*-*	*11.0*	*11.0*
bathroom	*6.0*	*6.0*	***6.9****	*6.9*	*8.4*	*8.4*
toilet	*1.6*	*1.6*	***1.6****	*1.6*	*1.6*	*1.6*
utility room	*4.0*	*4.0*	***5.3****	*5.3*	*6.2*	*6.2*
hall	*6.1*	*6.1*	***9.0****	*9.0*	*11.1*	*11.1*
total	*63.9*	*63.6*	***85.2****	*84.9*	*109.3*	*108.9*

**Selected apartment*

Table 4-2. Envelope construction packages (outside to inside)

Construction	materials	λ [W/m.K]	c [J/kg.K]	Density ρ [kg/m³]	Thickness d[m]
**External Wall (heavy)*	*PUR*	*0.02*	*1400*	*30*	*according to energy performance level*
	Brick	*0.54*	*1000*	*1550*	*0.14*
	Plaster Finish	*0.52*	*1000*	*1300*	*0.01*
External Wall (medium)	*PUR*	*0.02*	*1400*	*30*	*according to energy performance level*
	Brick	*0.54*	*1000*	*1000*	*0.14*
	Plaster Finish	*0.52*	*1000*	*1300*	*0.01*
External wall (light)	*Wood Fibre-board*	*0.50*	*2100*	*24*	*0.018*
	Mineral wool (MW)	*0.04*	*1030*	*50*	*according to energy performance level*
	OSB	*0.13*	*1700*	*650*	*0.015*
	Gypsum board	*0.24*	*1000*	*750*	*0.012*
Internal Wall (heavy)	*Plaster finish*	*0.52*	*1000*	*1300*	*0.01*
	Brick	*0.54*	*1000*	*1250*	*0.14*
	Plaster finish	*0.52*	*1000*	*1300*	*0.01*
Internal Wall (medium and light)	*Gypsum board*	*0.24*	*1000*	*750*	*0.01*
	Gypsum board	*0.24*	*1000*	*750*	*0.01*
	Mineral wool (MW)	*0.04*	*1030*	*50*	*0.04*
	Gypsum board	*0.24*	*1000*	*750*	*0.01*
	Gypsum board	*0.24*	*1000*	*750*	*0.01*
**Intermediate floors (heavy)*	*Hollow core concrete slab*	*0.92*	*840*	*1500*	*0.12*
	Heavy concrete	*1.7*	*1000*	*2400*	*0.04*
	PUR	*0.03*	*1400*	*30*	*0.03*
	Screed	*1.35*	*840*	*1000*	*0.1*
	Tiles	*1.2*	*840*	*2300*	*0.02*
	Gypsum board	*0.24*	*1000*	*750*	*0.012*

Intermediate floors (light)	*Mineral wool (MW)*	*0.06*	*1030*	*50*	*0.2*
	Wood fiber-board	*0.0555*	*2100*	*24*	*0.03*
	OSB	*0.13*	*1700*	*650*	*0.018*
	Linoleum	*0.623*	*1400*	*1200*	*0.003*
**Common Walls*	*Mineral wool (MW)*	*0.04*	*1030*	*50*	*0.05*
	Brick	*0.54*	*1000*	*1550*	*0.09*
	Plaster Finish	*0.52*	*1000*	*1300*	*0.01*
**Roof (heavy)*	*Bitumen*	*0.23*	*1000*	*1100*	*0.002*
	PUR	*0.03*	*1400*	*30*	*according to energy performance level*
	Light concrete	*0.32*	*1000*	*1050*	*0.1*
	Heavy concrete	*1.7*	*1000*	*2400*	*0.15*
	Plaster finish	*0.52*	*1000*	*1300*	*0.01*
Roof (light)	*EPDM*	*0.25*	*1000*	*1150*	*0.002*
	OSB	*0.13*	*1700*	*650*	*0.018*
	Mineral wool (MW)	*0.04*	*1030*	*50*	*according to energy performance level*
	Gypsum board	*0.24*	*1000*	*750*	*0.012*

**Default construction packages for the reference apartment with heavy thermal mass*

Table 4-3. Characteristics of the reference apartment

Net floor are (m2)	*85.2*
Floor height (m)	*2.55*
Orientation of living room and bedroom	*South*
Airtightness (h^{-1})	*1.47*
Thermal mass (according to EN ISO 13790 [72])	*Heavy*
External envelope U-value (W/m^2K) (Brick wall with external insulation)	*0.24*
Glazing (double glazed windows)	*U-value = 1.00 W/m^2K, g-value = 0.56*
WWR	*25%*
Window to floor ratio for operable window	*4%*
No of occupants- Internal gains (people, and equipment) calculated according to EN 16798-1 [217].	*3*

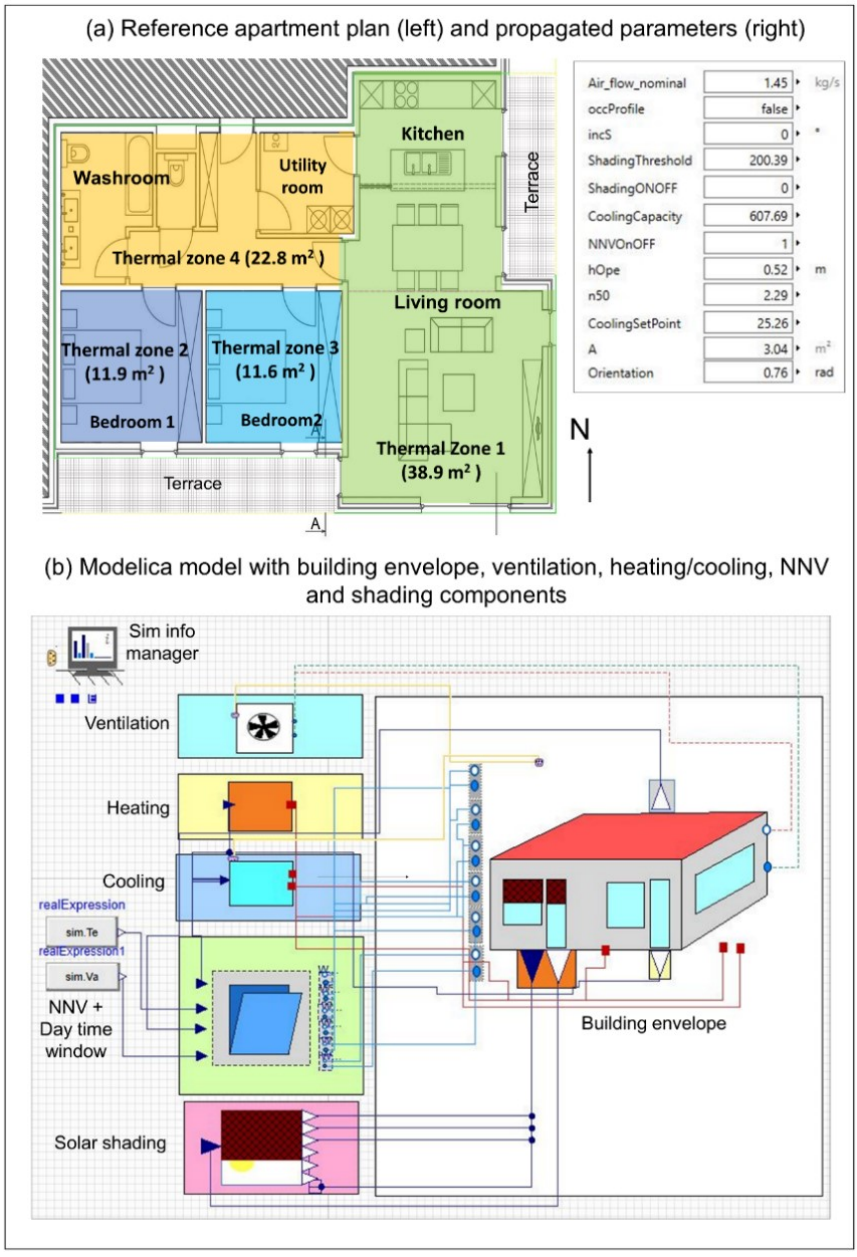


Figure 4-5. (a) Floor plan of reference apartment and parameters propagated to vary in simulations to assess thermal resilience performance and (b) Modelica model of the reference apartment building

4.3.3.3 Modelica model and model inputs of the reference apartment

Figure 4-5(b) illustrates the simulation model of the apartment. The simulation model of the reference apartment was developed in Modelica within the Dymola environment (version 2023x) [218]. The "Integrated District Energy Assessment by Simulation" (IDEAS) library (Version v3.0.0) [219] was utilized as part of the modelling framework. Modelica, an open-source, equation-based modelling language, was chosen for its versatility, and IDEAS library offers a collection of validated sub-models. These sub-models can accurately capture the dynamics of the building, HVAC systems, control strategies, and various events with different time scales. This integrated approach ensures a comprehensive and reliable representation of the simulated apartment. represents the energy balance in the zone model:

$$V_{ra} C_{p,a} \frac{dT_{ra,z}}{dt} = \dot{Q}_{a,z} C_{p,a} \left(T_{supp} - T_{ra,z}\right) + \dot{Q}_{inf,z} C_{p,a} \left(T_{oa} - T_{ra,z}\right) + Q_i + Q_{envelope}(t) \qquad \textit{(Equation 4-3)}$$

The left side of the equation represents the transient heat stored in the thermal zones (TZs). The 1st term on the right-hand side represents the net convective heat flow due to the building ventilation and energy systems, where $C_{p,a}$ is the specific heat of air, T_{supp} is the supply air temperature and $T_{ra,z}$ is the zone temperature. The second term is the convective heat flow due to infiltration through the building envelope (i.e., cracks, window opening), where T_{oa} is the dry bulb temperature. The 3rd term on the right-hand side Q_i is the sensible heat generation by occupants and lighting. $Q_{envelope}(t)$ is the convective heat exchange between the lecture room and its envelope.

Figure 4-5 shows the TZs of the apartment were modelled using the zone component, envelope components to model the external and internal envelope and window component to model the glazing. The building envelope uses equations described in [180] to compute the varying radiative heat exchange and convective heat transfer coefficients between the components and the zone air temperature. The zone model was considered to achieve well-mixed and homogeneous conditions of temperature. The separating floors and common walls connecting the apartment to common space were modelled as adiabatic. Since the apartment was situated between upper and lower floors, the floors and ceilings were considered airtight. The sensible and latent heat generation from occupants were 75 W and 45 W respectively during sedentary activities [182]. For lighting, 300 lux of illuminance in each bedroom, 400 lux in kitchen and living room, 70 lux in corridor and 500 lux in washroom and toilet [220]. The lights turned on when occupants were present and were considered off during night time (10.00 p.m. to 7.00 a.m.). In Modelica, a balanced mechanical ventilation system is modeled using components such as fans, ducts, and air handling units, which are interconnected to simulate air flow and heat exchange, ensuring that the mass and energy balances are maintained throughout the system.

Using the IDEAS library, the heating and cooling system was modelled with inputs such as indoor and outdoor temperatures, relative humidity and PI controls blocks to regulate the systems' operation. Since the focus of this study was to assess thermal

resilience to overheating during heatwaves, the heating system was modeled as an ideal heater in Modelica, using a prescribed heat flow component and a PI control block. The prescribed heat flow sets the desired heating output, while the PI control adjusts the control signal based on the difference between desired and actual heat output, maintaining the desired heating level in a feedback loop. To test how design parameters impact thermal resilience performance, two main categories of cooling terminals-convective and radiant terminals were evaluated [221]. In Modelica using the IDEAS library, a convective cooling model is created by incorporating heat exchangers and air flow networks, combined with a PI control block that adjusts the cooling output based on the difference between the desired and actual temperatures, ensuring efficient temperature regulation. For the radiant floor cooling system a Type A (EN15377, EN1264) described in [222]is modeled using components such as heat exchangers and pipes. The pipes are typically modelled to be embedded within the floor structure (in the screed or concrete), covering the desired floor area (for living room and bedrooms) and are placed close to the surface of the floor to maximize heat transfer efficiency. The system involves circulating a chilled water through the pipes, which absorbs heat from the room and cools the floor surface, thereby cooling the room through both convection and radiation. A minimum surface temperature threshold of 19°C [223] was assumed to avoid surface condensation.

To test resilience of passive strategies, NNV and solar shading system were modelled. External solar shading, with a SHGC of 0.04, was modeled using a PI control system that activates when solar radiation on the window surface exceeds 250 W/m^2. A sampler component ensured that shading control status (ON/OFF) remained consistent for 15-minute intervals. To simplify the model, additional shading effects from elements such as window overhangs, trees, or neighboring structures were not considered.

The reference apartment is equipped with automated control for night natural ventilation (NNV), which activates motorized windows from 10:00 p.m. to 6:00 a.m. between April 1st and October 31st. The NNV system was modeled using operable window components connected to both the outdoor air and zone models. The outdoor air component calculates wind pressure coefficients on the building facades, while the operable windows regulate airflow based on pressure differences using power-law equations [181]. The total effective operable area of these windows is 4% of the floor area [216]. When opened, the windows remain open for at least 15 minutes if the following conditions are met simultaneously:

- Max (T_{TZ}) $\geq$ 22°C and $\geq$ than the outdoor air temperature T_{oa} by 2°C
- Maximum zone temperature (T_{TZ}) of the previous day $\geq$ 23°C
- T_{oa} >12°C
- Indoor RH < 70%
- There is no rainfall and the wind velocity < 10 m/s measured at a height of 10 m from ground level.

Weather data (extreme heatwave files) were input in the Sim info manager component to evaluate thermal resilience performance during different shocks. Yearly simulations were conducted with 15-minute time step to capture the temporal variations in the

thermal zones, operation of passive strategies such as NNV and solar shading, cooling and heating systems as time-step shorter than 15 min did not lead to a better precision but increased significantly the simulation time [224].
To solve for the system of non-linear equations, the RADAU II, an implicit Runge-Kutta method, with a tolerance of 10^{-8} was used. This method is a single-step method, where the solution at the current time step is only affected by information from the previous time step [184]. Thus, RADAU II solver was able to handle complex equations and compute the solutions with higher accuracy in comparison to methods such as Euler or Dassl. The models were run in the HPC infrastructure [225] utilizing high computing capabilities to reduce computational time.

4.3.4 Uncertainty and Sensitivity Analysis Technique

An uncertainty analysis was conducted to evaluate the variability in buildings' thermal resilience performance resulting from various design variations. Additionally, sensitivity analysis utilizing standard regression coefficients aimed to identify the most influential design parameters impacting the thermal resilience of buildings. Standard regression coefficients are derived by standardizing both dependent (in this case the three resilience aspects) and independent variables (building and system design parameters). This standardization facilitates comparison of coefficients on a consistent scale, irrespective of the original units of the variables, facilitating the assessment of predictors' relative importance on the dependent variable. While numerous techniques for conducting uncertainty and sensitivity analyses exist, each with its own strengths and limitations, a global method evaluating the overall impact of all input parameters was selected [226–229]. A Monte Carlo analysis was chosen due to its lower computational cost, preference for quantitative methods over qualitative ones, and its effectiveness in conducting both uncertainty and sensitivity analyses [230–232]. The following steps were undertaken: (a) defining a range and distribution for selected design parameters, (b) sample size determination and sample generation, (c) simulating the Modelica model for each sample element to determine different KPI values, (d) conducting uncertainty analysis to determine the uncertainty in a buildings' thermal resilience performance due to different design variations and (e) performing sensitivity analysis for-(i) three different heatwaves, (ii) two cooling strategies and (iii) three KPIs to determine which building and system parameter has the highest influence on thermal resilience.

4.3.4.1 Defining a range and distribution of selected parameters

Input selection for uncertainty and sensitivity analysis depends on the study's goals, with common sampling methods including simple random and Latin Hypercube sampling. In a study investigating the role of sensitivity analysis in building performance analysis, Pang et al. [231] listed that the most frequent input parameters impacting overheating in buildings includes weather, building design- envelope elements (e.g., walls, roof, windows), glazing properties, ventilation/infiltration, HVAC systems, and occupant behaviors, while key outputs are building/urban energy

consumption and occupants' thermal/visual comfort. This comprehensive list was the starting point to curate the input parameters utilized for this study. Design parameters are categorized into static (e.g., orientation, thermal mass, airtightness) and dynamic (e.g., window operation, shading) factors, both influencing thermal performance. Studies by Fuad et al. [196], Machard et al. [159], and others emphasized factors like thermal mass, WWR, natural ventilation, and solar control as critical for reducing overheating and enhancing thermal resilience. Another study [233] highlighted cooling setpoint temperature, window SHGC, and exterior surface properties as more critical than envelope thermophysical properties. Apart from these studies listing the parameters that impact overheating in buildings, several simulation and qualitative studies analyzed building and system design parameters that impact thermal resilience aspects. Building and system design parameters affecting thermal resilience performance in buildings is comprehensively listed in [234]. Zhang et al. [235] qualitatively reviewed resilient cooling strategies that impact thermal resilience performance-absorptivity, recovery speed. Two simulation based studies showed the impact of thermal mass[47] and operation of passive strategies [46] on thermal resilience performance. Ji et al.[39] showed optimal shading operation, window area, NNV and envelope properties aids in improving thermal resilience in buildings. The key parameters for this study were chosen based on these previous studies with two main selection criterion-(a) frequent usage of the parameters in assessing and improving buildings' overheating and thermal resilience and (b) parameters' impact on different thermal resilience aspects.

Table 4-4 summarizes the selected building and system design parameters and their ranges that were varied to evaluate the buildings' thermal resilience performance. The design parameter ranges were derived from standard building practices, existing studies and standards. The variation range is set high to explore the building's behavior under extreme conditions.

Table 4-4. Design parameters and their ranges

Category	Description/parameter	Range	Default value
Building Parameters	**Building Orientation*	*0°(default South)-360°*	*45°*
	**U-value of external wall*	*0.10-0.30 W/(m²K)*	*0.24 W/(m²K)*
	**WWR*	*25-70%*	*25%*
	+U -value of glazing	*U-value (0.6-2.9) W/(m².K)* *g-value (0.4-0.78)*	*1 W/(m².K)* *g=0.56*
	+Thermal mass	*Heavy-Medium -Light*	*Heavy*
	**Air tightness (ACH) n50*	*0.6-3 h⁻¹*	*1.47*
Occupancy pattern	*+Occupancy profile (occProfile),Default (0) and high occupancy (1)*	*0-1*	*Default (EN 16798-1)*
Solar Shading	*+External shading Control (ShadingONOFF)*	*0-1*	*1 (ON)*
	**Shading Threshold*	*100-300 W/m²*	*250 W/m²*
NNV	**Effective opening area of window expressed as % of floor area (Floor area)*	*1- 6%*	*4%*
	+Night Cooling control (NNVOnOFF)	*0-1*	*1 (ON)*
Cooling system	**Cooling set point*	*24°C- 28°C*	*26°C*
	**Cooling capacity of convective and radiant cooling*	*0-42 W/m²*	*22 W/m²*

** These parameters are continuous variables-uniform distribution was assumed*

+ These parameters were changed discretely- each value was given equal probability in sample

Building design parameters include-building orientation, thermal transmittance of the external envelope, WWR, glazing properties, thermal mass, and airtightness. Building orientation significantly affects solar gains and optimal wind conditions for NNV. Therefore, the facade orientation of the bedrooms and living room was varied between 0° (default South) and 360° to evaluate the impact of solar gains and wind directions on the facade. According to EPBD [211] guidelines, new buildings have U-value of 0.24 W/m^2K. Passive house standard states a U-value of 0.15 W/m^2K [236] for energy efficient buildings. However in practice, in new highly-insulated building external envelopes' thermal transmittance may be as lows as 0.10 W/m^2K [237]. To study the impact of thermal insulation on resilience performance, U-value of the external wall was discretely varied from 0.10 W/m^2K (highly insulated building) to 0.30 W/m^2K (slightly above EPBD guideline to represent older buildings with less insulation) [238–240]. WWR determines the solar gains in the building and previous studies often linked higher WWR with overheating risks in buildings [196]. Previous studies have determined the optimal size of windows [11] and investigated to determine relationship in thermally autonomous buildings between maximum WWRs and the ambient temperature amplitudes with different envelope thermal transmittance [241] to find a balance between solar gains and transmission losses. The default WWR of the apartment is 25% and WWR was varied between 25-70% to evaluate a low WWR and a highly glazed buildings' thermal resilience. Glazing properties impacts solar gains and thus impacts thermal resilience of buildings' to overheating. The reference apartment has a double glazing with U-value=1.0 W/m^2K and SHGC (g-value)=0.56. In order to test parameters such as U-value and g-value of glazing as separate independent variables, these variables had to be varied independently. However, in Modelica, as often seen in practice, combinations of U-value of glazing are associated with typical g-value and cannot be varied independently. To test different glazing strategies, 3 different glazing strategies were varied discretely with equal probability of occurrence- (a) the default glazing (b) a high performance triple glazing with U-value =0.6 W/m^2K and g-value =0.42 (c) uncoated double glazing (U- value= 2.9 W/m^2K and g =0.78) [242] to test performance of a low performance glazing. Thermal mass has plays significant role in absorbing and releasing heat gains. Thus, thermal mass was varied between heavy, medium and low by varying the construction packages (see Table 4-2). Thermal mass of default apartment was calculated using EN 13786-2017 [243]. Often airtight buildings are associated with higher overheating risks [244]. Airtightness (n50) was varied between 3 ACH i.e., upper limit for the infiltration rate n50 compliance with the minimum air tightness level required by NBN D50-001[213] and 0.60 ACH i.e., lower limit based on Passive house standard [236] and rate corresponding to the hygienic ventilation.

Based on EN 16798-1[119], three occupants are assumed for the reference apartment. Moreover, these occupants are assumed to be absent during office hours (default). To test thermal resilience buildings where 80 % of all occupants (i.e., 2 out of 3 occupants) are always present in the apartment during entire shock duration, an additional occupancy profile (high occupancy) has been adopted from the occupancy patterns

developed by Buttitta et al. [245]. In the sampling, each occupancy profile has 50% chance of occurrence.

To ensure optimal summer comfort, an automated external shading system was implemented using a simplified control strategy based on total solar radiation thresholds (ranging from 100-300 W/m^2) [246]. The shading control toggles between ON (1) and OFF (0), deploying the shading at different thresholds to assess its effectiveness. When closed, the blinds block 70% of the incoming solar radiation on the shaded surface. North-facing windows are excluded from shading.

For NNV, two key parameters affecting its operation and performance are varied: (a) the control state, toggled between ON (1) and OFF (0), and (b) the effective opening area of operable windows, expressed as a percentage of the floor area, ranging from 1% to 6% [14,176,247]. This should not be confused with the floor-to-window ratio; rather, it represents the proportion of operable window area that opens during NNV operation relative to the total floor area of the space.

For the convective and radiant cooling systems, cooling capacity and cooling setpoints impact their design and operation. Various standards, regulations and guidelines focus on the minimum requirements for thermal comfort and indoor temperature set-points [123,248]. The cooling set point was varied between 24-28°C to balance occupant comfort and energy efficiency. Lower set points (closer to 24°C) ensure greater comfort during hotter periods, while higher set points (closer to 28°C) reduce energy consumption and operational costs, reflecting a compromise between thermal comfort and sustainable energy use. Peak cooling load for the reference apartment was calculated as 22 W/m^2 according to VDI 2078 [215] for the hottest day in typical weather condition. To test resilience performance of systems with higher cooling capacity, its range was broadened. The REHVA guidebook for low temperature heating and high temperature cooling [222] indicates that the radiant floor cooling system has the highest sensible cooling capacity of 42 W/m^2 at the room design condition of 26°C. Thus, cooling capacity for both the system were varied between 0 W/m^2 (no cooling) to 42 W/m^2 (highest capacity).

4.3.4.2 Assumption of linearity between independent and dependent variables

To evaluate the relationship between each input and output variable, the default reference apartment was simulated by varying one input parameter at a time. Table 4-3 provides the default value of the building and system design parameters and Table 4-4 provides the ranges of the building and system design parameters that were varied. Figure 4-6 illustrates the relationship between the individual input parameters and SET-Dh. In the default case, no solar shading, no natural night ventilation (NNV) and no cooling strategy is assumed. For example, increase in window to wall ratio (WWR), shows linear increase in SET-Dh, whereas increase in air-tightness decreases the SET-DH linearly. For testing cooling setpoint, a constant cooling capacity of 22 W/m^2 was assumed. Increase in cooling setpoint, increases SET-Dh linearly. For testing cooling capacity, a cooling setpoint of 25°C was assumed and cooling capacity was increased. Increase in cooling capacity linearly decreases SET-Dh. For NNV operation, SET-Dh

decreases linearly if the operable area of window increases as the air-flow rate is increased. Thus, the independent variables individually have linear relationship with SET-Dh, proving the assumption of linearity between the independent and dependent variables.

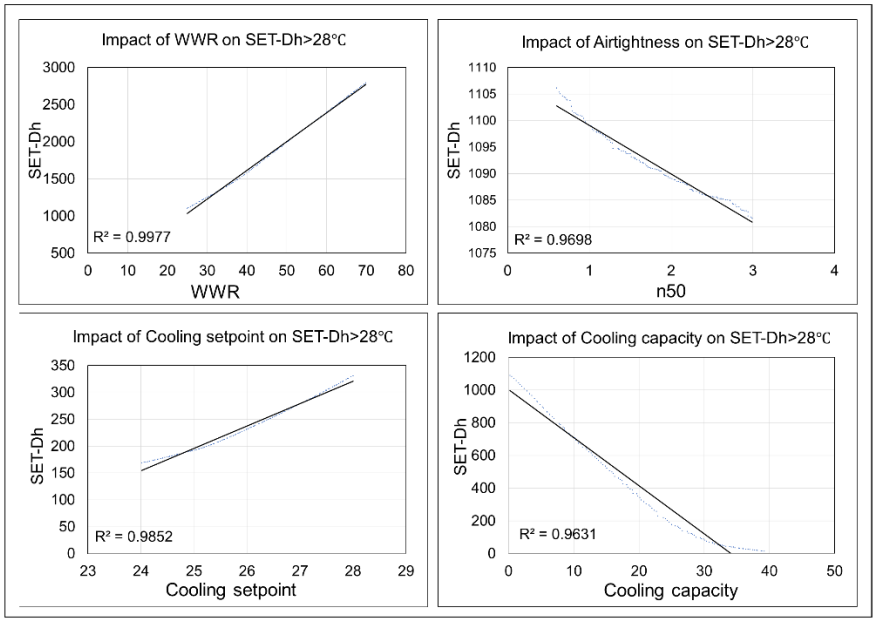


Figure 4-6.Relationship between input and output variables

4.3.4.3 Sample size determination and sample generation

The Latin Hypercube Sampling (LHS) method, facilitated by the pyDOE package [249], was employed for efficient exploration of input design parameters' ranges, ensuring even coverage and reduced clustering compared to traditional random sampling methods, thereby providing accurate estimates with fewer samples. A unform distribution of the input design parameters was assumed in LHS to ensure comprehensive and unbiased coverage of the input space, reduce variability, simplify the sampling process, and enhance the efficiency and robustness of the uncertainty and sensitivity analyses.

There are broadly two ways to determine the sample size-(a) full factorial or (b) generate a random sample that is representative of the full factorial sample ensuring convergence. While a full factorial method demands 2^n sample size (n being the number of parameters), this requires extensive computational power. Studies indicate that the quantity of calculations for reliable results in Uncertainty and Sensitivity Analysis is not solely dependent on the number of input parameters [232,250]. In current literature, there is no consensus of a methodology to determine the minimum number of samples so that the distribution of the output of a LHS method yields an output distribution that is statistically equivalent to the output distribution obtained from a full factorial design

without being computationally expensive. This means that the statistical properties, such as the mean, variance, and higher moments of the output, are preserved, ensuring that LHS is a reliable and efficient alternative to the full factorial design for sampling the input space. With 10 building parameters, the full factorial sample size is 2^{10} =1024. However, several studies have shown that the quantity of calculations required for obtaining reliable results is not contingent on the number of input parameters [7,8]. One such study shows 60 to 80 simulations are adequate, after which only marginal gains in accuracy are gained [9]. Another study conducting uncertainty and sensitivity analysis for building performance assessment, Hopfe conducted 200 simulations [10]. Since there was no consensus on the actual number of simulations to be carried out, one model variation was assessed once with full factorial design sample size and then gradually decreasing the sample size to reach the convergence- in this case with 128 samples. The output of these two cases were compared to determine if 128 samples were adequate.

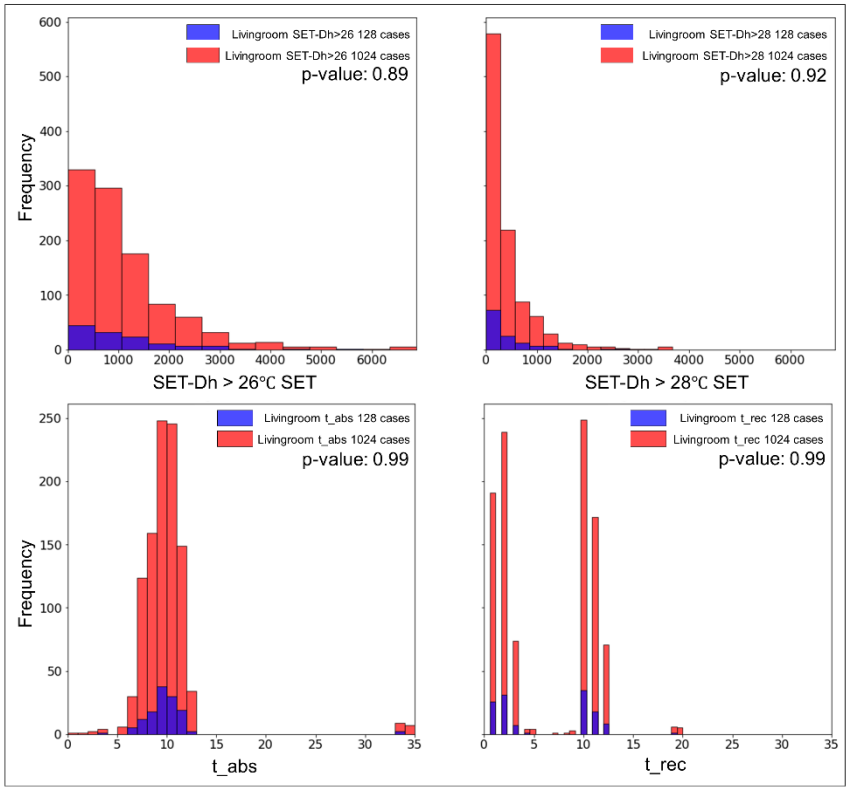


Figure 4-7. Distributions for SET-Dh, t_{abs} and t_{rec} for 128 and 1024 sample size for living room (TZ1)

Figure 4-7 illustrates the distribution of SET-Dh (for both above 26°C and 28°C) and t_{abs} and t_{rec} with full factorial 1024 and 128 samples. The reference apartment model (Table 4-3) was evaluated once with 1024 samples (full factorial) and then 128 samples (128

is the lowest sample size where the output is statistically significant with the full factorial output). The Kolmogorov-Smirnov (KS) test [11]- a well-established statistical method that compares two probability distributions based on their statistical significance was conducted. In KS test, the null hypothesis asserts that there is no significant difference between the cumulative distribution functions of two datasets (full factorial 1024 and 128 sample). The p-value indicates the probability of observing the test statistic under the assumption that the null hypothesis is true. If the p-value is less than a chosen significance level (typically 0.05), the null hypothesis is rejected, suggesting that the datasets come from different distributions. If the p-value is greater than the significance level (0.05), the null hypothesis is retained, indicating that the datasets are statistically consistent with being drawn from the same distribution. For TZ1, SET-Dh (26°C and 28°C), t_{abs} and t_{rec} were compared. The p-value is almost near unity suggesting the two distributions are the same and that 128 samples were representative of the final output distribution.

4.4 Results and discussion

4.4.1 Uncertainty in resilience performance assessment of different design variations

This section describes the uncertainty in SET-Dh, t_{abs} and t_{rec} respectively with three increasing levels of *doS* under a) convective and b) radiant cooling.

According to Figure 4-8 (a) and (b), 112, 62 and 27 i.e., 87%, 48% and 21% of the 128 variations of the apartments evaluated, have SET-Dh below the 230 [°.h] threshold for doS_{low}, doS_{medium} and doS_{high} respectively. For the default reference apartment design, SET-Dh is within the threshold of 230 [°.h] during all *doS* except for doS_{high}. As *doS* increases, the distribution of SET-Dh becomes more uniform and the peak SET-Dh increases by a factor of three for each *doS.* This demonstrates that current design trends for apartments have high thermal resilience to shock impact during shorter shocks but will fail to remain thermally resilient during longer or more-severe shocks which will be more common in future scenario. As seen in Figure 4-8, during doS_{high}, 35% of all cases exhibit SET-Dh exceeding 1000°C*h. For the three increasing *doS* levels, the reference apartment (with default design settings) has SET-Dh of 20.4 [°.h], 59.2[°.h], , and 326.8 [°.h] with convective cooling, and 38.2[°.h], 156.5[°.h], and 491.3[°.h], with radiant cooling.

Figure 4-8 shows that the typical apartment evaluated is not an ideal or "good" reference design, but rather the typical scenario of a reference Belgian apartment. It highlights the thermal resilience of the apartment under various shock scenarios, specifically focusing on the thermal response during different shock durations. Figure 4-8 shows that during the dos_{min} and dos_{medium} shock durations, the SET-Dh remains within the 230[°.h] threshold, which indicates that the apartment's thermal resilience is within acceptable limits. However, during a prolonged shock (45 days), the apartment's thermal resilience is significantly compromised. This reflects the real-world scenario where even buildings complying with legislations might struggle to maintain resilience during extended periods of extreme heat. Hence, the key takeaway here is not

necessarily the design of the apartment itself but rather how different shock durations and intensities impact its thermal resilience, especially over extended periods and the uncertainty in thermal resilience performance of different design variations.

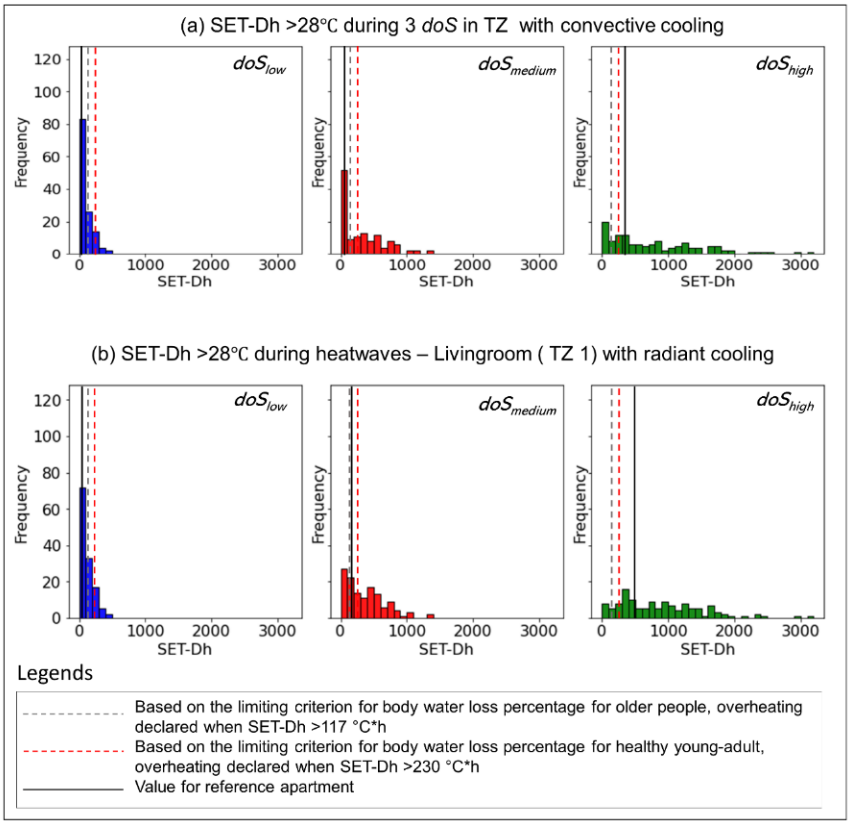


Figure 4-8. Uncertainty of SET-Dh in TZ1 for (a) convective and (b) radiant cooling for 3 doS.

According to Figure 4-9 (a) and (b), 53%, 43% and 9% of the 128 variations of the apartments evaluated absorbed the shock within three days (72h) for *doS low, doS medium* and *doS high* respectively. This includes the default apartment except for *doS medium* and *doS high*. 14% higher t_{abs} were noted for radiant cooling compared to convective cooling strategies for all *doS*. Current design variations have low t_{abs} during shocks when the peak outdoor temperature is reached faster, i.e., peak SET is also reached in shorter time (as seen in Figure 3-12 the temporal trend of outdoor temperature during different heatwaves). The results (in Figure 4-9) also demonstrate that t_{abs} is dependent on when peak SET is reached during each shock. This shows that with increasing *doS*, buildings keep accumulating heat and thus remain in an increasingly more severe overheated state for longer periods of time. As *doS* increases, the distribution of t_{abs} becomes skewed to the right with peak t_{abs} increasing by a factor of five from each *doS low* to *doS*

high. For three increasing *doS*, the reference apartment (i.e., with default design variation) has t_{abs} of 42h, 92h and 217h respectively with convective cooling. With radiant cooling, the default apartment has t_{abs} of 49h, 108h and 245h with three increasing *doS*.

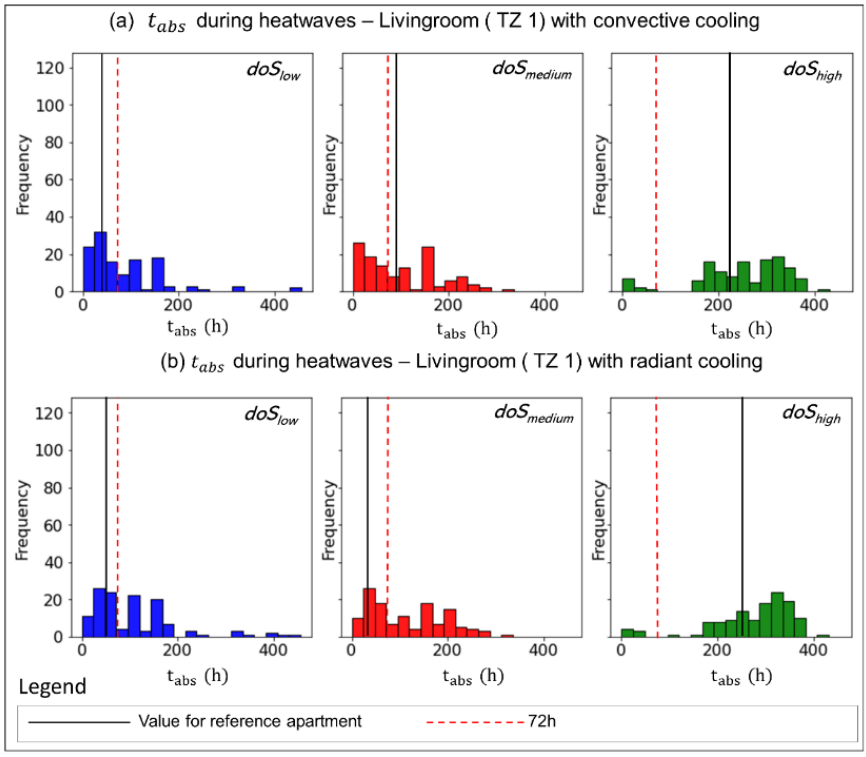


Figure 4-9. Uncertainty of t_{abs} in TZ1 for (a) convective and (b) radiant cooling for 3 doS.

According to Figure 4-10(a) and (b), 50%, 20% and 6% of the 128 variations of the apartments evaluated, recovered from the shock within three days (72h) for *doS low, doS medium* and *doS high* respectively. As *doS* increases, the distribution of t_{rec} becomes skewed to the right with peak t_{rec} increasing by factors of three for each *doS.* The longer and more severe shocks are, the longer it takes for the apartment to recover from the shock. Note that similar to t_{abs}, t_{rec} is also dependent on the shock type, i.e., when the peak of the shock occurs. For three increasing *doS*, the reference apartment (i.e., with default design variation) has t_{rec} of 11h, 198h and 217h respectively with convective cooling. With radiant cooling, the default apartment has t_{rec} of 25h, 218h and 245h with three increasing *doS*.

For both bedrooms, similar distribution trends of SET-Dh, t_{abs} and t_{rec} as of living room was observed during all three *doS.* Figure 4-12 and Figure 4-12 illustrates the distributions of the SET-Dh, t_{abs} and t_{rec} respectively with three increasing levels of *doS* for the bedroom1 (TZ2) convective and radiant cooling. For TZ2 with both cooling

system, during *doS low*, SET-Dh ranges between 0 [°.h] to a maximum of 500 [°.h] (89% of SET-Dh < 230[°.h]) whereas during *doS medium* and *doS high* maximum SET-Dh increases to 1400[°.h] (67% with convective cooling and 74% with radiant cooling of SET-Dh < 230[°.h]) and 3500[°.h] (46% with convective cooling and 58% with radiant cooling of SET-Dh < 230[°.h]) respectively. As *doS* increases, the peak SET-Dh increases due to significant increase in shock duration. SET-Dh increases 7x and 2.5x during *doS high* compared to the *doS low* and *doS medium* respectively as the shock duration and intensity increases. During *doS low* and *doS medium* SET-Dh is below 230 [°.h] . Similar results were seen for TZ1. Building design variations with no/low cooling capacity, non-operational solar shading and NNV shows higher SET-Dh for both cooling strategies. Since bedrooms are occupied during the night period (22:00 pm to 7 :00 am), operation of NNV has significant impact in the thermal resilience performance in bedrooms. During *doS high*, 25% of all cases have SET-Dh > 1000[°.h].

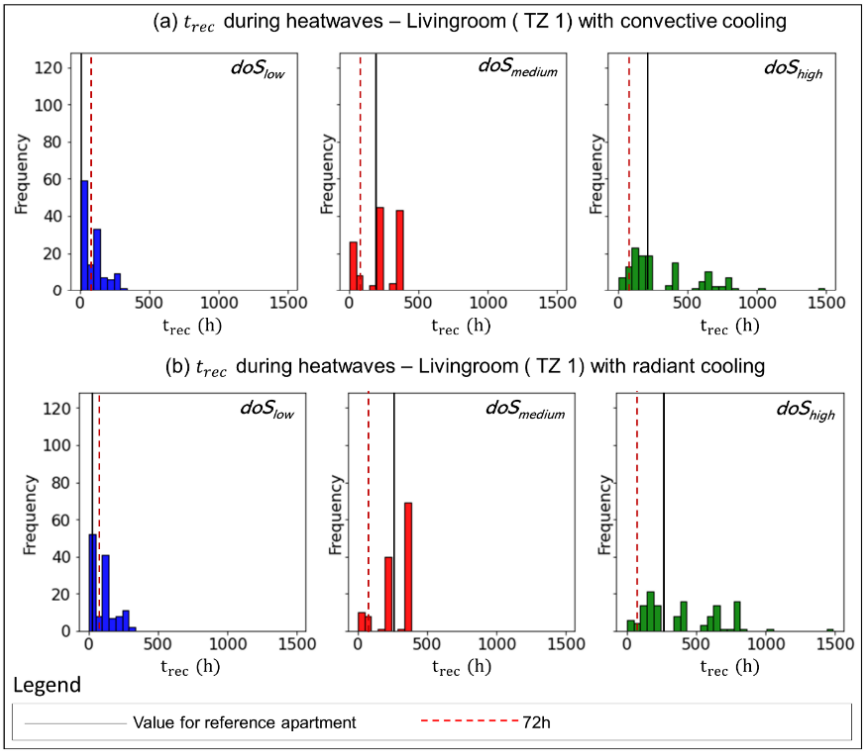


Figure 4-10. Uncertainty of t_{rec} in TZ1 for (a) convective and (b) radiant cooling during three doS.

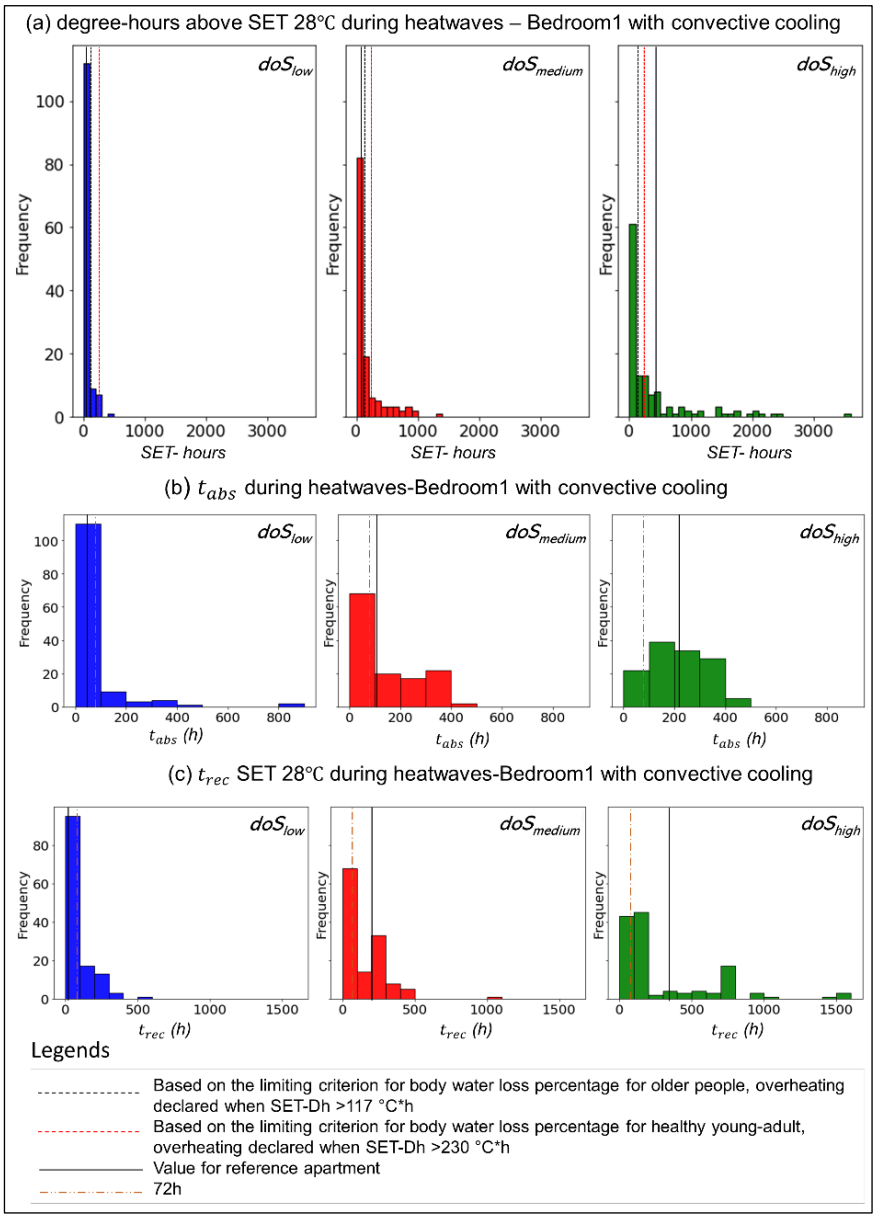


Figure 4-11. Uncertainty of SET-Dh, t_{abs} and t_{rec} in TZ2 (bedroom1) with convective cooling during 3 doS.

As illustrated in Figure 4-11(b) and Figure 4-12 (b), with both cooling systems, in TZ2 as *doS* increases the distribution of t_{abs} is more uniformly distributed than compared to positively skewed during lower *doS*. t_{abs} for *doS medium* and *doS high* ranges between 0 to 500h whereas for *doS low* , t_{abs} ranges between 0 to 900h. Higher t_{abs} (>800h) is associated with high WWR, no solar shading and no NNV design variations where high

heat gains increases the indoor temperature. With both cooling system, during *doS low*, t_{abs} ranges between 0h to a maximum of 900 h (80% of t_{abs}< 72h) whereas during *doS medium* and *doS high* maximum t_{abs} is 500h (66% of t_{abs}< 72h) and 500h (39% of t_{abs}< 72h) respectively. For three increasing *doS*, the reference apartment (i.e., with default design variation) has t_{abs} of 38h, 80h and 210h respectively with convective cooling. With radiant cooling, the default apartment has t_{abs} of 45h, 110h and 260h with three increasing *doS*.

As illustrated in Figure 4-11(c) and Figure 4-12(c), t_{rec} is significantly impacted by the duration of the shock. Longer duration of shock implies the building and its occupants in heat stress for longer duration. With both cooling systems, design variations with low cooling capacity, no solar shading, no NNV operation and high WWR ratio has t_{rec}. To be also noted that buildings with high thermal mass also have high t_{rec} as the absorbed heat takes longer to be flushed out especially without NNV operation. During *doS low* peak t_{rec} is 500h compared to 1600h (3.2x higher) during *doS high*. With both cooling system, during *doS low*, t_{rec} ranges between 0h to a maximum of 500 [°.h] (78% of t_{rec}< 72h) whereas during *doS medium* and *doS high* maximum t_{rec} is 1100h (58% of t_{rec}<72h) and 1600h (50% of t_{rec}<72h) respectively. Similar results were observed for TZ3 (bedroom2) as there were no significant difference in terms of building and system properties for both the bedrooms. For three increasing *doS*, the reference apartment (i.e., with default design variation) has t_{rec} of 9h, 200h and 330h respectively with convective cooling. With radiant cooling, the default apartment has t_{rec} of 27h, 210h and 350h with three increasing *doS*.

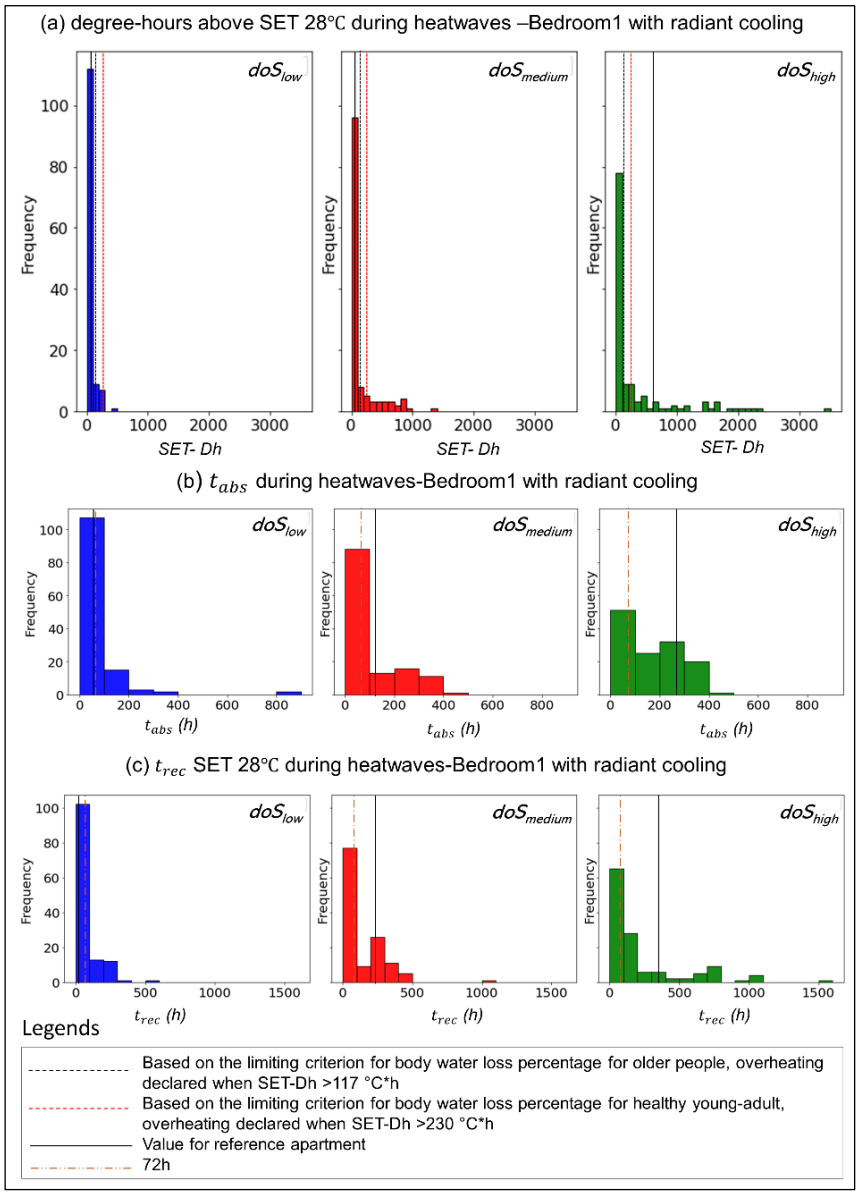


Figure 4-12. Uncertainty of SET-Dh, t_{abs} and t_{rec} in TZ2 (bedroom1) with radiant cooling during 3 doS.

4.4.2 Sensitivity Analysis: Impact of different input parameters on the resilience performance

This section presents the results of the sensitivity analysis to determine the most influential design parameters impacting thermal resilience performance during – (a)

three increasing *doS*, (b) for two different cooling strategies and, (c) three resilience performance aspects- SET-Dh, t_{abs} and t_{rec}.

4.4.2.1 Impact of parameters on the resilience performance for 3 increasing *doS*

Table 4-5 and Table 4-6 show the standardized regression coefficients for each parameters for 3 *doS* for convective and radiant cooling respectively. Figure 4-13 and Figure 4-14 illustrates the standardized regression coefficients for the building and system design parameters that impact the SET-Dh, t_{abs} and t_{rec} with convective cooling system during 3 increasing *doS.*

Table 4-5. Standardized regression coefficients for TZ1 with convective cooling

Convective cooling									
	SET-Dh			t_{abs}			t_{rec}		
doS	*doS high* **0.368**	*doS medium* **0.179**	*doS low,* **0.06**	*doS high* **0.368**	*doS medium* **0.179**	*doS low,* **0.06**	*doS high* **0.368**	*doS medium* **0.179**	*doS low,* **0.06**
WWR	*0.660*	*0.656*	*0.655*	*0.715*	*0.584*	*0.615*	*0.603*	*0.526*	*0.654*
Cooling Capacity	*-0.427*	*-0.396*	*-0.324*	*-0.394*	*-0.312*	*-0.335*	*-0.471*	*-0.440*	*-0.300*
NNVOnOFF	*-0.424*	*-0.427*	*-0.386*	*-0.294*	*-0.468*	*-0.324*	*-0.401*	*-0.170*	*-0.259*
ShadingONOFF	*-0.307*	*-0.323*	*-0.372*	*-0.279*	*-0.343*	*-0.366*	*-0.232*	*-0.324*	*-0.392*
Operable area NNV	*-0.130*	*-0.104*	*-0.089*	*-0.022*	*-0.032*	*-0.101*	*-0.116*	*-0.103*	*-0.115*
Glazing	*0.080*	*0.079*	*0.067*	*0.138*	*0.119*	*0.103*	*0.044*	*0.022*	*0.064*
Cooling Set Point	*0.077*	*0.070*	*0.093*	*0.153*	*0.051*	*0.127*	*0.078*	*0.192*	*0.148*
Thermal Mass	*-0.071*	*-0.079*	*-0.093*	*-0.065*	*-0.024*	*-0.114*	*0.002*	*-0.098*	*-0.154*
Orientation	*-0.053*	*-0.058*	*-0.065*	*-0.076*	*0.040*	*-0.106*	*-0.093*	*-0.072*	*-0.193*
Shading Threshold	*0.044*	*0.056*	*0.049*	*0.093*	*0.005*	*0.067*	*-0.003*	*0.164*	*-0.009*
Airtightness	*0.016*	*0.014*	*0.013*	*-0.011*	*0.023*	*0.079*	*0.027*	*0.017*	*0.054*
Occupancy	*0.013*	*-0.008*	*-0.019*	*0.110*	*-0.083*	*-0.040*	*-0.017*	*0.078*	*-0.024*
Envelope u-value	*0.001*	*-0.002*	*0.017*	*0.053*	*0.051*	*0.042*	*-0.025*	*0.082*	*-0.047*

Table 4-6. Standardized regression coefficients for TZ1 with radiant cooling

Radiant cooling									
	SET-Dh			t_{abs}			t_{rec}		
doS	*doS high* **0.368**	*doS medium* **0.179**	*doS low,* **0.06**	*doS high* **0.368**	*doS medium* **0.179**	*doS low,* **0.06**	*doS high* **0.368**	*doS medium* **0.179**	*doS low,* **0.06**
WWR	*0.713*	*0.719*	*0.699*	*0.751*	*0.620*	*0.689*	*0.712*	*0.563*	*0.663*
Cooling Capacity	*-0.385*	*-0.344*	*-0.260*	*-0.289*	*-0.315*	*-0.239*	*-0.411*	*-0.218*	*-0.153*
NNVOnOFF	*-0.410*	*-0.430*	*-0.380*	*-0.260*	*-0.350*	*-0.351*	*-0.343*	*-0.125*	*-0.224*
ShadingONOFF	*-0.318*	*-0.329*	*-0.391*	*-0.266*	*-0.155*	*-0.343*	*-0.245*	*-0.295*	*-0.444*
Operable area NNV	*-0.113*	*-0.087*	*-0.079*	*-0.048*	*-0.040*	*-0.047*	*-0.135*	*-0.072*	*-0.106*
Glazing	*0.103*	*0.102*	*0.083*	*0.164*	*0.132*	*0.098*	*0.081*	*0.076*	*0.094*
Cooling Set Point	*0.099*	*0.104*	*0.122*	*0.145*	*0.065*	*0.228*	*0.177*	*0.252*	*0.168*
Thermal Mass	*-0.075*	*-0.084*	*-0.103*	*-0.094*	*0.055*	*-0.071*	*-0.045*	*-0.062*	*-0.160*
Orientation	*-0.042*	*-0.039*	*-0.050*	*-0.034*	*-0.089*	*-0.083*	*-0.080*	*-0.021*	*-0.181*
Shading Threshold	*0.040*	*0.052*	*0.052*	*0.085*	*-0.016*	*0.083*	*0.041*	*0.066*	*0.017*
Airtightness	*0.023*	*0.024*	*0.014*	*-0.008*	*-0.042*	*0.058*	*0.016*	*-0.033*	*0.044*
Occupancy	*0.016*	*-0.013*	*-0.006*	*0.032*	*-0.053*	*-0.059*	*-0.026*	*0.056*	*0.038*
Envelope u-value	*0.024*	*0.021*	*0.026*	*0.089*	*-0.099*	*0.044*	*-0.039*	*0.106*	*0.016*

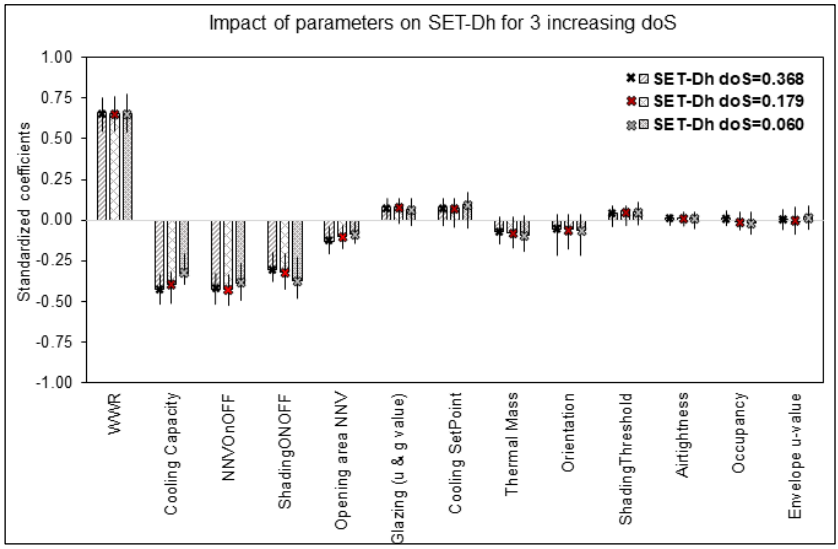


Figure 4-13. Impact of selected building and system parameters on SET-Dh for three increasing doS (TZ1 with convective cooling)

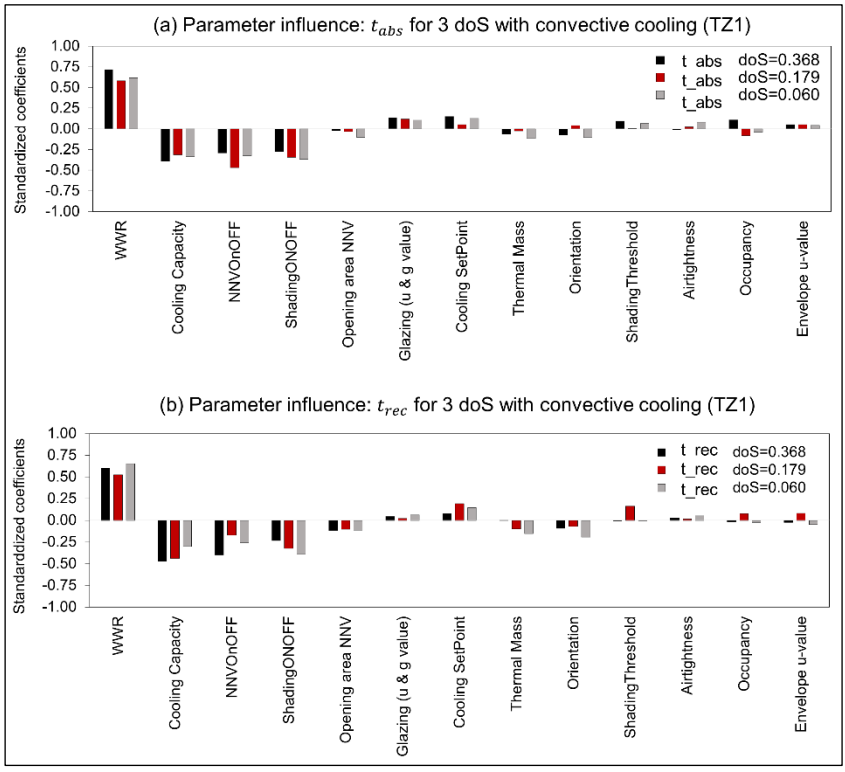


Figure 4-14. Impact of selected building and system parameters on t_{abs} and t_{rec} for three increasing doS (TZ1 with convective cooling)

Figure 4-13 shows that parameters such as WWR, cooling capacity, operation of NNV and solar shading have the highest impact on SET-Dh during all three *doS* for convective cooling strategy. Throughout all three *doS,* WWR has the same influence (1.01x increase during each *doS*) on SET-Dh. WWR dictates the solar gains and indoor heat accumulation and positively correlates with SET-Dh meaning as WWR increases, SET-Dh increases. Cooling capacity reduces SET-Dh, with its influence increasing by 1.07x and 1.22x during doS_{medium}, and doS_{high} compared to doS_{low} ,i.e., cooling capacity becomes more influential on buildings thermal resilience performance during longer duration shocks. NNV operation effectively lowers SET-Dh by removing heat at night, with its impact growing as *doS* severity rises (1.3x and 1.5x higher during doS_{high} compared to doS_{medium} and doS_{low}). Solar shading also reduces SET-Dh by mitigating solar gains, with a 1.2x higher impact at doS_{low} compared to doS_{medium} and doS_{high}. Other parameters have approximately 2x less impact on SET-Dh. Increasing the cooling set-point delays cooling, leading to higher heat accumulation, especially at doS_{low} (1.21x and 1.33x higher impact compared to doS_{high} and doS_{medium}). Impact of parameters such as effective opening area for NNV operation and n50 increases 1.25x and 1.13x as *doS* increases. Importance of parameters such as shading operation, thermal mass, shading threshold shifts during higher *doS*. These parameters become less relevant in impacting thermal resilience. The occupancy was varied by two different occupancy profile-(a) where at least one occupant was present in the apartment throughout the whole day and (b) no occupants during day time in the weekdays. The occupant density was not varied. The result of the sensitivity analysis suggests that occupancy has a relatively low impact on SET-Dh. This is likely because internal heat gains from the 1 occupant present during the day time compared to no occupants is relatively small compared to other heat sources, such as solar gains or outdoor air temperature fluctuations. Additionally, in well-ventilated or thermally heavy buildings, the impact of occupant heat gains is further moderated. The envelope U-value, which represents the level of insulation, also appears to have a limited impact on SET-Dh in the sensitivity analysis. This can be explained by the fact that insulation primarily affects long-term heat transfer rather than immediate overheating dynamics. In well-insulated buildings, the rate of heat exchange with the environment is reduced, but this does not necessarily mean that overheating will be significantly influenced, especially if ventilation and internal heat gains are the primary drivers. The immediate overheating risk in a building is often driven more by factors like direct solar radiation through windows, ventilation rates, and internal heat gains. These factors directly influence how quickly a space heats up and how effectively excess heat can be removed. The results of the sensitivity analysis thus demonstrates WWR, solar shading and NNV operation, and cooling strategies play a more dominant role in determining indoor thermal conditions under extreme heat conditions.

Figure 4-14 (a) and (b) shows similar trend- that parameters such as WWR, cooling capacity, operation of NNV and solar shading have the highest impact on t_{abs} and t_{rec} during all three *doS* for convective cooling strategy. For t_{abs} WWR has the highest influence during doS_{high} (1.2x compared to doS_{medium}). However, WWR has higher influence during doS_{low} compared to doS_{medium} (1.05x). WWR dictates the solar gains

and indoor heat accumulation and positively correlates with t_{abs} meaning as WWR increases, t_{abs} increases. For t_{rec} WWR has the highest influence during DoS_{low} . For shorter duration shock, the impact of WWR and the associated solar gains have higher impact for absorption and recovery from the shock. Cooling capacity reduces t_{abs} and t_{rec}. For t_{abs}, during DoS_{high} influence of cooling capacity increases by 1.26x and 1.17x during DoS_{medium}, and DoS_{low}, i.e., cooling capacity becomes more influential on buildings thermal resilience performance during longer duration shocks. Same trend can be seen for t_{rec}. For recovery from the shock, cooling capacity is more influential during longer shocks. NNV operation effectively lowers t_{abs} and t_{rec} by removing heat at night. For t_{abs} NNV has impact growing as DoS severity decreases. Passive strategy such as NNV aids in flushing out the heat gained during day-time, reducing the number of hours occupants are in heat-stress. The influence of passive strategies diminish during prolonged shock. For t_{rec} NNV has the highest influence during DoS_{high}. Solar shading also reduces t_{abs} and t_{rec} by mitigating solar gains, with a 1.2x higher impact at DoS_{low} compared to DoS_{medium} and DoS_{high}. Other parameters have approximately 2x less impact on t_{abs} and t_{rec}. Increasing the cooling set-point delays cooling, leading to higher heat accumulation. Impact of parameters such as thermal mass on t_{rec} increases 1.25x as DoS increases. Heavier thermal mass improves buildings' thermal response during shorter shocks where heat gains via envelope are slowed down by heavy thermal mass. However for longer shocks as the heat enters the building, heavy thermal mass have tendency to hold on to the heat for longer period, prolonging recovery from the shock.

Figure 4-15 illustrates the impact of building and system parameters on SET-Dh, t_{abs} and t_{rec} during all three DoS for radiant cooling strategy for TZ1. Building and system parameters impact SET-Dh during all three DoS similar to convective cooling. The ranking of parameters impacting t_{abs} and t_{rec} during all three DoS is same for convective and radiant cooling. With radiant cooling, only t_{abs} shows increase in influence of operation of NNV and solar shading, with highest impact during DoS_{medium} .

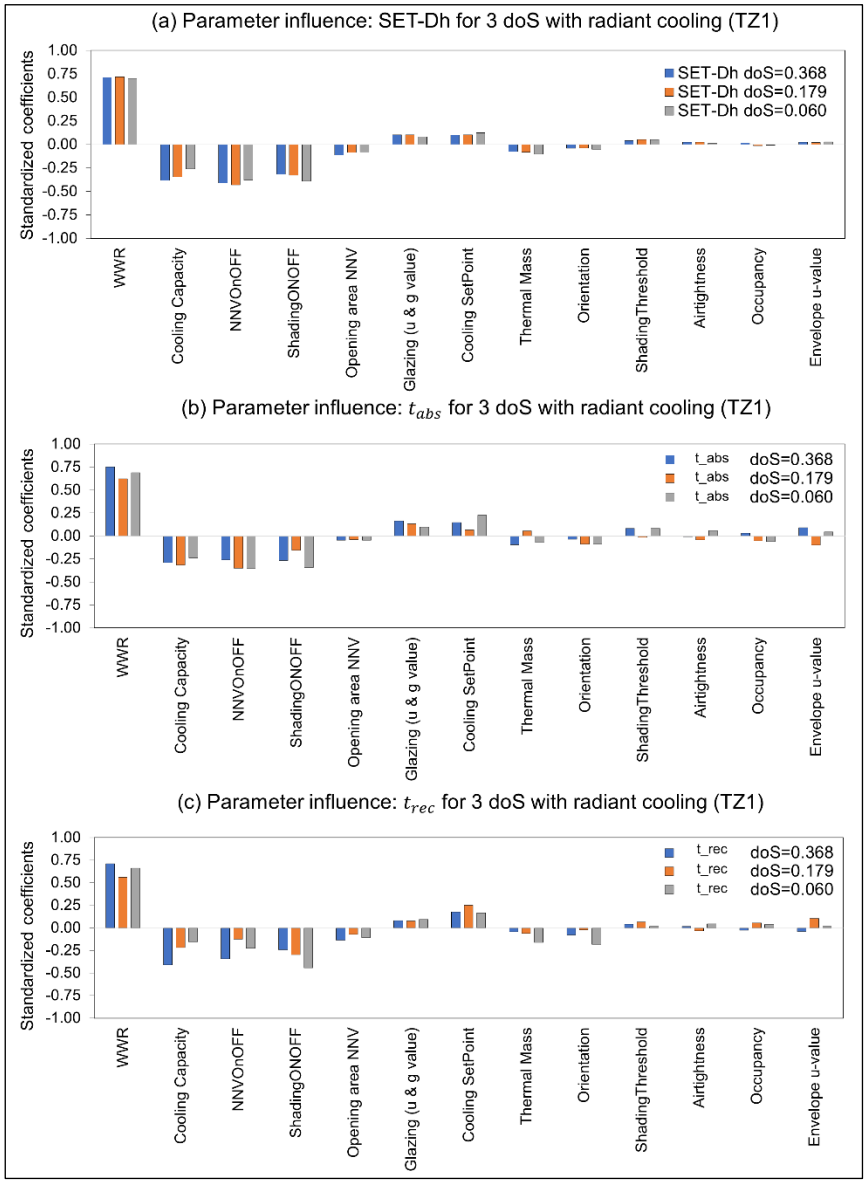


Figure 4-15. Impact of selected building and system parameters on SET=Dh, t_{abs} and t_{rec} for three increasing doS (TZ1 with radiant cooling)

4.4.2.2 Impact of parameters on the resilience performance for different KPIs

Figure 4-16 and Figure 4-17 illustrates design parameters' impact on the 3 aspects of resilience –(a)SET-Dh, (b)t_{abs} and, (c)t_{rec} with convective cooling system during all three *doS.*

The impact of design parameters on SET-Dh, t_{abs} and t_{rec} depends on the design parameters' ability for heat absorption, retention, and removal. Notably, for t_{abs} and t_{rec},

the impact of design parameters is significantly influenced by shock characteristics, such as duration, severity, and time to reach peak temperature.

During doS $_{high,}$ WWR as the highest impact on all three resilience aspects with a 1.08x and 1.19x greater influence on t_{abs} compared to SET-Dh and t_{rec}. This is due to WWR's effect on solar gains, which impact the heat absorption in the building. During doS_{medium} WWR has the highest impact on SET-Dh (on average 1.11x higher than t_{abs} and t_{rec}) whereas during doS_{low} WWR has the highest impact on t_{rec} (on average 1.01x higher than t_{abs} and SET-Dh. Apart from WWR, all three resilience aspects are impacted by cooling capacity and operation of NNV and shading in order of influence. Cooling capacity impacts t_{rec} 1.10x and 1.20x higher than SET-Dh and t_{abs} during doS_{high} and $doS_{medium.}$ During doS_{low} cooling capacity has the highest impact on t_{abs} as it determines how much cooling is provided to counter the sudden shock. Parameters that highly impacts t_{abs} on average more than SET-Dh and t_{rec} include glazing properties (2.4x), cooling set-points (1.9x), shading thresholds (2.1x), occupancy profiles (5x), and the envelope U-value (3x). High-performance glazing reduces solar heat gains, higher cooling set-points delay cooling deployment, higher shading thresholds delay shading activation, higher occupancy increases internal heat gains, and lower U-values indicate better insulation, reducing heat transfer. These parameters mainly focus on the absorption of solar and internal heat gains, thus impacting heat-built up duration in the indoor spaces. Parameters affecting t_{rec} on average more than SET-Dh and t_{abs} include cooling capacity (1.2x), building orientation (1.5x), and airtightness (2.1x). Higher cooling capacity allows for rapid heat removal, optimal building orientation mitigates solar heat gain and wind direction facilitates NNV operation, and improved airtightness reduces hot air infiltration, maintaining cooler temperatures and facilitating faster recovery through ventilation management to avoid prolonged heat retention in the building. These findings highlight the hierarchical significance of each parameter's impact on building thermal resilience, emphasizing the importance of strategic design choices during heatwaves.

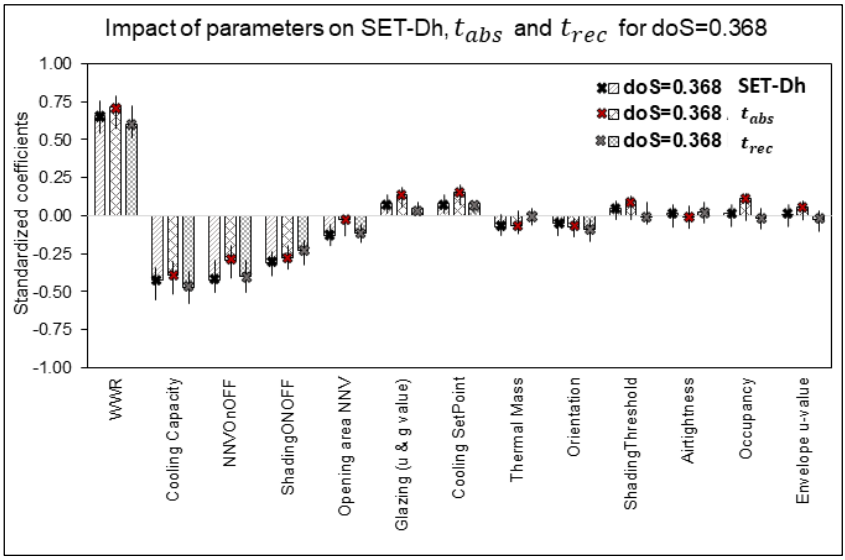


Figure 4-16. Impact of design parameters on SET-Dh, t_{abs} and t_{rec} during doS $_{high}$ for TZ1 with convective cooling

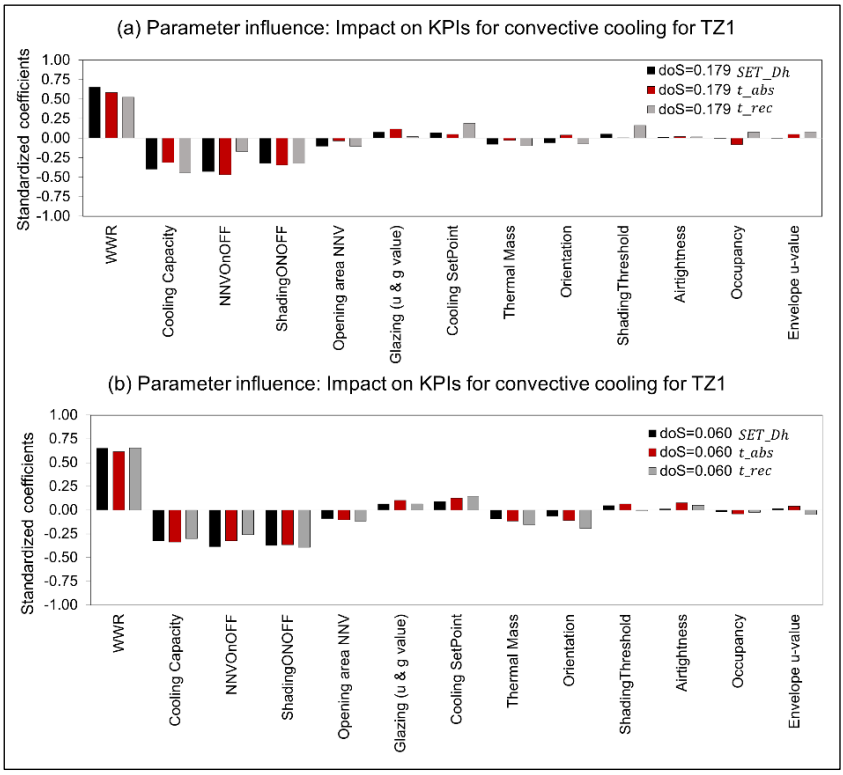


Figure 4-17. Impact of design parameters on SET-Dh, t_{abs} and t_{rec} during doS $_{medium}$ and doS $_{low}$ for TZ1 with convective cooling

Figure 4-18 illustrates the influence of building and system parameters on all three thermal resilience KPIs during three increasing *doS* for buildings with radiant cooling.

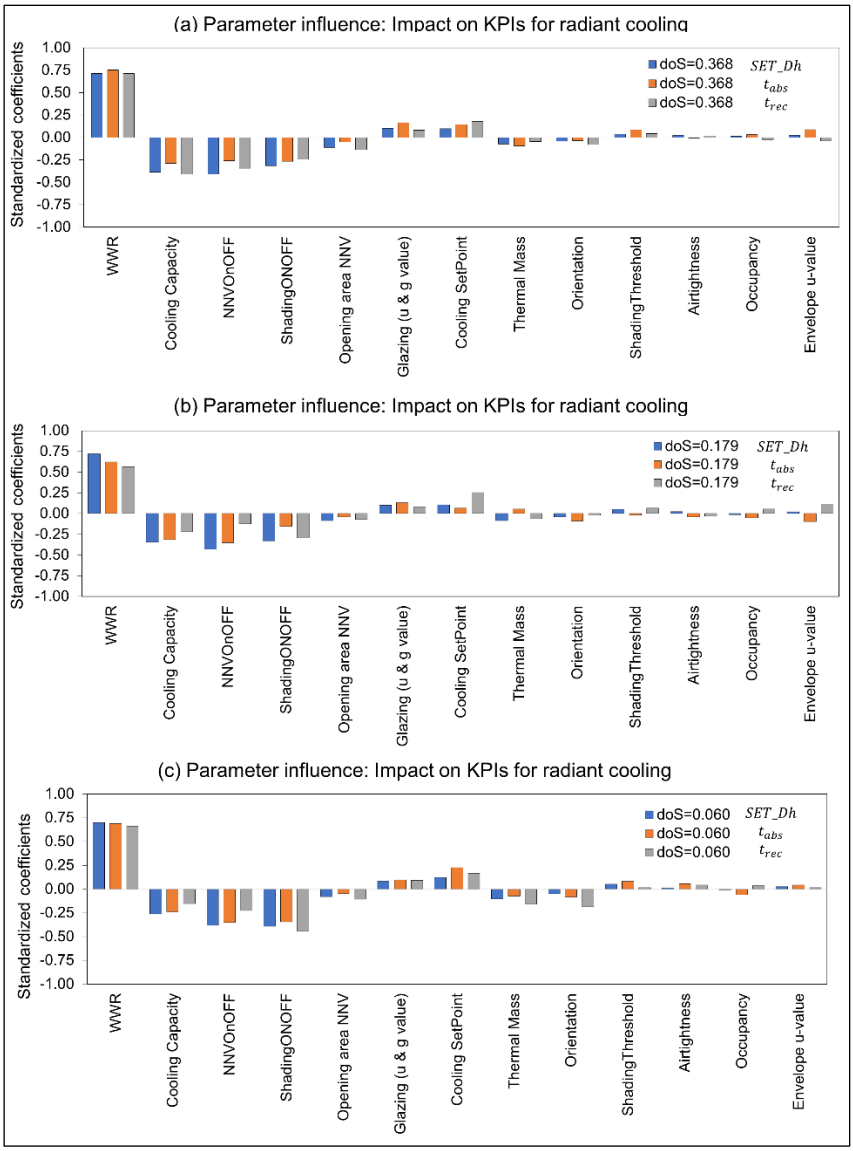


Figure 4-18. Impact of design parameters on SET-Dh, t_{abs} and t_{rec} during all three doS for TZ1 with radiant cooling

Similar to building with convective cooling, WWR has the highest impact on SET-Dh during doS_{low} and doS_{medium}. During doS_{high} the impact of parameters impacting SET-Dh, t_{abs} and t_{rec} are same for both convective and radiant systems. However, during doS_{medium} parameter such as cooling capacity, operation of NNV and solar shading has differences in the order of impact of the parameters. For example, cooling capacity has

highest influence on SET-Dh during doS_{medium} with radiant cooling whereas with convective cooling, cooling capacity has the highest impact on t_{rec}. This is due to the method of modelling these systems. In Modelica, radiant and convective cooling systems were modeled to analyze their responses to indoor condition changes. The convective system reacted quickly to fluctuations, while the radiant system accounted for building mass, making t_{rec} impacted by cooling capacity higher in convective cooling than radiant cooling. NNV operation impacted SET-Dh higher than t_{abs} and t_{rec} for radiant cooling during all three *doS*. Shading operation impacts SET-Dh highest with radiant cooling during doS_{high} and doS_{medium}. For doS_{low} shading operation impacts t_{rec}. With radiant cooling, parameters that have the highest impact during doS_{high} are operation of NNV, operation of solar shading, whereas t_{rec} is influenced by cooling capacity. Such case is also observed for convective cooling. During $doS_{medium,}$ parameters such as WWR, cooling capacity, operation of solar shading and NNV influences SET-Dh higher than t_{abs} and t_{rec}. Like convective cooling, for radiant cooling strategy, building and system parameters influence on SET-Dh, t_{abs} and t_{rec} depending on the parameters ability to absorb and flush out the heat during shocks. It also depends on the peak period of shock. For shorter shocks, parameters such as operation of passive strategies focus on mitigating heat absorption and removal whereas for longer shocks, cooling capacity gains higher impact on recovery from shock.

4.4.2.3 Impact of parameters on the resilience performance for different cooling strategies

Figure 4-19 and Figure 4-20 illustrates the impact of building and system design parameters on SET-Dh, t_{abs} and t_{rec} during periods of high demand (doS_{high}) with both convective and radiant cooling. The results show that the selected set of design parameters influence SET-Dh in similar pattern and magnitude for both cooling strategies. Furthermore, the correlation between design parameters and SET-Dh is consistent across both cooling methods, i.e., if an increase in a design parameter leads to an increase or decrease in SET-Dh for convective cooling, the same trend is observed for radiant cooling.

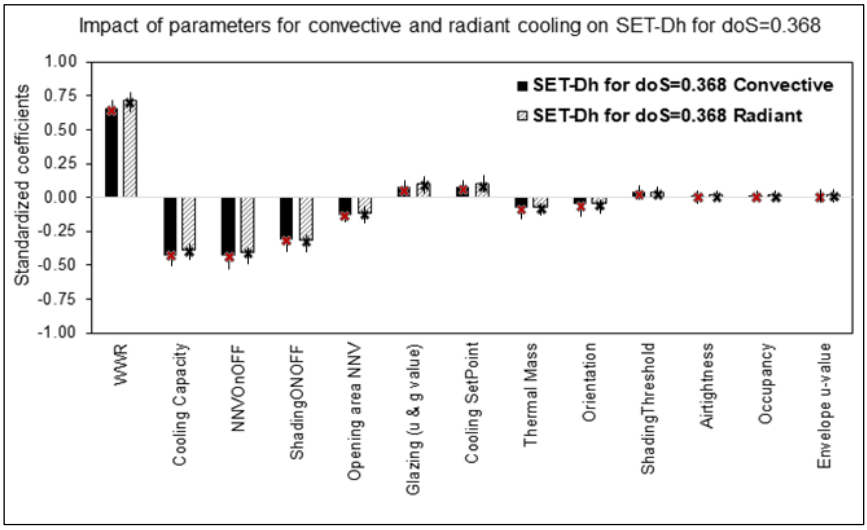


Figure 4-19. Impact of design parameters on SET-Dh during doS $_{high}$ for TZ1 with convective and radiant cooling.

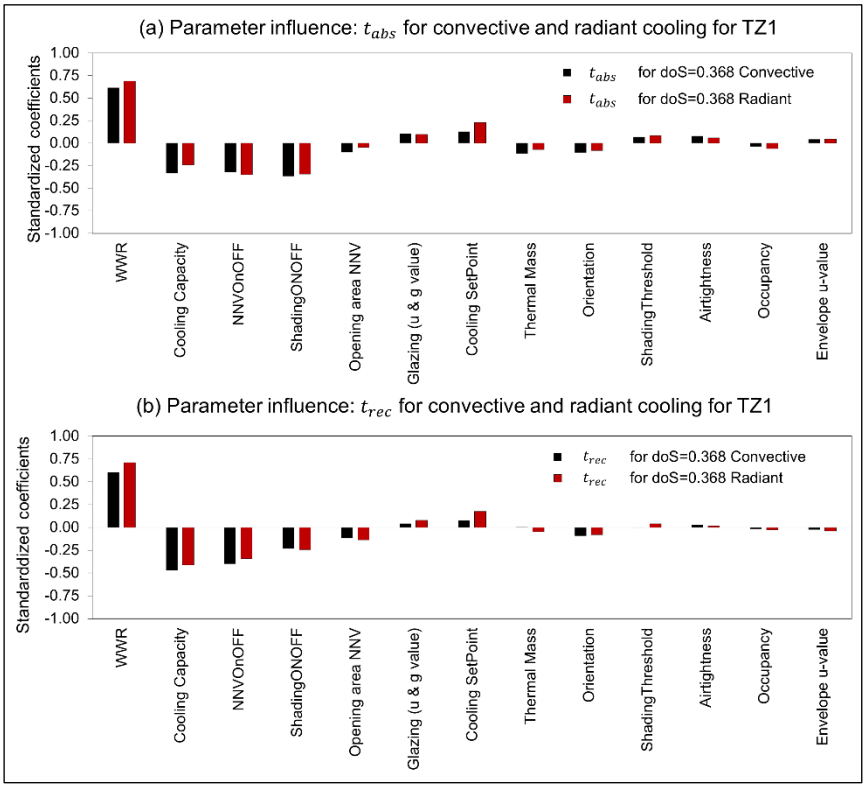


Figure 4-20. Impact of design parameters on t_{abs} and t_{rec} during doS $_{high}$ for TZ1 with convective and radiant cooling.

During *doS high*, for SET-Dh, convective cooling is impacted higher than radiant cooling by design parameters such as cooling capacity (1.2x), operation of NNV (1.1x), effective window opening area for NNV operation (1.3x), orientation (1.3x) and shading threshold (1.2x). Whereas parameters that impact radiant cooling higher than convective cooling are namely WWR (1.1x), shading operation (1.1x), Cooling set point and glazing U-value (1.3x), occupancy profile (1.2x), air tightness (1.5x), thermal mass and envelope U-value (1.2x). Convective cooling is more sensitive to building design parameters that influence air movement and temperature differentials within the space, whereas radiant cooling relies more on heat transfer mechanisms i.e., heat exchange coefficient and the temperature difference between surface and space. Previous studies have shown that the cooling capacity of a floor cooling system increases with increased direct solar gains on the floor surface [222]. Similarly another sensitivity study [221] identified the key parameters influencing the thermal performance of convective and radiant cooling system. It found that the air change rate, outdoor temperature, and air temperature stratification have the greatest impact on cooling requirements for maintaining a constant operative temperature while the differences between systems become noticeable for higher air change rates, however concluding that both systems achieve similar global comfort levels within recommended ranges. Thus, parameters such as WWR, operation of solar shading and control threshold, envelope and glazing properties and thermal mass have higher impact on radiant cooling compared to convective cooling. Similar trend is observed for both cooling strategies during *doS low* and *doS medium*. Figure 4-20 illustrates that during *doS high*, for t_{abs}, convective cooling is impacted higher than radiant cooling by design parameters such as cooling capacity(1.3x), operation of shading(1.2x), effective window opening area for NNV operation(1.3x), thermal mass(1.2x), orientation (1.1x) and airtightness(1.2x). Similarly, for t_{rec}, convective cooling is impacted higher than radiant cooling by design parameters such as cooling capacity (1.3x) and operation of NNV (1.3x). Parameters such as orientation, airtightness, envelope U-value has significant difference in impact between both the cooling strategies.

Figure 4-21and Figure 4-22 illustrates the impact of parameters on SET-Dh, t_{abs} and t_{rec} during *doS medium* and *doS low.* respectively. During *doS medium*, for SET-Dh, convective cooling is impacted higher than radiant cooling by design parameters such as cooling capacity(1.3x), operation of NNV and solar shading(1.1x) and effective window opening area for NNV operation (1.3x). Whereas for t_{abs} parameters that impact higher in buildings with convective system are operation of NNV and solar shading (1.3x). For t_{rec} the difference in impact on cooling capacity between convective and radiant cooling is 1.5x. Same for solar shading threshold. However, the results demonstrate that each resilience KPI is impacted similarly for both the cooling strategies with less than 10% difference in their order of impact.

While the results from the uncertainty analysis quantified the dynamic thermal resilience performance in todays' buildings due to design variations and different shocks, results from the sensitivity analysis identified the most influential design parameters that impact three thermal resilience aspects. The results showed that static design parameter WWR has the highest influence on all three resilience aspects.

Reducing WWR prevents potential heat gains, while dynamic strategies like operation and threshold of solar shading and cooling set-point address absorbed heat gains. Lower WWR, deployment of solar shading at lower threshold mitigates higher solar gains, thus impacting t_{abs}. Passive strategies such as NNV operation and adjusting size of opening for NNV operation and active cooling strategy impacts heat removal from the building. Increasing cooling capacity, operation of NNV and increase in openable window area impacts t_{rec}. These results are also supported by previous studies conducted on building and system design parameters affecting overheating risk in buildings. Those studies showed that limiting WWR mitigates overheating risks in buildings [196] and the use of passive strategies such as NNV and solar shading aids in improving thermal resilience of buildings during HWs [41]. While WWR and operation and control of shading, NNV and cooling strategies remain the most influential in controlling thermal resilience of buildings, other static building parameters such as airtightness, building orientation, envelope and glazing properties also influences thermal resilience. High-performance glazing consistently reduces solar gains across all *doS* levels. Heavy thermal mass buildings initially delay heat release into indoor but struggles during prolonged shocks, particularly as *doS* increases. This result also corresponds with recent studies on thermal resilience to overheating in buildings with heavier and lighter thermal mass [47]. South and southwest orientations increase solar gains, though this effect diminishes as *doS* rises. Airtightness increases SET-Dh, reducing thermal resilience, and higher occupancy and insulation levels further increase SET-Dh during higher *doS*. Airtight buildings are often associated with overheating risks [244]. Finally, comparing two cooling strategies, the order of impact of the parameters remains same for both cooling strategies. A recent study [251] compared resilience performance of radiant and air conditioners during heatwaves and power outages. This study demonstrated that during heatwaves, both systems maintained comfort, but heavyweight construction with radiant cooling offered better temperature control and faster recovery during power outages, indicating thermal greater resilience. The shift in the importance of these highly influential parameters during varying *doS* demonstrates the need to study different resilience aspects - t_{abs} and t_{rec} and during varying shocks to identify key building parameters to improve responsiveness of the buildings to shocks.

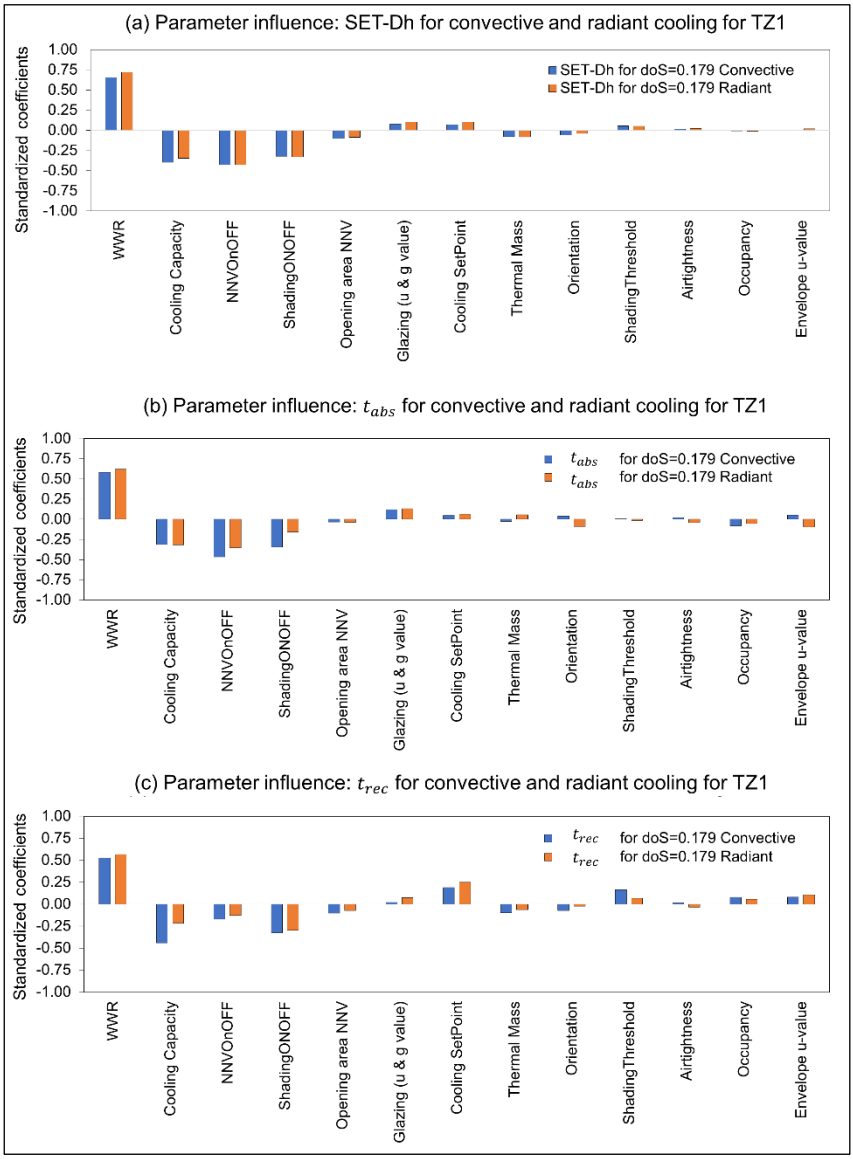


Figure 4-21. Impact of design parameters on SET-Dh, t_{abs} and t_{rec} during doS $_{medium}$ for TZ1 with convective and radiant cooling.

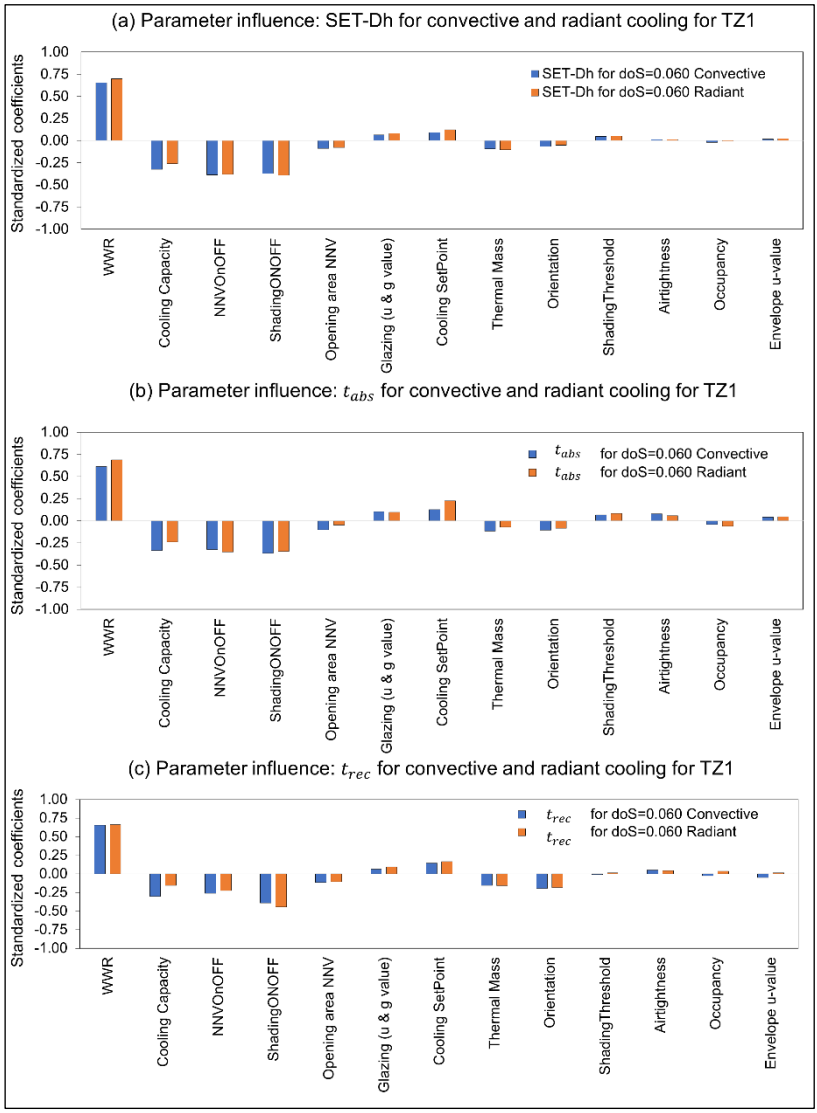


Figure 4-22. Impact of design parameters on SET-Dh, t_{abs} and t_{rec} during doS_{low} for TZ1 with convective and radiant cooling.

4.5 Conclusions

Given the lack of guidelines for targeting design parameters during early design phase to improve thermal resilience performance of buildings, this study employed uncertainty and sensitivity analysis method to determine the most influential design parameter that impacts buildings' thermal resilience performance during various types of heatwaves. The focus of the study was to determine the most influential design parameters that affect three thermal resilience aspects such as-the shock impact, time taken to absorb the shock and time taken to recover from the shock during three

increasing *doS* of heatwaves. This was applied for residential building equipped with convective and radiant cooling strategy. The main findings are as follows:

- Impact of peak period during *doS*: This study showed that the thermal resilience of buildings especially the heat absorption, and removal is not only affected by the severity and duration of the shock but also by the timing of peak outdoor temperatures during the shock period. This finding underscores the need for targeted strategies to enhance heat absorptivity and heat removal against different types of shock events.
- Uncertainty in thermal resilience: The study highlights substantial uncertainty in buildings' thermal resilience performance, primarily attributed to design variations and *doS*. The study pointed that even though current buildings are resilient to shorter shocks, the same design strategies have low thermal resilience during longer or more severe shocks. Buildings with suboptimal design significantly heightened impacts during severe shocks compared to those with optimized design. As *doS* escalates both in severity and duration, fewer buildings meet overheating criteria, signaling a diminishing thermal resilience trend. However, optimized design parameters (WWR, implementation of solar shading, NNV) mitigate this trend, resulting in markedly reduced absorption and recovery times during prolonged shocks. These findings underscore the critical importance of targeted design interventions to enhance building resilience against the mounting challenges posed by escalating climate change events.
- Identify most influential design parameters: The sensitivity analysis reveals that, out of the selected thirteen building and system design parameters, WWR dictating the solar gains and heat built-up has the highest impact during all three increasing doS. Parameters such as cooling capacity, operation of passive strategies such as NNV and solar shading dictates heat built-up, flushing out of the accumulated heat and temperature management strategy. Thus, during all three *doS* these four parameters have the biggest effect on all three aspects of thermal resilience performance of buildings with both cooling strategies. Parameters that influence the heat accumulation and absorption such as glazing properties, cooling set-points, shading thresholds, occupancy profiles, and envelope U-values affect $t_{abs.}$ Conversely, parameters that focus on heat removal such as cooling capacity, NNV operation and operable window area impacting ACH during NNV operation, thermal mass and airtightness significantly influence t_{rec}. The study also concludes that design parameters impact both cooling strategies in the same order of magnitude. Thus, cooling capacity of the system dictates thermal resilience performance more significantly than the type of cooling strategy used.
- Tradeoff between thermal resilience and energy efficiency: The results show that high cooling capacities and lower cooling setpoints prevent significant deviations in indoor conditions even during severe and long duration of shock. While these measures enhance thermal resilience performance, they increase building's energy use. Therefore, highlighting the necessity to study the trade-off between thermal resilience and energy efficiency.

- Dynamic strategies and refuge space for occupants to reduce heat stress: Even though occupant behaviour was not included as a sensitive parameter variable for this study, the results from this study can give an indication to the dynamic strategies (deployment of solar shading earlier than currently done, increasing openable area for ventilative cooling, etc.) that occupants can adapt and also indication of refuge space (rooms with lower WWR, on north-east direction, with potential of ventilative cooling during day and night time) to reduce heat stress.

Addressing the shortcomings of the current thermal resilience assessment methods, chapter 3 focused on quantifying shocks (with *doS*) that impact thermal resilience of buildings to overheating. Utilizing *doS* for most critical type of shock (heatwaves), chapter 4 focused on identifying the most influential design parameters in residential context that impact thermal resilience of buildings. Heat-stress indicator SET-Dh was utilized to assess thermal resilience. To develop and test a comprehensive thermal resilience assessment framework, it is crucial to explore design parameters' impact in diverse building types to evaluate if the currently identified design parameters have higher or lower impact on thermal resilience for other building types. To do this, parameters such as occupant density, internal gains, and ventilation rates should be evaluated as these parameters differentiate building types. In chapter 5, the methodology used in this chapter will be utilized and additional building type will be studied. Furthermore, building and system design parameters that show crucial impact in this study will be differentiated in a decision tree in Chapter 5 to systematically assess thermal resilience.

Chapter 5 : Determining and testing a comprehensive thermal resilience assessment framework

Chapter 5 focuses on the determining and testing of a comprehensive thermal resilience framework for assessing buildings' thermal resilience to overheating. Building on the findings of Chapter 4, which identified the most influential design parameters (related to both buildings and systems) impacting thermal resilience, these critical insights were leveraged to construct the framework (final step of the development of the assessment framework). The framework was designed using a decision tree approach, accommodating a range of *doS*, building typologies, construction types, and strategies. Its performance was evaluated through simulations of representative case study buildings, while its accuracy was rigorously validated using statistical methods.

+The simulations of the case study buildings described in section 5.3 and results shown in section5.6 are conducted by the project partners of ReCOver++: Improving thermal resilience of buildings to overheating [208]

5.1 Introduction

Chapter 5 introduces the development and evaluation of a comprehensive framework designed to assess and improve the thermal resilience of buildings to overheating. Building on the gaps identified in chapter 2, this framework integrates the impacts of thermal shocks, incorporates the heat-stress indicator SET-Dh to evaluate thermal resilience, and highlights critical design parameters influencing thermal resilience to overheating. Synthesizing insights from chapter 3, which quantified and compared the impacts of shocks with a focus on heatwaves, and chapter 4, which assessed SET-Dh and its key influencing parameters, the framework offers a robust and systematic approach to assessing and improving thermal resilience of buildings. This chapter primarily addresses RO4 to develop a comprehensive thermal resilience assessment framework to assess and improve buildings' thermal resilience to overheating

Section 5.2 outlines the methodology, including the formulation of the decision tree, selection and range definition of input parameters, definition of simulation scenarios, and the development and testing methods of regression models. Section 5.3 provides details of the three case study buildings used for validation, while Section 5.4 presents the thermal resilience assessments for these buildings. Section 5.5 discusses the development and testing of the regression models, culminating in the conclusions drawn in Section 5.6. Together, these sections establish a solid foundation for advancing the understanding and application of thermal resilience strategies in building design.

5.2 Method

To assess and enhance the thermal resilience of buildings, a step-by-step methodology was formulated to develop regression models. These models (TRI_{SET-Dh}) were specifically designed to predict the thermal resilience indicator, SET-Dh, offering a quantitative framework for evaluating how buildings respond to overheating. By systematically analyzing key factors influencing thermal resilience, the models provide valuable insights for improving building performance under thermal stress. To develop TRI_{SET-Dh} for aiding climate proofing of buildings during early design phase, a stepwise approach using building performance simulations, sensitivity analysis and development of multiple linear regression models was utilized. Figure 5-1 outlined the methodology for developing and testing TRI_{SET-Dh} to assess and improve buildings' resilience performance during heatwaves. First, a decision tree is formulated to assess thermal resilience for various *doS*, building typologies (residential and non-residential), construction type (thermal mass-heavy, medium and light) and different strategies. Evaluating the impact of selected building and system design parameters on SET-Dh by sensitivity analysis, regression models predicting thermal resilience indicator (TRI_{SET-Dh}) was developed for each branch of the decision tree. The developed TRI_{SET-Dh} is evaluated by comparing its predictions against simulations conducted on case study buildings, with its predictive accuracy rigorously tested by statistical method.

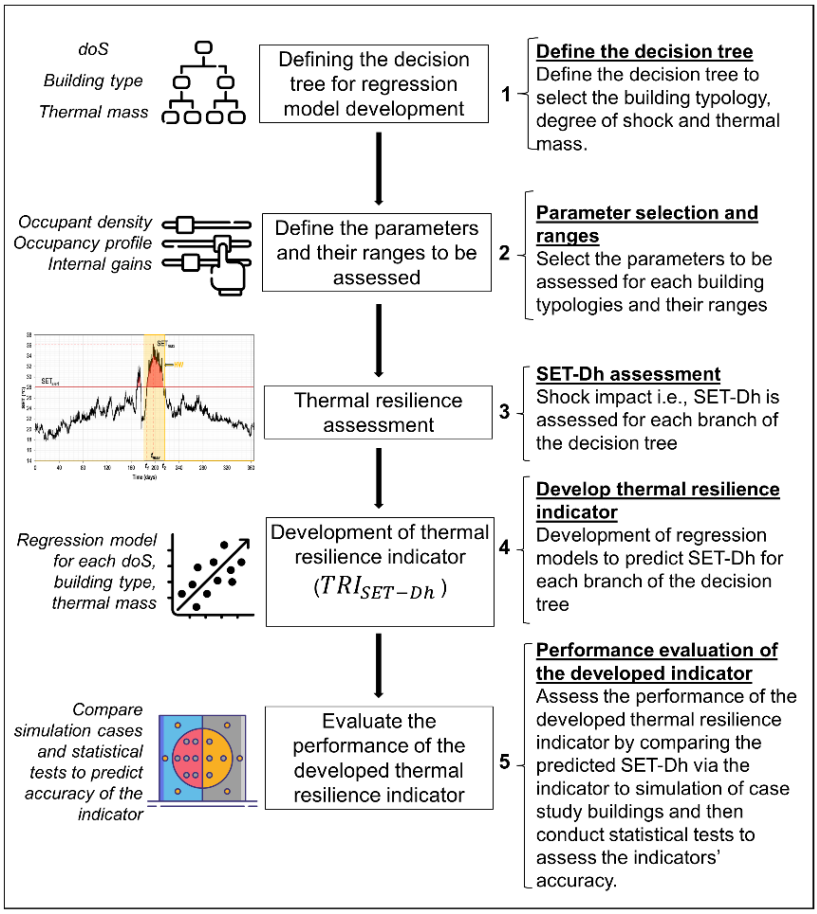


Figure 5-1. Methodology for the development and testing of TRI $_{SET-Dh}$

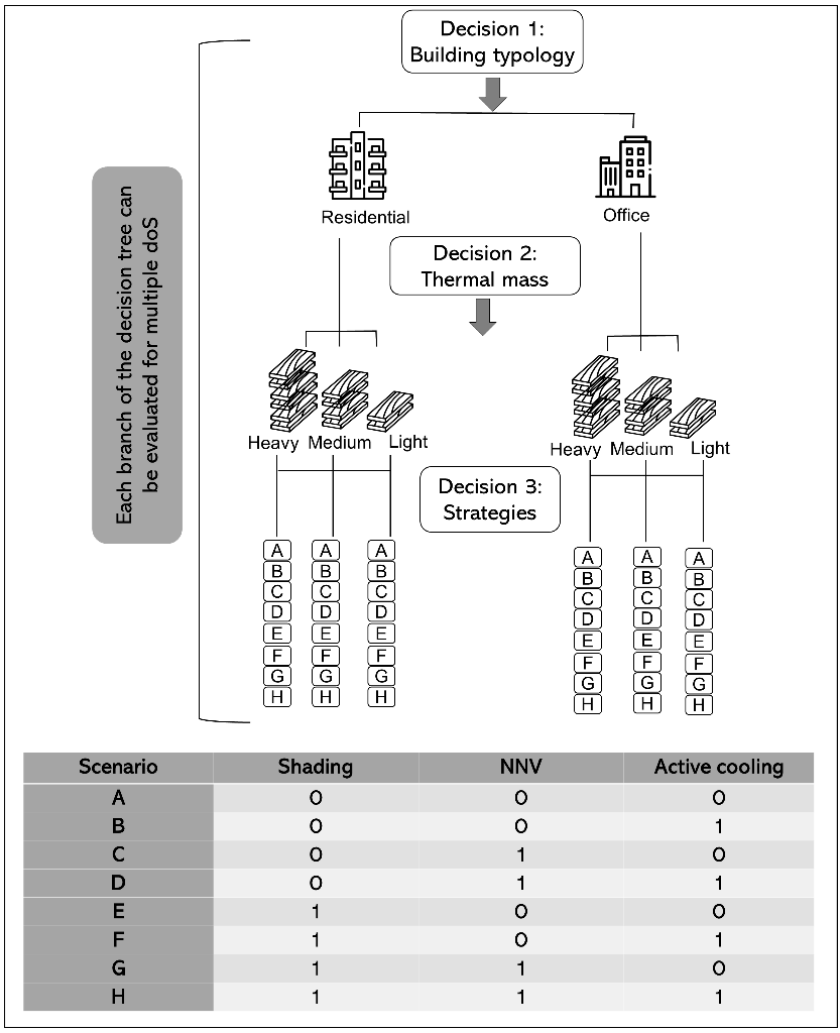
Figure 5-1 illustrates the steps taken in order to develop TRI_{SET-Dh}. They are listed as follows:

- In step 1, (as seen in Figure 5-1) a decision tree is generated for development of multiple TRI_{SET-Dh}, one for each branch of the decision tree The decision tree branched levels for different *doS*, building typologies, thermal mass, strategies and their combinations. Section 5.2.1 presents the development of the decision tree.
- In step 2, based on the identified impactful parameters that influence buildings' thermal resilience to overheating (chapter 4), building and system design parameters were iterated. The variation and range of these new parameters and their model inputs are defined in section 5.2.2.
- In step 3, samples for each branch of the decision tree (developed in step 1), are generated and building performance simulations are conducted for each sample to determine SET-Dh for each *doS*, building typology, thermal mass and strategies. Section 2.2.3 and 5.2.4 presents the simulation scenarios and simulation output.

- In step 4, linear regression models are developed (one for each branch of the decision tree). These regression models predicting TRI_{SET-Dh} describe the relationship between two or more explanatory variables (building and system design parameters) and a response variable (SET-Dh) by fitting a linear equation to observed data. Section 5.5 presents the regression models developed to predict TRI_{SET-Dh}.
- Finally, in step 5, the prediction accuracy of the regression models are evaluated by comparing the predicted SET-Dh from the developed TRI_{SET-Dh} to SET-Dh obtained from dynamic building performance simulations conducted on three different case study buildings by the project partners of ReCOver++ project [208]. Case study buildings and their thermal resilience assessment are presented in section 5.3 and 5.4 Testing the accuracy of the developed TRI_{SET-Dh} are presented in section 5.8.

5.2.1 Defining the decision tree for development of (TRI $_{SET-Dh}$)

A systematic development of a comprehensive thermal resilience assessment framework requires inputs of shock and design parameters (building and systems). In chapter 3, shocks were quantified and compared with *doS* and heatwaves were identified as the most critical shock. In chapter 4, the most influential design parameters were identified that impact thermal resilience of a reference two bedroom apartment during varying *doS* of heatwaves (input from chapter 3). However, in order to develop a comprehensive thermal resilience assessment framework it is crucial to extend the results of chapter 4 to include other building typologies. Residential (dwelling, apartments) and non-residential buildings (offices) differ from each other mainly due to difference in occupant density, occupancy pattern, internal gains and ventilation rates. To develop resilience assessment framework for different building types, it is crucial to systematically analyze and classify thermal resilience performance based on shock, on building type, thermal mass and strategies. As seen in Chapter 3 and 4, thermal mass influences thermal resilience (absorption of a shock and recovery from a shock) differently based on shock duration, leading to the decision to categorize buildings' thermal mass separately. A decision tree is developed to branch out systematically thermal resilience assessment based on levels of decision-(a) *doS*- which type of shock is the building tested against , (b) building type, (c) thermal mass and finally (d) strategies. Each branch represents a decision based on a specific attribute, leading to outcomes that show how different building types with different strategies might perform under different heatwave conditions. This approach allows for easy identification of patterns, enabling more informed decision-making in the design of buildings to improve thermal resilience to overheating. Figure 5-2 illustrates the decision tree for development of the various TRI_{SET-Dh}.



Scenario	Shading	NNV	Active cooling
A	0	0	0
B	0	0	1
C	0	1	0
D	0	1	1
E	1	0	0
F	1	0	1
G	1	1	0
H	1	1	1

Figure 5-2.Decision tree and the scenarios assessed to develop

As seen in Figure 5-2 , in the first level of the decision tree, building type is selected from three categories- (a)residential and (b)mid-size offices. These buildings differ from each other due to occupant density, occupancy patter, internal loads and ventilation rates. These crucial parameters are standardized from EN 16798-1[119].

In Chapter 4, even though the sensitivity analysis showed that WWR has the highest impact on SET-Dh,the choice to treat thermal mass as a separate category in the decision tree stems from the complexities thermal mass showed when tested for varying duration of shock. increase in thermal mass in buildings (from light to heavy), has variable effects under different shock duration. For shorter shocks, buildings with heavy thermal mass can be beneficial as it absorbs heat and delays the temperature increase inside the building. However, for longer shocks, buildings with heavy thermal mass, even though prolongs the absorption of heat, can retain heat for longer periods,

prolonging discomfort and delaying recovery. Thus thermal mass was treated separately to avoid oversimplifying its influence on thermal resilience to overheating against heatwaves. In level two, buildings' thermal mass is chosen, and categorized as (a) heavy, (b) medium, or (c) light. This categorization is based on EN ISO 13790 [72].
In the third level, three strategies and their combinations are selected to assess the resilience performance of the buildings: (a) solar shading, (b) NNV, and (c) cooling. To explore these strategies, a matrix of eight scenarios (labeled from A to H- see Figure 5-2) are developed. Using a (2^3) factorial matrix to analyze the combinations of three factors—(a) solar shading, (b) NNV operation, and (c) cooling—each factor has two levels: 1 (ON) and 0 (OFF). The matrix consisted of eight unique combinations (scenario A to H), with each scenario representing a specific configuration of these factors. For example, the combination (1, 0, 1) in scenario F indicated solar shading is ON, NNV is OFF, and cooling is ON. By systematically varying these factors across all possible combinations (e.g., from (0, 0, 0) to (1, 1, 1)), the matrix provided a comprehensive analysis of how each factor and their interactions influence buildings' thermal resilience. Finally, each branch of the decision tree is evaluated for two *doS* –(a) doS_{low}=0.060 and (b) doS_{medium} =0.179 to evaluate thermal resilience against a low and medium intensity heatwaves. The HW (ID = 3B, doS_{high}=0.368) with highest impact was not selected since it was predicted to occur in the long-term future scenario (2081-2100) when the current building stocks are assumed to have undergone improvement in construction techniques, energy efficient cooling strategies and other overheating mitigation strategies (see Table 3-3 for *doS* details).

5.2.2 Defining the parameters and ranges for development of TRI_{SET-Dh}

In order to vary the building type, the geometry of the reference apartment model was retained and parameters such as occupant density, occupancy pattern, internal gains and ventilation rates were altered according to EN 16798-1[119] to represent other building typologies. Design parameters such as building orientation, WWR, airtightness, envelope and glazing properties were tested for a wide range of variations that encompasses both building types-residential and non-residential, and thus the range of these parameters were unaltered. Previously, in section 4.3.3.3, simulation model of the reference apartment was developed in Modelica [218]. This reference apartment model in Modelica was altered (model inputs such as occupancy pattern, occupant density, internal gain and ventilation rate) retaining the geometry to develop simulation models for office (see Table 5-1). For example, TZ1 is assumed to be an open office in the office building which was the living room for the residential building. Likewise TZ2 and TZ3 were assigned different functions based on building type. TZ4 (the corridor and washroom) remains with the same function. shows thermal zones and inputs for each zone for both type of buildings (see Figure 4-5 for the TZs).

Table 5-2 defines all the input parameters and ranges for each building type. Three types of thermal mass -heavy, medium and lightweight were evaluated. For heavy thermal mass, buildings were defined as having heavy external walls, internal walls, and intermediate floors. For medium thermal mass, heavy external walls and intermediate

floors with light internal walls were considered, while for light thermal mass, heavy intermediate floors with light external and internal walls were used (defined in Table 4-2). Parameters such as occupant density, occupancy pattern and internal gains, were varied according to EN 16798-1[119] (see

Table 5-2). In accordance with EN 16798-1[119], the electrical equipment is assumed to have an electric power density of 3 W/m^2 and 12 W/m^2 for residential and offices respectively with a radiant fraction of 20%. The lighting schedules are also in line with EN 16798-1[119]. For offices, an illuminance level of 500 lux is required for working space, for the circulation hall this is 100 lux. For residential buildings, 300 lux of illuminance is required in each bedroom, 400 lux in the kitchen and living room, 70 lux in the corridor and 500 lux in the washroom and toilet was assumed [220]. Based on EN 16798-1[119], the ventilation rate is defined as :

$$q_{tot} = n * q_p + A_R * q_b \qquad \textit{(Equation 5-1)}$$

q_{tot} =total ventilation rate for the breathing zone [l/s]
n= design value for the number of persons in the room [person]
q_p= ventilation rate for occupancy per person [l/(s. person)]
A_R= floor area [m^2]
q_b= ventilation rate for emissions from building [l/(s. m^2)]

For category II air quality, ventilation rate between 30 to 54m^3/ hour per person was assumed for offices [252] and for the residential building ventilation rates were determined according to NBN D50-001 [213]. For the residential building the ventilation rates were not varied. Ideal heating system was assumed for space heating to simplify the technical building systems. Thermal environment is controlled based on the operative temperatures in the different zones. Heating is available between beginning of October and end of April while cooling and NNV between beginning of April and end of September. For offices, a heating setpoint of 21°C when occupied and 18°C during unoccupied period with a setback temperature of 16°C is used. For cooling, a convective system was modelled. Cooling set points and cooling capacity vary within the given ranges.

Table 5-1. Thermal zone and functions for different building types

	TZ1	TZ2	TZ3
Office	*Open office*	*Meeting room*	*Individual office*
Residential	*Living room*	*Bedroom 1*	*Bedroom 2*

Table 5-2. Parameters and their ranges for different building types

Parameters		Office	Residential
Parameters that vary for each building typology	*Occupant density [m^2/pers]*	*10*	*28.3*
	Occupancy profile	*9h-18h Weekdays*	*24*7 (at least 1 occupant at daytime)*
	Ventilation rate [m^3/h] per person (V)	*30-54*	*30*
	Internal gains-appliances [W/m^2]	*12*	*3*
Parameters that are varied in same range for all building typologies	*Orientation [°] (ϑ)*	*0-360°*	
	Window to wall ratio [%] (WWR)	*25-80%*	
	Airtightness n50 [1/h] (n50)	*0.6-3*	
	U-value external wall [W/m^2K] (U_{wall})	*0.10-0.30*	
	U-value glazing [W/m^2K] ($U_{glazing}$)	*0.6-1.0*	
	Shading threshold [W/m^2] (S_t)	*100-300*	
	Effective window opening (for NNV) as [%]of floor area (A_{window})	*1-8%*	
	Cooling setpoint [°C] ($T_{set-cool}$)	*24-28°C*	
	Cooling capacity [W/m^2] (Q_c)	*0-40*	

5.2.3 Building performance simulation (BPS)

Latin hypercube sampling (LHS) was conducted and 128 random samples were generated for each scenario (see section 4.3.4.3 for sampling method and size selection). BPS were conducted in Modelica. For each building type, the medium thermal mass (heavy external walls and intermediate floors with light internal walls) was selected due to its prevalence as the most commonly used construction technique [253]. Thus, for a single *doS*, for 1 type of building (example residential), 1 type of thermal mass (example medium), and for 8 scenario (see Figure 5-2) with 128 samples, a total of 1024 simulations (1 thermal mass x 8 scenario x 128 samples) were conducted. BPS was conducted for two types of buildings-(a) residential and (b) office buildings.

5.2.4 Simulation output-Thermal resilience assessment

In Section 4.3.2 three aspects of thermal resilience –(a) shock impact (SET-Dh), absorptivity time (t_{abs}) and recovery time (t_{rec}) were presented and sensitivity of design parameters on these three aspects were quantified in section 4.4.2. To evaluate the impact of thermal shocks and develop regression models (TRI_{SET-Dh}), SET-Dh was selected as the predictive output. Thresholds for SET-Dh have already been defined by previous research conducted on heat stress on occupants [88]. Low thermal resilience was considered if SET-Dh exceeds 230 [°C*h] for young, healthy adults, and 117 °C*h for older adults [36]. Such thresholds are yet to be defined for t_{abs} and t_{rec}. Without a clearly defined threshold, using such an indicator to assess thermal resilience is unreliable, as it lacks the necessary benchmarks to accurately evaluate the thermal resilience under varying conditions

5.2.5 Development of the linear regression models (TRI $_{SET-Dh}$)

Linear regression is a statistical method used to predict the value of a dependent variable based on one or more independent variables [254]. In building performance assessments, linear regression analysis is often utilized to investigate the effects of various design parameters such as building construction, weather, HVAC system, lighting system etc. on the energy performance of buildings [255–258]. It has also been utilized for developing simplified energy equations and design tools for building energy standards [259,260]. In this study, linear regression models (TRI_{SET-Dh}) were developed for each branch of the decision tree to predict the dependent variable (SET-Dh) based on several independent variables (design parameters such as WWR, building orientation, airtightness, envelope and glazing properties, resilient cooling strategies, and their control thresholds-see

Table 5-2). TRI_{SET-Dh} was developed and optimized by fitting a hyperplane to data points, representing the relationship between design parameters and SET-Dh. The objective was to find the best-fitting hyperplane that explained the relationship between SET-Dh and the building and system design parameters. Once TRI_{SET-Dh} generated predicted SET-Dh, these predictions were compared to the actual observed values (derived from simulations of the Modelica models), and the difference between them was referred to as the residual or error. To optimize TRI_{SET-Dh} the goal was to minimize the residual sum of squares (RSS) using the least squares method, ensuring

that the chosen hyperplane minimized the squared residuals/errors. This approach gave more weight to larger errors, improving the overall model's accuracy in predicting SET-Dh. By reducing the total error, the developed TRI_{SET-Dh} improved in capturing the underlying relationship in the data.

5.2.5.1 Regression models with and without interaction terms

Regression models were developed and tested in two stages- (a) multiple regression without interaction terms (see (Equation 5-2) and (b) multiple regression with interaction terms (see (Equation 5-3).

$$TRI_{SET-Dh} = \beta_0 + \beta_1 \times x_1 + \beta_2 \times x_2 + \cdots + \in$$

(Equation 5-2 without interaction terms)

$$TRI_{SET-Dh} = \beta_0 + \beta_1 \times x_1 + \beta_2 \times x_2 + \beta_3 \times (x_1 \times x_2) + \cdots + \in$$

(Equation 5-3 with interaction terms)

In both the regression models β_0 represents the intercept. The intercept β_0 is the predicted value of the TRI_{SET-Dh} when all the design parameter values are set to zero providing baseline estimate for TRI_{SET-Dh} when no other independent variables (like x_1, x_2 which represented building and system design parameters such as WWR, orientation, airtightness etc.) are affecting the outcome. The coefficients β_1, β_2, etc. in the regression equation provided quantitative insights into the relationship between TRI_{SET-Dh} and building and system design parameters. Each coefficient represented the change in TRI_{SET-Dh} for a one-unit change in the corresponding design parameter value, holding all other parameter values constant. $\in$ was the error term accounting for any variability not explained by the model. Positive coefficient indicated that TRI_{SET-Dh} increases as the parameter value increased and vice versa.

For the regression model with interaction terms (() coefficient β_3 represents the interaction between x_1 and x_2. β_3 quantifies how the effect of x_1 on TRI_{SET-Dh} depends on x_2 , and vice versa. A positive β_3 indicates that the effect of x_1 on TRI_{SET-Dh} increases as x_2 increases, while a negative $\beta_3$means the effect decreases. Theoretical insights and domain expertise were used to hypothesize potential interactions. For example the WWR and orientation were potential parameters that might have an interaction and impact the TRI_{SET-Dh}. Correlation analysis is conducted to determine relationships between variables that might have warranted interaction terms. Model performance is evaluated using fit statistics such as adjusted coefficient of determination R^2, aiding in eliminating non-essential interaction terms. Thus, the final model balanced interpretability, retaining only those interactions that contributes in improving the model's predictive power.

5.2.5.2 Evaluation methods for linear regression model performance

For both types of regression models, the coefficient of determination (R^2) measured how well the independent variables explained the variance in the dependent variable. R^2 ranged from 0 to 1 where a value of 1 indicated that the model perfectly explained the variability in the dependent variable, while a value of 0 meant the model explained none of the variability. R^2 does not account for model complexity or the possibility of

overfitting. A model with a high R^2 might still be overfitted, especially if the number of predictors is high relative to the sample size. It's important to consider other model validation metrics (such as the residual analysis, root mean square error (RMSE), mean percentage error (MPE) and F-test [261].

Residuals represent the differences between observed and predicted values and provide insights into the model's accuracy. For a reliable linear regression, residuals should have been normally distributed and displayed constant variance (homoscedasticity) across all levels of the independent variables. When dealing with multiple independent variables, multicollinearity could occur when two or more independent variables were highly correlated, potentially causing instability in coefficient estimation. RMSE measures the average magnitude of the residuals or prediction errors, providing a measure of how well the model's predictions match the actual data (see Equation 5-4.).

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - y_j\right)^2} \qquad \textit{(Equation 5-4)}$$

Where y_i is the actual value, y_j is the predicted value and n is the number of observations. Lower RMSE indicates a better fit of the model as the predictions are closer to the actual value and vice versa. However, RMSE should be considered relative to the range or scale of the dependent variable. For example, an RMSE of 5 might be excellent if the dependent variable ranges from 0 to 1000, but it would be poor if the variable ranges from 0 to 10. Thus it is crucial to check the mean percentage error. Mean percentage error (MPE) provides error estimation, with positive or negative values indicating whether the model under- or over-predicted SET-Dh (see Equation 5-5). MPE measures the average percentage error between predicted and actual values in a regression model. It expresses the error as a percentage.

$$MPE = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - y_j}{y_i}\right) \times 100 \qquad \textit{(Equation 5-5)}$$

In regression analysis, the F-test serves as a statistical tool to evaluate whether the overall model significantly accounts for the variability in the dependent variable, as opposed to a model with no predictors. This test assesses the null hypothesis- none of the independent variables exhibit a statistically significant relationship with the dependent variable. Result of the F-test with a low p-value (commonly less than 0.05), suggests that the model is statistically significant. In this context, statistical significance implies that at least one predictor variable meaningfully contributes to the explanation of variance in the outcome variable. The F-test fundamentally compares the proportion of variance explained by the model (due to the predictors) to the residual or unexplained variance. dependent value is not standardized, thus standardized threshold of p-value is evaluated to test the models.

$$F = \frac{Explained\ Variance\ (SSR)/(k-1)}{Unexplained\ variance\ (SSE)/(n-k)} \qquad \textit{(Equation 5-6)}$$

Where, SSR: Sum of Squares due to Regression (variance explained by the model), SSE: Sum of Squares for Error (variance unexplained by the model), k: Number of parameters (including the intercept),n: Number of observations.

Developing TRI_{SET-Dh} for each building type, thermal mass, and strategies (scenario A to H, see Figure 5-2) will aid in understanding how each building and system design parameter contributes to the prediction of SET-Dh. TRI_{SET-Dh} can be used to make predictions for new data within the parameter ranges that are used to develop these models. While the developed TRI_{SET-Dh} will be useful for prediction, it was important to validate these model with new or test data to ensure its predictive power beyond the original dataset used for model training.

5.2.5.3 Accuracy evaluation of TRI $_{SET-Dh}$

An F1 test is utilized to evaluate the overall performance of the developed regression models. The F1 score has been a crucial metric in the field of machine learning and statistics, especially when evaluating the performance of classification models [262]. It provides a balance between precision and recall, offering a single metric that accounts for both false positives and false negatives. It ranges from 0 to 1, where 1 indicates perfect precision and recall. There are 4 components of an F1-test-true positives (TP), true negatives (TN), False positive (FP) and False negatives (FN). A true positive had referred to the instances where the model correctly predicted the positive class. For example instances where the SET-Dh is higher than the threshold limit of 230[°C*h] and the developed TRI_{SET-Dh} could predict the same. A true negative had denoted the instances where the model correctly predicted the negative class. A false positive had occurred when the model incorrectly predicted the positive class. This meant that the model had identified an instance as higher than 230 [°C*h] when it was actually negative. And finally, a false negative had been the instances where the model failed to predict the positive class, incorrectly labeling it as negative.

$$F1\ score = 2x\frac{Precision\ X\ Recall}{Precision+Recall} \qquad \textit{(Equation 5-7)}$$

Precision measures the proportion of true positive predictions among all positive predictions made by the model. Whereas, Recall (or sensitivity) refers to the proportion of actual positive observations that are correctly identified. Specificity measures the proportion of true negative predictions among all actual negative instances. Accuracy measures the proportion of correct predictions (both true positives and true negatives) among all predictions.

$$Precision = \frac{TP}{TP+FP} \qquad \textit{(Equation 5-8)}$$

$$Recall = \frac{TP}{TP+FN} \qquad \textit{(Equation 5-9)}$$

$$Specificity = \frac{TN}{TN+FP} \qquad \textit{(Equation 5-10)}$$

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \qquad \textit{(Equation 5-11)}$$

5.3 Case study buildings for testing the performance of the developed TRI $_{SET-Dh}$

In order to evaluate the performance of each of the developed TRI_{SET-Dh}, the predicted SET-Dh were compared to simulation results of three different case study

buildings. Figure 5-3 and Table 5-3 gives an overview of the case study buildings simulated to evaluate the performance of the developed TRI_{SET-Dh}. Table 5-4 illustrates the parameter values and the simulation scenarios for each case study building.

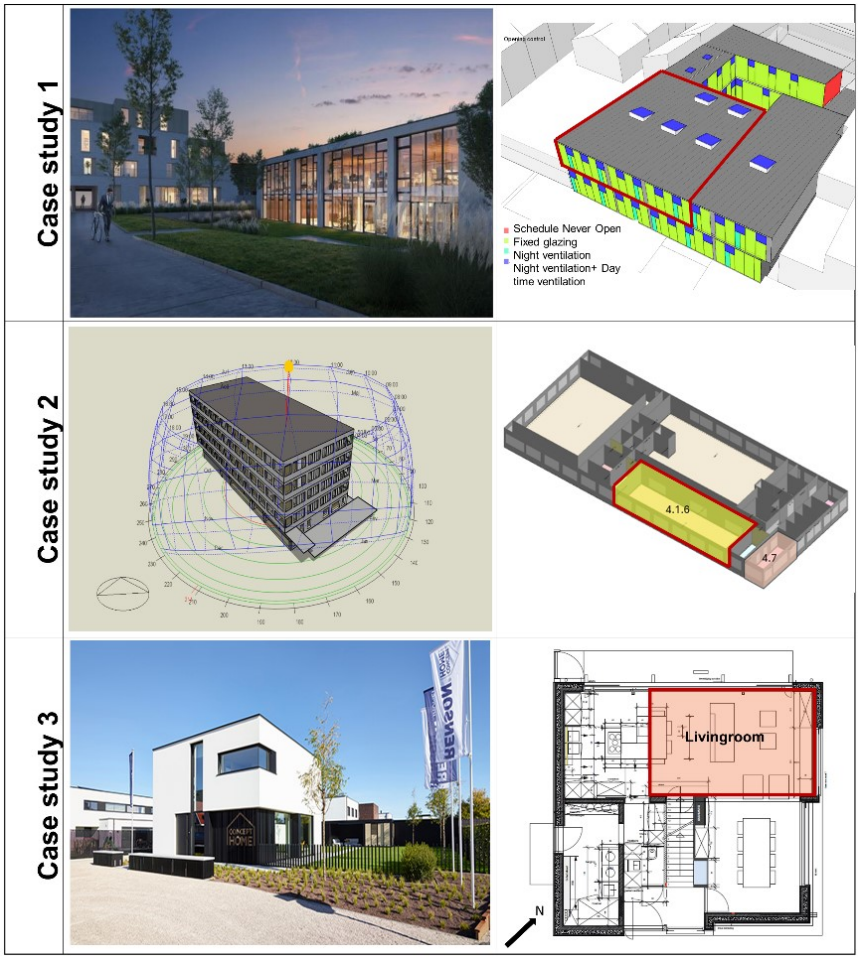


Figure 5-3. Three case study buildings and the critical zones (in red) [263,264]

5.3.1 Case study 1: Office designed by architectural firm Archipelago Architects, Belgium

Case study 1 is an office under construction in Leuven, Belgium. The building's design focuses on minimizing operational energy needs and technical installations by incorporating climate-responsive strategies such as automated natural ventilative cooling, exposed thermal mass, active solar shading, and a balanced glazing ratio. The office consists of two levels, connected internally by a central void. Both levels are arranged around a patio and linked to the adjacent park. The park-facing facade (northwest orientation) and the patio facades are the only glazed sections, while

skylights provide additional daylight and potential for stack ventilation. The design team extensively used thermal dynamic simulations during the design process to optimize the building for reducing overheating. The building, its design process, and its performance under future Typical Meteorological Year (TMY) conditions are described in detail in [263,264]. A simulation model of the case study building was developed in the Building Energy Simulation software IDA ICE v5.0 as a multi-zone model [265] . The building has an area of 940 m^2. Two open offices were evaluated—(a)+0 open office and (b)-1 open office.

The building is equipped with external solar shading on the roof and patio-side windows, which activate when solar radiation on the window surfaces exceeds 100 W/m^2. The heating setpoint is 20°C with a setback to 15°C. Mechanical ventilation is provided with a heat exchanger (efficiency = 83%) and summer bypass. Active cooling (ceiling panels) with cooling capacity of 30 W/m2 and ventilative cooling (both daytime and nighttime) are also evaluated. Ventilative cooling is activated during:

- Weekdays (8h-18h)
- Cooling season (21/04 to 21/09)
- When $T_{outdoor}$> 14°C
- When T_{indoor}> $T_{outdoor}$+1°C
- When T_{indoor}> 23°C
- It is deactivated when the T_{indoor}falls below 18°C (from mid-May to mid-September) or below 21°C (for the rest of the year).

NNV is activated under the following conditions:

- At night (22h-7h)
- During the cooling season (21/04 to 21/09)
- When $T_{outdoor}$> 8°C
- When T_{indoor}> $T_{outdoor}$ +1°C
- Cooling season: when T_{indoor}> 21°C
- Outside the cooling season: when T_{indoor}> 22°C

5.3.2 Case study 2: Office designed by consultancy firm Arcadis, Belgium

Case study 2 is an office and laboratory in Ghent, Flanders. The building has an area of 630 m^2. The building is designed to minimize energy use and increase thermal comfort of occupants. Two open offices on the top floor of the building-(a) office 4.1.6 and (b) office 4.7,were evaluated. The simulation model was developed using DesignBuilder based on the EnergyPlus simulation engine [266]. The building envelope adheres to passive house standards, compliant with EPB regulatory requirements. Solar shading with external screens was modelled for the southwest-facing windows in both rooms, activated when solar radiation exceeded 250 W/m^2 on the window surface. The shading assumes a solar reflectance of 50% and a transmittance of 10%. The double glazing used in the model has a Solar Heat Gain Coefficient (SHGC) of 49.4%. The mechanical ventilation system includes a pre-heating coil to warm incoming fresh air to 20°C. The thermal zones are equipped with convector heaters that maintain the winter setpoint temperature, powered by a heat pump that supplies both the heating coil and the

convectors. The heating setpoint is 21°C, with a setback temperature of 16°C. Various cooling strategies with different capacities were evaluated, including three modes:

- COOL: the fan coil was auto-sized to deliver unlimited cooling.
- RAD: the fan coil had limited cooling capacity.
- VC: mechanical free cooling via the AHU bypass.

Mechanical free cooling is activated when:

- $T_{outdoor}$>15°C
- T_{indoor}> 24°C
- T_{indoor} - $T_{outdoor}$ >2°C

The AHU is equipped with additional features to maintain relative humidity between 40% and 60%. Both heating and cooling are powered by a reversible heat pump, which can operate in both modes simultaneously. The Coefficient of Performance (COP) for heating and cooling is 3.3 and 3.2, respectively. The cooling setpoint is 26°C, with a setback to 32°C. NNV was modelled for pre-cooling. Night Natural ventilation NNV for pre-cooling of the office's rooms during night time. Two types of NNV are evaluated: (1) C- all windows to be opened (7.9% of floor area) (2) C2-a limited number of windows to be opened corresponding to 5.2% of floor area. NNV is activated from 8:00 p.m. to 6:00 a.m., if all the following conditions are met simultaneously:

- T_{indoor}> 24°C
- T_{indoor} - $T_{outdoor}$ >2°C

5.3.3 Case study 3: Single family dwelling, Renson Concept Home, Belgium

Case study 3 is a single-family dwelling located in Waregem, Flanders. The two storied, two-bedroom residential building was modelled in Modelica within the Dymola environment (version 2023x) [218]. The dwelling was modelled as eleven thermal zones and two most critical zone (living room and a bedroom) was evaluated. The building envelope adheres to the current EPBD regulations. Mechanical ventilation is provided to extract air from the kitchen and bathroom whereas the supply of fresh air is modeled through vents provided in living rooms and bedrooms. The building is equipped with a heat pump for heating and cooling. Floor heating is designed to heat the conditioned spaces. Heating set point is 20°C and cooling setpoint is 26°C. Automated external screens are provided on windows on the south and west facades. They are deployed when the radiation on the window surface exceeds 300 W/m^2.

Table 5-3.Details of the three case study buildings

	Office, Archipelago		Office, Arcadis		Renson Concept home	
Thermal zones	+0 open office	-1 open office	Office 4.1.6	Office 4.7.	Living room	Bedroom Garden side
Area [m^2]	225.1	315.3	187.1	38.7	20.2	22.4
Zone height [m]	2.93	3.3	4.09	4.09	2.8	2.8
Zone level	Top floor	Ground floor	Top floor	Top floor	Ground	Top floor
Windows/ glazing positions	2 Parallel surfaces	2 Parallel surfaces	1 Facade	2 Adjacent facade	1 Facade	1 Facade
Number of walls of the zone exposed to external environment	3 (adjacent)	2 (parallel)	1	2 (adjacent)	2 (adjacent)	2 (adjacent)
Occupant density [m^2/person]	9.8	10.2	10.4	10.4	22.9	31.9
Occupancy profile	Flexible hours 7-18 with lunch break, 0.5 in July, 0 during weekends & holidays		8h-18h weekdays No occupants on weekends and holidays		7h-22h All days	22h-7h All days
Internal gains [W/m^2]	Appliance= 12 Light=5		Appliance =12 Light= 7		Appliance=2.4	
Shading	Automatic External Screens (g tot=0.15)		Automatic External Screens (g tot=0.78)		Automatic External Screens (g tot=0. 56)	
NNV	Automated windows for NNV		Automated windows for NNV		No	
Type of cooling	Climate ceiling		Depending on the scenario tested		Air system	

Table 5-4. Parameter values for the different buildings used for the simulations

	Office, Archipelago Architects		**Office, Arcadis**		**Renson Concept home**	
Thermal zones	*+0 open office*	*-1 open office*	*Office 4.1.6*	*Office 4.7.*	*Living room*	*Bedroom*
Ventilation rate [m^3/h per person]	*42.14*	*54*	*40*	*40*	*Supply=25 Exhaust=75 m^3/h*	*Supply=30*
Thermal mass	*Medium*		*Medium*		*Medium*	
Orientation[°] (0°=South)	*165*	*165*	*181.8*	*45*	*219*	*129*
WWR [%]	*41.3*	*35.8*	*60*	*58*	*28.7*	*21.35*
Airtightness n50 [1/h]	*0.5*	*0.5*	*1*	*1*	*3*	*3*
Solar shading threshold [W/m^2]	*100*	*100*	*250*	*250*	*300*	*300*
Window opening area [%]of floor area)	*4.1*	*2.7*	*C=7.9 C2=5.3*	*C=19.3* C2=5.2*	*No*	*No*
Cooling set point [°C]	*25*	*25*	*26*	*26*	*26*	*26*
Cooling capacity [W/m^2]	*29.3*	*21.3*	*B_RAD= 38 B_VC=24 D_RAD= 33 D_VC= 26 F_COOL=37 F_VC=23 H_COOL=27*	*B_RAD= 38 B_VC=31 D_RAD= 33 D_VC= 31 F_COOL=42* F_VC=33 H_COOL=36*	*30*	*25*

			H_RAD=21 H_VC=27	H_RAD=21 H_VC=33		
U-value of external wall [W/m²K]	*0.16*	*0.16*	*0.14*	*0.13*	*0.15*	*0.15*
U-value of glazing [W/m²K]	*1.06*	*1.06*	*1*	*1*	*1*	*1*
doS simulated	*doS $_{low}$ =0.060* *doS $_{medium}$ =0.179+*		*doS $_{low}$ =0.060* *doS $_{medium}$ =0.179+*		*doS $_{low}$ =0.060*	
Simulation scenarios (see Figure 5-2)	*Scenario-A, B, F, G, H*		*Scenario-A to H*		*Scenario-A, B, F*	

**Parameter value out of range*

+Scenario A (for all three case study buildings) and H (for both offices) simulated during doS=0.179

5.4 Thermal resilience assessment of the case study buildings

Dynamic building performance simulations of three case study buildings were conducted by the project partners as part of the VLAIO flux50 ICON research project ReCOver++: Improving resilience of buildings to overheating. Building performance simulations followed the simulation framework developed within IEA EBCs' Annex 80-Resilient cooling of buildings [110]. The three case study buildings (described in Section 5.3) were simulated for two varying doS (see Table 3-3) -(a) $doS_{low} = 0.060$ and (b) $doS_{medium} = 0.179$, and for different simulation scenarios (depending on the resilient strategies used in the buildings, see Table 5-4). Yearly simulations were conducted and the heatwave period was analyzed to evaluate the hourly SET variations. Thermal resilience assessment is shown for the most critical thermal zones (marked in red in Figure 5-3) for each case study building.

5.4.1 Thermal resilience during $doS_{low}=0.060$

Figure 5-4 illustrates the hourly variation of SET in the +0 open office (Case Study 1,) during a 6 days-long heatwave (Table 3-3). Located on the top floor, the +0 open office was identified as the most critical zone due to its susceptibility to overheating from higher solar gains. The designers chose to simulate five different simulation scenarios—A(no strategy), B (cooling), F(shading +cooling), G (shading +NNV), and H (shading +NNV + cooling) (see Figure 5-2 for all the scenario details).

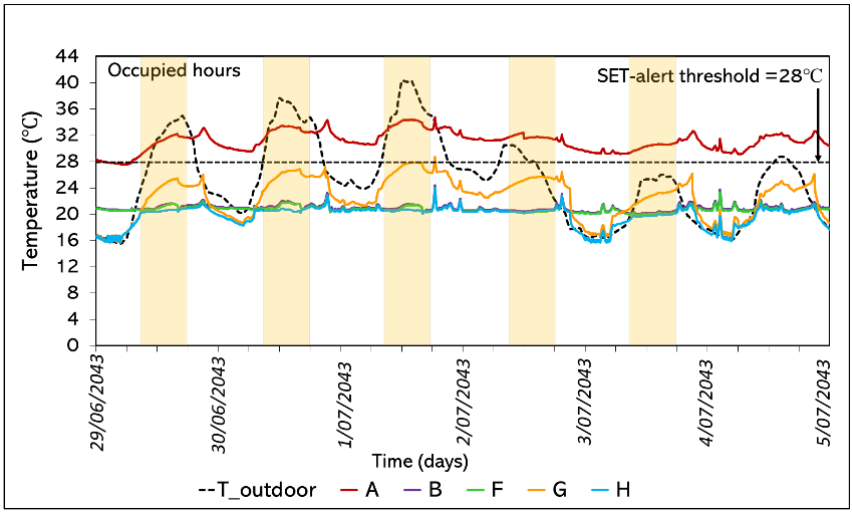


Figure 5-4. Hourly $T_{outdoor}$ and T SET variations for scenario A, B, F, G and H during doS=0.060 in office +0

Figure 5-4 shows that only in scenario A, which lacked any resilient strategies, the office space experiences SET values exceeding the SET_{alert} threshold of 28.1°C. In other simulation scenarios (scenario B,F,G and H) the building was designed to prioritize reducing operational energy consumption through climate-responsive features such as automated natural ventilation, thermal mass, active solar shading, and an optimized

WWR. These measures effectively kept the most critical zone below the SET_{alert} threshold even during this short-intense heatwave. In scenario A, the absence of any resilient cooling strategies led to low thermal resilience, with the SET_{alert} threshold being exceeded throughout the occupied period during the heatwave. In scenario B, where ceiling panel cooling with a capacity of 29.3 W/m^2 was implemented, and in scenario F, which included both ceiling panel cooling and solar shading deployed when solar radiation on the window surface exceeded 100 W/m^2, the office space maintained SET below 24°C during all occupied hours. In scenario G, which utilized passive strategies such as solar shading and NNV, SET increased by 4°C during the occupied period. On the hottest day of the heatwave, when the $T_{outdoor}$ reached 41°C, SET only exceeded the SET_{alert} threshold at 19:00 (outside occupied hours). Due to the passive strategies, the peak daytime temperature was moderated by reduced solar radiation absorption and heat flushing during the night. Even without active cooling, the office demonstrated good thermal resilience during the heatwave. In scenario H, which included cooling, solar shading, and NNV, the peak SET was 24°C on the hottest day. Overall, the building's optimized design, incorporating factors like WWR, thermal mass, a lower threshold for solar shading deployment, and NNV operation, exhibited thermal resilience during a short and intense heatwave.

To be noted that in scenario B, the indoor T_{set} is approximately 20°C during a heat wave, despite the cooling setpoint being 25°C, is due to the dynamics of thermal comfort and the SET. The cooling setpoint of 25°C is a target for the operative temperature, but it does not directly translate to the occupants' perceived thermal comfort or the indoor T_{set}. During a heat wave, the indoor environment might still experience significant radiant heat gain and other factors that can influence the perceived temperature. As SET accounts for not only air temperature but also radiant temperature, humidity, and air movement, it is possible for the indoor SET to be lower than the cooling setpoint due to these combined factors. In scenario G, the same can be explained by considering the thermal dynamics of NNV and the role of air velocity in influencing SET. During NNV operation, the outdoor temperature is typically lower than the indoor temperature, and when the building is ventilated, cooler outdoor air is introduced into the indoor space. However, the relationship between indoor temperature and outdoor temperature isn't always direct, as factors like air velocity, the thermal mass of the building, and the effectiveness of ventilation can lead to fluctuations in the indoor environment. If the air velocity is sufficiently high, it can enhance the cooling effect on the occupants, lowering the perceived SET below the outdoor air temperature. Additionally, the building's thermal mass can absorb and release heat in ways that temporarily cause the indoor temperature to remain lower than the outdoor temperature, particularly during nighttime cooling.

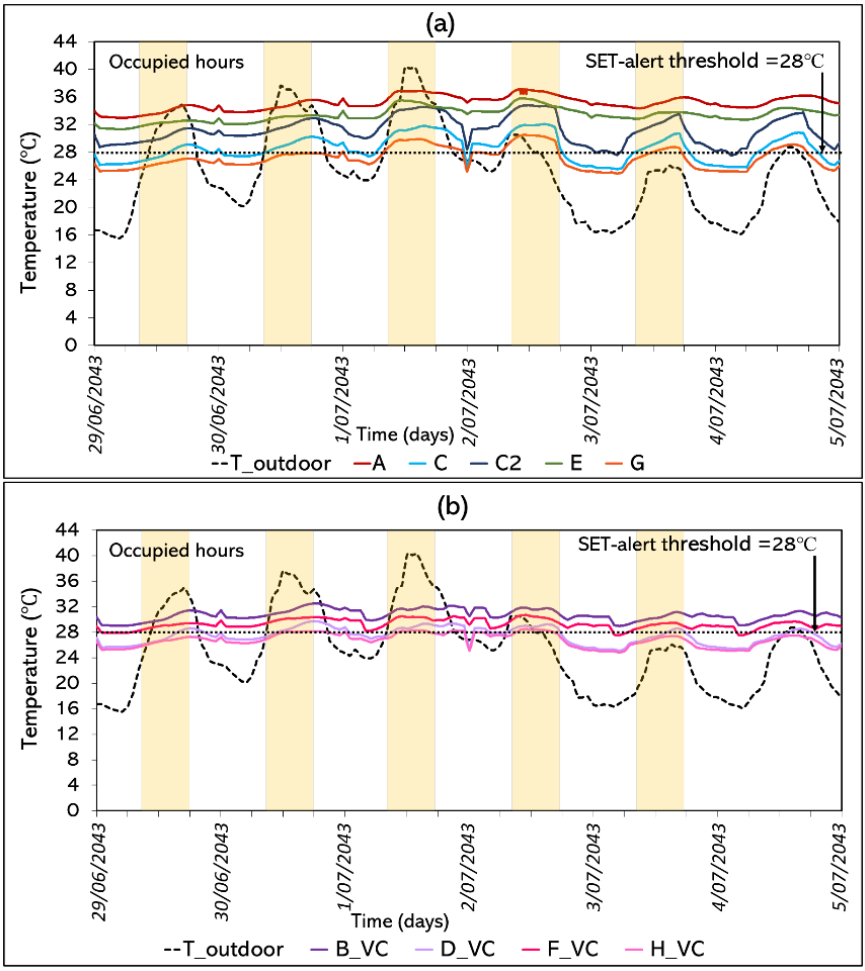


Figure 5-5. $T_{outdoor}$ and T SET variations for scenario A to H during doS=0.060 in office 4.1.6

Figure 5-5 (a) illustrates the hourly variation of SET in the 4.1.6 open office (Case Study 2,) during a 6 days-long heatwave (Table 3-3) with scenario A, C, C2, E and G. These scenarios represent passive measures and combinations of these measures. Figure 5-5 (b) illustrates the SET variation with cooling strategies and combinations (B, D,F and H). All the scenarios (A to H) were evaluated for the most critical thermal zone- office 4.1.6 (southwest on the top floor). In scenario A, which lacked any resilient cooling strategies, the office space violated the SET_{alert} threshold during occupied hours, with a peak SET of 36°C observed after the hottest day of the heatwave. The operation of NNV improved thermal resilience compared to scenario A by reducing the daytime peak SET through heat flushing at night. However, in scenario C2 (5.3% of floor area), where only a limited number of windows were operable compared to scenario C (7.9% of floor area), the cooling potential of NNV was reduced. The operation of solar shading alone

(scenario E) had less impact on thermal resilience improvement compared to NNV individually (C and C2). Solar shading was deployed at a higher threshold (250 W/m^2), and its effectiveness was limited when used individually without any combination of other strategies during the heatwave. When passive strategies were combined, as seen in scenario G, the SET_{alert} threshold was only violated after 48h since the start of the heatwave. A combination of resilient cooling strategies - solar shading+ NNV (scenario H) or cooling + solar shading (scenario F) significantly improved thermal resilience in the critical zone, and the SET_{alert} threshold was not violated. Conversely, in cases with only cooling (scenario B), the office showed low thermal resilience due to limited cooling capacity. High WWR, and insufficient solar shading led to high solar gains. Without combined strategies, the critical thermal zone violated the SET_{alert} threshold for the entire duration of the heatwave, leading to heat stress for occupants.

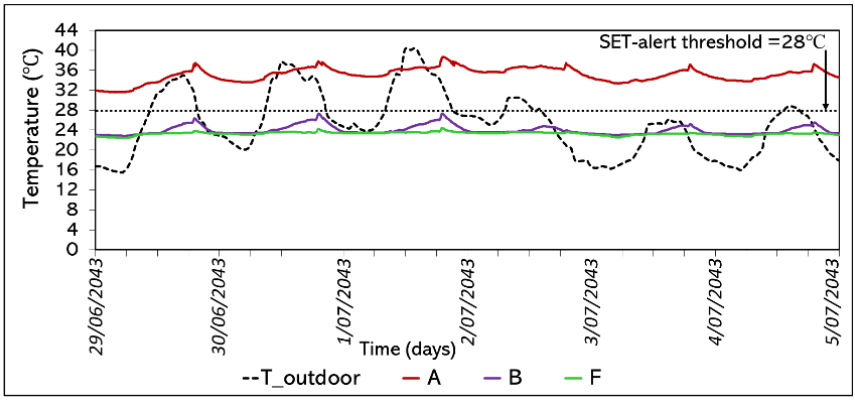


Figure 5-6. $T_{outdoor}$ and T SET variations for scenario A, b and F during doS=0.060 in the living room

Figure 5-6 illustrates the hourly SET variations in the living room of the single family dwelling (Case Study 3). The living room was identified as the most critical zone due to its occupancy during peak temperature periods of heatwaves and its higher glazing ratio. In scenario A, which lacked any resilient cooling strategies, the living room exhibited low thermal resilience, with the SET_{alert} threshold being violated throughout the heatwave. In scenario B, where only cooling was used, and in scenario F, which combined cooling with solar shading, the living room demonstrated good thermal resilience. The integration of solar shading with cooling reduced daytime peak SET by preventing solar gains through shading operation. In scenario B, a peak SET of 27°C was observed on the hottest day of the heatwave, indicating that the cooling capacity might have been insufficient to maintain the setpoint during the heatwave. However, the combination of solar shading and cooling in scenario F resulted in minimal variation in SET throughout the heatwave period, highlighting an effective strategy for maintaining thermal comfort.

5.4.2 Thermal resilience during $doS_{medium} = 0.179$

Figure 5-7 illustrates the hourly SET variations for the most critical zones in all three case study buildings during 14 days-long severe heatwave ($doS_{medium}=0.179$) for scenario A. Without any resilient measures, these buildings are unable to have good thermal resilient during a longer and severe shock.

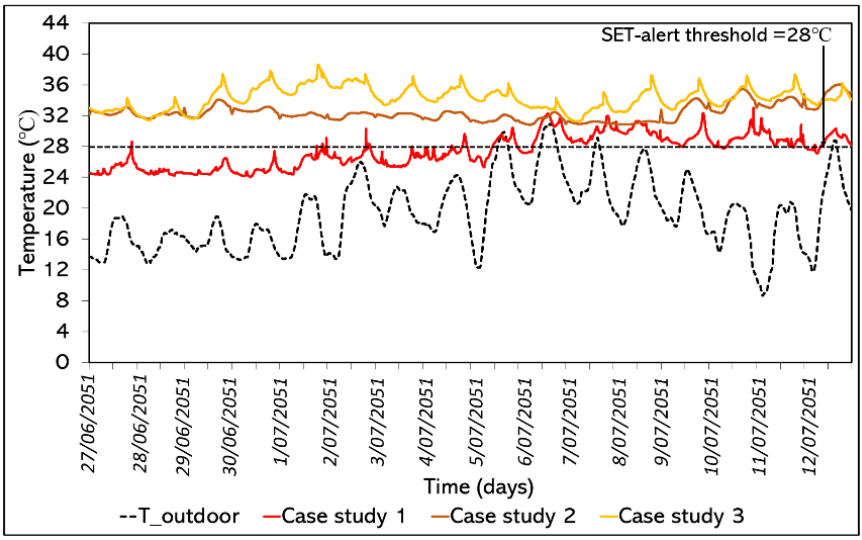


Figure 5-7. $T_{outdoor}$ and T SET variations for scenario A during doS=0.179 in critical zone of all the three case study (no resilient strategies)

When $T_{outdoor}$ reached the peak of 43.6 °C, the T_{indoor} for the case study 2 office and 3 had a SET peak of 37°C (higher that $SET_{emergency}$ threshold). However, case study 1 with this optimized WWR, thermal mass and favorable building orientation performed better than the other two case study buildings. During the long and severe heatwave, the case study 1 with optimized building design without any cooling strategies reached a peak SET of 33°C. During prolonged shocks, without any mechanism to flush out the excess heat from the buildings, there is significant heat accumulation within the buildings that leads to low thermal resilience. Thus during longer shocks, implementation of cooling measures or a combination of strategies such as solar shading and NNV are crucial to maintain habitable conditions for occupants in the building.

5.4.3 Insights from thermal resilience assessment of three case study buildings

The thermal resilience assessment of the three case study buildings offered insights into how different strategies influence thermal resilience during varying *doS*. These assessments demonstrated the importance of resilient cooling strategies, particularly in critical zones in the buildings (vulnerable to solar gains and internal loads). Strategies such as NNV, solar shading, and the use of thermal mass, were instrumental

in improving thermal resilience, particularly in buildings that were designed with these features in mind. These strategies rely on the building's ability to manage heat gains and losses offering a more energy-efficient approach. However, the effectiveness of these strategies vary depending on factors such as building orientation, WWR, and operation of solar shading. In Case Study 1, the combination of solar shading and NNV maintained comfortable indoor temperatures during occupied hours. Solar shading mitigated solar gains, while NNV removed heat during cooler nights, effectively limiting overheating. In Case Study 2, passive strategies reduced peak temperatures. The reduced number and operability of windows limited NNV potential, and due to high WWR, solar shading alone was not sufficient to mitigate impact of heatwave. This highlights the importance of proper calibration and deployment of passive strategies for optimal performance. In Case Study 3, cooling alone was moderately successful but struggled during peak heatwave conditions, especially when large WWR allowed for significant solar heat gains. While cooling systems ensures comfort during peak heat during heatwaves, it is energy intensive. In buildings with higher solar gains or insufficient shading, cooling alone struggles to maintain comfort during extreme heat. However, when combined with solar shading, cooling strategies performed better, with minimal temperature variation, demonstrating that a hybrid approach is more effective in improving thermal resilience. These case studies collectively highlight that the integration of resilient cooling strategies are critical for improving thermal resilience. During prolonged shocks, even buildings with optimized designs (WWR, orientation, thermal mass) are not able to maintain good thermal resilience without aid of a combination of resilient cooling strategies.

5.5 Determination of the thermal resilience indicator (TRI $_{SET\text{-}Dh}$)

To simplify thermal resilience assessment and make thermal resilience an actionable concept for architects and designers during early design stage, linear regression models (TRI_{SET-Dh}) were developed. Regression models shared all the following assumptions of correlation: linearity of relationships, the same level of relationship throughout the range of the independent variable, interval or near-interval data, absence of outliers, and data whose range is not truncated. Two types of regression models were developed and tested- (a) linear regression models without interaction terms and (b) linear regression models with interaction terms.

5.5.1 Linear regression models without interaction terms

First, linear regression models without interaction terms were developed (higher explain ability and simple to implement). One regression model for each branch of the decision tree (Figure 5-2) was developed. This branching was based on building type, thermal mass, scenarios (depending on resilient cooling strategy applied) and varying *doS*. Table 5-5 shows the regression models without interaction terms for each building type and scenarios for $doS_{low}=0.060$ and $doS_{medium}=0.179$. Models were evaluated by R^2, RMSE, MPE, and p-value as described in Section 5.2.5.2. The regression models presented in Table 5-5 can be applied to residential and office buildings as long as the parameter values fall within the range shown in the table below. These ranges are quite

wide and reflect current building design and practice, ensuring that the regression equations remain valid within this context. However, it is important to acknowledge that certain parameters, such as the floor-to-wall ratio, have not been explicitly evaluated, which presents a limitation of the current models. Further research is needed to incorporate additional parameters such as building volume, floor height, number of exposed walls, and the position of the thermal zone (top, middle, or ground floor). Including these factors in future studies will lead to more robust and comprehensive regression models that better capture variations in building performance.

Table 5-5.The developed multiple regression models without interaction terms for office and residential building during doS=0.060 for scenarios A to H

Office _ Medium thermal mass _ *doS*=0.060					
Scenario	TRI_{SET-Dh}	**R^2**	**RMSE**	**p value**	**MPE (%)**
A	$= 37.05 - 0.01 \times \theta + 3.66 \times WWR - 4.7 \times n50 - 75.79 \times U_{wall} + 36.12 \times U_{glazing} - 0.94 \times \dot{V}$ *(Equation 5-12)*	*0.80*	*37.2*	*0.05*	*4.28*
B	$= 34.55 - 0.02 \times \theta + 3.78 \times WWR - 5.49 \times n50 - 0.68 \times T_{set-cool} - 2.70 \times Q_c - 67.80 \times U_{wall} + 43.07 \times U_{glazing} - 0.48 \times \dot{V}$ *(Equation 5-13)*	*0.81*	*28.2*	*0.04*	*3.75*
C	$= 7.72 - 0.01 \times \theta + 2.65 \times WWR - 3.90 \times n50 - 28.69 \times A_{window} - 72.65 \times U_{wall} + 47.89 \times U_{glazing} - 0.06 \times \dot{V}$ *(Equation 5-14)*	*0.82*	*18.3*	*0.04*	*4.94*
D	$= -12.2 + 0.01 \times \theta + 2.52 \times WWR - 3.66 \times n50 - 5.02 \times A_{window} - 0.06 \times T_{set-cool} - 1.57 \times Q_c - 53.67 \times U_{wall} + 49.03 \times U_{glazing} + 0.16 \times \dot{V}$ *(Equation 5-15)*	*0.84*	*17.8*	*0.03*	*0.64*
E	$= -43.74 - 0.01 \times \theta + 2.82 \times WWR - 0.85 \times n50 + 0.25 \times S_t - 43.16 \times U_{wall} + 21.98 \times U_{glazing} - 0.66 \times \dot{V}$ *(Equation 5.-16)*	*0.82*	*27.0*	*0.02*	*3.81*
F	$= -90.9 + 0.01 \times \theta + 2.45 \times WWR - 1.10 \times n50 + 0.28 \times S_t + 1.34 \times T_{set-cool} - 2.51 \times Q_c - 18.89 \times U_{wall} + 25.63 \times U_{glazing} - 0.08 \times \dot{V}$ *(Equation 5-17)*	*0.86*	*17.9*	*0.02*	*0.2*
G	$= -39.97 + 0.04 \times \theta + 1.76 \times WWR - 0.39 \times n50 + 0.16 \times S_t - 5.76 \times A_{window} - 27.53 \times U_{wall} + 28.57 \times U_{glazing} \mp 0.56 \times \dot{V}$ *(Equation 5-18)*	*0.89*	*10.0*	*0.01*	*0.69*

H	$= -63.90 + 0.01 \times \theta + 1.38 \times WWR - 0.51 \times n50 + 0.15 \times S_t - 2.70 \times A_{window} + 0.81 \times T_{set-cool} - 1.37 \times Q_c - 8.71 \times U_{wall} + 24.62 \times U_{glazing} - 0.27 \times \dot{V}$ *(Equation 5-19)*	*0.88*	*10.22*	*0.01*	*1.42*
	Offices _ Medium thermal mass _ doS=0.179				
Scenario	$\mathbf{TRI_{SET-Dh}}$	**R^2**	**RMSE**	**p value**	**MPE (%)**
A	$= 565.17 - 0.03 \times \theta + 7.58 \times WWR - 10.66 \times n50 - 135.46 \times U_{wall} + 85.10 \times U_{glazing} - 1.88 \times \dot{V}$ *(Equation 5-20)*	*0.82*	*45.1*	*0.05*	*0.55*
H	$= -107.16 + 0.08 \times \theta + 8.70 \times WWR - 2.03 \times n50 + 0.77 \times S_t - 14.14 \times A_{window} + 3.67 \times T_{set-cool} - 10.81 \times Q_c - 14.55 \times U_{wall} + 132.42 \times U_{glazing} - 0.26 \times \dot{V}$ *(Equation 5-21)*	*0.88*	*41.1*	*0.01*	*1.86*
	Residential _ Medium thermal mass _ doS=0.060				
A	$= 89.52 - 0.04 \times \theta + 7.64 \times WWR - 8.45 \times n50 - 113.67 \times U_{wall} + 69.02 \times U_{glazing}$ *(Equation 5-22)*	*0.83*	*75.1*	*0.04*	*2.41*
B	$= -90.97 - 0.03 \times \theta + 8.36 \times WWR - 11.80 \times n50 + 4.41 \times T_{set-cool} - 9.52 \times Q_c - 52.38 \times U_{wall} + 108.14 \times U_{glazing}$ *(Equation 5-23)*	*0.88*	*58.1*	*0.03*	*0.01*
C	$= 21.85 - 0.01 \times \theta + 6.18 \times WWR - 6.58 \times n50 - 19.42 \times A_{window} - 85.99 \times U_{wall} + 99.30 \times U_{glazing}$ *(Equation 5-24)*	*0.85*	*37.7*	*0.03*	*1.66*
D	$= -12.44 - 0.02 \times \theta + 6.30 \times WWR - 7.39 \times n50 - 11.26 \times A_{window} - 0.27 \times T_{set-cool} - 4.57 \times Q_c - 61.15 \times U_{wall} + 107.12 \times U_{glazing}$ *(Equation 5-25)*	*0.84*	*37.6*	*0.04*	*1.49*
E	$= -86.00 - 0.03 \times \theta + 6.44 \times WWR - 1.88 \times n50 + 0.54 \times S_t - 75.30 \times U_{wall} + 44.56 \times U_{glazing}$ *(Equation 5-26)*	*0.83*	*57.1*	*0.03*	*2.23*

F	$= -316.37 - -0.03 \times \theta + 5.17 \times WWR - 6.45 \times n50 + 0.61 \times S_t + 9.23 \times T_{set-cool} - 8.87 \times Q_c - 8.21 \times U_{wall} + 60.74 \times U_{glazing}$ *(Equation 5-27)*	*0.88*	*46.8*	*0.04*	*0.25*
G	$= -104.91 - -0.01 \times \theta + 4.63 \times WWR - 0.19 \times n50 + 0.42 \times S_t - 14.48 \times A_{window} - 40.29 \times U_{wall} + 64.20 \times U_{glazing}$ *(Equation 5-28)*	*0.91*	*22.9*	*0.02*	*0.55*
H	$= -120.64 - -0.02 \times \theta + 3.61 \times WWR - 1.31 \times n50 + 00.41 \times S_t - 6.40 \times A_{window} + 1.48 \times T_{set-cool} - 4.25 \times Q_c - 12.29 \times U_{wall} + 54.21 \times U_{glazing}$ *(Equation 5-29)*	*0.90*	*25.7*	*0.03*	*1.54*
	Residential _ Medium thermal mass _ doS $_{medium}$ =0.179				
A	$= 784.89 - 0.08 \times \theta + 15.54 \times WWR - 20.42 \times n50 - 245.88 \times U_{wall} + 145.90 \times U_{glazing}$ *(Equation 5-30)*	*0.83*	*81.2*	*0.04*	*1.05*

Note: Orientation (θ) in [°], Ventilation rate ($\dot{V}$)in$[\frac{m3}{h} per\ person]$, Airtightness (n50) in [1/h], Window to wall ratio (WWR) in [%], U-value of external wall (U_{wall}) in [W/m²K] , U-value of glazing ($U_{glazing}$) in [W/m²K] , Effective opening area for NNV expressed as % of floor area (A_{window}) in [%], shading threshold (S_t) in [W/m²], Cooling setpoint ($T_{set-cool}$) in [°C] and Cooling capacity (Q_c) in [W/m²]

The models developed were assessed based on R^2, RMSE, MPE, and p-value. As seen in Table 5-5 , the R^2 of all the developed models for both office and residentials for all scenarios (A to H) were acceptable (>80%). An R^2 -value of 80% indicated that 80% of the variance in the dependent variable is explained by the independent variables included in the model, suggesting a strong relationship between the predictors and the outcome. This implied that the model fitted the data well and the predictors are effective in capturing the variation in the dependent variable. p-value lower than 0.05 showed that the regression models are statistically significant. Regression models developed for the offices have an average MPE of 2.26% while for the residential building it is 1.24%. As it can be seen, for each scenario (A to H), there are several variables (depending on the strategy of resilient cooling), which showed how SET-Dh increased or decreased. The negative coefficients specify an inversely proportional relationship between the SET-Dh and the design parameters. These models generate an analytical approach, equipped to capture the intricate interactions between building parameters and SET-Dh.

5.5.2 Linear regression models with interaction terms

To develop regression models with interaction terms, it is crucial to determine correlation between input parameters. Figure 5-8 illustrates the Pearson correlation matrix between the input parameters.

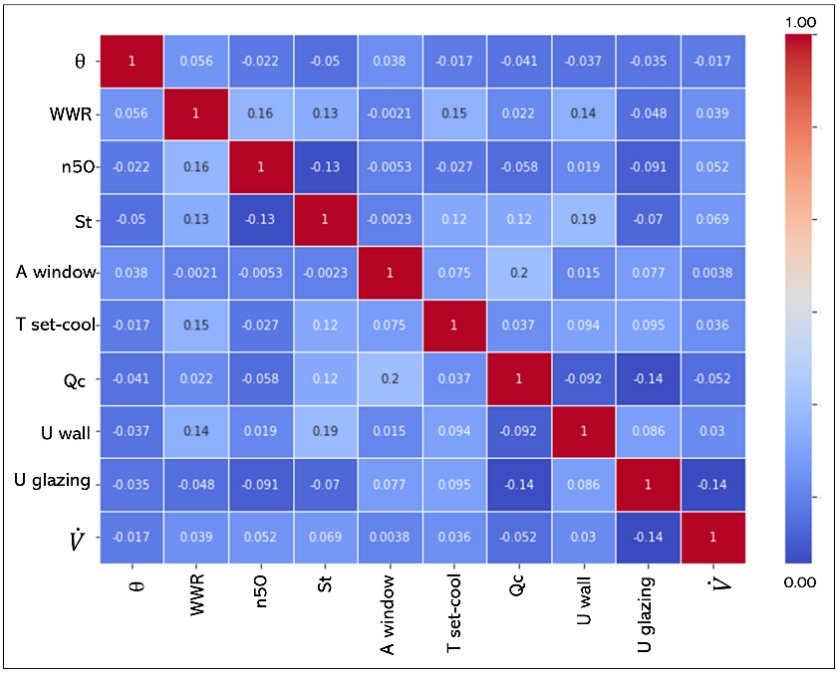


Figure 5-8. Pearson correlation matrix of the independent variables

To determine the interaction terms, a Pearson correlation coefficient analysis was first conducted. Pearson correlation is ideal for linear regression as it measures the strength

of linear relationships of the input parameters, aligning with the model's assumptions. It also helps detect multicollinearity, where highly correlated variables can destabilize regression coefficients, ensuring a more robust and interpretable model. The Pearson correlation matrix provides an easy way to compare correlations and identify clusters of variables that co-varied. The value of the correlation coefficient ranged from -1 to 1. A correlation coefficient of 0.3 or higher is considered indicative of a moderate or stronger correlation. A value higher than 0.5 indicated strong correlation [267]. Figure 5-8 shows the correlation between the independent variables. The correlation matrix results indicates that, there is no moderate or strong correlation among the parameters under study (<0.3). To evaluate if the regression models with interaction terms led to higher precision in predicting SET-Dh, regression models were developed with interaction terms.

To evaluate regression equations with interaction terms, domain knowledge and backward elimination methods were employed. Domain knowledge was utilized to identify parameters that could plausibly interact and influence the dependent variable, while eliminating interactions that were not theoretically justified. Backward elimination involved initially including all potential interaction terms and subsequently removing pairs based on model performance metrics. For scenario A, the interaction between parameters such WWR and orientation was analyzed. This interaction referred to how the window area and the building's directional exposure together influenced Thermal resilience. The impact of WWR varied depending on the orientation, as different orientations received varying levels of sunlight, affecting thermal resilience during a heatwave period. In scenarios with cooling, interactions between cooling setpoint and cooling capacity were evaluated. The interaction between cooling setpoint and cooling capacity referred to how the chosen temperature for cooling system to operate and the system's ability to remove heat together influenced the building's thermal resilience. A higher cooling setpoint required a higher cooling capacity to maintain the temperature, affecting thermal resilience. For scenarios involving solar shading, the interaction between WWR and solar shading threshold was assessed as this interaction describes how shading activation and window area together influence solar heat gain. A higher WWR increased solar exposure, while the shading threshold determined when shading were deployed preventing excess heat gains. Similarly, for scenarios with NNV, the interactions between window orientation and openable area were examined. The interaction between NNV and orientation referred to how building orientation and window placement influenced airflow and ventilation efficiency. Wind direction played a key role, as windows aligned with prevailing winds enhanced ventilation, while misalignment reduced airflow, affecting cooling effectiveness. Linear regression models with interaction terms were developed for $doS_{low} = 0.060$ for office buildings for scenario A, B, F, G and H. Table 5-6 shows the developed models with the interaction terms.

Table 5-6. The developed multiple regression models with interaction terms for offices during doS=0.060 for scenarios A,B, F, G and H

Offices_ Medium thermal mass _doS=0.060					
Scenario	TRI_{SET-Dh}	R^2	*RMSE*	*p value*	*MPE (%)*
A	$= 34.50 - 0.17 \times \theta + 3.78 \times WWR - 0.004 \times \theta \times WWR - 5.9 \times n50 - 99.02 + 0.87 \times WWR \times U_{glazing} - 99.02 \times U_{wall} - 2.8 \times U_{glazing} - 0.84 \times \dot{V}$ *(Equation 5-31)*	*0.78*	*38.1*	*0.05*	*4.82*
B	$= 95.33 - 0.02 \times \theta + 3.79 \times WWR - 5.39 \times n50 - 3.06 \times T_{set-cool} - 6.03 \times Q_c + 0.13 \times T_{set-cool} \times Q_c - 67.80 \times U_{wall} + 42.51 \times U_{glazing} - 0.45 \times \dot{V}$ *(Equation 5-32)*	*0.80*	*29.4*	*0.05*	*3.90*
F	$= 20.16 + 0.03 \times \theta + 1.35 \times WWR - 1.57 \times n50 - 0.03 \times S_t - 1.03 \times T_{set-cool} - 5.82 \times Q_c + 0.006 \times WWR \times S_t + 0.12 \times Q_c \times T_{set-cool} - 0.0009 \times \theta \times WWR - 19.41 \times U_{wall} + 27.32 \times U_{glazing} - 0.03 \times \dot{V}$ *(Equation 5-33)*	*0.79*	*18.8*	*0.05*	*0.81*
G	$= -70.34 + 0.02 \times \theta + 2.49 \times WWR - 0.29 \times n50 + 0.15 \times S_t - 1.24 \times A_{window} - 0.00005 \times \theta \times WWR - 0.14 \times WWR \times A_{window} - 22.28 \times U_{wall} + 27.15 \times U_{glazing} - 0.04 \times \dot{V}$ *(Equation 5-34)*	*0.86*	*11.2*	*0.03*	*0.74*
H	$= 2.47 - 0.004 \times \theta + 0.62 \times -0.75 \times n50 - 0.02 \times S_t - 2.59 \times A_{window} - 0.42 \times T_{set-cool} - 3.00 \times Q_c + 0.06 \times Q_c \times T_{set-cool} - 6.86 \times U_{wall} + 25.31 \times U_{glazing} + 0.29 \times \dot{V} - 0.00000004 \times \theta \times WWR + 0.003 \times S_t \times WWR$ *(Equation 5-35)*	*0.84*	*11.1*	*0.02*	*1.86*

Note: Orientation (θ) in [°], Ventilation rate ($\dot{V}$) in $[\frac{m3}{h} per\ person]$, Airtightness (n50) in [1/h], Window to wall ratio (WWR) in [%], U-value of external wall (U_{wall}) in [W/m²K] , U-value of glazing ($U_{glazing}$) in [W/m²K] , Effective opening area for NNV expressed as % of floor area (A_{window}) in [%], shading threshold (S_t) in [W/m²], Cooling setpoint ($T_{set-cool}$) in [°C] and Cooling capacity (Q_c) in [W/m²]

As seen in Table 5-6 the regression models with interaction terms on average have 2% decrease in the R^2 value compared to the regression models without the interaction terms. Additionally the RMSE and the MPE increases in the models with the interaction terms. The interaction terms have significantly small coefficients (<0.1) which do not lead to any significant improvement in the regression models' prediction accuracy yet decreases the explain ability of the models.

5.6 Evaluation of the TRI $_{SET-Dh}$ without interaction terms

5.6.1 Evaluation of the TRI $_{SET-Dh}$ with simulation results for *doS $_{low}$*

Regression models (TRI_{SET-Dh}) were developed for each scenarios (A to H) for the offices and for the residential building with medium thermal mass (see Table 5-5). SET-Dh predicted from these regression models is compared to the SET-Dh obtained from the simulations of the different cases described in Table 5-3 and in Section 5.3. Thermal resilience performance was categorized into two levels-(a) resilient, if SET-Dh <230 [°.h] (b) non resilient if SET-Dh >230[°.h]. These levels indicates whether the building and system can effectively manage and mitigate heat accumulation, with values exceeding the limit of 230[°C*h] suggesting insufficient thermal resilience.

5.6.1.1 Case study 1-Office

Figure 5-9 illustrates the comparison between the predicted and the simulated SET-Dh for two thermal zones (+0 open and -1 office) in case study 1.

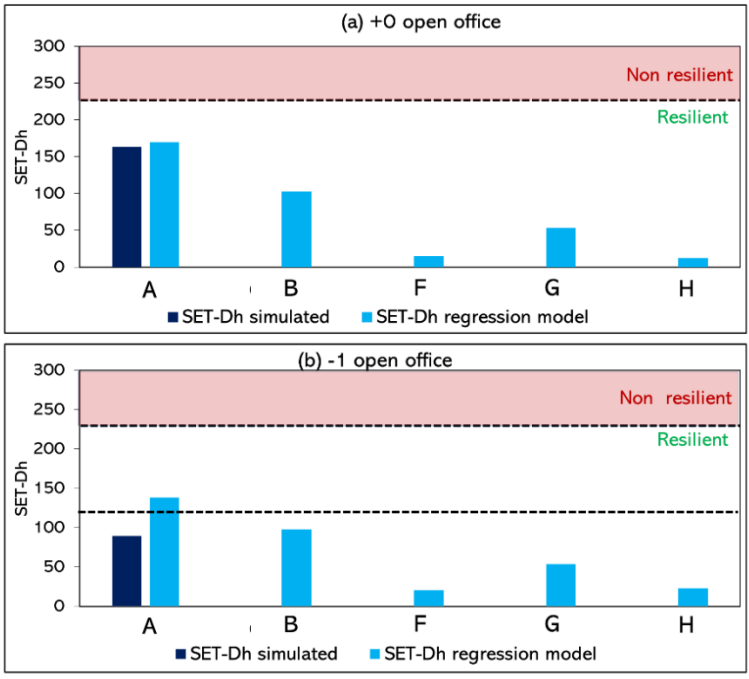


Figure 5-9. Predicted and simulated SET-Dh during doS=0.060 in the office building (case study 1) for (a) +0 and (b)-1 open office

As seen in section 5.3, the optimized design and control of the resilient cooling strategies resulted in good thermal resilience performance of the office during the short and intense heatwave period. In scenario A, in +0 office, without any resilient cooling strategy, there were 163.2 [°.h].The regression model predicted 169.7 [°.h]. Similarly for scenario A, in -1 office, both the predicted and simulated SET-Dh were below the 230 [°C*h] threshold. For both the office spaces (+0 and -1), in scenario B, F, G and H simulation results showed that there were zero SET-Dh as the cooling capacity was sufficient to maintain the office spaces below the cooling setpoint. In scenario B, F, G and H, for both office spaces, the regression model predicts SET-Dh values higher than zero. However these values are below the 230 [°.h] threshold. Thus, by both the prediction and simulation, all the cases showed good agreement on the thermal resilience performance of the office building.

5.6.1.2 Case study 2-Office

Figure 5-10 illustrates the comparison between the predicted and the simulated SET-Dh for two office spaces (4.1.6 and 4.7) of case study 2.

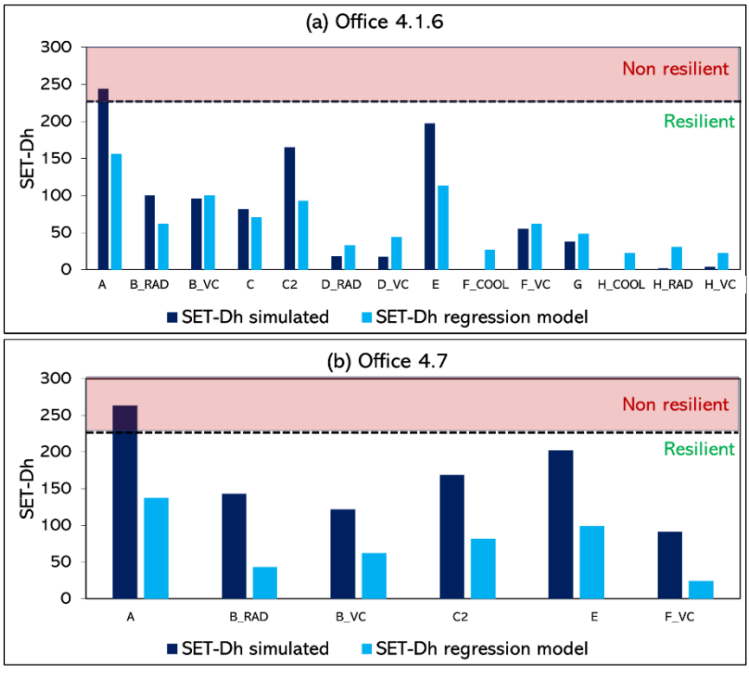


Figure 5-10. Predicted and simulated SET-Dh during doS=0.060 in the office building (case study 2) for (a) office 4.1.6 and (b) office 4.7

In scenario A, in both the office spaces (4.1.6 and 4.7), without any resilient cooling strategy, there were 244.2 and 263.7[°.h] The regression model predicted 157 and 139 [°.h]. The regression model underestimated SET-Dh due to a difference in glazing g-values: 0.56 in the reference building used for model development versus 0.78 in the simulation. The g-value was not an independent parameter in the regression model, as

it could not be changed independently in Modelica, though SHGC is a key factor in solar gains and overheating. In scenario E, for both thermal zones, thermal resilience improves with solar shading, and simulation results show 197[°.h] and the regression models predicts 106.1[°.h].With solar shading, there was a 19.2% decrease in SET-Dh. To be noted here that for solar shading, the SHGC is also an important parameter that impacts the amount of solar gains within the room. Due to limitations of the Modelica's applicability, SHGC was not used as a sensitivity parameter while developing the regression models. The simulated case study building has a SHGC 60% higher than the value used for the development of the regression model. This leads to a higher relative difference between the absolute values of the predicted and simulated cases with solar shading. However, both simulated and predicted values indicate that the office spaces were within the 230 [°C*h] limit. With NNV, two different window opening areas were evaluated. NNV aided in dampening the peak temperature during day time. In office 4.1.6. with bigger window opening area for NNV operation (scenario C) higher ventilation rate at night leads to a 66.8% decrease in SET-Dh and with 2.6% less window opening area in scenario C2 leads to 32.2% decrease in SET-Dh compared to case without any strategy. For scenario B_VC with limited cooling capacity, the relative difference between predicted and simulated case is 9.8%. For scenario H, varying cooling capacity were evaluated. With a combination of cooling and resilient capacity, both the zones showed high thermal resilience (<230 [°.h])as predicted by the regression model.

To be noted here that for office 4.7, several cases could not be compared as the simulations were run with parameter values that are out of the parameter ranges used to develop the regression models and thus the regression models cannot be applied. It is important to note that the order of both simulated and predicted SET-Dh values is the same, indicating consistency between the model's predictions and the simulation results. Both the regression model and the simulations showed the offices had lower thermal resilience when no strategies were applied (scenario A). Scenario E (with only solar shading) demonstrated overheating, yet stayed within the 230 [°.h] limit. The order of SET-Dh values in both simulated an predicted cases, remained consistent.

5.6.1.3 Case study 3-Single family dwelling

Figure 5-11 illustrates the comparison between the predicted and the simulated SET-Dh for the living room and bedroom (garden side) (case study 3).

In scenario A, in both living room and bedroom, without any resilient cooling strategy, there are 513.5 [°.h] and 465.1[°C*h]. The regression model predicted 326.6 [°.h] and 273.9 [°.h] respectively. Both the predicted and simulated results showed that the living room and the bedroom were not thermally resilient during the heatwave and violated the 230 [°C*h] threshold. For both the zones, in scenario B, and F, simulation results showed that there were zero SET-Dh as the cooling capacity was sufficient to maintain the spaces below the cooling setpoint. The regression model predicts SET-Dh values higher than zero. However these values are below the 230 [°.h] threshold. Thus by both the prediction and simulation, all the cases show good agreement on the thermal resilience performance of the living room and bedroom. Even with higher relative

differences, the developed regression models could predict whether the 230 [°C*h] limit was violated or not. This shows that the developed regression models can predict if a building is able to maintain habitable conditions for occupants during heatwaves.

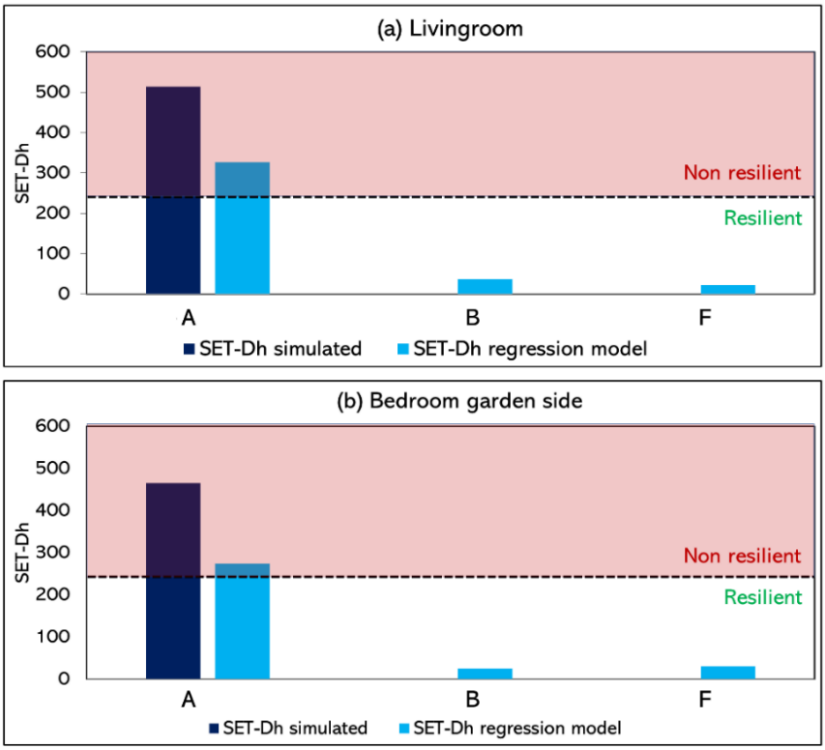


Figure 5-11. Predicted and simulated SET-Dh during doS=0.060 in the single family dwelling (case study 3) for (a) living room and (b) bedroom

5.6.2 Evaluation of the TRI $_{SET\text{-}Dh}$ with simulation results for *doS* $_{medium}$

Figure 5-12 illustrates comparison of simulation results to the SET-Dh predicted by the linear regression models without interaction terms developed for scenario A and scenario H for office buildings, and only for scenario A for residential building, for *doS* $_{medium}$ *=0.179*

As shown in Figure 5-12, during a prolonged thermal shock, the thermal resilience performance of all buildings declined significantly. Without the implementation of resilient cooling strategies (scenario A), both office and residential buildings exhibited high SET-Dh values, exceeding the critical threshold of 230 [°.h].This trend was accurately captured by the developed regression models, which predicted similar outcomes. For the office buildings, additional regression models were developed for scenario H, which incorporated a combination of resilient strategies, including cooling, solar shading, and NNV. The results demonstrated that even during extended thermal shocks, the critical zones of both offices maintained high thermal resilience, staying

well within the acceptable comfort range. The regression model predictions for these scenarios closely aligned with the simulated outcomes, indicating a high degree of accuracy in forecasting thermal resilience when resilient cooling strategies were applied.

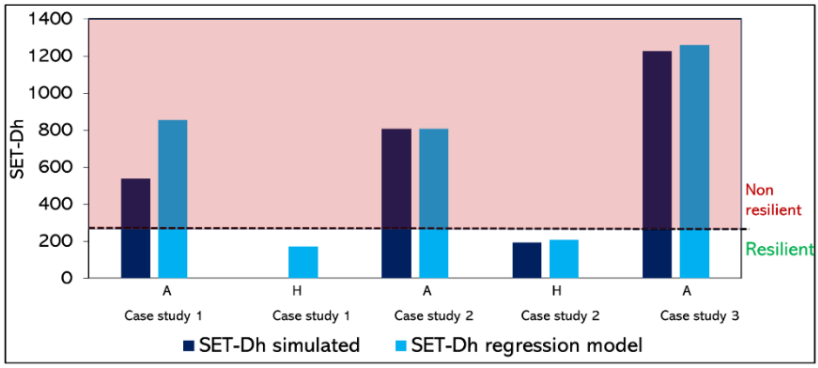


Figure 5-12. Predicted and simulated SET-Dh during doS=0.179 in the all three case study buildings

While Figure 5-9, Figure 5-10, Figure 5-11 and Figure 5-12 does show a relative difference between predicted and simulated SET-Dh values for different residential and office typologies, the primary objective was to assess the buildings' thermal resilience using the SET-Dh threshold. The analysis is more focused on a categorical prediction of thermal resilience rather than a precise numerical match between predicted and simulated SET-Dh values. The goal was to determine whether the buildings meet the resilience threshold, and the figures show that, in nearly all cases, the predicted and simulated results align in terms of categorizing the buildings as resilient or not. Therefore, the small differences in values do not detract from the overall assessment, as the key outcome is the consistency in categorizing the buildings' resilience.

5.7 Evaluation of the TRI $_{SET\text{-}Dh}$ with interaction terms

Figure 5-13 illustrated the performance of the regression models with interaction terms for scenario A, B, F, G, and H in Case Study 1. Regression models with interaction terms were developed for offices with medium thermal mass and doS_{low}. The results indicated that incorporating interaction terms did not significantly improve the adjusted R^2 of the regression models. In fact, the inclusion of interaction terms slightly reduced the explanatory power of the models. On average, there was a 1.6% increase in MPE across all cases when interaction terms were added, and the RMSE also increased, indicating reduced model accuracy.

Furthermore, the models with interaction terms failed to clearly explain how one interaction impacted the other within the regression equation, thereby increasing the complexity of interpretation. This added complexity made it difficult to derive meaningful insights into how interactions between variables contributed to the dependent variable, ultimately reducing the model's usefulness in practical applications. Given the absence of significant multicollinearity and the higher errors

associated with the interaction models, the linear regression model without interaction terms was selected as the most appropriate approach for further testing. This simpler model provided better alignment with the actual data, with lower MPE and RMSE, and a higher adjusted R^2, demonstrating its suitability for accurately predicting outcomes with less complexity.

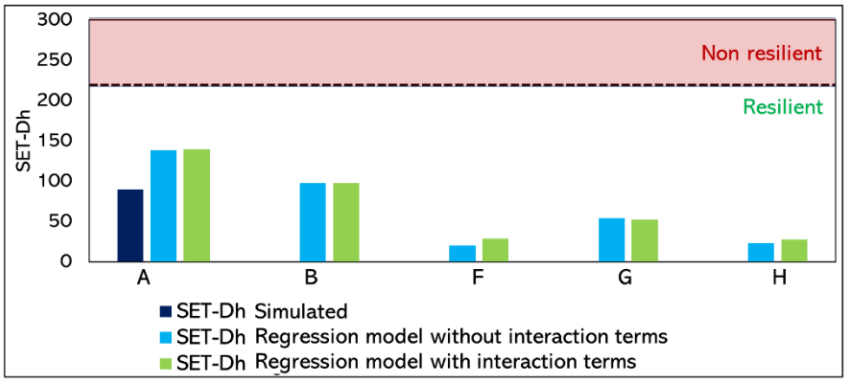


Figure 5-13. Performance of the regression models with and without regression equation

5.8 Prediction accuracy of the regression models

The main aim of the developed regression models (TRI_{SET-Dh}) was to predict whether the building is thermally resilient or not during heatwaves (violating the threshold of 230 [°C*h]). Apart from comparison between the predicted and simulated SET-Dh values, it was crucial to perform statistical tests to quantify the models' accuracy and precision. Figure 5-14 shows the F1-test conducted comparing all the instances of predicted and simulated SET-Dh for all three case study buildings for $doS_{low}=0.060$ and $doS_{medium}=0.179$. The results showed F1 score =0.89, Precision=0.8, Recall=1.00, Specificity=0.96 and Accuracy=0.97 and this demonstrates that the model predicted with 97% accuracy if the SET-Dh is "Above 230 [°.h] " and "Below 230 [°.h] " classes. High precision and recall values are high (demonstrates the model accurately identified with high certainty the positive and negative instances while balancing between precision and recall.

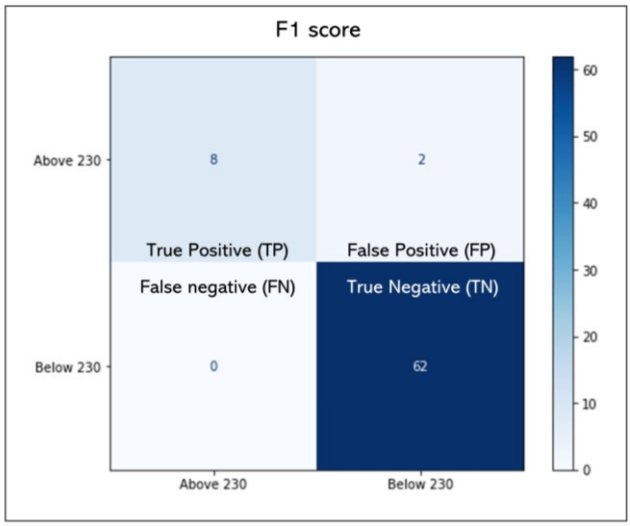


Figure 5-14. F1-score for evaluation of the developed TRI $_{SET-Dh}$

However, while these scores evaluate the performance of the developed TRI_{SET-Dh}, there is an imbalance in the data, i.e., the instances where the SET-Dh was above the threshold of 230 [°C*h] is 83.8% less than the instances where 230 [°C*h] was not violated. More parametric simulation cases are required for further evaluations of each TRI_{SET-Dh}. A balanced data set with higher data points will lead to more comprehensive testing of the developed TRI_{SET-Dh}.

5.9 Conclusions

In conclusion, this chapter demonstrated the method to develop and evaluate comprehensive thermal resilience indicator, TRI_{SET-Dh}, to assess and improve thermal resilience of buildings to overheating during heatwaves. The methodology employed combined dynamic building performance simulations, sensitivity analysis, and development of linear regression models to systematically analyze the factors influencing a building's thermal resilience. By integrating resilient cooling strategies and considering different thermal mass variations, TRI_{SET-Dh}, provided a comprehensive approach to predict how different buildings would perform under extreme temperature conditions. The main findings are as follows:

- Thermal resilience assessment of case study buildings: Assessment of three case study buildings revealed that buildings that are optimized during design phase (balanced WWR, thermal mass, implementation of solar shading and ventilative cooling strategies) tend to have higher thermal resilience during shocks. In scenarios without any strategies, both offices and dwellings have low thermal resilience even during shorter heatwaves. Combining resilient strategies is the most effective approach for improving thermal resilience during heatwaves. Passive measures like solar shading, when paired with active cooling, stabilized indoor temperatures and reduced overheating, particularly in buildings with large WWR. This hybrid approach also

minimized energy consumption, offering a more sustainable solution than relying solely on active cooling. Application of cooling strategy, even with limited cooling capacity aided in maintaining good thermal resilience in the office spaces, even with high WWR and internal gains during heatwave periods. With cooling there was a 60% improvement in thermal resilience compared to the buildings without any strategy, however this is energy intensive. In offices with solar shading there is 10% improvement in thermal resilience compared to the building without solar shading. With ventilative cooling thermal resilience improved by 36.1%. Combination of strategies (cooling + ventilative cooling + shading) leads to a 70% improvement in thermal resilience compared to buildings without any resilient cooling strategy. During longer shocks, a combination of strategies are critical in maintaining habitable conditions in the buildings.

- Development of decision tree for systematic thermal resilience assessment : Building upon the results from Chapter 4, where the most influential building and system design parameters affecting the thermal resilience of a Belgian reference apartment were identified, this chapter extends the analysis to other building typologies. This broader exploration provides insights into the applicability of the findings across different building types. A decision tree is developed for systematic selection of shock severity, building type, thermal mass and strategies. Branching of doS aids in selecting which type of shock (short, intense or severe) the building will be assessed against. Building type selection aids in pre-selection of standardized (from EN 16798-1) inputs such as occupancy profile, occupant density, internal gains and ventilation rates. Furthermore to account for the heat storage and release properties of different buildings, the study evaluated three levels of thermal mass (heavy, medium, and light). This was crucial because thermal mass influences a building's ability to absorb, store, and slowly release heat, thereby impacting its overall resilience to thermal shocks. Additionally, the cooling strategies were segregated as categorical cases to evaluate the impact of each strategy comprehensively. Based on these categorizations, a decision tree was developed and the TRI_{SET-Dh} was formulated for each branch of the decision tree.
- Developed thermal resilience assessment framework: The developed TRI_{SET-Dh}, provides an analytical framework to predict the relationship between building design parameters and thermal resilience performance of buildings during various shocks. Within the scope of the *doS*, building type, strategies and design parameter values, the developed TRI_{SET-Dh}, offers a simplified but robust method to predict a building's thermal resilience, making the assessment process more accessible to architects and designers during the early stages of building design. For thermal resilience assessment of buildings in early design stage, the following steps can be taken –(1) selection of the level of *doS* (2) selection of the decision tree branch based on design choices (example building type, thermal mass, strategies (3) input parameter values (such as building orientation, envelope and glazing properties, etc.) (4) based on the decision tree branch and parameter value attain a TRI_{SET-Dh} value to assess thermal resilience (5) iteratively change design input values to , i.e., repeat step 4 to reach desired thermal resilience level.

- Scope of the developed regression models: While the development of TRI_{SET-Dh}, provided a comprehensive approach for assessing thermal resilience, the study identified some limitations that need to be addressed in future research. One, the developed TRI_{SET-Dh}, is applicable to the building typology evaluated, the parameters studied and specifically only holds true for the ranges of each parameters used for the development of the TRI_{SET-Dh}. Additionally, the regression models were developed for two increasing *doS* and will not be able to predict the thermal performance of buildings during longer or even more severe shocks. To overcome this limitation, doS should be included as a sensitivity parameter which was outside the scope of this study. Another limitation was the exclusion of some design parameters, such as type of shading(external, internal, overhang and even the solar heat gain coefficients, from the regression models. Including these parameters in future studies could enhance the accuracy of the regression models and provide a more detailed understanding of how different design strategies impact thermal resilience. Furthermore for current testing of the TRI_{SET-Dh}, there was an imbalance in the dataset with a significantly lower number of cases where the SET-Dh exceeded the critical threshold of 230 [°.h]. To improve the robustness of the regression models, future studies should include a more balanced dataset with a higher number of parametric simulation cases.

Chapter 6 : Research Valorization

Chapter 6 outlines the valorization plan, focusing on leveraging the research findings to-(a) explore contract research and (b) provide recommendations to support policy makers.

6.1 Introduction

Current building design practices and regulations primarily address typical climatic conditions, failing to account for the increasing severity and frequency of heatwaves expected in the coming decades. While substantial efforts have been directed toward improving energy efficiency in buildings (such as the European Green Deal, EU Renovation Wave and the Flemish Government's Energy and Climate Plan)[268][269], they often overlook the urgent need to enhance thermal resilience. Studies predict that by 2050, up to three times more Europeans could be exposed to extreme heat[4], highlighting the critical need to include thermal resilience in design methods and building regulation policies and standards.

The economic argument for improving buildings' thermal resilience to overheating during heatwaves is also compelling. Heatwaves in Europe not only led to high mortality rate but also resulted in productivity losses due to heat-related workforce impacts, with non-resilient office and commercial buildings. According to the European Environment Agency (EEA), economic losses from extreme weather events, including heatwaves, amounted to €450–520 billion from 1980 to 2020 across EEA member countries (which include the EU and several neighboring states)[270]. Projections suggest that these losses could grow to exceed 1% of GDP annually in Europe by the 2060s due to increasing frequency and severity of heatwaves[271][272]. Thus, improving thermal resilience to overheating not only protects public health but also enhances productivity, reduces cooling energy demand, and lowers the operational costs of buildings. To ensure sustainable growth and resilient infrastructure-(a)EU building stakeholders must embed climate-proofing measures into the design and renovation processes and (b) policy makers must include thermal resilience in current building regulation standards and codes. By aligning thermal resilience strategies with energy efficiency goals, the building sector can achieve a dual impact: safeguarding occupant well-being and preparing Europe's building stock to withstand future climate challenges.

This research offers actionable pathways to bridge the gap between scientific insights and real-world applications in the building sector. The key outcomes focus on equipping professionals, policymakers, and industry stakeholders with strategies, and frameworks needed to enhance thermal resilience to overheating in buildings. The research findings can be valorized–(a)by exploring contract research with companies such as architectural firms, engineering companies, HVAC designers and manufacturers and (b) by recommendations to support policy makers.

6.2 Valorization plan exploring contract research

Exploring contract research may include –(i) Developing new design methods for specific professional groups (architects, designers, HVAC engineers) to enable professionals to incorporate thermal resilience into building and system design (ii) Creating a resilience toolbox for decision-making at the early design or renovation stage to equip stakeholders (including designers, developers, and building owners) with tools to assess and enhance the thermal resilience of buildings during preliminary phases.

6.2.1 Development of new design and assessment methods

Traditional building and HVAC design often prioritizes average conditions, neglecting resilience to extreme events such as heatwaves or power outages. Integrating the thermal resilience assessment framework through contract research offers a practical pathway for real-world application. This approach allows researchers to collaborate with clients such as architectural firms, and HVAC companies to address specific challenges, such as designing for extreme climates, retrofitting existing systems, or complying with evolving regulations. The proposed valorization path includes development of new design method that integrates the evaluation of extreme conditions early in the design process. By assessing the performance of buildings and systems under normal operations as well as shock scenarios, such as prolonged heatwaves, architects, designers, and engineers can optimize thermal resilience alongside energy efficiency, ensuring robust solutions for a changing climate. Incorporating thermal resilience assessments during the early design phase empowers stakeholders to identify vulnerabilities and implement targeted improvements. This proactive approach includes simulating shock scenarios to assess a building's thermal performance and identify weaknesses. Designers can then strategically integrate solutions such as passive cooling measures, advanced HVAC controls, and thermal energy storage systems, effectively mitigating risks. Companies adopting this methodology will not only enhance the environmental and safety standards of their projects but also gain a competitive edge in markets increasingly shaped by stringent climate policies and regulations. Example architectural firms can integrate step-by-step method to assess their design choices' impact on thermal comfort, energy efficiency, overheating and thermal resilience to overheating in early design stage. Implementation of such design process will position themselves ahead in the market to offer an integrated solution through design. To implement this framework, several key steps are proposed:

- Educate Stakeholders: Raise awareness about the importance of resilience in building and system design. Emphasize why evaluating resilience in every project is essential, the expected outcomes, and the potential gains from such analyses. This will ensure widespread understanding and commitment to resilience goals.
- Streamline Early-Stage Assessments: Establish efficient workflows for conducting resilience assessments during the early stages of design. This ensures resilience considerations are embedded from the outset, enabling timely and cost-effective decision-making.
- Integrate Resilience Metrics in Design Tools: Incorporate resilience indicators or scales into commonly used building energy simulation (BES) software. This enables the seamless calculation and iterative refinement of resilience to overheating, aligning with standard workflows for architects and engineers.
- Refine System Solutions for Extremes: Enhance integrated system solutions to address shocks such as extreme weather. This includes revising system scopes, adjusting system sizing and redefining guarantee conditions.
- Develop Practical Design Guidelines: Formulate and disseminate rules of good design that can be easily applied during the architectural and system design processes.

These guidelines will provide a low-effort yet effective approach to enhancing building resilience, ensuring practical applicability in diverse projects.

6.2.2 Creation of a Resilience Toolbox

Follow-up projects could focus on developing a toolbox designed to assess and enhance thermal resilience against various overheating shocks across different climates. The development of a resilience toolbox builds upon the thermal resilience assessment framework to create a structured collection of tools, strategies, and resources aimed at improving the thermal resilience of buildings against overheating. Such a toolbox enables stakeholders (including architects, designers, developers, and even building owners) to evaluate and enhance thermal resilience through a multi-dimensional approach that considers building regulations, building and HVAC design, thermal resilience assessment methods (frameworks, indicators and simulation tools), material choices, occupant behavior, costs and socio-environmental factors.

Architects can leverage the toolbox to explore innovative design methods and assessment techniques to integrate resilience into their projects. Developers and building owners can use it to identify cost-effective strategies to enhance building performance and long-term value. Urban planners can rely on the toolbox for insights into guidelines and regulations that promote climate-adaptive designs, ensuring alignment with policy goals. HVAC designers can optimize system performance to support thermal resilience while minimizing energy use. The emphasis is broader, targeting decision-makers involved in both technical and strategic planning. The toolbox includes guidelines, performance standards, and assessment tools, making it applicable beyond technical design to project planning. A resilience toolbox can also serve as a key outcome of research valorization, translating theoretical assessment frameworks into actionable tools for companies and building stakeholders. Building and system designers and stakeholders could benefit from this thermal resilience toolbox in the decision-making process. The outcomes would not only safeguard occupant health during extreme heat events but also provide a compelling value proposition for building owners by enhancing the economic and environmental performance of their assets. By bridging research and practice, such a toolbox could significantly contribute to sustainable and resilient building design on a global scale.

In US, organizations have begun integrating resilience frameworks into urban and architectural planning. For example, the Penn State's Thermal Resilience Framework incorporates multiple factors, including building materials, socio-demographic data, and occupant behaviors, to evaluate and enhance thermal resilience [273]. It focuses on integrating these variables to provide a holistic view of resilience and sustainability for different building types and climates. Testing includes advanced simulations and material assessments, with insights shared with city planners and decision-makers in cities like Philadelphia to guide retrofitting and planning efforts.Countries like Netherlands have implemented resilience assessment tools such as Climate-Adapt[274]. These framework assist companies and municipalities in designing climate-resilient buildings, showcasing the global adoption of such toolkits. For example, the

Netherlands' Delta Plan for Climate Adaptation allocated €1.2 billion to resilience projects, highlighting the financial opportunities in this space [275].

6.2.3 Target group

The contract research on improving thermal resilience of buildings to overheating primarily targets-architectural firms, engineering consultancies and HVAC designers. Innovative architectural firms and engineering consultancies involved in high-performance building projects are expected to be key early adopters. These firms, already focused on sustainable design and energy efficiency, will see value in integrating thermal resilience strategies into their offerings to meet emerging climate challenges. HVAC designers, developers and manufacturers: HVAC designers and manufacturers play a crucial role in enhancing indoor thermal resilience. By adopting this research, designers can create systems that meet energy efficiency standards while ensuring buildings can withstand extreme heat events, while manufacturers can develop robust products and technologies that maintain system functionality and occupant comfort during such critical conditions.

6.2.4 Market estimate and positioning

The Belgian market alone presents substantial opportunities, with an annual market size of approximately 10,000 non-residential projects (new and retrofit) and 53,000 residential projects (new and retrofit)[276]. The Belgian building industry comprises a robust ecosystem of stakeholders, including approximately 500 architectural firms, 30-50 medium to large engineering firms, and numerous manufacturers of ventilation systems (50), solar shading solutions (30), and heat pumps (100)[277][278]. This proactive approach enables industry stakeholders to differentiate themselves in the competitive market by taking a leadership role in developing overheating-resilient buildings, enhancing their reputation and market position while contributing to a more sustainable and climate-adaptive built environment.

6.2.5 Economic value

Research co-operation with companies will provide them with the insights and tools needed to deliver comprehensive advice on designing resilient, futureproof buildings and systems. These results enable tailored business models for each sectors (architectural firms, engineering and consultancy firms and HVAC designers, manufacturers and product developers), maximizing their economic and operational impacts. The economic impacts are projected as follows:

Architectural Firms: Architectural firms that adopt thermal resilience in their design approach can significantly differentiate themselves in a competitive market. By integrating resilience checks early in the design process, these firms can offer efficient solutions that enhance thermal comfort, energy performance, and overall building resilience. This approach enables firms to provide comprehensive client advice while maintaining cost efficiency, thereby boosting market competitiveness and client satisfaction. Incorporating resilience assessments can also expand project scope and

justify higher design fees, leading to cumulative employment growth equivalent to several full-time employees (FTEs) and significant financial returns through enhanced client offerings and efficiency in project delivery. By positioning themselves as leaders in climate-adaptive design, firms can secure premium projects aligned with sustainability goals. Additionally, resilience-focused retrofit projects can increase the value of existing buildings, tapping into the growing retrofitting market. These strategies highlight the economic potential of integrating thermal resilience into building design and operations.

Engineering and Consultancy Firms: Consultancy services can leverage their expertise to offer thermal assessments, retrofitting advice, and strategy implementation for building owners, developers, and policymakers, similar to environmental design services. Companies can generate revenue by promoting and implementing a resilience toolbox that enhances thermal resilience in buildings through multiple avenues. Improved analysis capabilities are expected to increase project numbers, creating additional employment and revenue opportunities. This approach also strengthens competitive advantages through science-based methods. Subscription or licensing models for software and simulation tools that assess thermal resilience present another avenue, targeting architects and engineers and tapping into the growing green building software market. The European green building market, valued at around €179.5 billion in 2023, is projected to grow at a compound annual growth rate (CAGR) of 11.19%, reaching €342 billion by 2029 [279]. This growth is driven by demand for sustainable construction, energy-efficient systems, and policies like the European Green Deal and EPBD, which require nearly zero-energy buildings (nZEB). Training and certification programs, inspired by initiatives like LEED or BREEAM, offer additional revenue by educating professionals on resilience strategies. Companies can also leverage policy support and advocacy, partnering with governments to shape building codes and secure public contracts, mirroring opportunities under initiatives such as the EU Renovation Wave.

HVAC designers, manufacturers and product developers: Apart from architects, HVAC designers, manufacturers and product developers can leverage this knowledge to improve their designs and guarantee safety in extreme conditions, potentially increasing their market position., By incorporating resilience insights into product offerings—such as integrated comfort systems or shading solutions—companies can increase market differentiation. Even modest percentage increases in turnover from improved product advice and performance guarantees could translate into substantial cumulative employment and financial growth. By adopting resilience-focused strategies, companies can also innovate within their existing product portfolios and integrated comfort solutions, ensuring that their products are scientifically validated and capable of addressing both current and future climate challenges. In the rapidly evolving HVAC market, where differentiation based on product features alone is becoming increasingly difficult, thermal resilience provides a key opportunity for innovation. By investing in resilience research and embedding these concepts into products and services, companies can position themselves as leaders in climate

resilience, offering not only high-performance systems but also solutions that anticipate and mitigate the risks of extreme weather events. This proactive approach can help companies build stronger relationships with stakeholders, position their products as essential for future building designs, and foster brand loyalty. Product innovation in advanced materials like dynamic insulation or passive cooling technologies can generate revenue through direct sales or partnerships, as seen in the growth of smart shading systems.

From an environmental perspective, enhancing thermal resilience to overheating in buildings significantly reduces reliance on energy-intensive cooling systems during heatwaves, offering distinct advantages over conventional approaches that depend heavily on high-energy HVAC solutions. These strategies not only decrease energy consumption but also contribute to lowering greenhouse gas emissions, aligning with sustainability goals. However, the widespread adoption of thermal resilience measures remains hindered by the absence of centralized policy frameworks and limited integration into existing building regulations. Despite these challenges, key stakeholders in the building industry are actively pursuing innovative approaches to address overheating risks. Emerging technologies, such as smart shading systems, thermal energy storage, and dynamic insulation materials, are gaining traction for their potential to provide effective, sustainable solutions to improve thermal performance while reducing environmental impact.

6.3 Valorization plan exploring for recommendations to support policy makers

The second pathway for research valorization includes leveraging the research results to provide recommendations to support policy makers. The absence of standardized procedures for addressing thermal resilience in buildings highlights a critical gap in current building codes, especially when compared to established standards for other resilience dimensions, such as seismic safety or fire protection. However, efforts to integrate climate resilience into building practices are gaining traction globally. For example, the RELi™ Rating System [280], integrated into the resilience-focused credits of the LEED certification[281], provides a structured approach to designing buildings that can withstand climate extremes. RELi emphasizes innovative design and planning for resilience, including strategies for extreme heat and weather events. Similarly, the United Nations Environment Programme (UNEP)[282] has published guidelines aimed at designing climate-resilient infrastructure and communities, offering actionable measures to address risks like overheating. On a broader level, international standards like ISO 37123[283], focuses on indicators for resilient cities and aim to provide a cohesive framework for integrating resilience across urban infrastructure, including the building sector. These initiatives help translate the concept of resilience into practical guidelines and standards, enabling stakeholders to systematically incorporate thermal resilience into building design, construction, and retrofitting.

The EPBD [284], established by the EU, aims to improve the energy efficiency of buildings across member states. Despite the EPBD's focus on energy efficiency, thermal

resilience to overheating is not explicitly addressed in the current policies. This omission creates a critical gap, particularly in light of the increasing frequency and severity of heatwaves due to climate change. There are no specific guidelines or requirements to assess or enhance a building's ability to remain thermally comfortable during extreme heat events. This research on improving thermal resilience of buildings to overheating can provide the scientific basis and practical strategies for filling this gap. By offering guidelines, thermal resilience thresholds (such as maximum allowable indoor temperatures as part of the EPBD compliance calculations), performance metrics, and innovative design solutions, the findings can inform policymakers on how to integrate thermal resilience into EPBD frameworks. Encouraging policymakers to incorporate thermal resilience as a mandatory requirement for new build and major renovations, particularly for vulnerable building typologies such as elderly care homes, schools, and densely populated residential areas. Integrating thermal resilience into EPBD compliance frameworks can significantly reduce the societal and economic impacts of heatwaves, such as heat-related illnesses, loss of productivity, and infrastructure degradation. This research can position EU regulations in adapting building policies to address climate resilience. Incorporating thermal resilience into guidelines and standards aligns with the EU Green Deal [285] and the Renovation Wave [269] strategy, emphasizing the importance of making Europe's building stock not only energy-efficient but also climate-resilient.

6.3.1 Trajectory towards policy implementation

The trajectory towards policy implementation outlines a strategic approach for transforming scientific findings into actionable policy recommendations and practical solutions. By focusing on knowledge transfer, pilot programs, and policy integration, this trajectory ensures the research influences building codes and supports climate-resilient development across Europe.

- Knowledge Transfer and Collaboration: Initiate partnerships with EU bodies, national governments, and NGOs to disseminate findings through white papers, workshops, and stakeholder consultations. This step ensures alignment with existing regulatory frameworks and priorities.
- Policy Integration: Develop practical tools and guidelines for policymakers, including thermal resilience thresholds, performance metrics, and design solutions tailored for EU-specific climate challenges. Advocate for integrating these into EPBD updates and related national frameworks.
- Pilot Programs and Demonstrations: Collaborate with municipalities or public institutions (e.g., schools, elderly care facilities) to test and demonstrate resilience-focused building designs. Showcase these projects as scalable solutions for broader adoption.
- Advocacy and Awareness Campaigns: Use conferences, policy forums, and academic publications to highlight the economic and societal benefits of thermal resilience. Link the research to pressing issues like health impacts from heatwaves, infrastructure longevity, and productivity losses to strengthen its policy appeal.

- Standardization Efforts: Engage with international bodies such as UNEP[282] and ISO[283] to align research findings with global frameworks like ISO 37123[283], further enhancing its credibility and facilitating adoption across EU member states.

Through these research valorization trajectory, this research can influence policy at multiple levels, ensuring its findings have widespread and long-lasting impacts on Europe's efforts to combat climate-related challenges.

Chapter 7 : Conclusions

Chapter 7 presents the conclusions drawn from this research on developing a comprehensive framework for assessing and improving thermal resilience to overheating in buildings. These conclusions are aligned with the research objectives outlined in Chapter 1. Additionally, the chapter discusses the study's limitations, and highlights potential directions for future research.

7.1 Conclusions

Climate change and extreme heat challenge the building sector to balance energy efficiency with thermal resilience. Rising temperatures and frequent heatwaves demand buildings that prevent overheating, reducing serious health risks like heat stress and mortality. This dissertation introduced a comprehensive framework for assessing and improving thermal resilience to overheating in buildings, focusing on impact of shock, occupants' health and targeting critical building and system design parameters.

7.1.1 RO1: Gain insight into thermal resilience to overheating in buildings

In chapter 2, a thorough review of state-of-the-art gave insight into the concept and definition of buildings' thermal resilience to overheating and the existing assessment methods. Furthermore, dynamic building performance simulations were utilized to assess thermal resilience of case study buildings with existing overheating and thermal resilience indicators in the moderate Belgian climate. This analysis revealed two key aspects: (a) it provided valuable insights into the thermal resilience performance of buildings and its cooling systems and (b) it highlighted significant shortcomings in the current methods used for thermal resilience assessments.

Thermal resilience assessment of buildings with existing methods

Thermal resilience assessment for different types of buildings (dwellings, apartments, offices, and educational) equipped with various cooling strategies (such as solar shading, natural night ventilation, indirect evaporative cooling, floor cooling, and mechanical cooling) revealed significant variations in thermal resilience across building types, influenced by design features, cooling strategies, and shock types. Passive strategies like solar shading and natural night ventilation significantly enhance thermal resilience in dwellings, reducing recovery time by 89%. Apartments with optimized design—lower window-to-wall ratios and heavy thermal mass—perform better during shorter heatwaves, though passive cooling like night ventilation is less effective in extreme conditions. Offices and educational buildings heavily depend on mechanical cooling, with offices reaching unsafe temperatures (30°C in 8 hours) during power outages, while lighter thermal mass in schools improves recovery speed by 11% during longer heatwaves. To sum it up, all building types showed that airtight, highly insulated buildings with large window-to-wall ratio exhibited low thermal resilience, while heavy thermal mass prolonged recovery after long heatwaves. Solar shading and reduced WWR significantly improved thermal resilience, whereas natural night ventilation and other cooling strategies (such as indirect evaporative cooling, ground source cooling), though beneficial, were less effective during extreme or prolonged shocks.

Evaluation of existing thermal resilience assessment methods

Additionally, this assessment exposed significant gaps in existing thermal resilience indicators and assessment frameworks. Current indicators focus on deviations in indoor conditions and the duration of such deviations but often fail to link these with heat stress thresholds, which are essential for understanding impacts on occupants' health. While qualitative indicators provide insights into various resilience phases—adaptive

capacity, absorptive capacity, restorative capacity, and recovery speed—these are not standardized, and many lack defined thresholds for evaluating thermal resilience performance. On the other hand, quantitative indicators such as passive survivability, and overheating escalation factor either lack clear thresholds or are applied inconsistently. Moreover, resilience assessments often depend on complex building performance simulations or detailed data analyses, making them inaccessible during the early design phase. Existing frameworks fail to integrate shock-specific resilience metrics, limiting their ability to assess how buildings perform under varying shock intensities or types, such as heatwaves or power outages. Additionally, these frameworks do not explicitly link resilience performance to critical design parameters, leaving designers unable to evaluate the impact of interventions without conducting new simulations. This fragmented approach underscored the need for a comprehensive framework that includes health-linked thresholds, accommodates various shock types, and connects resilience with design choices to guide more effective building strategies.

Selection of heat stress indicator for thermal resilience assessment

The heat-stress indicator SET-Dh[128] was used to assess thermal resilience and the impact of shocks on occupants' health. Derived from Gagge et al.'s[127] two-node physiological model and endorsed by ASHRAE Standard 55[123], the Standard Effective Temperature (SET) is a widely accepted metric linking indoor conditions to thermal sensation [129][200]. SET thresholds—comfortable (24.1°C), alert (28.1°C), and emergency (32.1°C) [39]—reflect habitable and critical health conditions. Overheating occurs when SET-Dh exceeds 230 ± 42 [°C·h] for young adults or 117 ± 30 [°C·h] for older adults, with higher values indicating lower thermal resilience [88].

7.1.2 RO2: Identify and quantify shocks that impact buildings' thermal resilience performance.

Chapter 3 identified and quantified the shocks affecting buildings' thermal resilience to overheating. Additionally, to compare their impacts, a systematic approach to shock quantification was adopted. Shock events can be classified into two categories depending on how they impact the thermal environment: (i) external shocks e.g., heatwaves that affect the outdoor thermal conditions increasing the heat gains through the envelopes and reducing the efficiency of mechanical and ventilative cooling strategies and (ii) system shocks which correspond to partial or complete failure in the operation of building systems.

Development of degree of shock (doS) to quantify shocks

This study employed the concept of normalized degree of shock (*doS*) to define and measure the severity and characteristics of various shock types. By utilizing *doS*, the analysis enabled consistent comparisons across different shocks, facilitating a clearer understanding of their relative influence on buildings' thermal resilience. This approach provided a standardized basis for assessing and contrasting the impact of diverse (independent of climatic context) shocks on buildings' thermal resilience to overheating. Including *doS* in the thermal resilience framework will bridge the missing

gap of assessing thermal resilience of a building during different shock severity in a single framework that is currently evaluated to be missing.

Development of weather files

To test *doS* for external shocks such as heatwaves, future weather (typical meteorological year and heatwave) dataset was developed for three periods –(a) historical 2010s (2001-2020) (b) mid-term future 2050s (2041-2060) and (c) long-term future 2090s (2081-2100) using CORDEX climate projections under the Representative Concentration Pathway (RCP) 8.5 scenario, with bias correction applied using observational data from Ghent (2015–2020) for ASHRAE climate zone 4A [153]. RCP 8.5 was chosen as the primary scenario, representing a high greenhouse gas emissions pathway and extreme future climate change projection. This methodology to develop future weather data sets are adaptable for global use and relies on relative thresholds to define heatwave duration and severity. Such as Belgium's criteria: temperatures exceeding 25°C for 5 days, with at least 3 days above 30°C. From the developed heatwaves for Ghent for each period extreme heatwaves -most severe, most intense and longest were selected to test thermal resilience of buildings.

Identifying the most critical shock type

A parametric study varying (a) shock type (heatwaves, system failure, power outage.), (b) increasing the intensity (severity and duration) of *doS* for each shock type, (c) varying the time of occurrence of shocks and (d) combining shock types. Results showed system failures starting at the start of operational period and occurring for the full duration of the operational period are 1.2x more critical than system shock starting later in the day as it allowed for significant heat build-up. Failure to flush out heat gains (i.e., natural night ventilation failure) is more critical than solar shading and cooling strategy failure. However, the impact of heatwaves is 20x to 93x more critical than the worst system shock. Thus, heatwaves in future climate scenarios can be considered as the most critical shock. Heatwaves increase outdoor temperatures, solar gains, and reduce diurnal temperature differences, rendering cooling strategies less effective. Assessing buildings' resilience to different types of heatwaves—such as short, intense versus long, severe ones—is crucial, as longer and more intense heatwaves will impact thermal resilience and cooling strategies' performance. Additionally, in future climate scenarios a combination of heatwaves and power outages will become frequent. A combination of a full day of cooling strategy and shading failure occurring on the hottest day of a short heatwave, is 10% more critical than an individual longer heatwave. Finally, three representative *doS* of heatwaves (high, medium and low) were selected (a short heatwave, an intense heatwave, and a long-severe heatwave) to identify the most influential design parameters that impact thermal resilience to overheating.

Impact of increasing doS of heatwave on thermal resilience

As the severity and duration of *doS* of heatwaves increases, fewer buildings meet thermal resilience criteria, indicating a decline in thermal resilience to overheating. For same building and system design, the shock impact during highest *doS* (45 days long

heatwave) is 4´ more critical than the lowest *doS (6 days long)*. For *doS* $_{low}$ 87% of design variations have SET-Dh <230°C*h, for *doS* $_{medium}$ = 48% and for *doS* $_{high}$ =21%. Buildings' thermal resilience, particularly heat absorption, retention, and removal, is influenced not only by shock severity and duration but also by the timing of peak outdoor temperatures during shock.

7.1.3 RO3: Identify the most influential design parameters of buildings and systems that impact thermal resilience of buildings to overheating

In order to identify the most influential design parameters that impact thermal resilience to overheating in buildings, an uncertainty and sensitivity study was conducted on a reference apartment in Belgian context varying different design parameters (Chapter 4). 13 parameters were selected from literature based on their influence in overheating risk in buildings and were varied within ranges defined by current building standards and knowledge and experience of designers.

Uncertainty in thermal resilience performance

The uncertainty in thermal resilience varies significantly across *doS*. For SET-Dh, values range from 0–400 for low, 0–1400 for medium, and 0–3000 for high *doS*, indicating a 700% increase from low to high. Similarly, absorptivity time ranges show a 77% increase from low (24–448 hours) to medium (24–360 hours) and a 98% increase to high (0–424 hours). Recovery times exhibit the greatest variability, spanning 24–448 hours for low *doS*, 24–400 hours for medium, and escalating to 24–1500 hours for high, reflecting a 235% increase from low to high. Buildings that perform well under shorter shocks struggle during longer or more severe shocks. Buildings with suboptimal designs experience 6xhigher shock impact during severe shocks compared to those with optimized designs.

Identify the most influential design parameters

Sensitivity analysis showed that for a reference Belgian apartment across three escalating *doS*, parameters such as WWR, cooling capacity, and the operation of passive strategies like natural night ventilation and solar shading exerted 2´ more influence compared to building orientation, envelope and glazing properties, occupancy pattern, airtightness, and cooling set-point. This is true for all three evaluated resilience aspects (shock impact, absorptivity and recovery time), and two cooling systems (convective and radiant). WWR dictating the solar gains and the heat-built up has the highest impact on thermal resilience. As *doS* increases, the impact of cooling capacity increases as building and passive strategies are unable to improve thermal resilience. Shock impact (SET-Dh) is impacted by building and system features that dictate the heat absorption as well as flushing of heat from the building. Parameters such as cooling capacity and set-points are impactful in improving shock impact during prolonged shocks. While increasing the cooling capacity and lowering the cooling set-point enhance thermal resilience, they reduce the building's energy efficiency, underscoring the need to study the trade-off between thermal resilience and energy efficiency. Absorptivity time is predominantly affected by parameters that influence the heat accumulation and

absorption such as glazing properties, cooling set-points, shading thresholds, occupancy profiles, and envelope U-values. Conversely, parameters that focus on heat removal such as cooling capacity, NNV operation and operable window area impacting ACH during NNV operation, thermal mass and airtightness significantly influence recovery time. The findings contribute to linking design parameters to thermal resilience assessment and in developing a comprehensive thermal resilience assessment framework, which will help architects and designers improve building resilience during the early design and renovation phases.

7.1.4 RO4: Develop a comprehensive thermal resilience assessment framework to assess and improve buildings' thermal resilience to overheating

In order to develop a comprehensive assessment framework that aids in assessing and improving buildings' thermal resilience to overheating addressing the shortcomings of the current resilience assessment methods, a varied range of building type, shocks, thermal mass, resilient cooling strategies and design parameters were evaluated.

Thermal resilience assessment scenarios across shock, building types, construction characteristics, and strategies

Assessing thermal resilience of various building types (as seen in Chapter 2) showed difference in response of different buildings and associated strategies due to various crucial factors-(a) occupant density (b) occupancy pattern (c) construction type. A decision tree is crucial to categorize thermal resilience performance. These categorization is based on factors such as – *doS* level (high, medium and low), building typology (residential and non-residential), construction type (thermal mass heavy, medium or light) and with individual or combination of strategies (solar shading, natural night ventilation, cooling). Apart from these categorization, impactful design parameter inputs (such as orientation, WWR, envelope and glazing properties, airtightness etc.) are crucial for thermal resilience assessment.

The methodology for the development of a comprehensive thermal resilience assessment framework consists of seven key steps. First, shocks are identified and quantified using the degree of shock (*doS*) to evaluate impacts, with heatwaves identified as the most critical. Second, SET-Dh is selected as the heat-stress indicator to assess thermal resilience during shocks, with thresholds defined for habitable conditions. Third, dynamic building simulations measures SET-Dh under varying *doS* scenarios. Fourth, sensitivity and uncertainty analyses identifies influential design parameters affecting thermal resilience. Fifth, a decision tree is utilized to classify buildings based on typology, construction, and system strategies for tailored resilience assessments. Finally, regression models are created and validated to predict resilience outcomes across diverse scenarios, enhancing the framework's adaptability and reliability.

Thermal resilience assessment framework

Figure 7.1 illustrates the comprehensive thermal resilience assessment framework. The framework integrates shock quantification, occupants' health-focused heat-stress indicators, and addressing f key design parameters that influence thermal resilience.

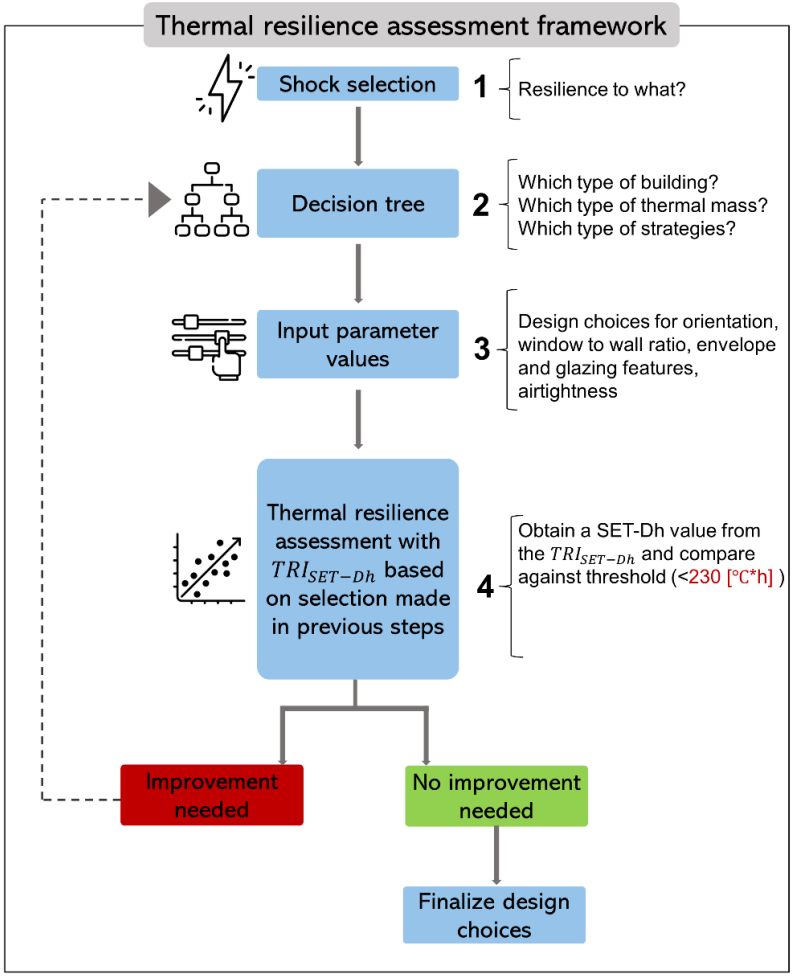


Figure 7.1. Overview of the thermal resilience assessment framework

The first step for thermal resilience assessment is to answer –'Resilience to what ? Step 1 is to select the specific type of shock against which the building's thermal resilience will be evaluated. Step 2 involves selecting a branch of the decision tree by defining the building type, thermal mass, and resilient cooling strategies to be tested. The building type determines parameters such as occupancy profile, occupant density, and internal gains, while thermal mass is chosen based on construction typology. At this stage, one or more resilient cooling strategies can also be selected. Based on these choices, the

appropriate regression model (TRI_{SET-Dh}) is applied. In Step 3, design parameter values are input into the model to calculate SET-Dh. If the resulting SET-Dh falls within the acceptable threshold, the design choices can be finalized. Otherwise, an iterative process of adjusting resilient cooling strategies or modifying design parameters is undertaken until the desired thermal resilience level is achieved. This comprehensive thermal resilience assessment framework offers a simplified yet robust framework for architects and designers to assess thermal resilience in the early stages of design.

7.2 Limitations and future research

This subsection offers critical reflections on research limitations and provide suggestions for future research and potential enhancements to the proposed methodology.

Building typology, design parameters and climate context

In this research, the focus was primarily on residential and small non-residential buildings, as these building types are particularly vulnerable to overheating due to factors such as higher occupancy densities, larger glazing areas, absence of mechanical cooling and often inadequate passive design strategies. The study was conducted within the context of the Belgian climate (ASHRAE climate 4A), which experiences moderate but increasing temperatures, particularly during summer heatwaves. However, this framework and methodology could be extended to include other building typologies, such as large commercial or institutional buildings (large offices, hospitals, elderly care-homes) and applied to different climatic contexts, where the specific challenges and solutions for thermal resilience may vary. Future research could explore adapting the framework for diverse regions with varying temperature extremes, humidity levels, and seasonal variations, offering broader applicability across different climates. Additional design parameters such as window-to-floor ratio, floor height, geometry and volume of the space, material choices, and static shading elements etc. could be evaluated and included in the framework.

Inclusion of weather scenarios

For the development of weather files (typical meteorological year and heatwave files), Representative Concentration Pathway (RCP) 8.5 was used as the primary climate scenario, as it represents a high greenhouse gas emissions pathway and is considered the most extreme projection of future climate change[4]. This scenario was selected to capture the potential impacts of extreme heat events under a "worst-case" climate scenario, providing valuable insights into the upper limits of thermal resilience under future conditions. However, other RCPs, such as RCP4.5 or RCP2.6, can be used in future research to evaluate the thermal resilience of buildings under lower emissions scenarios. Using these alternative RCPs might lead to more moderate projections of temperature increases, possibly resulting in different conclusions regarding the extent and need for thermal resilience measures. It is important to note that this assessment focused solely on extreme heatwaves, as resilience to these events can demonstrate the ability of buildings to withstand other, less intense heatwaves. However, it is also crucial

to study a broader range of heatwaves, including moderate and frequent ones, as their cumulative effects can also impact building resilience over time.

Impact of UHI

The impact of urban heat island (UHI) effect was not included during development of the weather data, though it could significantly amplify indoor temperatures, especially in urban areas. Incorporating UHI would likely have increased the severity of overheating scenarios, emphasizing the need for additional resilience measures in cities. While our research focused on building-level resilience and did not assess the collective impact of buildings' cooling strategies on the wider urban context, it is essential to recognize that improving one building's thermal resilience could potentially lead to unintended consequences for neighboring buildings. For instance, enhanced cooling strategies in one building could result in additional heat transfer to neighboring buildings, thereby reducing their resilience and creating a reinforcing loop of increased thermal stress in the surrounding environment. This system's thinking approach highlights the importance of considering the broader context when designing buildings to mitigate overheating and improve thermal resilience. We acknowledge that this is an area that warrants further investigation in future research, specifically exploring how individual building design decisions can impact not only the building itself but also its surrounding environment, leading to potential positive or negative feedback loops in terms of thermal resilience. Future research could also consider inclusion of UHI in climate scenarios to more accurately reflect the full scope of thermal risks buildings may face in urban environment, surrounding context of the building (urban or rural impacting urban heat island effect, shading due to adjacent building, presence of a water-body impacting relative humidity and surrounding building form that impact wind velocity for natural night ventilation).

Other shock types and their risk of occurrence of the shocks

This study focused on critical shocks, particularly heatwaves, and their impact on building thermal resilience. While heatwave development models predicted the number of heatwaves in different time periods (2010s, 2050s, and 2100s), the likelihood of these extreme events occurring was not assessed. Similarly, other shocks like system failures and power outages, though less critical than heatwaves, can still significantly threaten a building's thermal resilience, especially during prolonged durations. One limitation of the current study is the inability to quantify the probability of occurrence of these shocks, including their frequency and regional distribution. For example, data from Fluvius in Belgium [286] on power outages could be leveraged to create probabilistic models predicting the occurrence, duration, and geographic impact of power disruptions. By integrating these models with climate projections, we could better understand when and where such disruptions might occur and how they affect building resilience. This approach could be incorporated into the regression models developed in this study, with the degree of shock (*dos*) as a variable, allowing for a more comprehensive assessment of thermal resilience. Future research could focus on developing probabilistic models that combine climate data, historical trends, and future

projections to quantify the risk of thermal shocks like power outages and heatwaves. This would improve the ability to identify vulnerable buildings and tailor resilience measures to the most critical scenarios, ultimately enhancing thermal resilience assessments in future studies.

Degree of shock quantification

In this research *doS* was not quantified for combined shocks. The study also showed that apart from the severity and duration of shock, the occurrence of peak temperature during the shock has an impact on the thermal resilience aspects such as absorptivity and recovery time. Future research on *doS* quantification could also focus on (a) quantifying the *doS* of combined shocks and (b) incorporating aspects such as minimum temperature (night-time temperature) and occurrence of the peak temperature in *doS* quantification. *doS* quantification of combined shocks (such as power outage, or power outage during heatwaves) can be done by evaluating individual impact of each shock type and their nature of interaction. To address this, Sobol sensitivity analysis can be applied to determine the contribution of individual shocks as well as their interaction effects. Sobol's method allows for the decomposition of variance in *doS*, identifying first-order effects (individual shock contributions), second-order effects (pairwise interactions), and higher-order effects (multi-shock interactions). By applying this method to a surrogate model or energy simulations, data-driven weights can be extracted for each shock and their interactions, refining the formulation of *doS* to be more globally applicable for shock combinations. Moreover, in future research, implementing the time of occurrence of minimum and maximum temperature during heatwaves in the *doS* quantification can give further insight into the absorptivity and recovery aspects of thermal resilience assessment.

Impact of occupant behavior

Current research focused on the critical design parameters. These were selected based on their impact on overheating risk in buildings. As occupants are reactive to changing conditions in the built environment rather than passive, they will react by modifying clothing, opening windows, deploying shading etc.[287–290]. However, these occupant behaviors are stochastic and subject to certain probabilities such as how many occupants will modify clothing, how many times or for how long the windows will be opened or shading deployed. Since the focus of this study was identifying building and system design parameters impacting thermal resilience, occupant behavior was not evaluated. Future research could also consider impact of occupant behavior for day-time window opening, and control strategies for NNV and solar shading. In future research, the aim is to delve deeper into the separate effects of the U and g values of glazing on thermal resilience. This will provide a more comprehensive understanding of how each of these parameters influences buildings' resilience performance, overcoming the limitations posed by the current inability to independently adjusting these variables in Modelica.

Quantify other aspects of thermal resilience

SET-Dh was selected as the heat-stress indicator to assess thermal resilience. Additionally there is already pre-defined threshold for SET-Dh to asses occupants' health based on body dehydration rate [207]. While SET-Dh serves as a useful heat-stress indicator, additional thermal resilience metrics such as absorptivity and recovery time are essential to capture key aspects of thermal resilience. Quantifying thresholds for these aspects (absorptivity time and recovery time) requires additional research. For instance, experiments could track body dehydration rates, heart rate, and core body temperature under various thermal conditions to determine threshold limits for absorptivity and recovery time. These metrics would then need to be integrated into the existing models to assess how building design and environmental factors influence the body's ability to handle prolonged heat exposure. Moreover, data collection on vulnerable populations, such as the elderly, children, or individuals with pre-existing health conditions, would be necessary to refine these thresholds. Once these thresholds are defined, they could be incorporated into a more comprehensive framework that evaluates not just the immediate heat-stress response, but also the recovery and long-term effects on occupants' health and well-being.

Quantify climate and occupant based threshold for SET-Dh

The current approach adopted to determine the SET-Dh threshold by Laouadi et al. [88] is based on per-occupant, whereas evaluating zones rather than individuals is more suited for building performance assessments. While the current BES model doesn't account for occupant-specific conditions, this is due to the software's limitations, which provides a uniform environmental condition for the entire zone. Laouadi et al.[88]'s threshold, based on body dehydration and applicable to both healthy and sick occupants, was adopted here but will need adaptation if applied in extreme climates where the adaptation levels are differ for occupants. While the 230°h SET-DH threshold provides a useful baseline for thermal resilience assessment, its universal application should be reconsidered. Thus recommendations for future work would be-(i) the 230 [°.h] limit should be recalibrated for different climates based on occupant adaptation and building characteristics (ii) a combination of occupant-level and zone-level evaluations could balance thermal comfort and building design insights to get more insight on localized discomfort and (iii) future studies should explore adaptive comfort models that adjust T_{set} dynamically based on seasonal conditions and local adaptation patterns.

Tradeoff between thermal resilience, energy efficiency and cost

Buildings with air-conditioning or high cooling capacity proves to be the most resilient buildings to overheating as the indoor temperature do not have any deviation during shocks such as heatwaves. However, , it is crucial to evaluate the tradeoff between thermal resilience, energy efficiency and costs. The challenge lies in achieving a balance where energy efficiency does not undermine a building's ability to withstand extreme thermal conditions. And the cost of implementing features to enhance thermal

resilience must be This requires an integrated approach that considers both energy performance and the capacity for passive and active thermal resilience strategies. Future research could incorporate multi-objective optimization to more effectively balance energy efficiency and thermal resilience. By considering both operational energy consumption and the ability to maintain safe indoor conditions during extreme heat, such optimization approaches could identify solutions that minimize costs while maximizing resilience. This would involve developing algorithms that optimize multiple factors—such as insulation, ventilation, cooling systems, and renewable energy integration—ensuring both energy efficiency and thermal resilience. In this way, building designs could be developed that are not only energy-efficient but also resilient to future climate challenges, addressing both operational and thermal resilience goals in a holistic manner.

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List of publications

Journal paper as first author

*1. **A. Sengupta**, D. Al, O. Berk, J. Shinoda, H. Breesch, M. Steeman, Building and system design's impact on thermal resilience to overheating during heatwaves : An uncertainty and sensitivity analysis, Build. Environ. 265 (2024) 112031. https://doi.org/10.1016/j.buildenv.2024.112031.*

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*3. **A. Sengupta**, J. Kerckaert, M. Steeman, H. Breesch, Assessing thermal resilience to overheating in a Belgian apartment: impact of building parameters, International Journal of Ventilation (2024). https://doi.org/10.1080/14733315.2024.2384715.*

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*1.**A.Sengupta**, D. Al Assaad, M. Steeman, H. Breesch, Thermal resilience to overheating assessment in a Belgian educational building with passive cooling strategies during heatwaves and power outages, E3S Web Conf. 396 (2023). https://doi.org/10.1051/e3sconf/202339601018.*

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*4. D. Al-Assaad, **A. Sengupta,** H. Breesch, Evaluation of the spatial aspect of building resilience in classrooms equipped with displacement ventilation, E3S Web Conf. 362 (2022) 1–8. https://doi.org/10.1051/e3sconf/202236214006.*

Statement for use of Generative AI

Generative AI assistance tools were not used during the research process of the thesis, except for language assistance.

ChatGPT was used for error solving and debugging the programming code. In this thesis, codes were programmed for-(a) making scripts to run jobs in the HPC (high performance computer) to sum Modelica simulations for the entire sample sets. Additionally codes was developed to post-process the simulation data and analyze the results. However no Generative AI tools were used to generate these codes. Chat-GPT was used to de-bug the codes and solve errors.

The text/code/ images in this thesis are my own (unless otherwise specified) and generative AI has only been used in accordance with the KU Leuven guidelines and appropriate references have been added. I have reviewed and edited the content as needed and I take full responsibility for the content of the thesis.

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